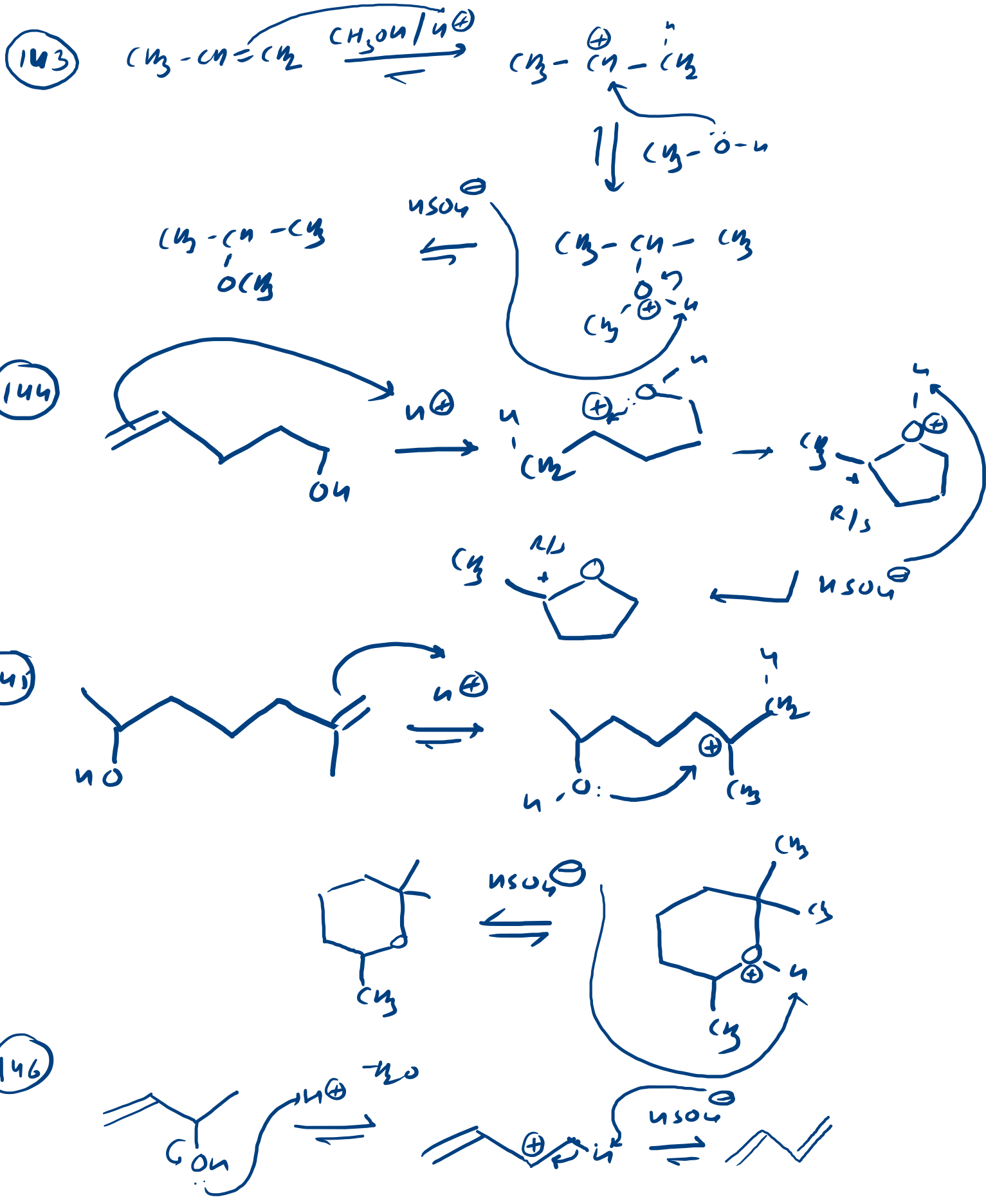
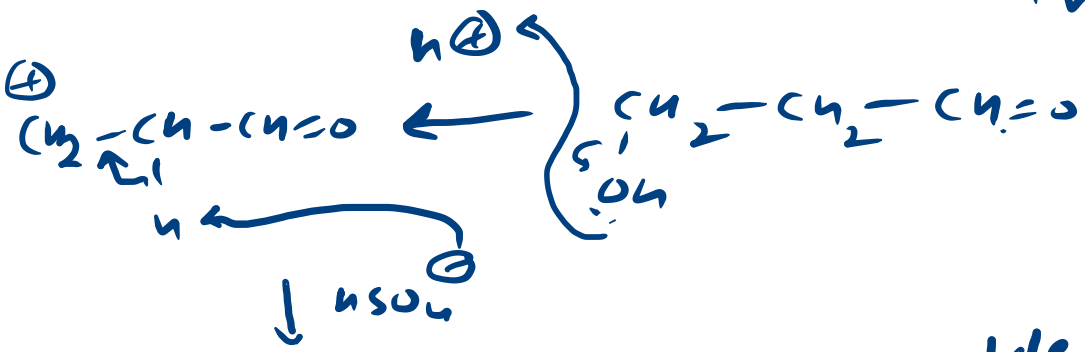


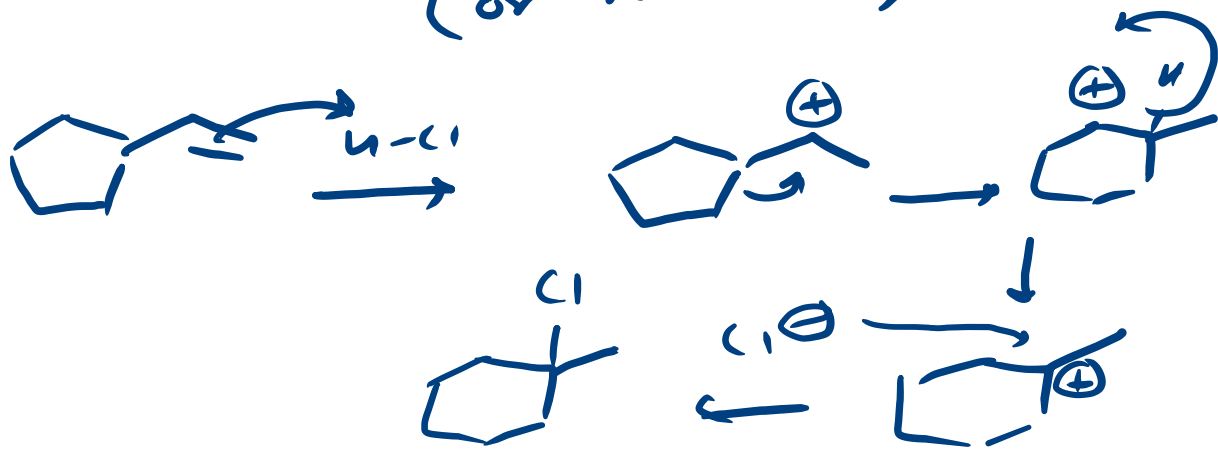
(122)



$CH_2=CH-CH=O$ Acryl aldehyde

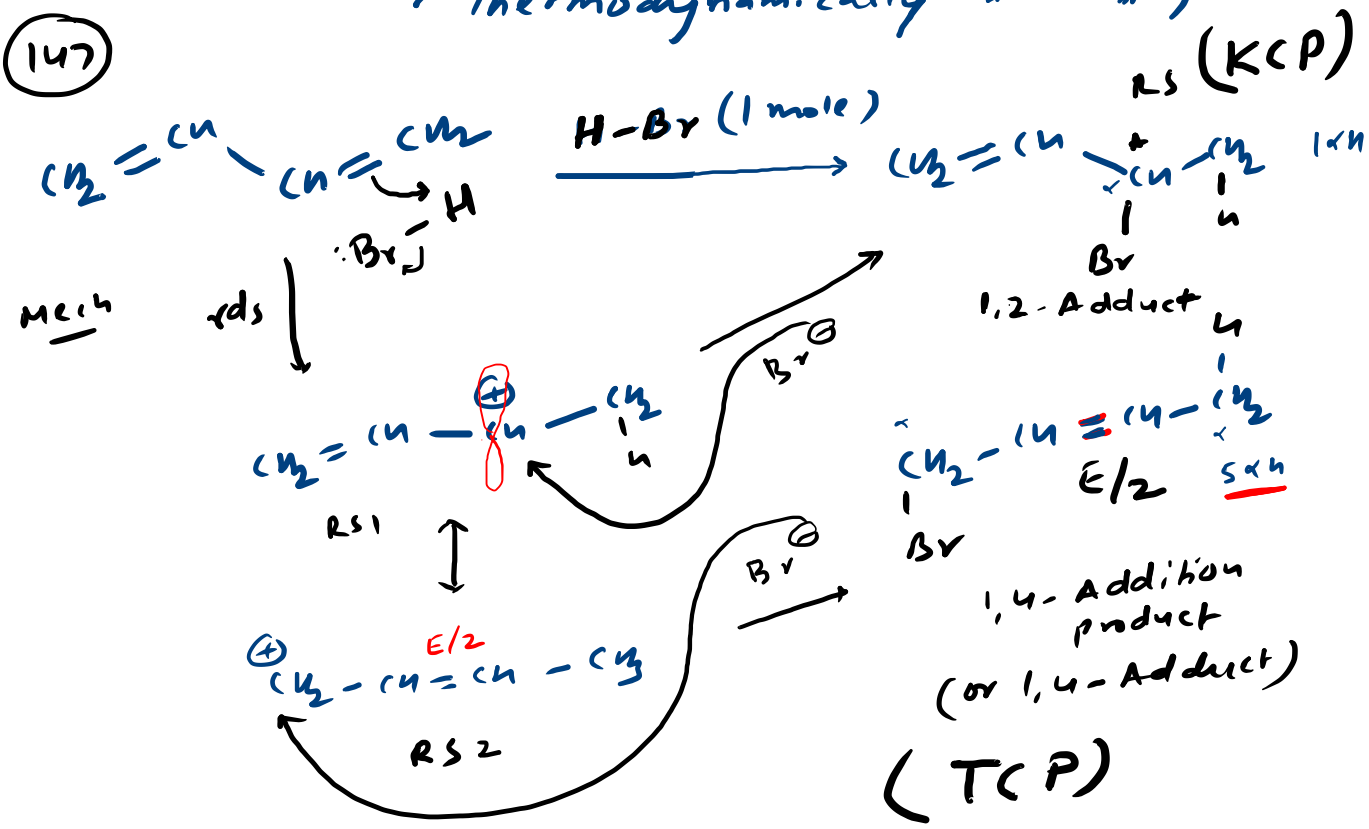
(or Acrolein)

(134)



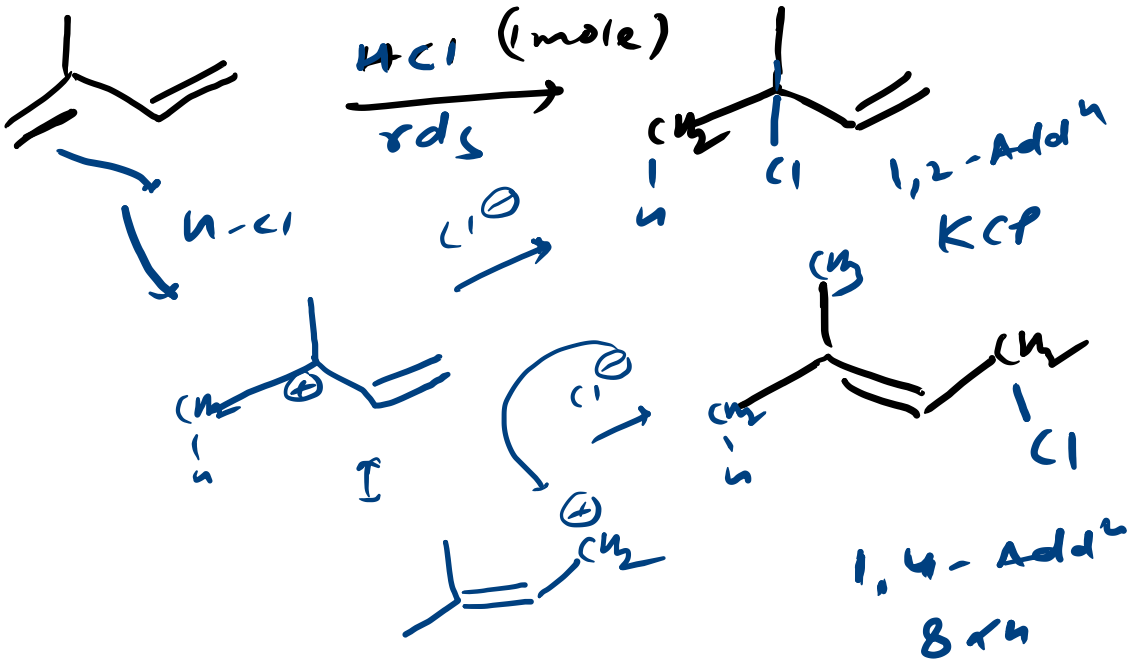
KCP / TCP (Kinetically controlled product / Thermodynamically " ")

(147)

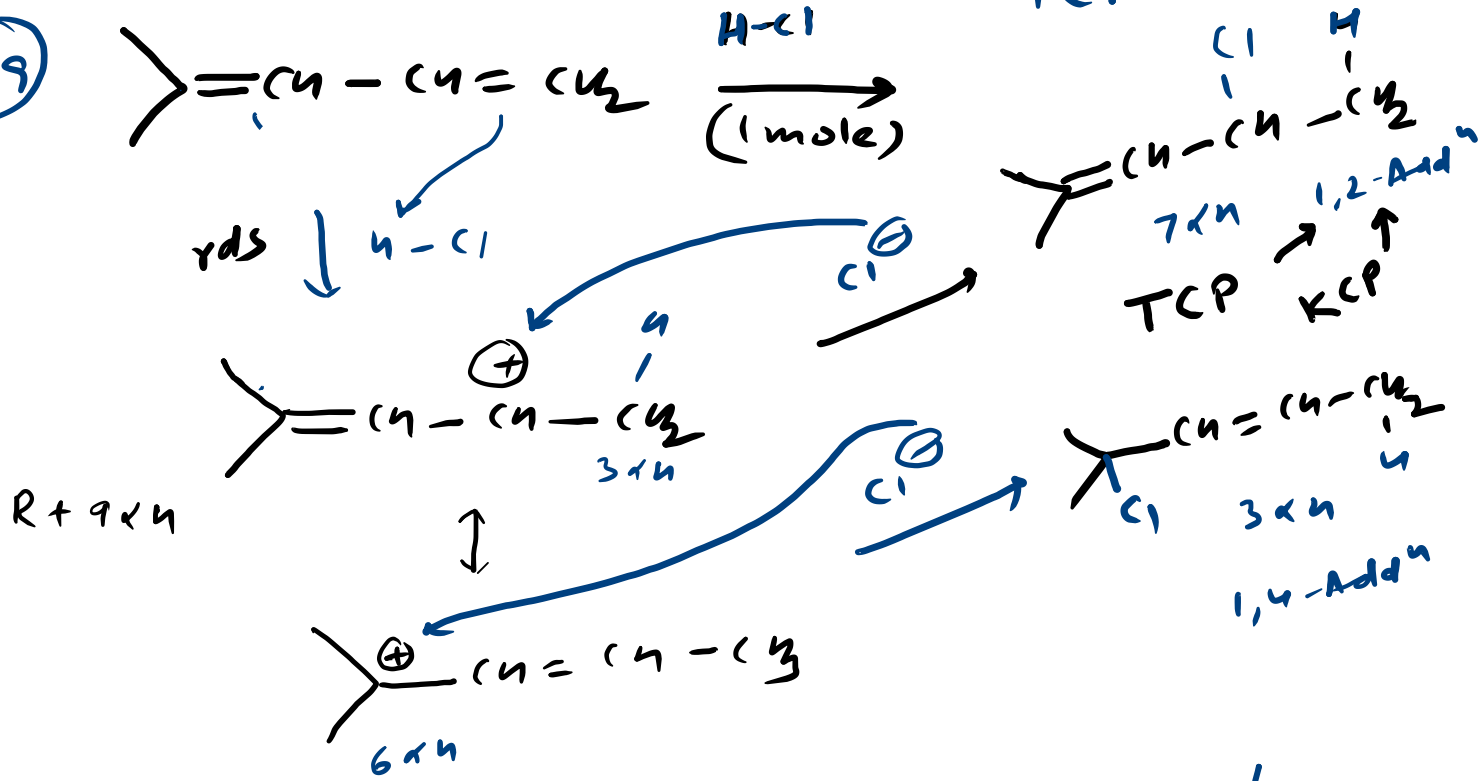


KCP	TCP
# KCP is formed as Major product at low temp (-80°C to 5°C)	TCP is formed as major product at higher temp (15°C to 80°C)
# KCP is formed by more collision probabi- -lity (i.e 1,2-Addition)	# TCP is most stable product (or may 1,4-Addition product or 1,2-Addition product.
# It depend on path i.e Ea (Activation Energy) or Energy barrier ∴ It is formed by lower Energy barrier.	# It do not depend on path.

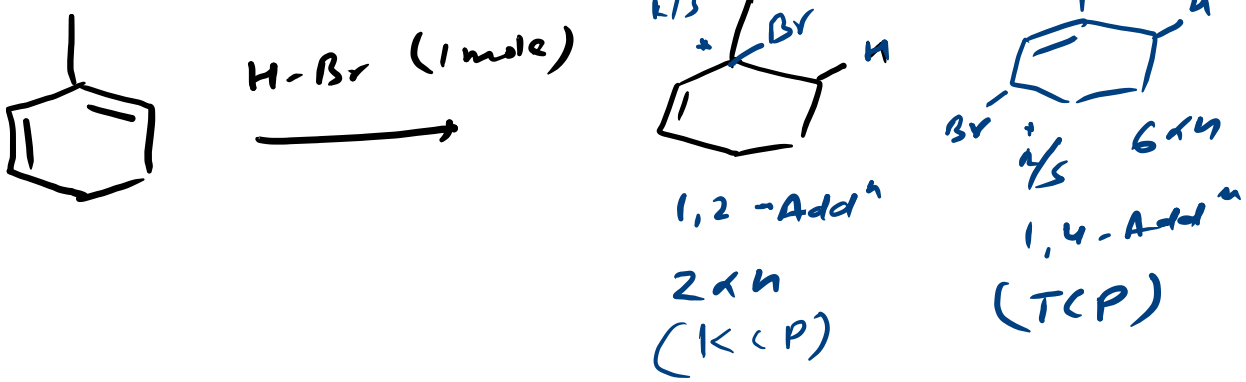
148


H-Cl (1 mole)
rds
1,2-Addⁿ
KCP
H-Cl
Cl⊖
I
1,4-Addⁿ
8 atm
TCP

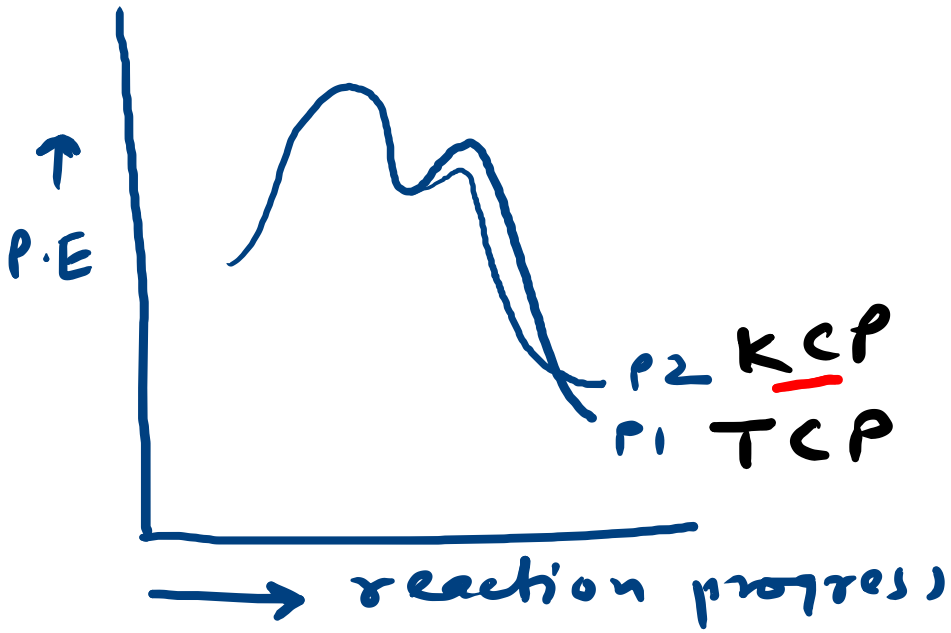
149


H-Cl
(1 mole)
rds
H-Cl
7 atm
TCP
1,2-Addⁿ
KCP
3 atm
R + 9 atm
3 atm
1,4-Addⁿ
6 atm

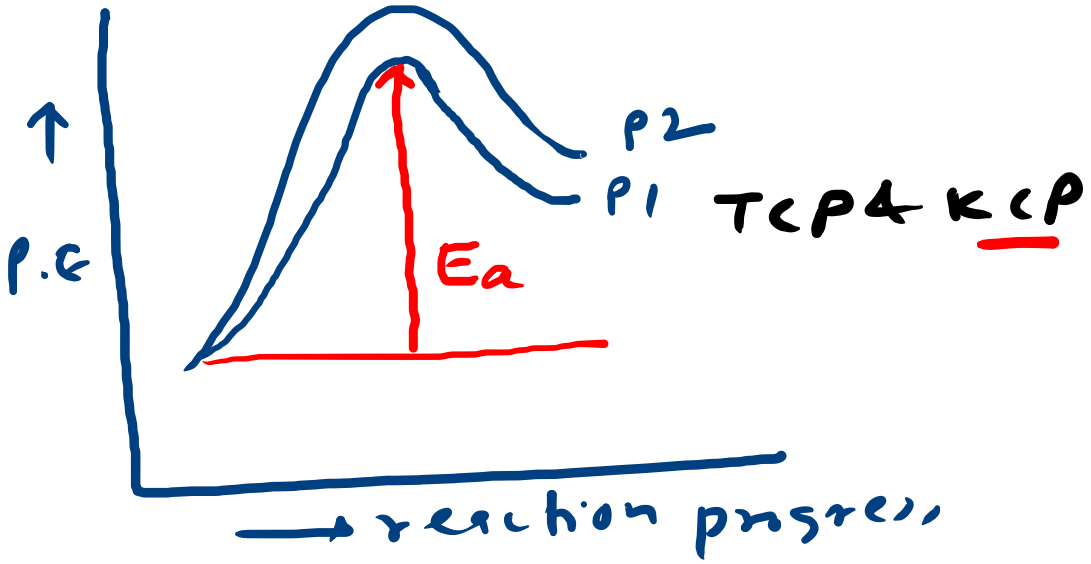
150


H-Br (1 mole)
r/s
1,2-Addⁿ
2 atm
(KCP)
r/s
6 atm
1,4-Addⁿ
(TCP)

151

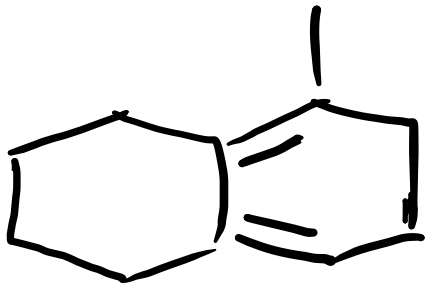


152



H.W

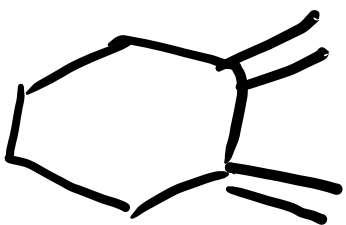
153



H-Cl (1 mole) ⟶

H.W

154



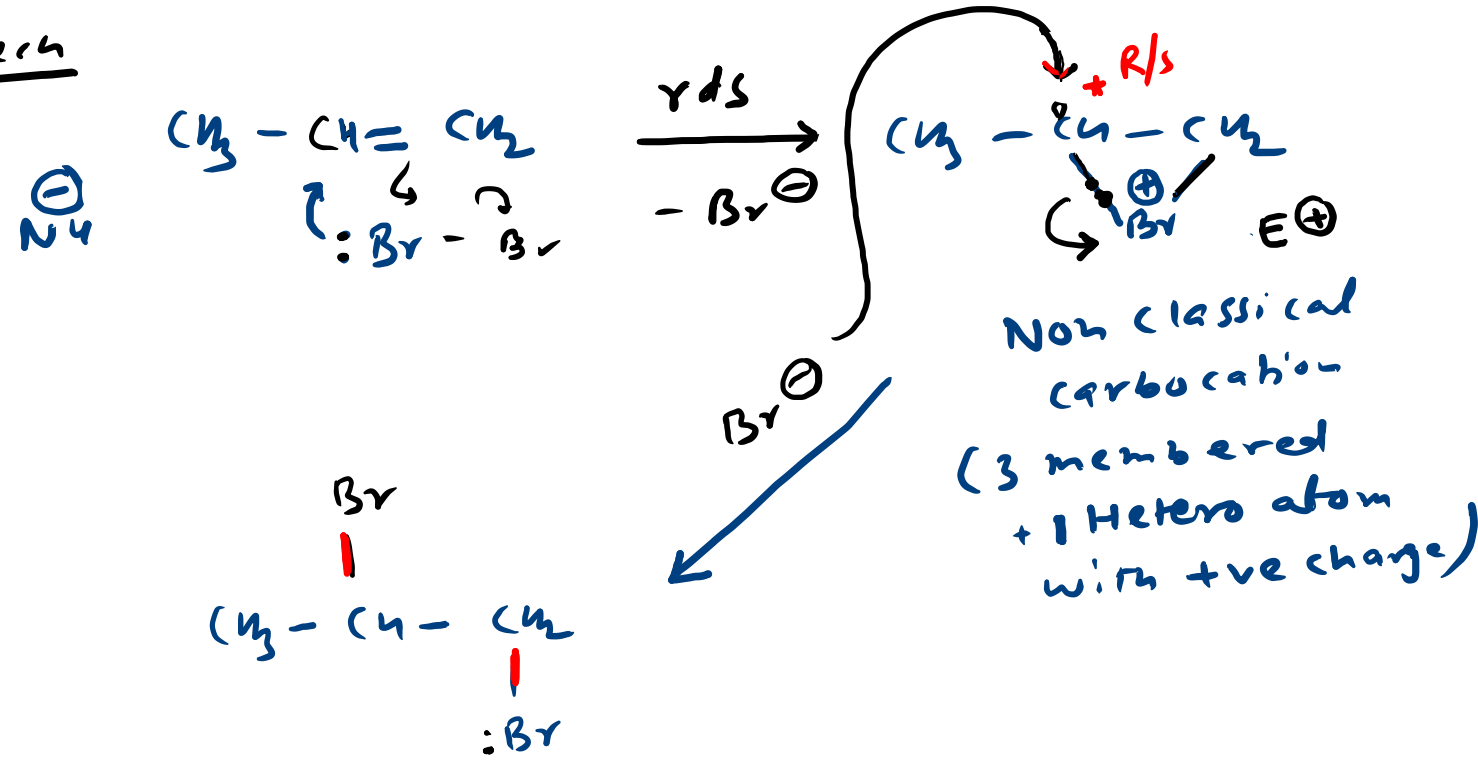
H-Br (1 mole) ⟶

?

Electrophilic Addition Reaction by NCC⊕ or EAR by Non classical carbocation.

(155) $CH_3-CH=CH_2 \xrightarrow[CCl_4\ (solvent)]{Br_2}$ $CH_3-\underset{R/S}{CH}-CH_2$ (Br on CH, Br on CH_2)

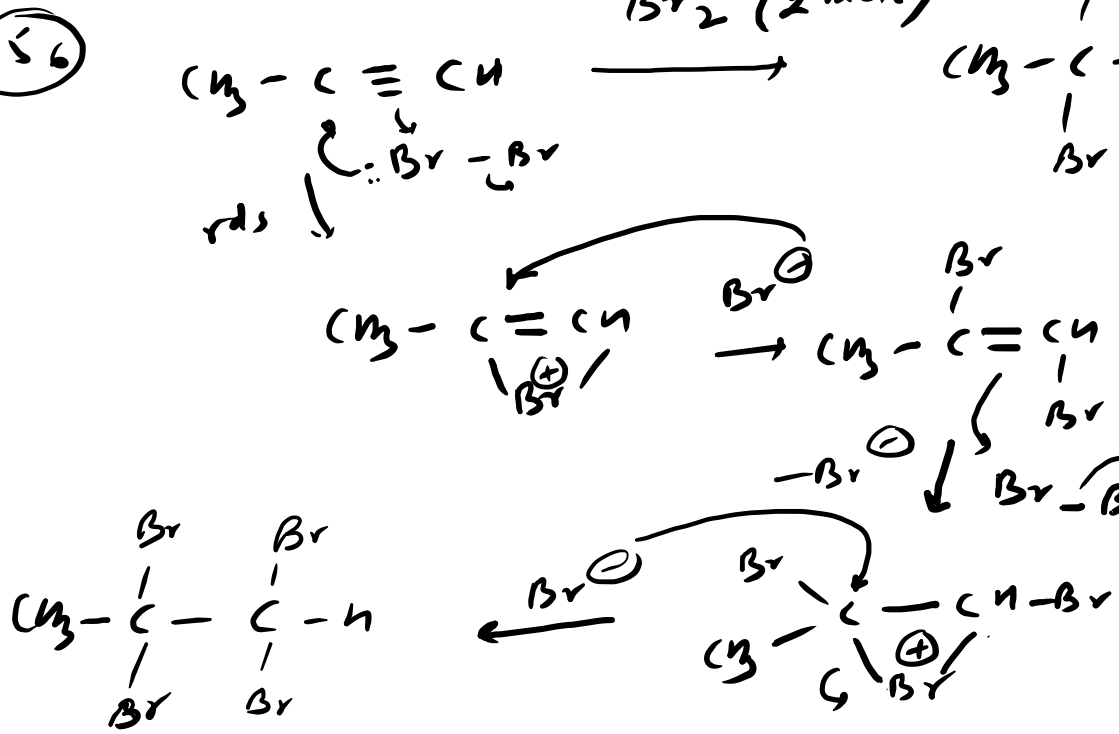
Mech



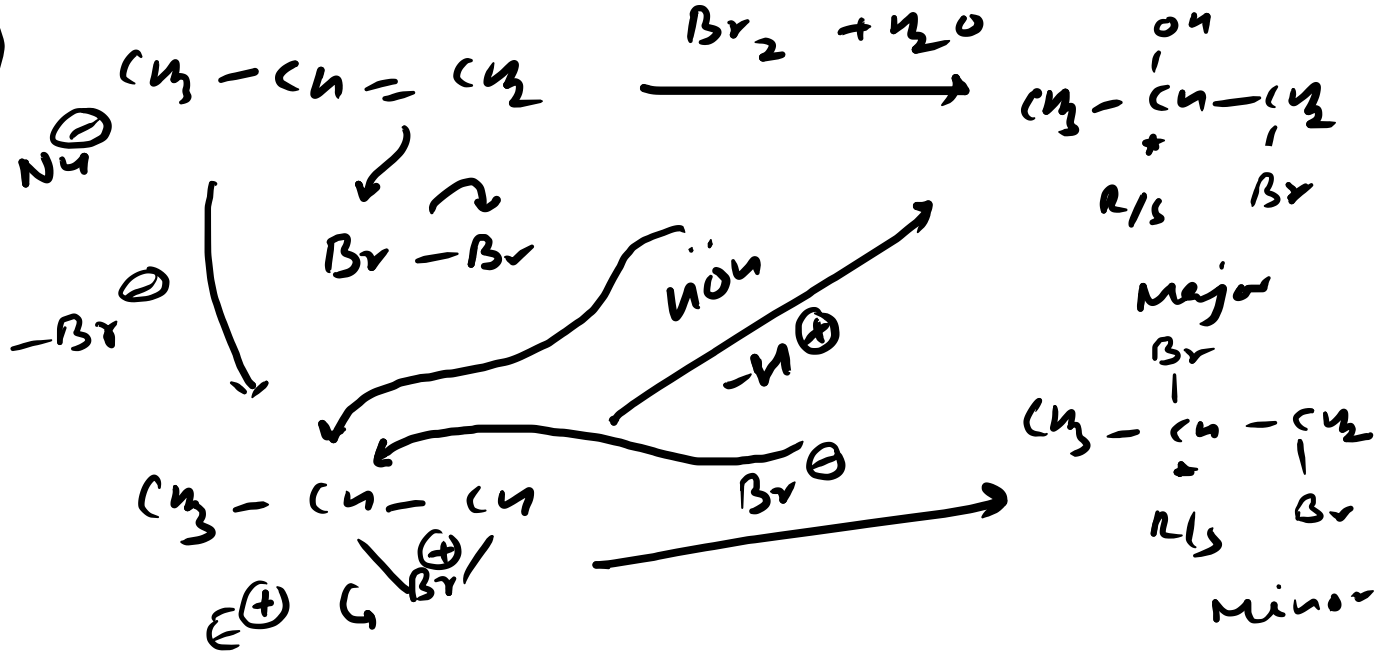
$CH_3-CH(Br)-CH_2(Br)$

* Reaction of Alkene/Alkyne to X_2 is EAR by NCC⊕

(156) $CH_3-C\equiv CH \xrightarrow{Br_2\ (2\ mole)}$ $CH_3-C(Br)_2-CH(Br)_2$

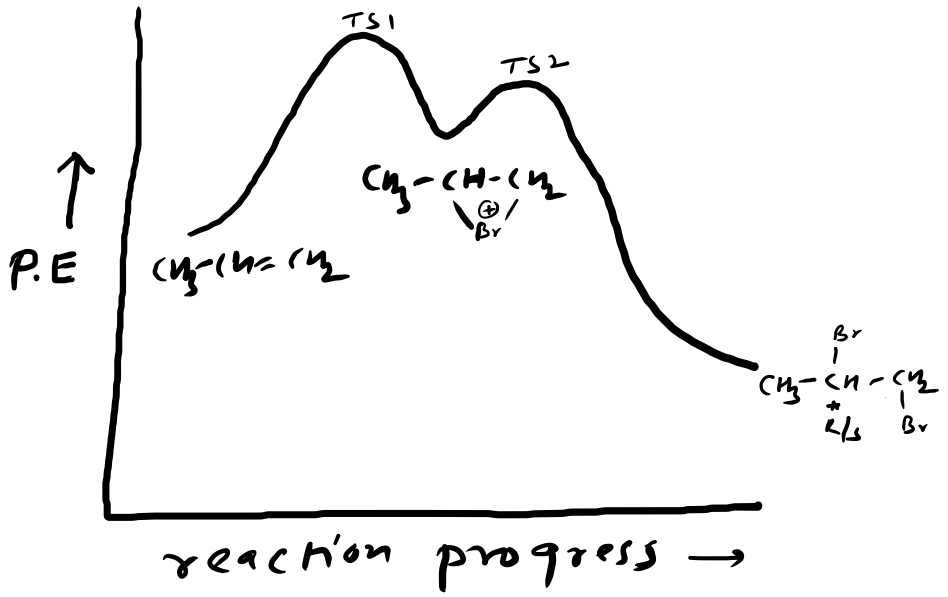


(157)



VIP Brown colour of Br_2 water disappear with alkene or Alkyne, so it is also called Br_2 water test.

Graph of (155)



Characteristics of reaction

(1) It is two step reaction

Step -I formation of $NCC^{\oplus}$, which is RDS

Step -II Attack of $Nu^{\ominus}$

(2) Rate of EAR by $NCC^{\oplus}$ $\propto$ Nucleophilicity of alkene

or Rate of EAR by $NCC^{\oplus}$ $\propto$ Nucleophilicity of X_2

Order of EAR $Cl_2 > Br_2 > I_2$

(3) Rate of EAR by $NCC^{\oplus} \propto \left[\begin{matrix}\text{Alkene} \\ Nu^{\ominus}\end{matrix}\right]\left[\begin{matrix}X_2 \\ E^{\oplus}\end{matrix}\right]$

$\therefore$ It follow 2^{nd} order kinetics.

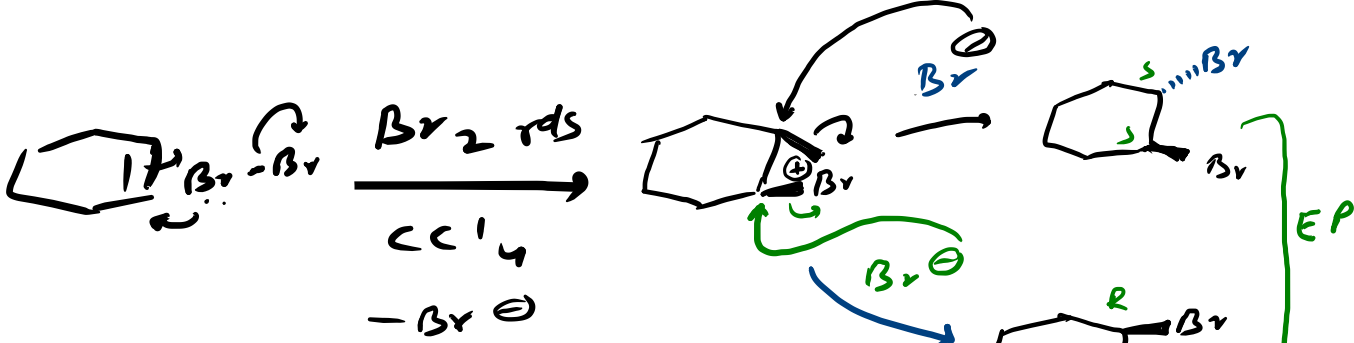
(4) There is NO Rearrangement in $NCC^{\oplus}$ because carbon has $\oplus$d charge.

(5) $Nu^{\ominus}$ Attack to that carbon of $NCC^{\oplus}$ which suppose to give more stable carbocation

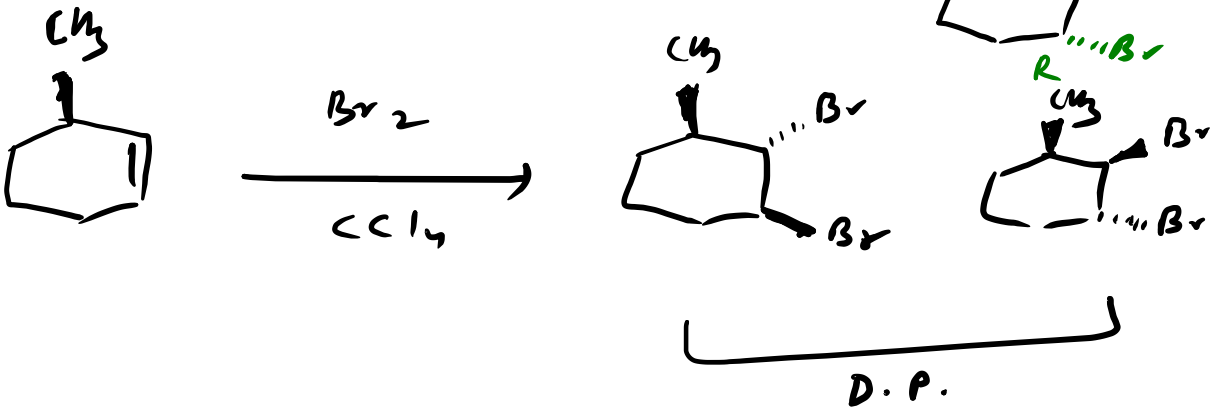
(6) All Available $Nu^{\ominus}$ Attack to $NCC^{\oplus}$ and gives the products.

(7) Overall This Addition is the example of anti Addition

158

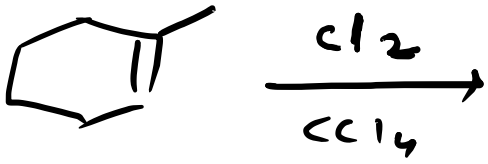


159



HW

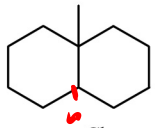
160

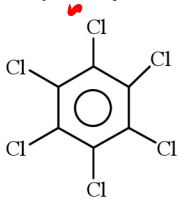


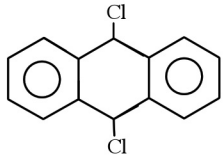
RACE # 17 ORGANIC CHEMISTRY

M.M. : 40 TIME : 30 Min.

1. **Column-I** **Column-II** **[8, 0]**

(A) 

(B) 

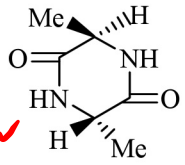
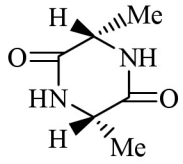
(C) 

(P) Compound can show optical isomerism

(Q) Compound can show geometrical isomerism

(R) Compound can show positional isomerism

(S) Structure may have centre of symmetry in one of its form

2.  (A)  (B) **[3, –1]**

Statement-1 : A is optically inactive and B is optically active.

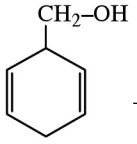
Statement-2 : A has centre of symmetry as well as plane of symmetry.

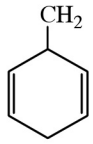
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

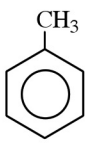
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

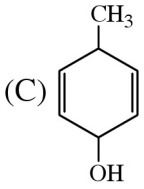
(C) Statement-1 is true, statement-2 is false

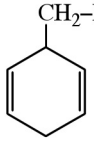
(D) Statement-1 is false, statement-2 is true

3.  $\xrightarrow[\Delta]{\text{Conc.}H_2SO_4}$ Major Product is **[3, –1]**

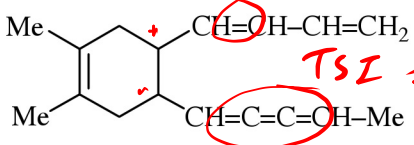
(A) 

(B) 

(C) 

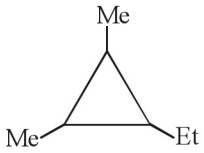
(D) 

4. How many stereoisomers are possible for the following : **[3, –1]**

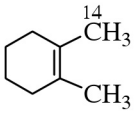


(A) 16 (B) 4 (C) 6 (D) 8

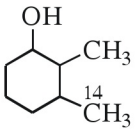
5. Identify total number of stereo isomers for following compound : **[3, –1]**

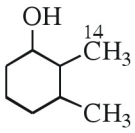


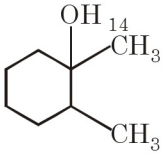
(A) 2 (B) 4 (C) 6 (D) 8

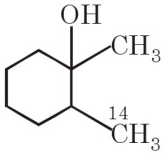
6. (X) Alcohol $\xrightarrow[\Delta]{H^{\oplus}}$  **[4, –1]**
(Major product)

(X) may be

(A) 

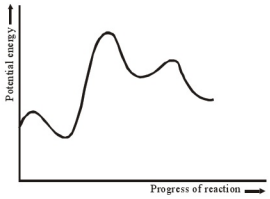
(B) 

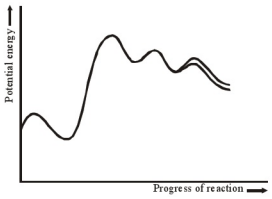
(C) 

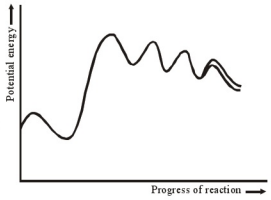
(D) 

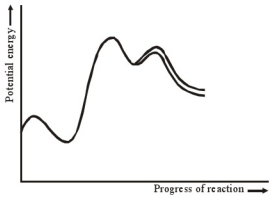
Paragraph for Q.7 to Q.9

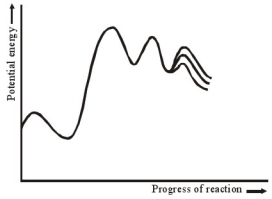
Dehydration of alcohol is an elimination reaction, proceeds via formation of carbocation intermediate. The energy profiles for such reaction may follow different patterns, some of them are given here.

(I) 

(II) 

(III) 

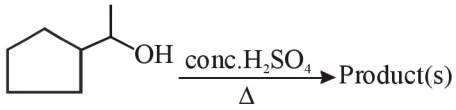
(IV) 

(V) 

7.

Which of the energy profiles given in the paragraph can represent the above reaction **[3, –1]**

(A) II (B) III (C) IV (D) V

8. 

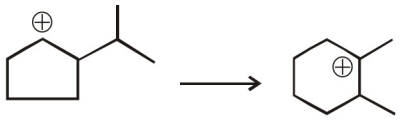
Which of the energy profiles given in the paragraph can represent the above reaction **[3, –1]**

(A) I (B) II (C) III (D) IV

9. Which of the following energy profiles given in the paragraph represent product formation without rearrangement - **[3, –1]**

(A) I, III (B) I, IV (C) II, IV (D) II, IV

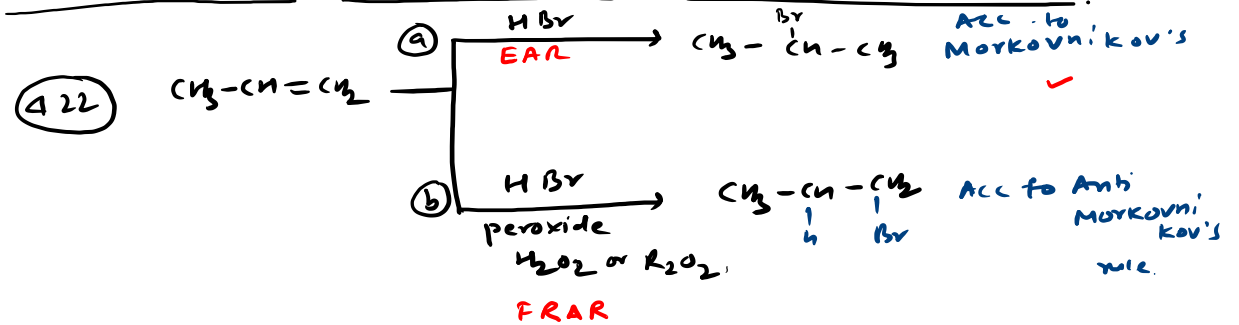
10. How many 1,2-shifts are involved in the following rearrangement of carbocaiton ? **[4, 0]**



FILL THE ANSWER HERE

1.	A (P) (Q) (R) (S) (T) B (P) (Q) (R) (S) (T) C (P) (Q) (R) (S) (T) D (P) (Q) (R) (S) (T)	**2.**	(A) (B) (C) (D)	**3.**	(A) (B) (C) (D)	**4.**	(A) (B) (C) (D)
5.	(A) (B) (C) (D)	**6.**	(A) (B) (C) (D)	**7.**	(A) (B) (C) (D)	**8.**	(A) (B) (C) (D)
9.	(A) (B) (C) (D)	**10.**	(0) (1) (2) (3) (4) (5) (6) (7) (8) (9)				

peroxide effect / kharasch effect / Anti Markovnikov's Add^n



Mech of 422 (b)

Step 1 Initiation step

$H-O-O-H \xrightarrow{\text{low temp}} \dot{O}H + \dot{O}H$

$\dot{O}H + H-Br \longrightarrow H_2O + Br^{\cdot}$

Step 2 propagation step (chain reaction)

$CH_3-CH=CH_2 + Br^{\cdot} \xrightarrow{RDS} CH_3-\dot{C}H-CH_2Br$ 2° FR

$CH_3-\dot{C}H-CH_2Br + H-Br \longrightarrow CH_3-CH(H)-CH_2Br + Br^{\cdot}$

Step 3 Termination step

FR + FR $\longrightarrow$ Neutral molecule.

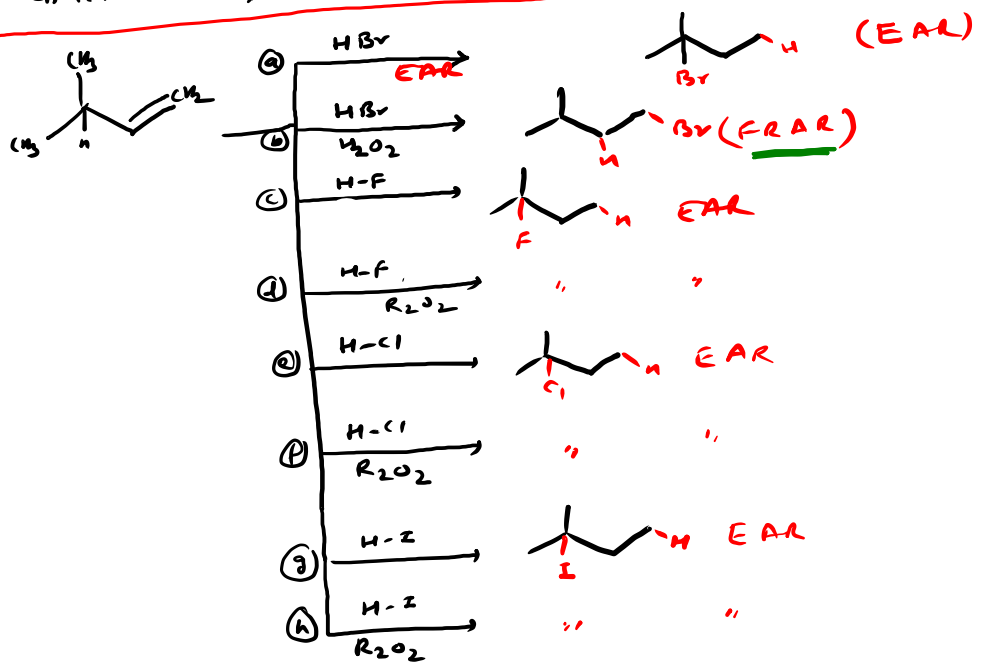
Note: Only HBr gives Anti markovnikovs Add^n (FRAR) in presence of peroxide

H — F (129 K.cal)]
H — Cl (103 K.cal)] Higher B.E (Initiation step is endothermic)

H — I] $I^{\cdot}$ is more stable, which is unable to cleave π bond in RDS step.

* केवल H-Br, परॉक्साइड के साथ फरार (FRAR) होती है.

(423)



(424)

$$CH_3-CH=CH_2 + BrCCl_3 \xrightarrow{R_2O_2} CH_3-\underset{Br}{CH}-\underset{CCl_3}{CH_2}$$

$H^{\cdot} < Br^{\cdot}$

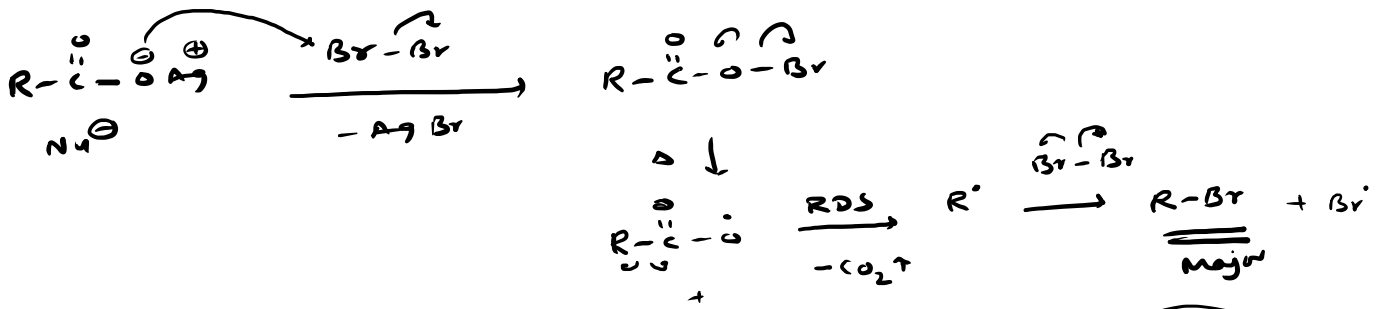
$Br^{\cdot} < {}^{\cdot}CCl_3$

Hunsdiecker Reaction

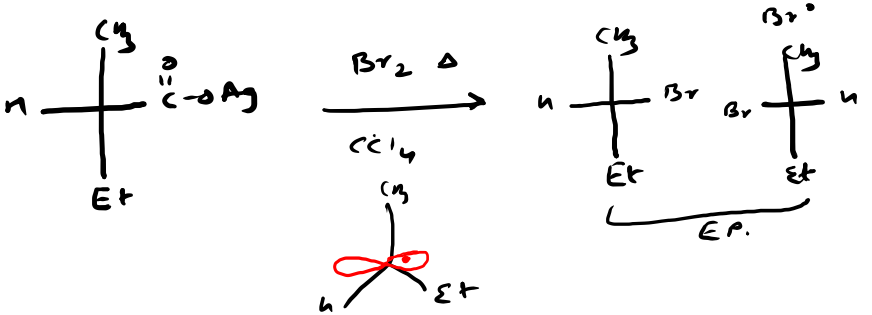
(425) $R-\overset{O}{\overset{\|}{C}}-O-Ag \xrightarrow[CCl_4]{Br_2\ \Delta} R-Br + CO_2 + AgBr$

(solvent: CCl_4)

FRSR Mech



(426)



$R-\overset{O}{\overset{\|}{C}}-O-R$ minor

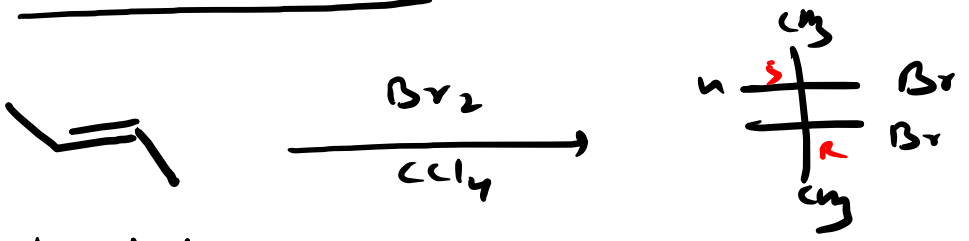
(427) $R-\overset{O}{\overset{\|}{C}}-O\,Ag \xrightarrow[CCl_4]{Cl_2\ \Delta} R-Cl + CO_2 + AgCl$ white ppt

(428) **Birn Baum Simonini**

$R-\overset{O}{\overset{\|}{C}}-O\,Ag \xrightarrow[CCl_4]{I_2\ \Delta} R-\overset{O}{\overset{\|}{C}}-O\,R + AgI$ yellow ppt.

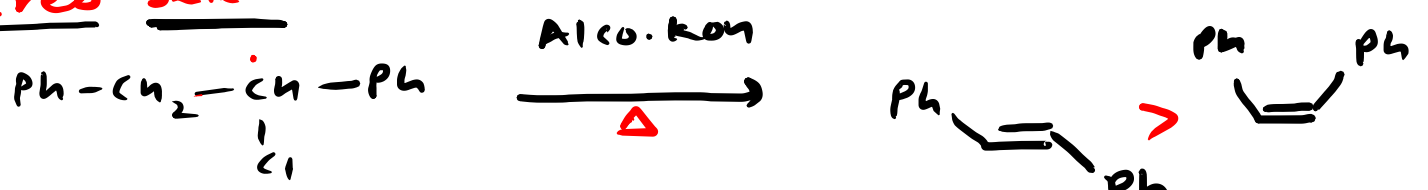
major (ester)

(429) **Stereo specific reaction**



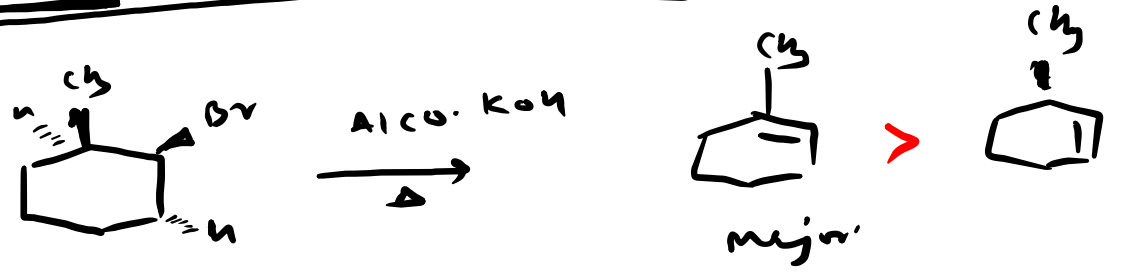
(2R 3S) 100%

(430) **stereo selective reaction**



structural

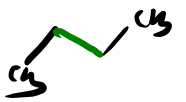
(431) **Regio selective reaction**



FR combination Reaction

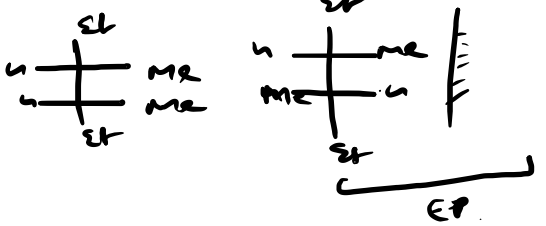
(432) $\dot{R} + \dot{R} \longrightarrow R-R$

(433) $\dot{C}H_3 + \dot{C}H_3 \longrightarrow CH_3-CH_3$

(434) $\dot{C}H_3 + CH_3-\dot{C}H_2 \longrightarrow CH_3-CH_3$ $\quad CH_3-CH_2-CH_3$ 

(435)

RR
SS
RS



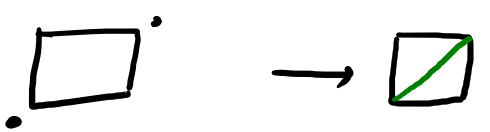
(436)

(437)

(438)

(439)

(440)

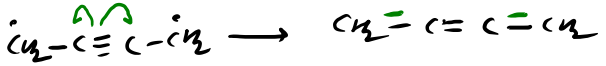
(441) 

(442) $\dot{C}H_2-\overset{\curvearrowright}{CH}=\overset{\curvearrowright}{CH}-\dot{C}H_2$

$\downarrow$

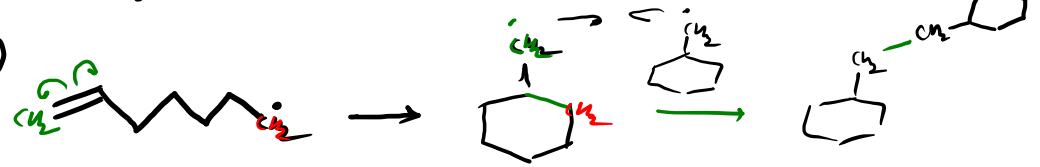
$CH_2=CH-CH=CH_2$

(443)

(444) 

(445)

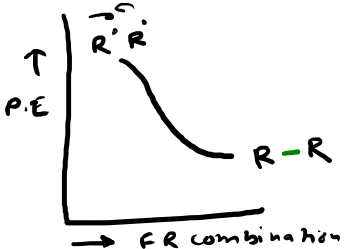
(446)

(447) 

(448)

* Intramolecular reaction faster Than intermolecular reaction upto formation of 6 membered ring, after 6 membered ring intermolecular combination takes place which gives linear polymer

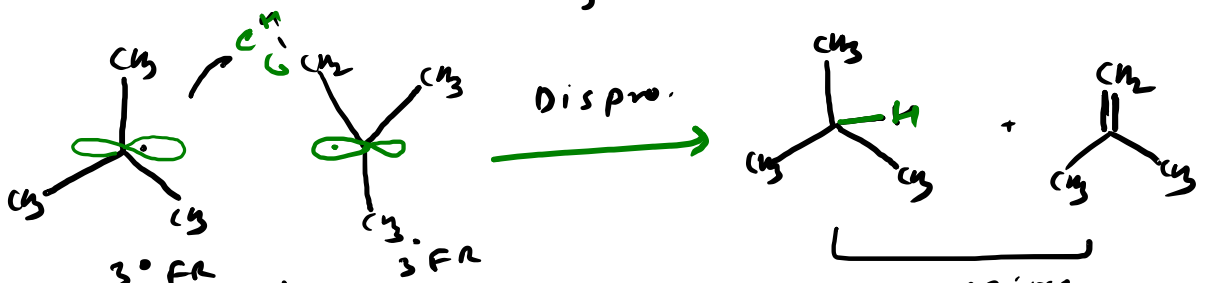
* Note: The Activation Energy (E_a) of FR combination reaction is zero $E_a = 0$



Disproportionation Reaction of FR (Redox)

(1)

(2)



1° FR + 1° FR
1° " + 2° "
1° " + 3° "
2° " + 2° "
2° " + 3° "

→ combination product as major

3° FR + 3° FR → Disproportionation As Major.

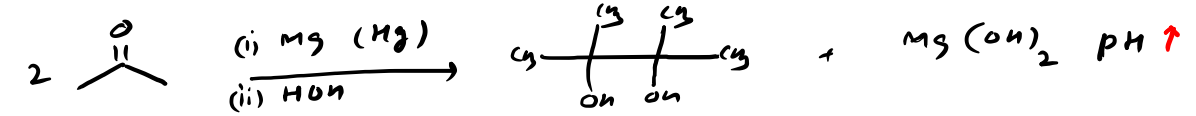
↑ steric hindrance

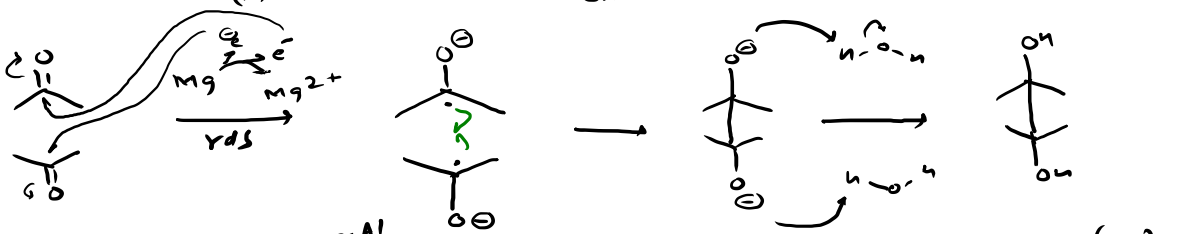
+ Ea of Disproportion reaction is not zero

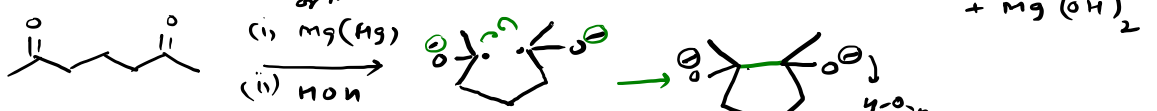
$Ea \neq 0$

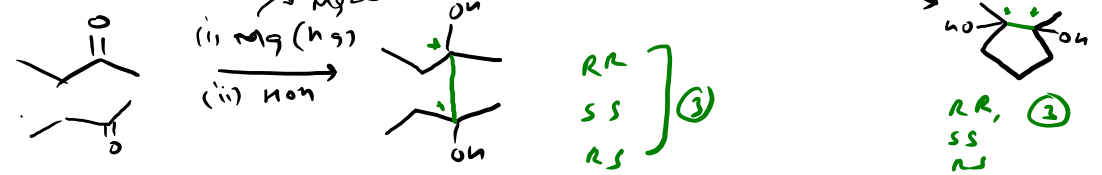
Combination Reaction

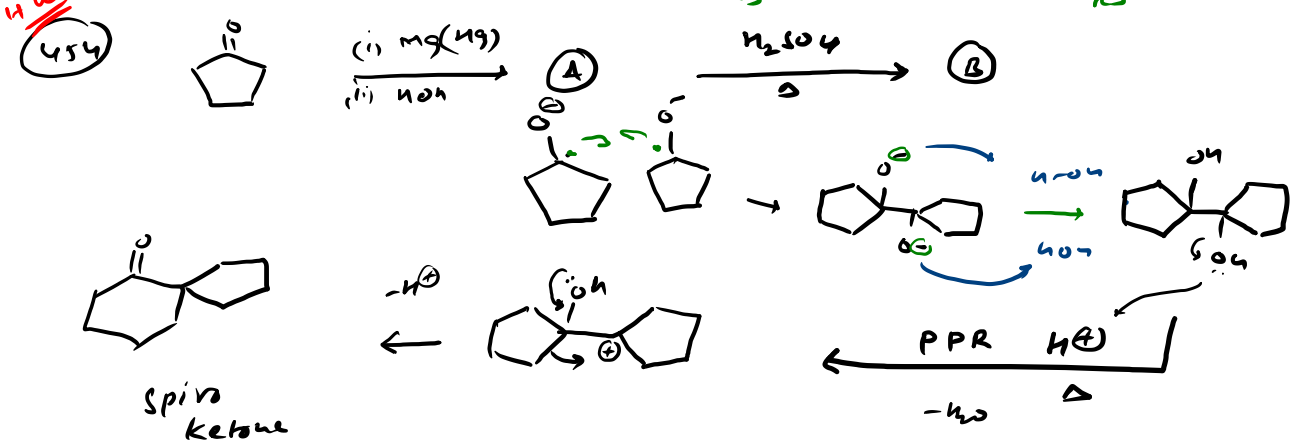
(1) # Pinacole formation

451 

Mech 

452 

453 

HW 454 

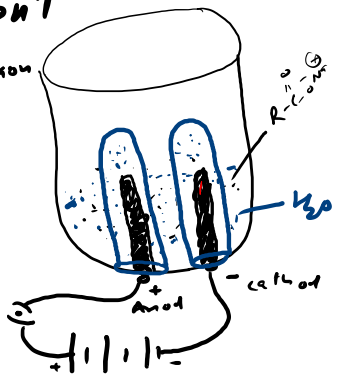
spiro ketone

DBE = 3.

Kolbe Electrolysis (K.E)

(455) $R-\overset{O}{\overset{\|}{C}}-O^{\ominus}Na^{\oplus} \xrightarrow{K.E} \underset{\text{Anod}}{R-R} + \underset{\text{Anod}}{CO_2\uparrow} + \underset{\text{cathod}}{H_2\uparrow} + NaOH$

pH↑



Mech $R-\overset{O}{\overset{\|}{C}}-O^{\ominus}Na^{\oplus} \xrightarrow{NaOH} R-\overset{O}{\overset{\|}{C}}-\ddot{O}: + Na^{\oplus}$

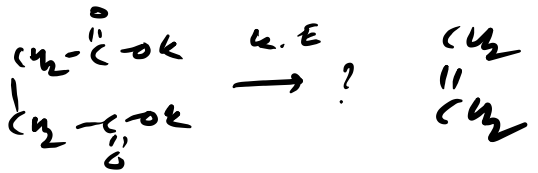
Anod $\downarrow -e^{-}$

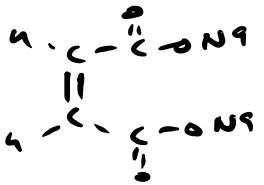
$R-\overset{O}{\overset{\|}{C}}-\ddot{O}: \xrightarrow{rds} R^{\bullet} \; \overset{R^{\bullet}}{\longrightarrow} \; R-R$ (Anod)

$Na^{\oplus} \xrightarrow{H-O-H} NaOH + H^{\oplus} \xrightarrow[+e^{-}]{\text{cathod}} H^{\bullet} \; \overset{H^{\bullet}}{\longrightarrow} \; \underset{\text{cathod}}{H_2\uparrow}$

(456) $\xrightarrow{K.E}$?

(457) $CH_3-\overset{O}{\overset{\|}{C}}-OK + CH_3-CH_2-\overset{O}{\overset{\|}{C}}-OK \xrightarrow{K.E}$? CH_3-CH_3

(458)  $\xrightarrow{K.E}$?

(459)  $\xrightarrow{KE}$?

460 $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-ONa \xrightarrow{K.E}$ products

products is/are ?

$CH_3-CH_2-\dot{C}-\ddot{O}\cdot \xrightarrow{CH_3\dot{C}H} CH_3-CH_2-\overset{O}{\overset{\|}{C}}-CH-CH_3$

A) ~

B) CH_3-CH_3

C) $CH_2=CH_2$

D) $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-O-CH_2-CH_3$

yield of products

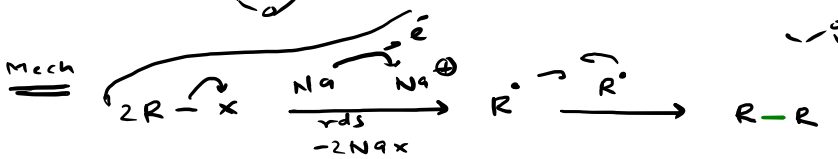
$a > d > b = c$

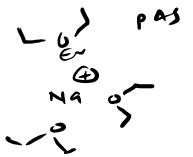
461 $H-\overset{O}{\overset{\|}{C}}-O^{-}\,\overset{\oplus}{Na} \xrightarrow{K.E} H_2$ (Anod) $+$ H_2 (cathod) $+ CO_2$ (Anod) $+$ NaOH

pH ↑

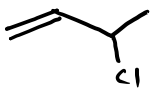
Wurtz process or Wurtz reaction

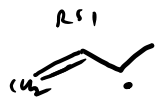
462 $2\,R-X \xrightarrow[Et-O-Et \text{ or } THF]{Na} R-R + 2NaX$



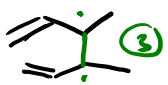


463 $CH_3-Cl \xrightarrow[Dry\ Ether]{Na}$?

464  $\xrightarrow[Dry\ ether]{Na}$?



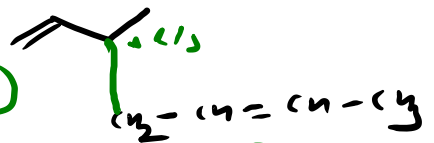
$\leftrightarrow$ $\dot{C}H_2-CH=CH-CH_3$ (RS2, E/Z) (10)

RS1 RS1 →  (3) RR, SS, EZ

RS2 + RS2 → $CH_3-CH=CH-CH_2-CH_2-CH=CH-CH_3$

(3)	Z	E
	E	E
	Z	E

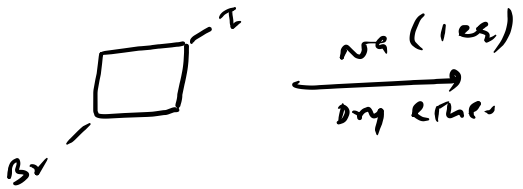
RS1 + RS2



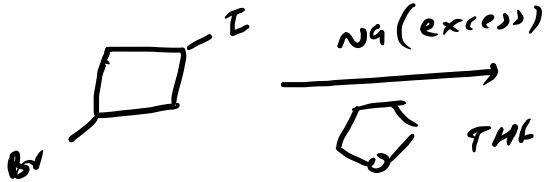
(4) $CH_2-CH=CH-CH_3$ E/Z

Z	R
Z	S
E	R
E	S

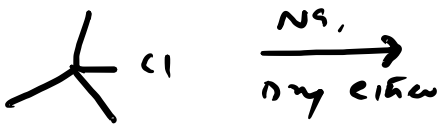
HW 465



HW 466

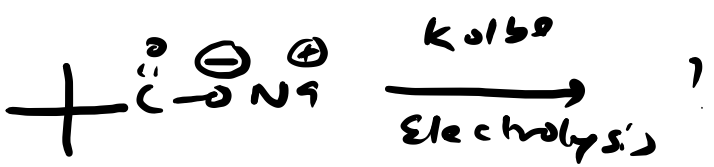


HW 467

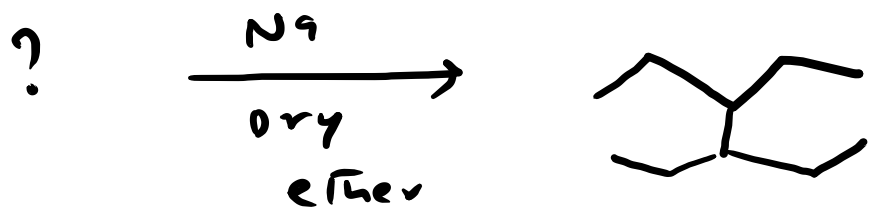


HW 468

HW 469



HW 470



Note ① CH_4 can't synthesize by FR combination in wurtz process.

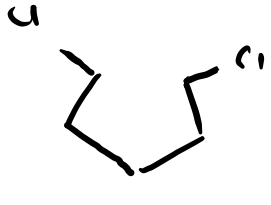
Note ② It is not good process for preparation of open chain alkane having odd no. of carbons

eg. (471) CH_3-CH_2-Cl + Cl $\xrightarrow[DE]{Na}$ $CH_3-CH_2-CH_2-CH_2-CH_3$ (D.P) ← Desired product

$CH_3\dot{C}H_2$ + .

UDP. (undesired product)

(472)

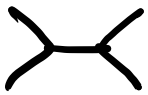


$\xrightarrow[DE]{Na}$

Note ③ wurtz process is not good process for preparation of unsymmetrical open chain alkane eg (474)

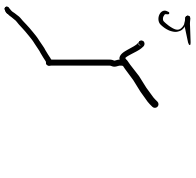
(473)

$\xrightarrow[Dry\ ether]{Na}$

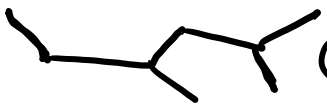


Symmetrical

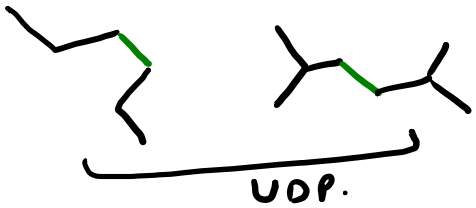
(474)



$\xrightarrow[Dry\ ether.]{Na}$



(DP)



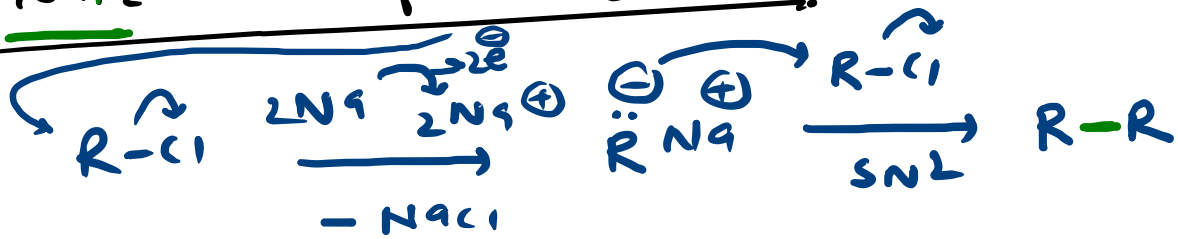
UDP.

(475) $2\ R-Cl \xrightarrow[\text{Dry ether}]{Na,\ O_2} R-R + 2NaCl$

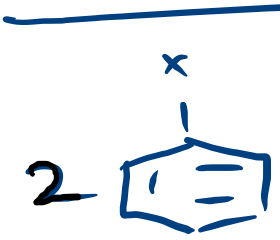
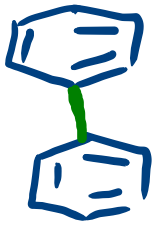
$R^{\cdot} + \overset{\curvearrowright}{O}=\overset{\curvearrowright}{O} \longrightarrow R-O-\dot{O}$

O_2 is (FR scavenger) added to reaction mixture retard the FR mech then also R-R Alkane is formed (it is due ionic mechanism)

ionic mech of wurtz process



Fittig reaction

(476)  $\xrightarrow[\text{Dry ether}]{Na}$  $+ 2NaX$

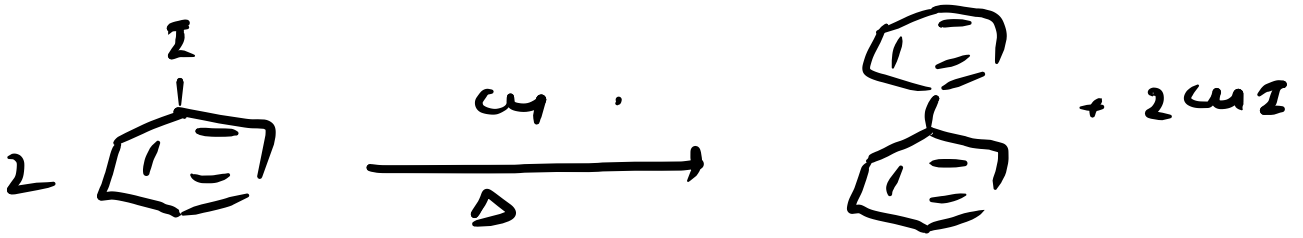
FR Mech

Wurtz - Fittig Reaction

(477) $R-X +$ $\xrightarrow[\text{Dry ether}]{Na}$ $R-R,$ $+$

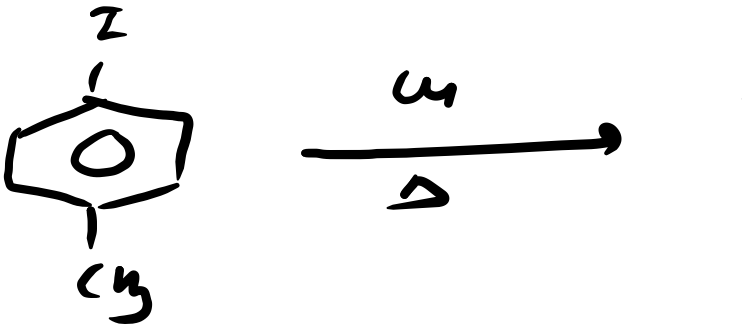
Ulmann Reaction

478

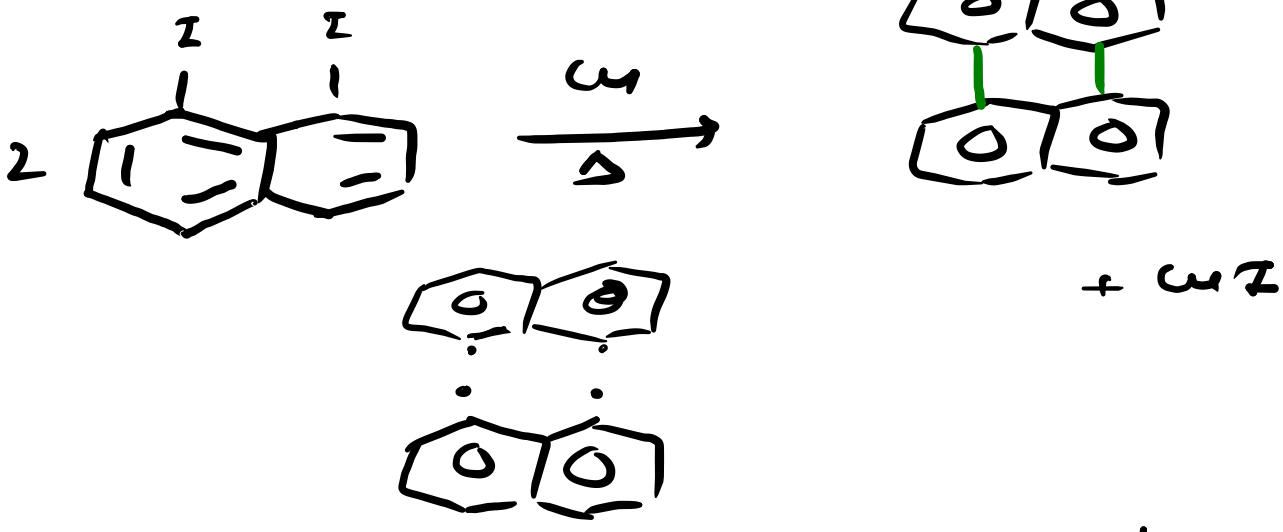


FR mech

479



480



The end.

part

Remaining / alcohol & ether

Grignard reagent — $\overset{\ominus}{R}\overset{\oplus}{Mg}X$

organometallic compound

$\overset{\ominus}{R}\overset{\oplus}{Mg}X$ (G.R) — 52% ionic character, 48% covalent

R_2CuLi (Gilman Reagent)

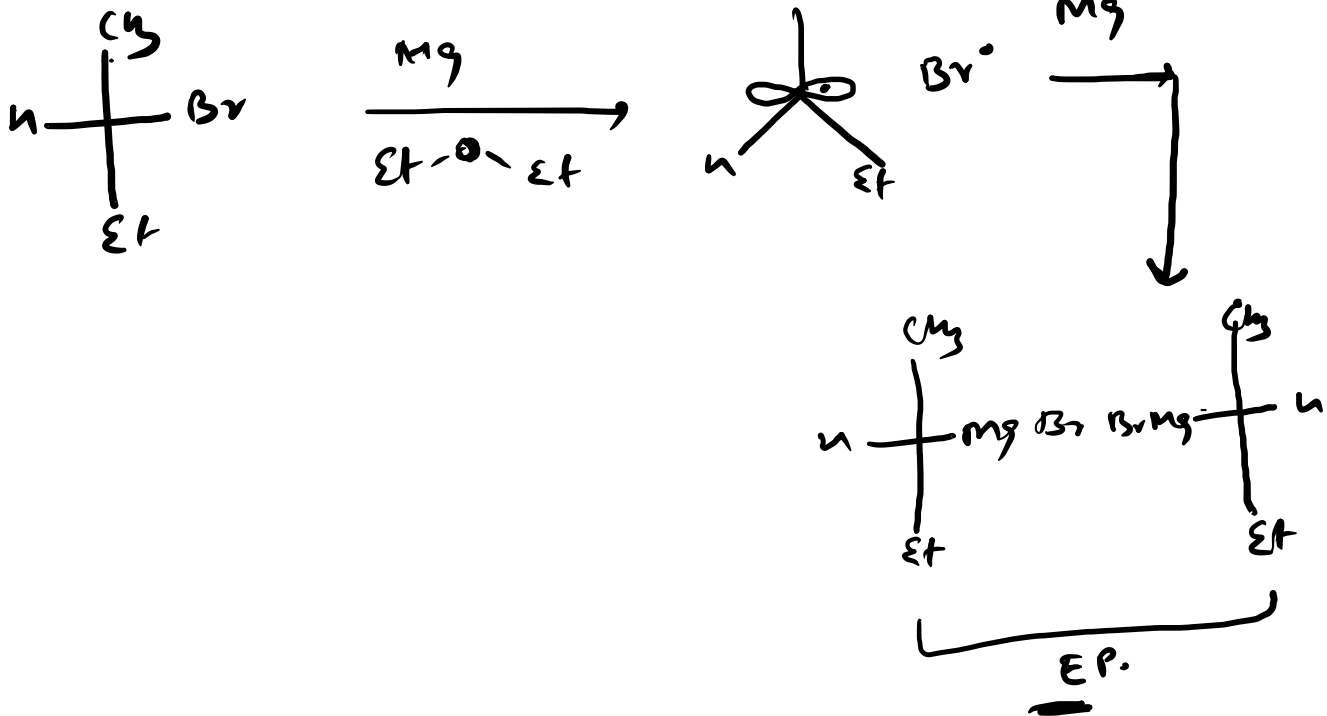
R_2Cd Di alkyl cadmium

$\overset{\ominus}{R}Li^{\oplus}$ Alkyl Lithium

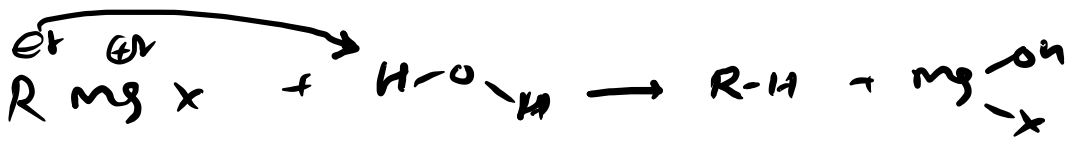
$\overset{\ominus}{R}Na^{\oplus}$ Alkyl sodium

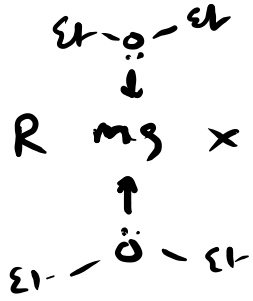
preparation of G.R

(1) $R-X \xrightarrow[Et-O-Et]{Mg} R^{\bullet}\ X^{\bullet} \xrightarrow{{}^{\uparrow}Mg} R\text{-}Mg\text{-}X$

(2) 

Note GR is highly reactive, which can react with moisture itself.

③ 

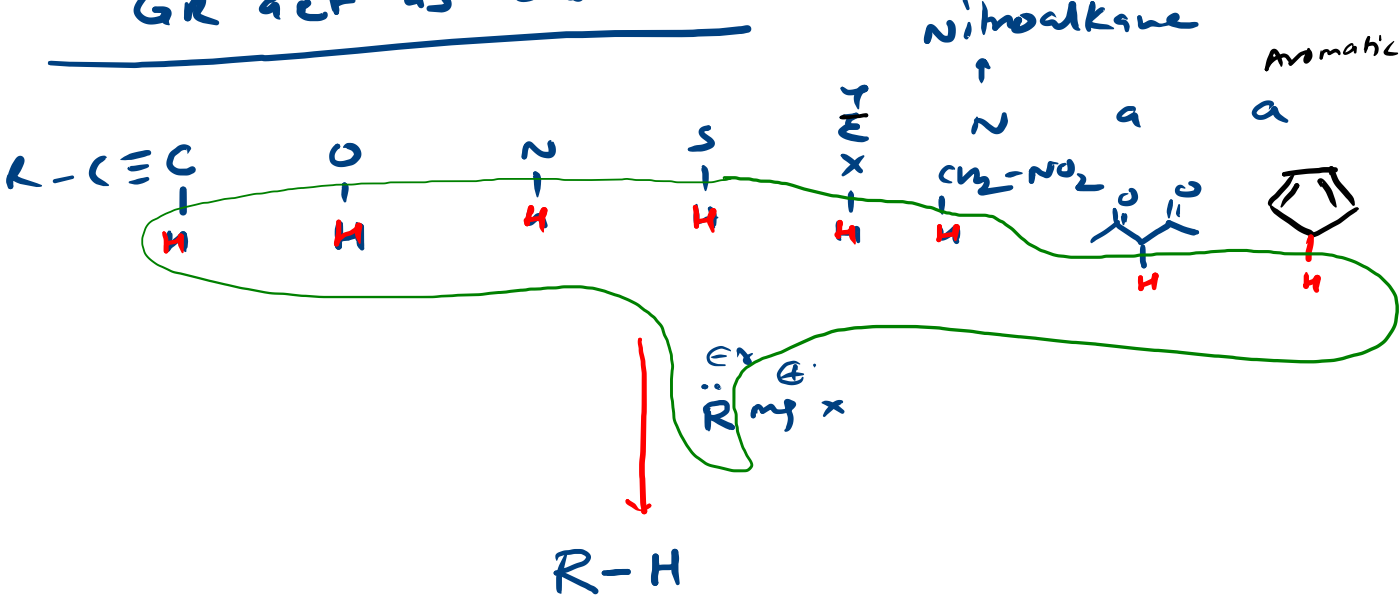


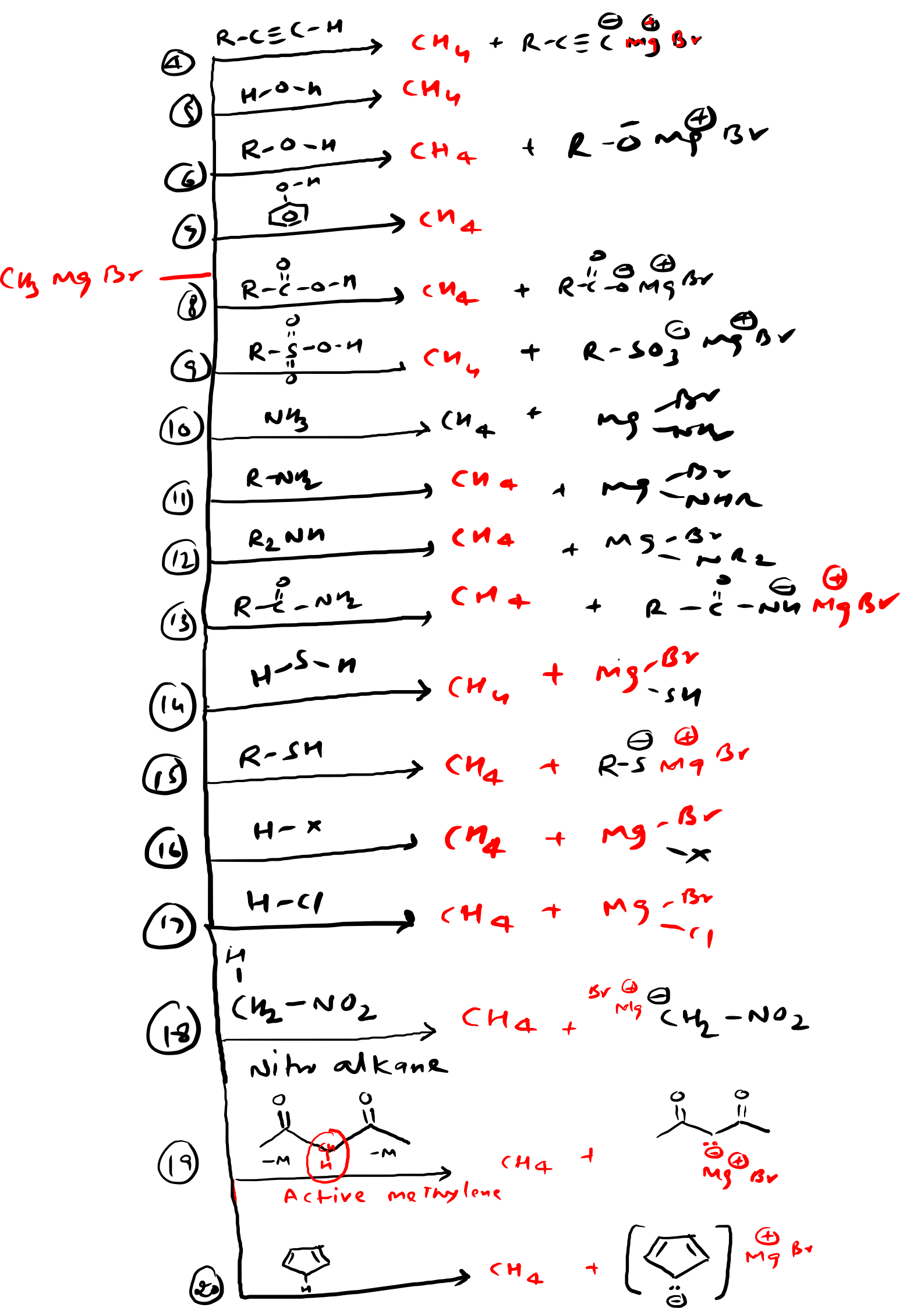
Ether stabilize the GR and decreases its reactivity.

GR shows Acid base reactions and it also show $E^{\oplus}/Nu^{\ominus}$ reactions.

Generally acid base reactions are faster than $E^{\oplus}/Nu^{\ominus}$.

Acid base Reactions of GR in which GR act as base



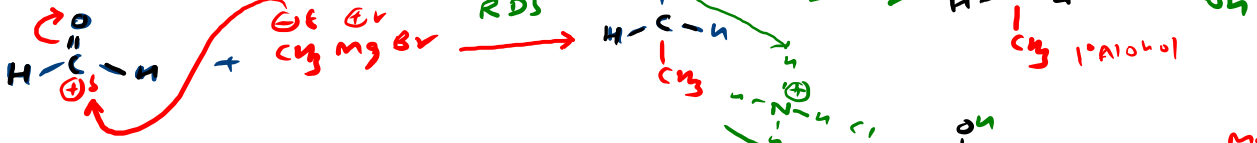

CH3 Mg Br
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
Nitro alkane
Active methylene

$E^{\oplus}/Nu^{\ominus}$ Reactions of GR in which GR act as $Nu^{\ominus}$

GR $Nu^{\ominus}$ →
- NAR (Nucleophilic Addition Reaction)
- SN^{AE} (Nucleophilic substitution Addition-Elimination type)
- SN^2

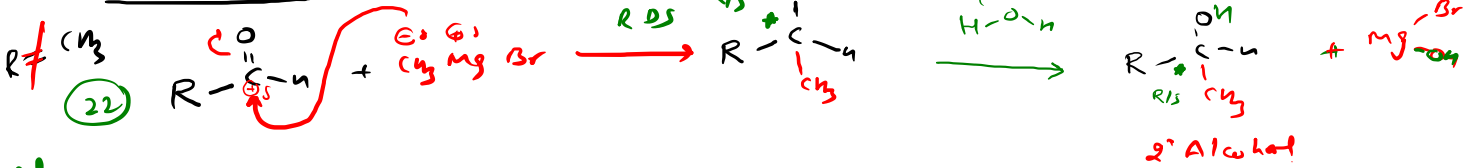
NAR ① Reaction of G.R with carbonyl compound (Aldehyde, Ketone)

NAR (a) with $H-\overset{O}{\overset{\|}{C}}-H$

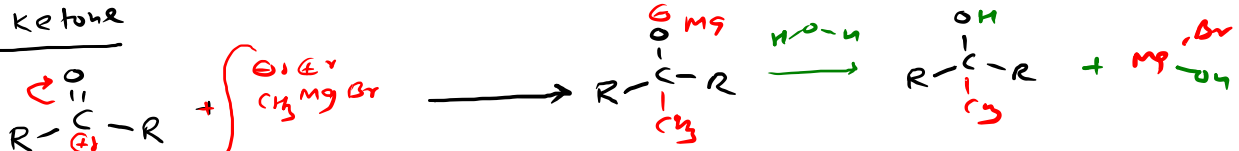
(21) 

NAR (b) with $R-\overset{O}{\overset{\|}{C}}-H$

$R \neq CH_3$

(22) 

NAR (c) with Ketone

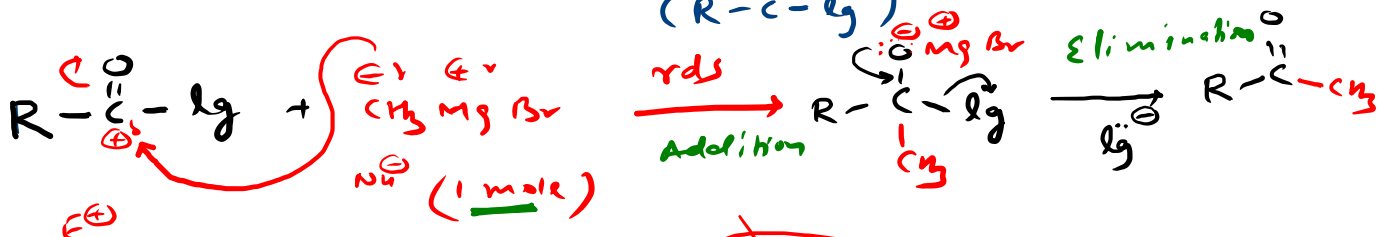
(23) 

rate of reaction $r_{21} > r_{22} > r_{23}$

SN^{AE}

(2) Reactions of GR with Acid derivative ($R-\overset{O}{\overset{\|}{C}}-lg$)

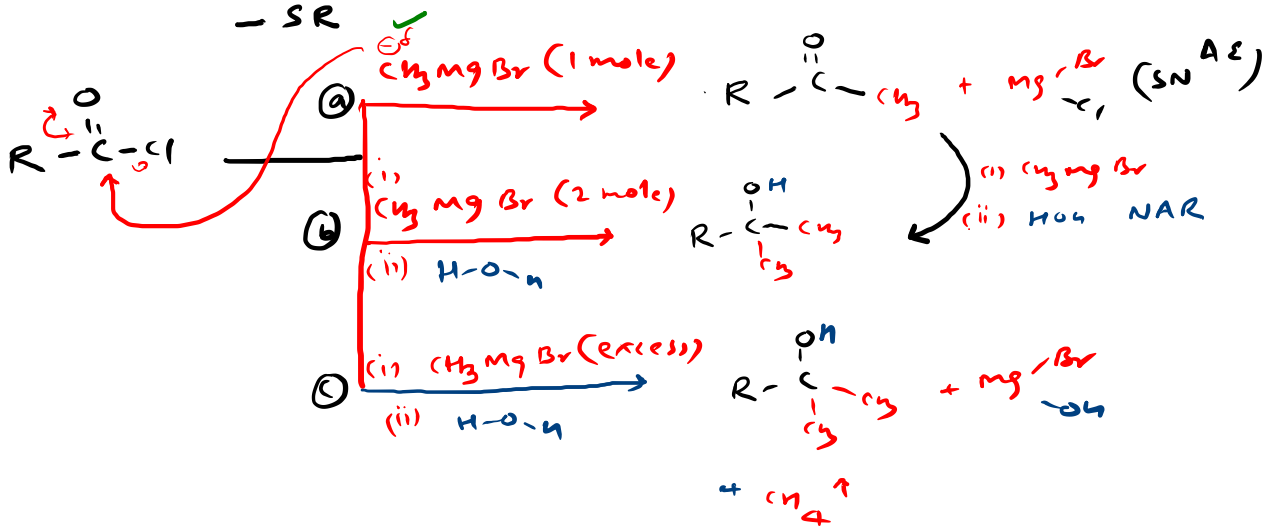
(24) SN^{AE}



lg = $-Cl$ ✓, $-Br$ ✓, $-OR$ ✓, $-O-\overset{O}{\overset{\|}{C}}-R$ ✓, $-NR_2$ ✓, $-SR$ ✓

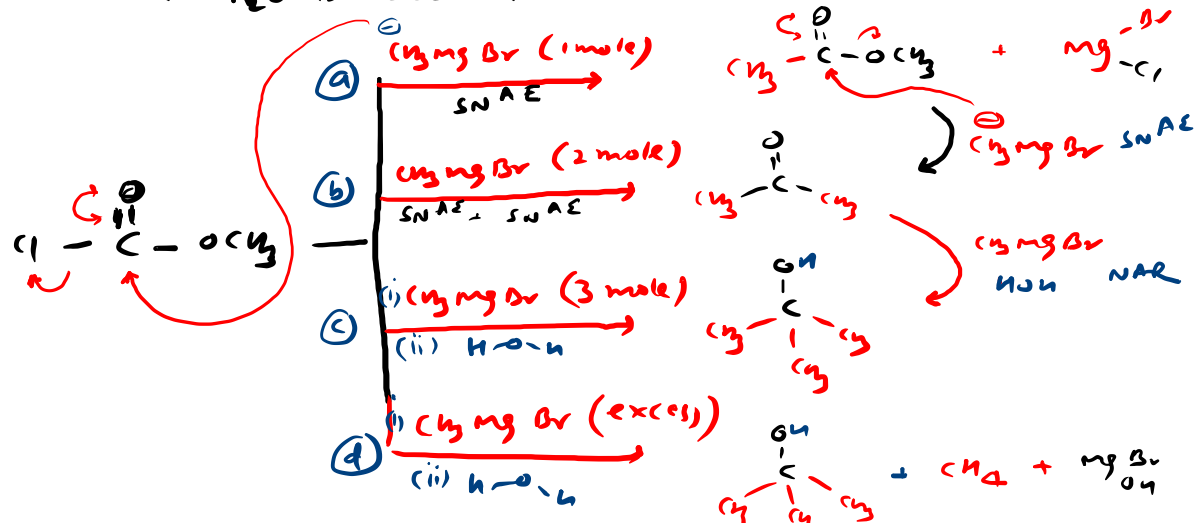
lg = $-OH$, $-NH_2$, $-NHR$, $-SH$ ← They gives acid base reactions

(25)



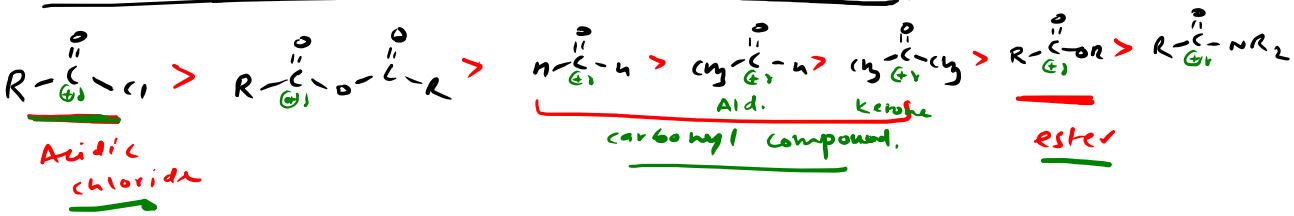
Note Remaining GR (in 25(c)) is destroyed first as H_2O is added to the reaction mixture.

(26)

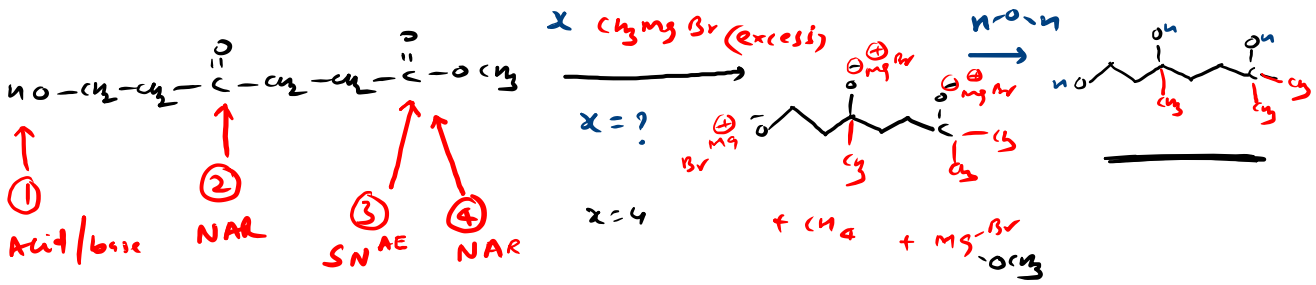


SN^{AE}, SN^{AE}, NAR, Acid base.

Rate of attack of GR (or order of Electrophilicity)

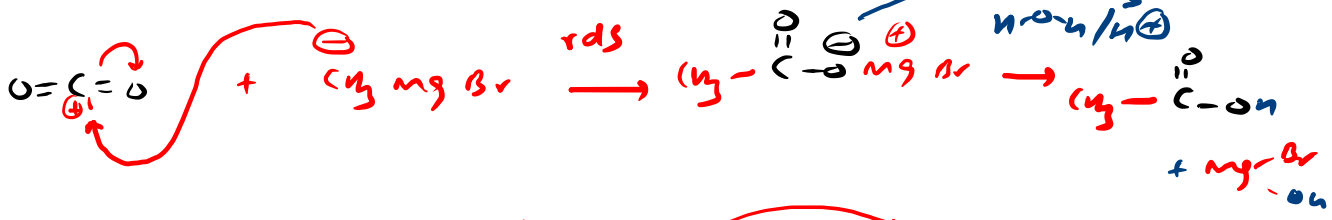


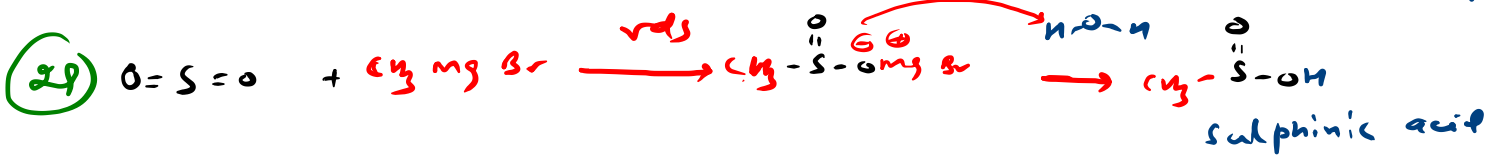
(27) How many moles of GR are consumed with following compound.

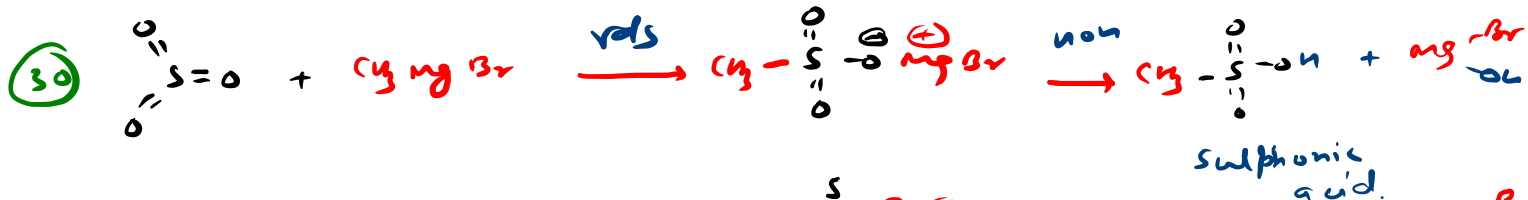


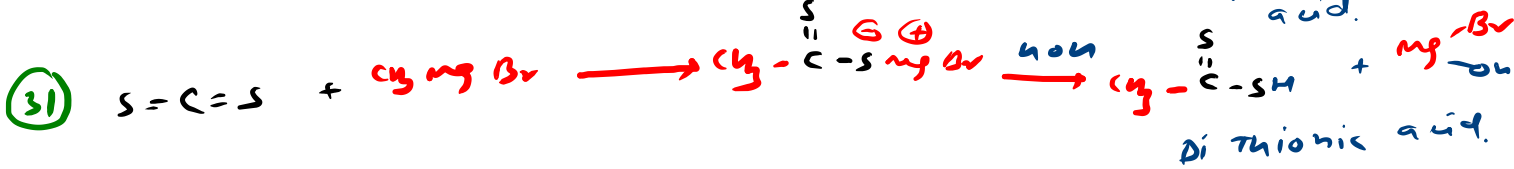
(3) Reactions of GR with oxides (CO_2, SO_2, SO_3) a sulphide (CS_2)

NAR

(28) 

(29) 

(30) 

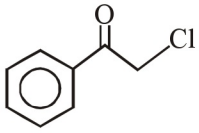
(31) 

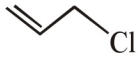
RACE # 22 — ORGANIC CHEMISTRY

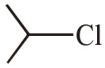
M.M. : 37 — TIME : 25 Min.

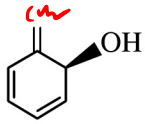
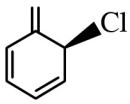
1. **Column - I (Compound)** — **Column - II (Rate of reaction with NaI acetane)** **[8, –0]**

(A) Me–Cl — (P) 0.02

(B)  — (Q) 100000

(C)  — (R) 200

(D)  — (S) 79

2. **Statement-1:** Reaction of $SOCl_2$ with  gives  as a major product.

Statement-2: Major product of above reaction is given by S_Ni mechanism. **[3, –1]**

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

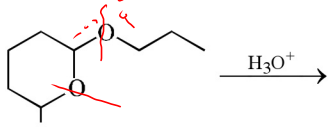
(D) Statement-1 is false, statement-2 is true.

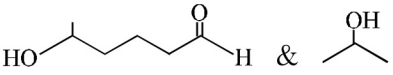
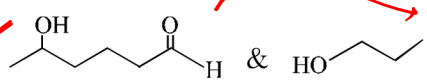
3. An unknown alcohol is treated with the "Lucas reagent' to determine whether the alcohol is primary, secondary or tertiary. Which alcohol reacts fastest and by what mechanism :- **[3, –1]**

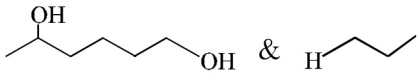
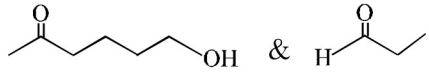
(A) Secondary alcohol by SN^1 (B) Tertiary alcohol by SN^1

(C) Secondary alcohol by SN^2 (D) Tertiary alcohol by SN^2

4. What are the most likely products of the reaction shown below? **[3, –1]**



(A)  (B) 

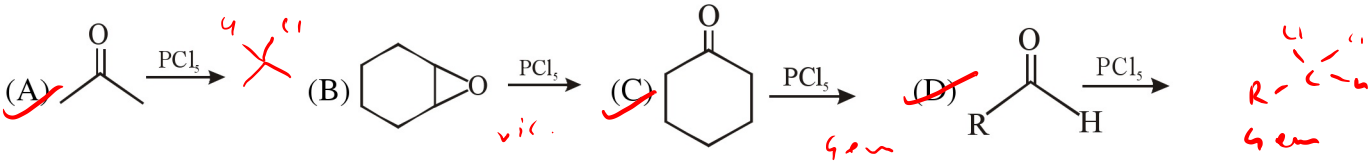
(C)  (D) 

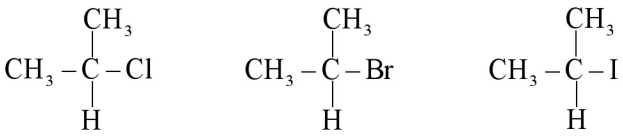
5. Glycerol $\xrightarrow[\Delta]{KHSO_4}$ Product (P) : **[3, –1]**

P is :

(A) $CH_2=CH_2$ (B) $CH_2=CH-CH=O$ (C) $CH_2=CH-CH_3$ (D) $CH_2=CH-\overset{\overset{O}{\|}}{C}-CH_3$

6. In which of the following reaction gemdichloride will formed:- **[4, –1]**



7.  **[4, –1]**

From left to right correct statements are :

(A) Rate of S_N1 mechanism increases in polar protic solvent

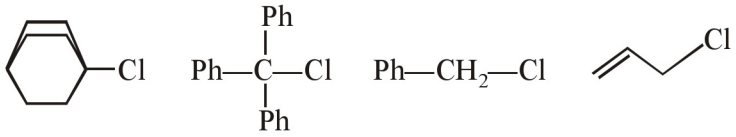
(B) Rate of S_N2 mechanism increases in DMSO

(C) Rate of E_2 mechanism increases

(D) Rate of E_1 mechanism increases

Paragraph For Q.no. 8 & 9

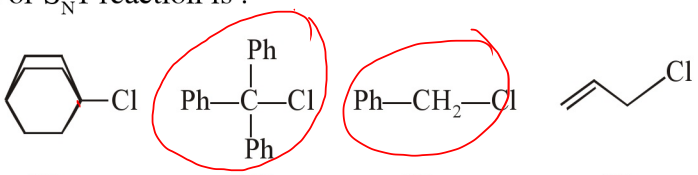
In an S_N1 reaction mechanism, rate of S_N1 reaction mechanism depends on stability of carbocation and also related to the case of heterolysis of C–Lg bond.



(P) (Q) (R) (S)

(A) P > Q > R > S (B) Q > R > S > P (C) P > R > Q > S (D) R > S > Q > P

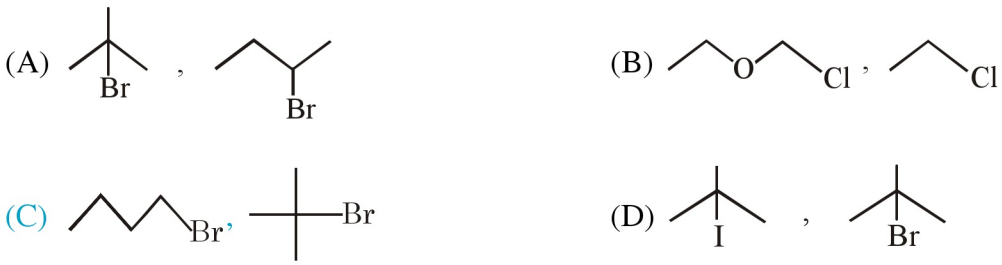
8. Rate of S_N1 reaction is : **[3, –1]**



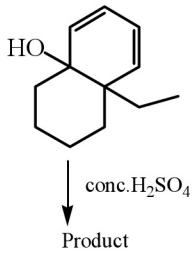
(P) (Q) (R) (S)

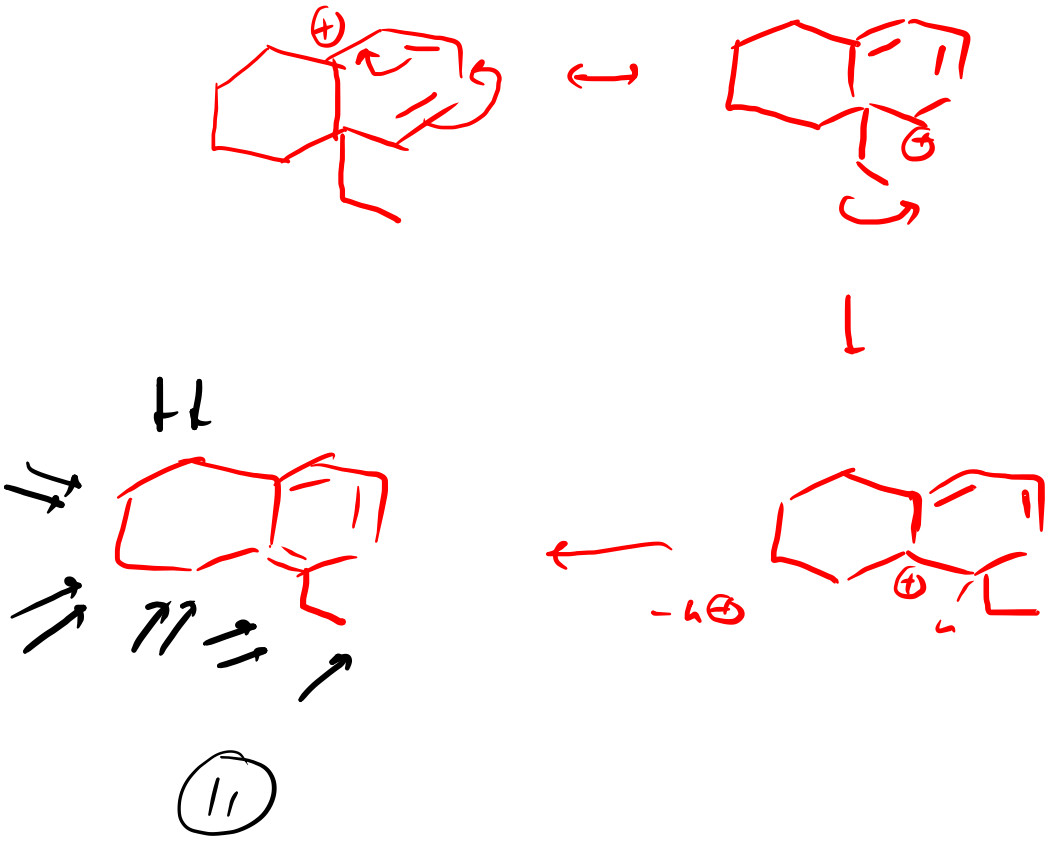
(A) P > Q > R > S (B) Q > R > S > P (C) P > R > Q > S (D) R > S > Q > P

9. Among the following pairs, in which pair (II) is more reactive than (I) for SN^1 reaction : **[3, –1]**



10. How many monocloroderivatives (with Cl_2,hv) are possible for the major product in the following reaction : **[3, –1]**

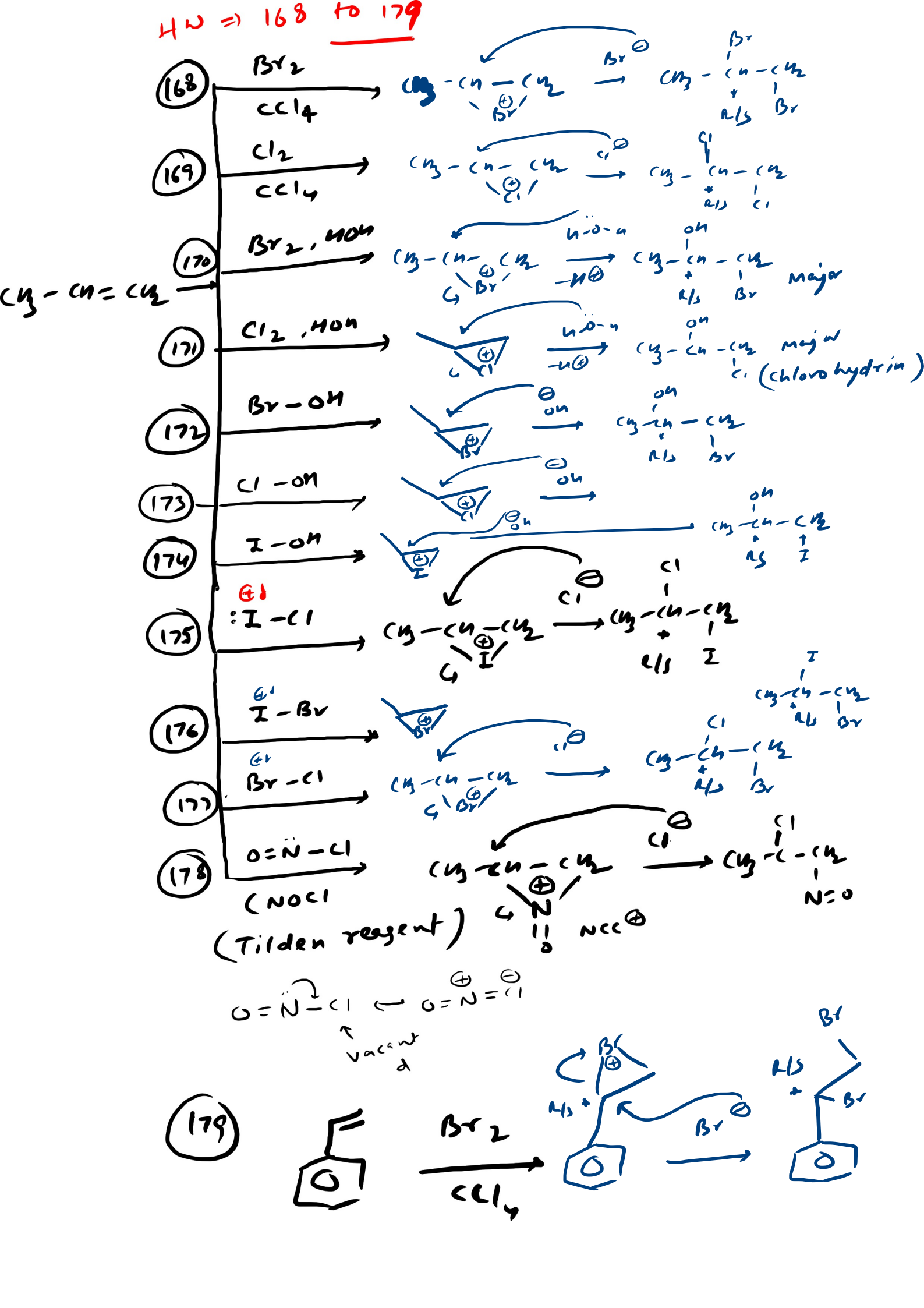




FILL THE ANSWER HERE

1.	A (P) (Q) (R) (S) B (P) (Q) (R) (S) C (P) (Q) (R) (S) D (P) (Q) (R) (S)	**2.**	(A) (B) (C) (D)	**3.**	(A) (B) (C) (D)	**4.**	(A) (B) (C) (D)
5.	(A) (B) (C) (D)	**6.**	(A) (B) (C) (D)	**7.**	(A) (B) (C) (D)	**8.**	(A) (B) (C) (D)
9.	(A) (B) (C) (D)	**10**	(0) (1) (2) (3) (4) (5) (6) (7) (8) (9)				

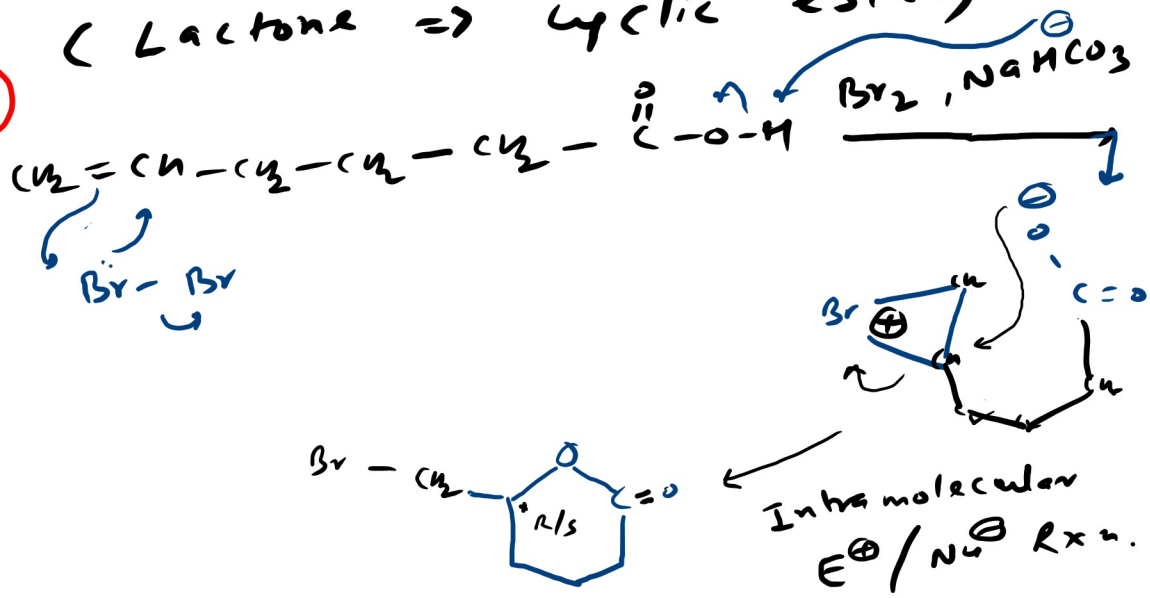
HW ⇒ 168 to 179


168
Br2
CCl4
169
Cl2
CCl4
170
Br2, H2O
171
Cl2, H2O
172
Br-OH
173
Cl-OH
174
I-OH
175
:I-Cl
176
I-Br
177
Br-Cl
178
O=N-Cl
(NOCl
(Tilden reagent)
CH3-CH=CH2
Major
(chlorohydrin)
vacant d
179
Br2
CCl4

Bromolactonisation

(Lactone ⇒ cyclic ester)

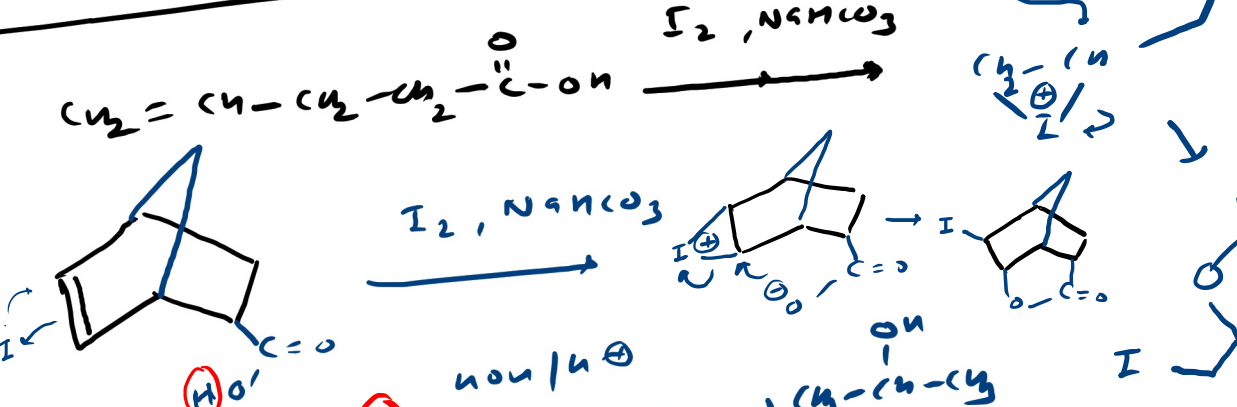
180



Iodolactonisation

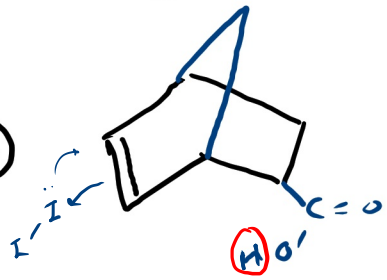
HW

181

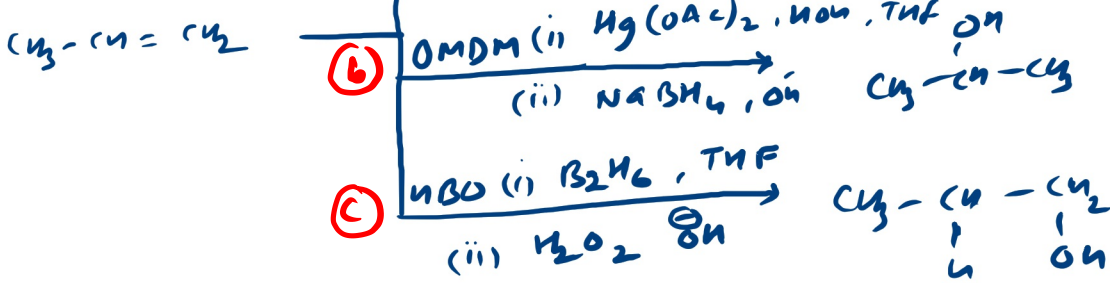


HW.

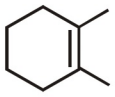
182

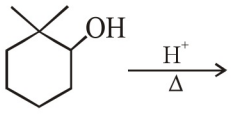
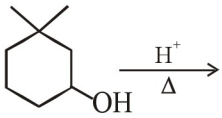
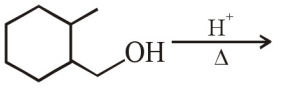
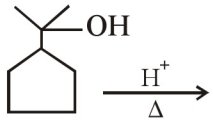


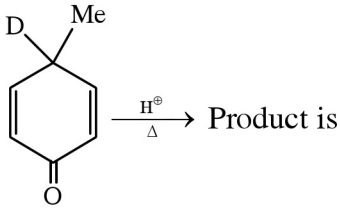
183

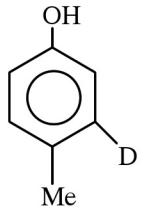
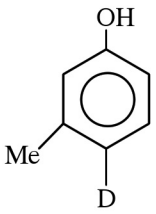
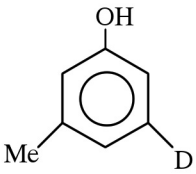
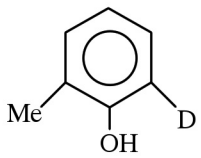


4. Which of the following reaction will not produce given alkene as major product ? **[3, –1]**

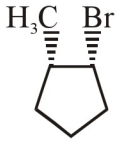
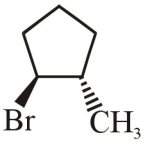


(A)  (B)  (C)  (D) 

5.  Product is **[3, –1]**

(A)  (B)  (C)  (D) 

6. Find out the correct statements about (I), (II), (III) & (IV)- **[4, –1]**

(I) (II) (III)  (IV) 

(A) I, III are diastereomer

(B) I & II are enantiomer

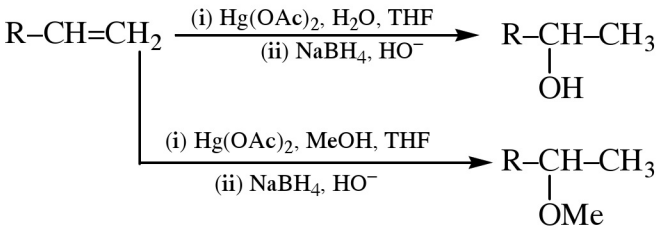
(C) II, III are identical

(D) II & IV are identical

Read the following paragraph and answer the questiosn 7 to 9 given below :

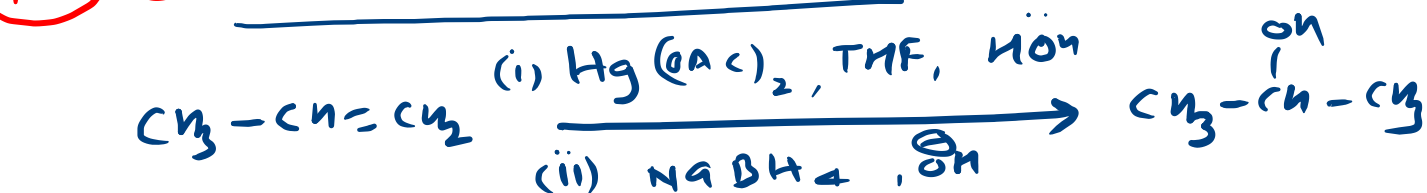
Oxymercuration-demercuration (OMDM) :

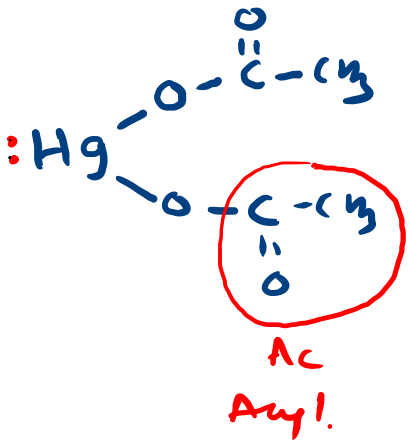
OMDM is a hydration process of alkene according to Markovnikov's rule with no rearrangement of cyclic mercuinium ion. In oxymercuration, the alkene is treated with mercuric acetate in aqueous tetrahydrofuran (THF). When reaction with that reagent is complete, sodium borohydride and hydroxide ion are added to the reaction mixture



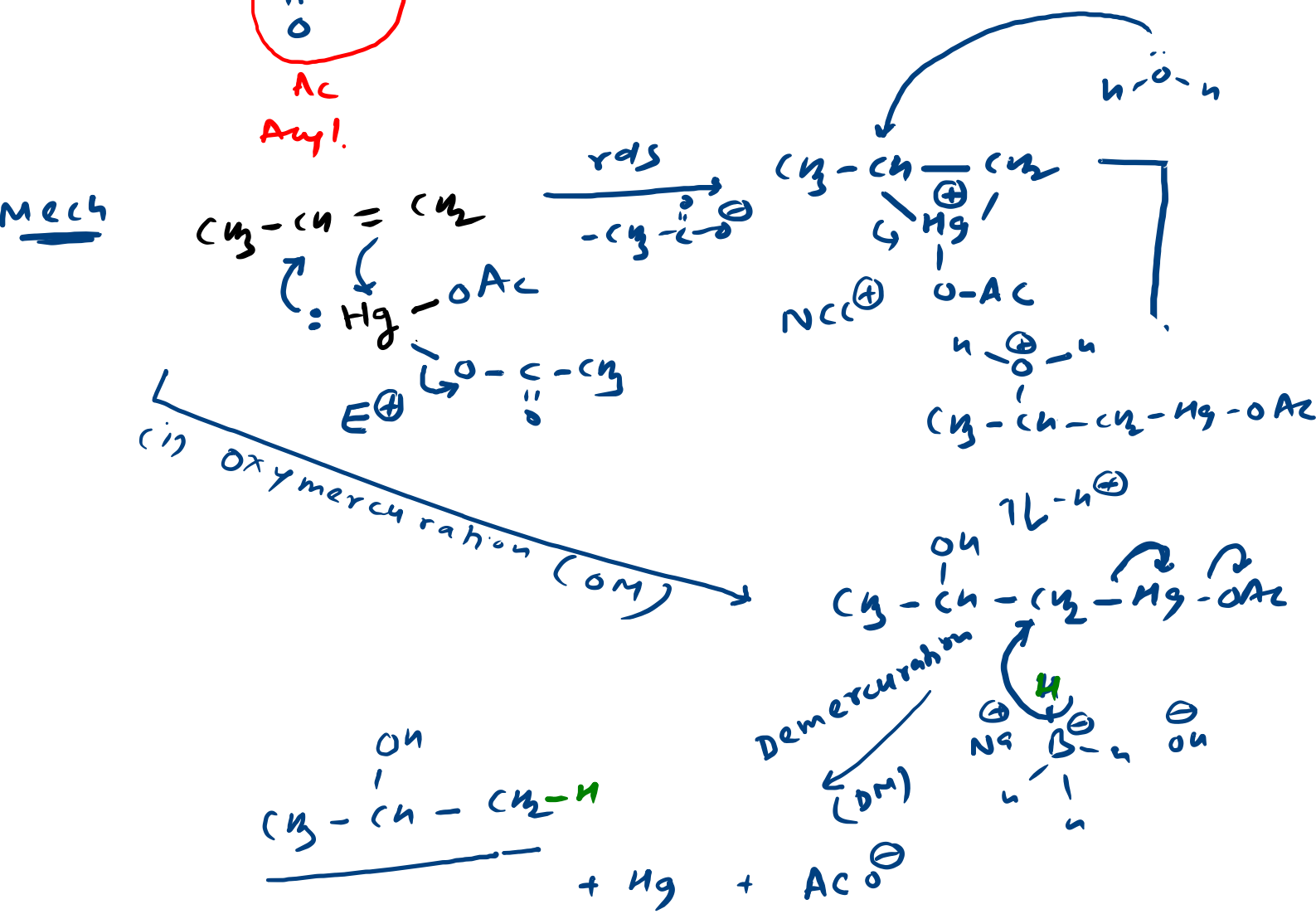
$AcO^- = CH_3COO^-$

183 (6) (OMDM) Oxymercuration, Demercuration





Mech



+ Hg + $AcO^{\ominus}$

Note OMDM is Add^{n} of HOH/ROH Acc. to Markovnikov's, without rearrangement

(184) $CH_3-CH=CH_2$ —(i) $Hg(OAc)_2$, THF, ROH (solvent) / (ii) $NaBH_4$, $\bar{O}H$→ $CH_3-CH(OR)-CH_2-H$ Ether

(185) $CH_3-C\equiv CH$ —(i) $Hg(OAc)_2$, THF, HOH / (ii) $NaBH_4$, $\bar{O}H$→ $CH_3-C(OH)=CH-H$ enol ⇌ Tauto $CH_3-C(=O)-CH_3$

---

(183) (c) Hydroboration-oxidation (HBO)

3 $CH_3-CH=CH_2$ —(i) B_2H_6, THF / (ii) H_2O_2, $\bar{O}H$ (oxidation)→ 3 $CH_3-CH(H)-CH_2(OH)$

Syn Addn

Mech

$CH_3-CH=CH_2$ → $CH_3-CH(H)-CH_2-B(H)-H$ → $CH_3-CH(H)-CH_2-B(CH_2-CH=CH_3)_2$

$CH_3-CH=CH_2$ → $[CH_3-CH(H)-CH_2]_3B$

Syn

K → Dil $KMnO_4$ (Syn hydroxylation)

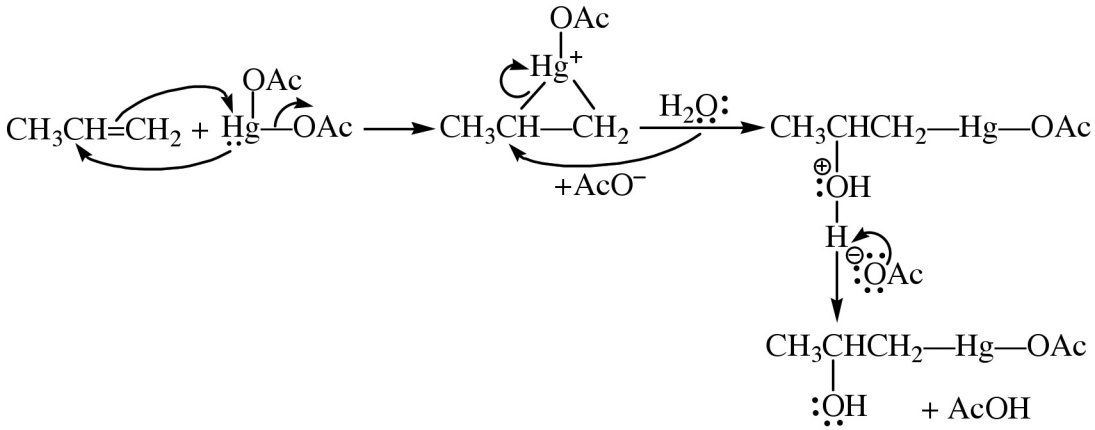
O → OsO_4

H → HBO, HBR, H_2, Ni, Pt, Pd (Direct hydrogenation), NH_2-NH_2, H_2O_2, Δ (Indirect hydrogenation)

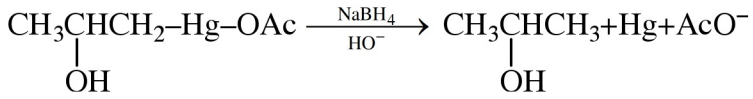
Note HBO is Syn Addn of HOH

Acc. to Anti markovnikov's without rearrangement.

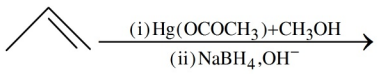
Mechanism for oxymercuration :

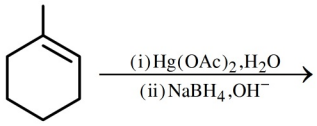


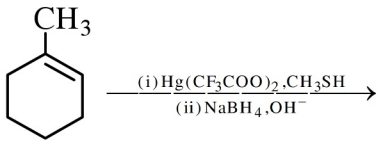
Sodium borohydride ($NaBH_4$) converts the carbon-mercury bond into a carbon-hydrogen bond. Because the reaction results in the loss of mercury, it is called demercuration.

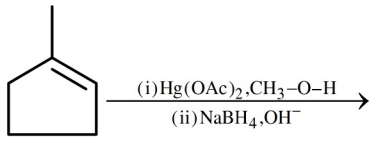


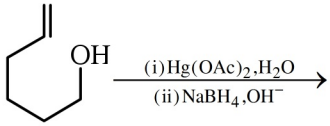
7. **[6, 0]**

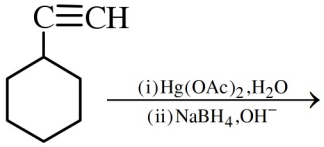
I. 

II. 

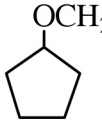
III. 

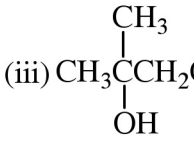
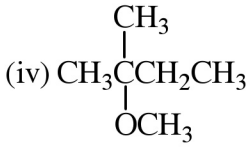
IV. 

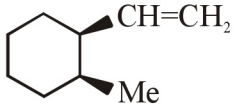
V. 

VI. 

8. How could each of the following compounds be synthesized from an alkene by OMDM ?

(i) (ii) 

(iii)  (iv)  **[4, 0]**

9.  Product mixture (P). **[4, 0]**

(a) Number of alcohol product formed in the above reaction.

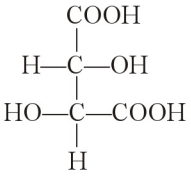
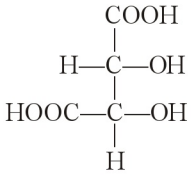
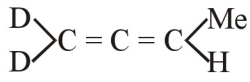
(b) Number of stereoisomers theoretically possible for the major product

(c) Number of enantiomeric pairs formed in the product mixture

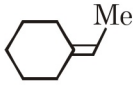
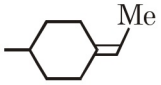
(d) Number of fractions obtained by fractional distillation of product mixture

Write answer of part (a), (b), (c) & (d) in the same order and present the four digit number as answer in OMR sheet. For example : If all these answer are 9 then fill 9999 in OMR sheet.

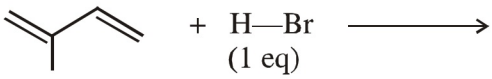
3. Identify optically active compound among following : **[3, –1]**

(A) 
(B) 
(C)
(D) 

4. Which of the following compound can show geometrical isomerism as well as optical isomerism? **[3, –1]**

(A)
(B) 
(C) 
(D)

5. Correct statement regarding product of reaction : **[3, –1]**



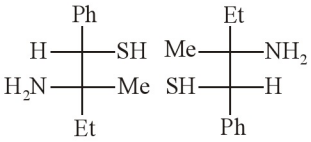
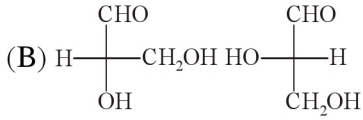
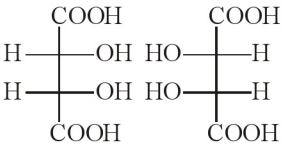
(A) Kinetic controlled product is a secondary bromide whereas thermodynamically controlled product is a tertiary bromide

(B) Kinetic controlled product is a tertiary bromide whereas thermodynamically controlled product is a primary bromide

(C) Both kinetic & thermodynamic controlled product is secondary bromide

(D) Both kinetic & thermodynamic controlled product is a tertiary bromide

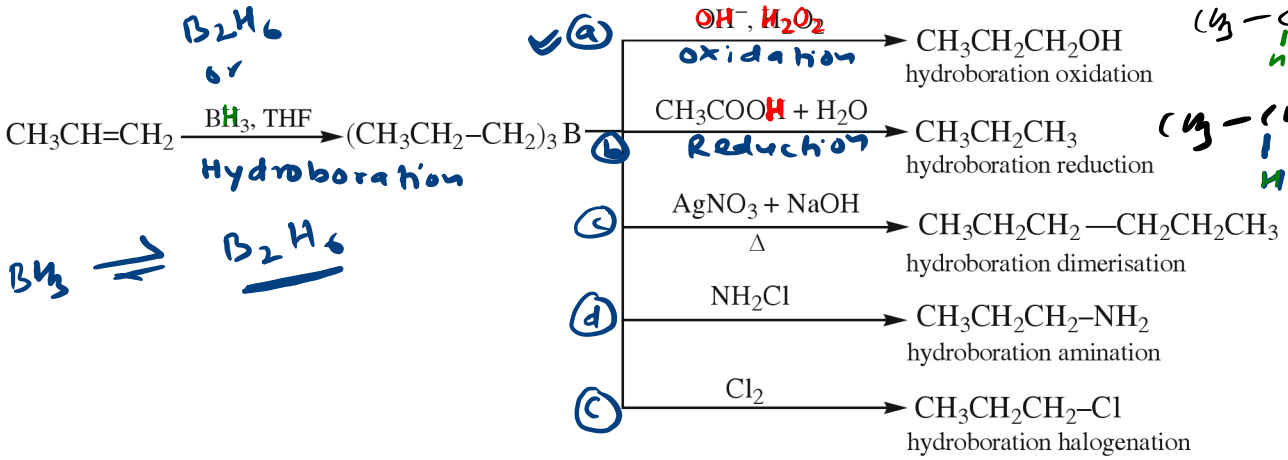
6. Which of the following is/are incorrect in the given isomeric pair : **[4, –1]**

(A)  (Enantiomer)
(B)  (Identical)
(C)  (Enantiomer)
(D) (Distereomer)

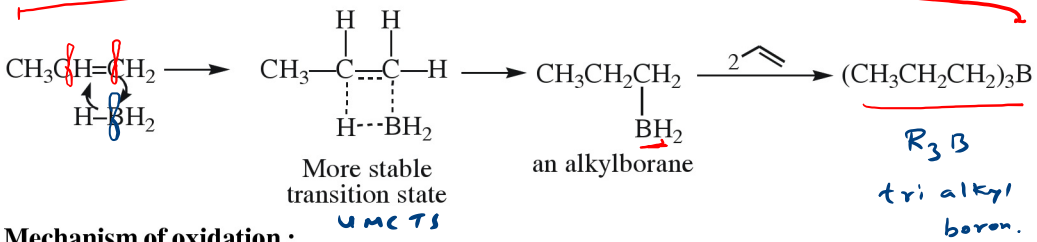
Read the following paragraph and answer the questiosn 7 to 9 given below :

HYDROBORATION-OXIDATION

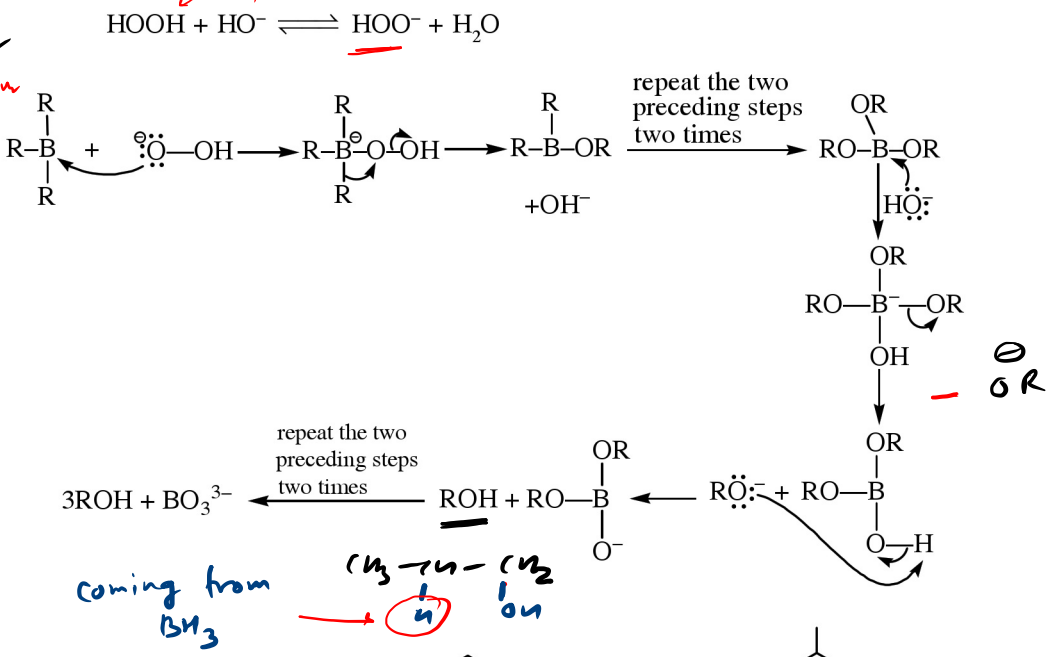
Hydroboration has been developed by brown as a reaction of tremendous synthetic utility because alkyl boranes are able to undergo a variety of transformation. Hydroboration is a one step, four centre, cis addition process in accordance with Markovnikov's rule but after oxidation it seems to be appear to violate Markovnikov's rule.

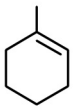


Mechanism of Hydroboration :



Mechanism of oxidation :

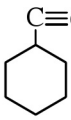
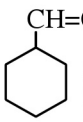
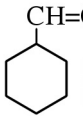


7. **I.** $CH_2=CH_2 \xrightarrow[(ii)HO^-,H_2O_2]{(i)BH_3}$ **II.** $\xrightarrow[(ii)H_2O/AcOH]{(i)BH_3,THF}$ **III.**  $\xrightarrow[(ii)HO^-,H_2O_2]{(i)BH_3}$ **[5,0]**

IV. $\xrightarrow[(ii)H_2O/AcOH]{(i)BH_3,THF}$

V.
$\xrightarrow{(i)BD_3,THF}$ $\xrightarrow{(ii)CH_3COOH,H_2O}$
$\xrightarrow{(i)\ BH_3,\ THF}$ $\xrightarrow{(ii)\ CH_3COOD,D_2O}$
$\xrightarrow{(i)\ BT_3,\ THF}$ $\xrightarrow{(ii)\ CH_3COOD,D_2O}$

8. **[5, 0]**

I.  $\xrightarrow[(ii)H_2O_2/HO^{\ominus}]{(i)BH_3,THF}$ **II.**  $\xrightarrow[(ii)NH_2Cl]{(i)BH_3,THF}$ **III.**  $\xrightarrow[(ii)AgNO_3,\Delta]{(i)BH_3,THF}$

IV. $\xrightarrow[(ii)Cl_2]{(i)BH_3,THF}$ **V.** $CH_3-C\equiv CH \xrightarrow[(ii)AgNO_3+NaOH,\Delta]{(i)BH_3,THF}$

Kucherov Reaction

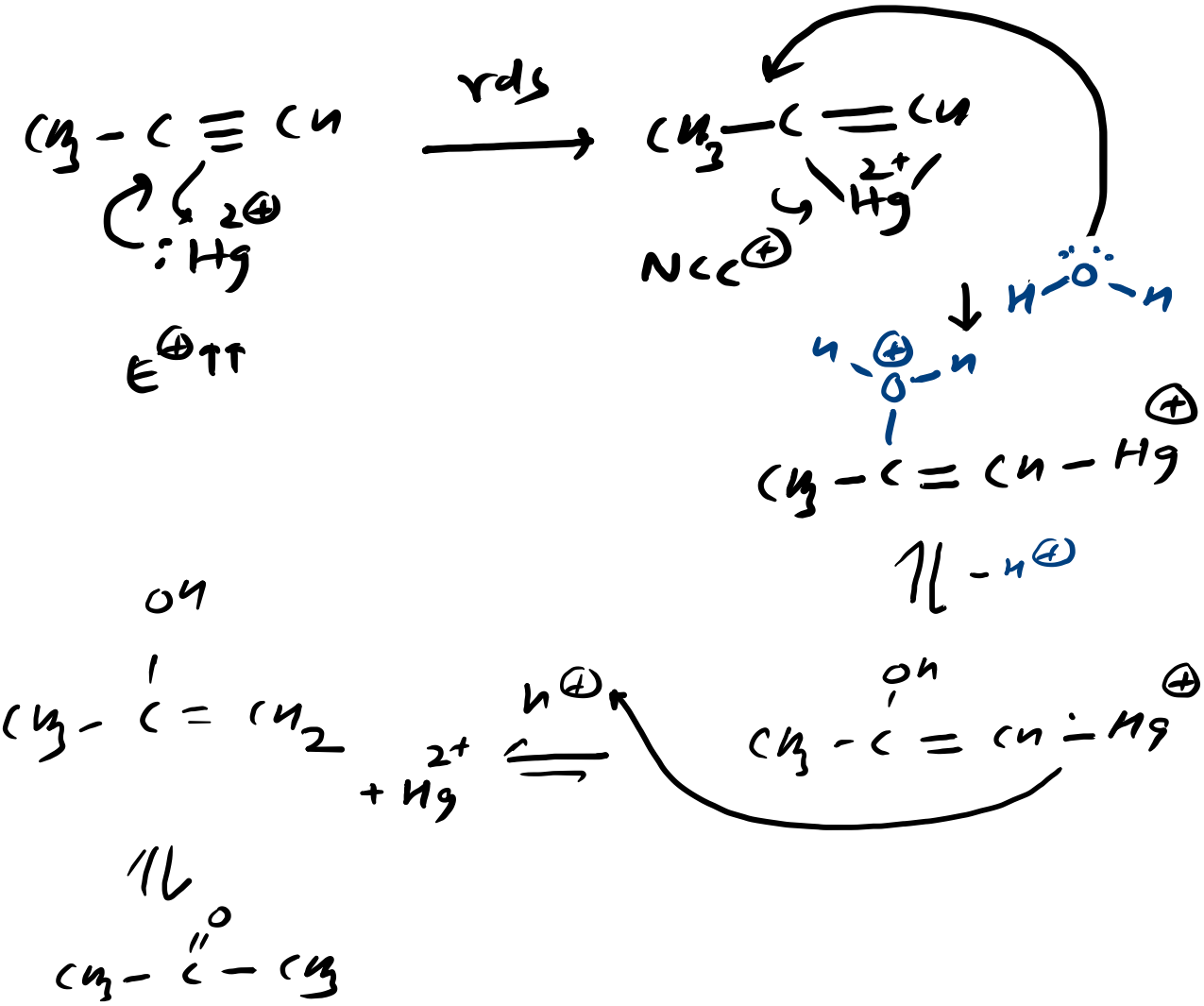
Q. (185) $CH_3-C\equiv CH \xrightarrow{dil\ H_2SO_4 + HgSO_4} CH_3-\underset{\underset{OH}{|}}{C}=CH_2$

$Hg^{2\oplus}$ is far better $E^{\oplus}$ than $H^{\oplus}$

1° Tautomerisation

$CH_3-\overset{O}{\overset{||}{C}}-CH_3$

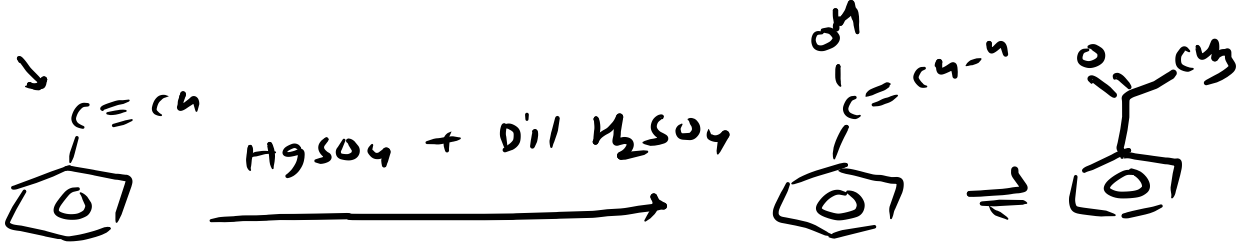
Mech



TRICK Attach OH to that tripple bonded carbon, which suppose to give more stable carbocation, and attach $H^{\oplus}$ to another tripple bonded carbon, finally Tautomerise it.

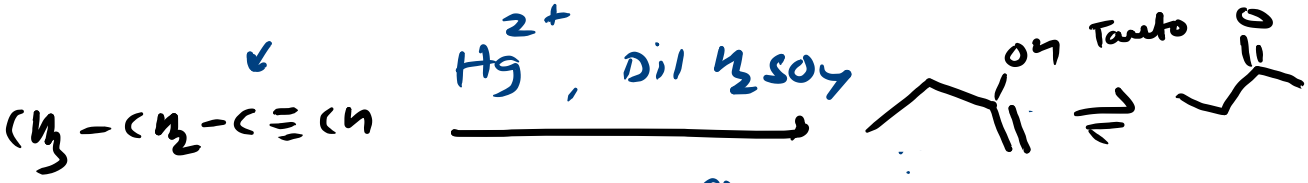
** कुचेरॉव मरकरी वाला एल्काइन पर वारिश करवाते हैं।

187

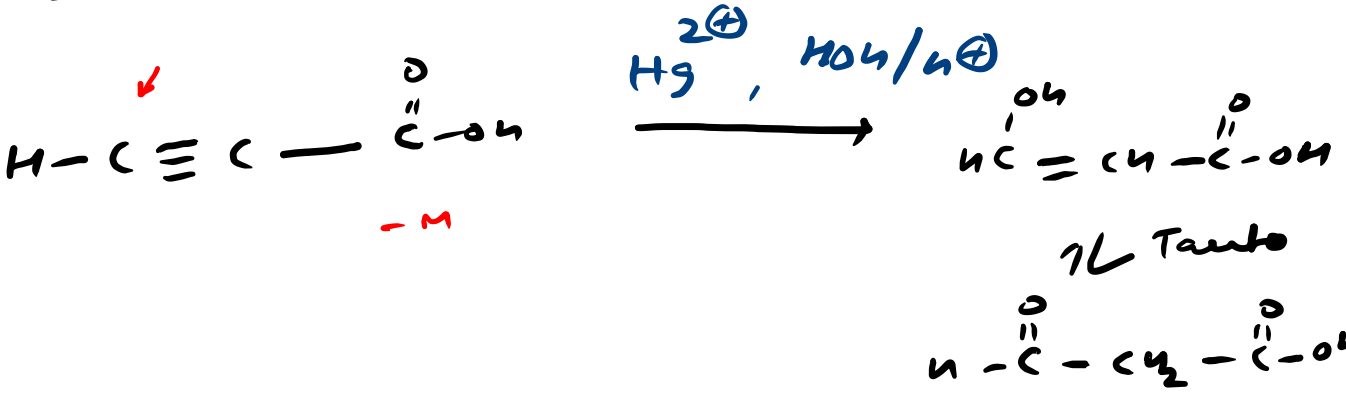


188

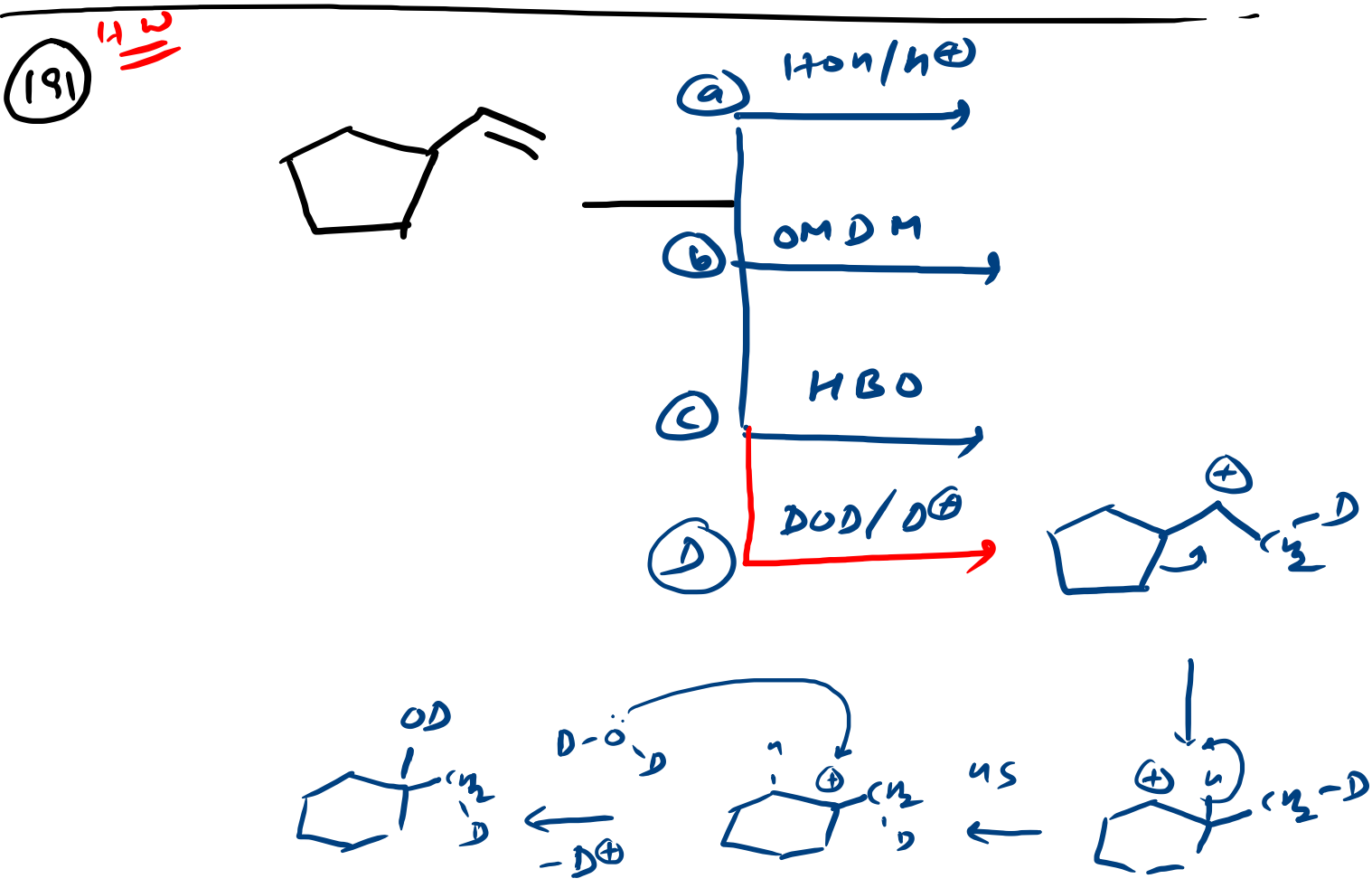
189



190



191

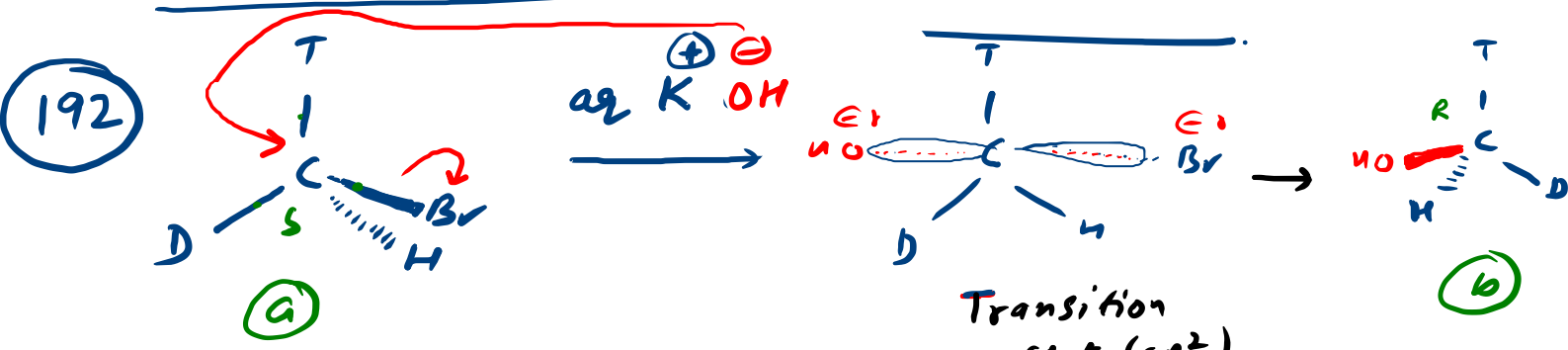


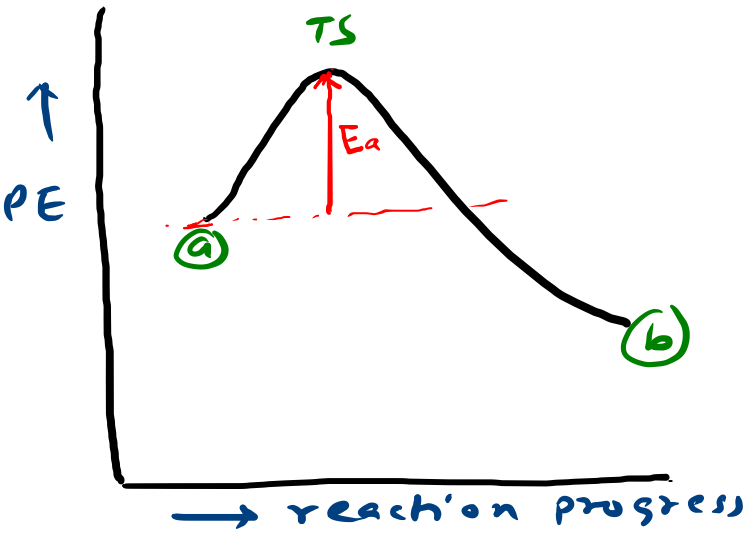
Nucleophilic Substitution Reactions

1. SN^1
2. SN^2
3. SNi
4. SN^{NGP}
5. SN^{AE}
6. SN^{EA}
7. $ArSN^1$
8. $ArSN^2$
9. SN^{1-}
10. SN^{2-}

SN^2 (Nucleophilic Substitution Bimolecular)

192





Characteristics of SN^2

1. It is one step reaction proceed via TS
2. Rate of $SN^2 \propto$ Stability of TS

   or Rate of $SN^2 \propto \frac{1}{\text{Steric hindrance of R-X or R-lg}}$
3. Rate of $SN^2 \propto [\underset{R-X}{\text{substrate}}][Nu^{\ominus}]$

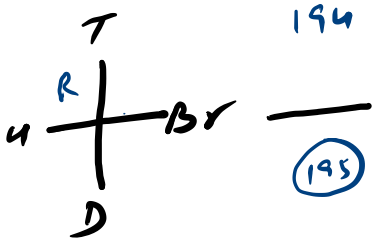
   $\therefore$ It follow 2nd order kinetics

   $\therefore$ called SN^2
4. Molecularity = 2
5. Walden inversion (Inversion of bonds) takes place in SN^2

   Note R convert into S and S convert into R configuration only when the priority of lg and attacking $Nu^{\ominus}$ is same acc. to CIP sequence Rule.
6. There is No Rearrangement, because No intermediate is formed.
7. $Nu^{\ominus}$ attack from back side of lg in SN^2
8. Generally Nucleophilicity of $Nu^{\ominus}$ is the driving force in SN^2
9. factors which affect SN^2
   - (a) Effect of substrate
   - (b) Effect of lg
   - (c) Effect of $Nu^{\ominus}$
   - (d) Effect of solvent.

Some examples of SN^2

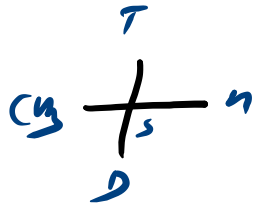
HW



193 aq KOH or aq NaOH

194 $Na^{\oplus}\,{}^{\ominus}S-R$

195 CH_3MgBr



196 $CH_3-C\equiv C^{\ominus}\,Na^{\oplus}$

197 $LiAlH_4$

198 $K^{\oplus}\,{}^{\ominus}CN$

199 Ag-CN

200 NaCN

201 $K^{\oplus}\,{}^{\ominus}O-N=O$ or $NaNO_2$

202 Ag-O-N=O

203 $CH_3-\overset{\overset{O}{\|}}{C}-O^{\ominus}\,Na^{\oplus}$

204 $Na^{\oplus}\,N_3^{\ominus}$

205 $Na^{\oplus}\,{}^{\ominus}OR$ (WES)

Swarts 206 KF / NaF / AgF, DMF

Finkelstein 207 NaI, Acetone

208 NH_3

HALOGEN DERIVATIVES

EXERCISE # O-I (MAINS ORIENTED)

1. Identify set of electrophiles :

(A) CO_2, $\overset{\oplus}{C}H_3$, $:\overset{\square}{C}H_2$, Br_2

(B) HOH, SO_3, $:\overset{\square}{C}Cl_2$, $Cl^{\oplus}$

(C) SO_2, $CH_3-\ddot{O}H$, $\overset{\oplus}{N}O_2$, $:\overset{\square}{C}(Cl)F$

(D) $H-\overset{\oplus}{C}=O$, $\overset{\oplus}{N}=O$, $Ph-\overset{\oplus}{C}H_2$, $\ddot{N}H_3$

2. Identify set of nucleophiles :

(A) $\overset{\ominus}{Cl}$, $\overset{\ominus}{O}H$, $\overset{\ominus}{R}$, $:\overset{\square}{C}H_2$

(B) $CH_3-\overset{O}{\overset{\|}{C}}-\overset{\ominus}{O}$, $\overset{\ominus}{N}_3$, $H_3O^{\oplus}$, $\overset{\ominus}{S}H$

(C) $CH_2=CH_2$, $CH_3-\ddot{N}H_2$, $CH_3-CH_2-\ddot{O}H$,

(D) $\ddot{H}^{\ominus}$, $:\overset{\ominus}{C}N$, CS_2,

3. Which of the following will form 2° carbocation?

(A)

(B)

(C)

(D)

4 Incorrect statement about carbocation is :

(A) It is lewis acid

(B) It has 6 electrons in valency shell

(C) It is electrophile

(D) It is always trigonal planer

5 Which of the following carbocation is most stable ?

(A)

(B)

(C)

(D)

6. Which carbocation is least likely to be formed as an intermediate ?

(A) $(C_6H_5)_3\overset{\oplus}{C}$

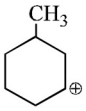
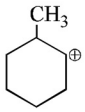
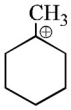
(B)

(C)

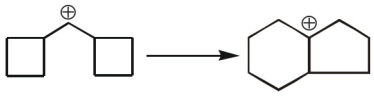
(D) $CH_3-\overset{\oplus}{C}H_2$

node06\B0BO-BA\Other center\JEE(Advanced)\Nurture\Physics\Module#01\Eng\01-Unit & Dimension, Basic Maths & Vector\01_Theory & Ex

7. Which one of the following carbocation would you expect to rearrange :

(A)  (B)  (C) (D) 

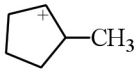
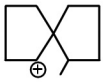
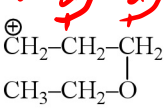
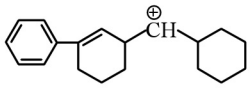
8. How many 1,2-shifts are involved during the course of following reaction :



(A) 1 (B) 2 (C) 3 (D) 4

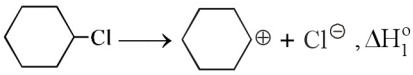
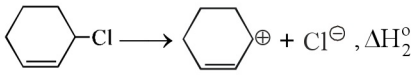
9. How many following carbocation undergo re-arrangement -

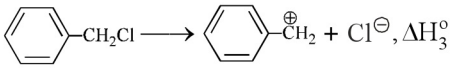
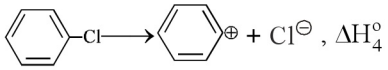
(a) $CH_3CH_2CH_2^+$ (b) $(CH_3)_2\overset{+}{C}HCH_3$ (c) $(CH_3)_3\overset{+}{C}CHCH_3$ (d) $(CH_3CH_2)_3CCH_2^+$

(e)  (f)  (g)  (h) 

(A) 5 (B) 8 (C) 6 (D) 7

10 For the reactions

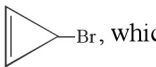
(I)  (II) 

(III)  (IV) 

The correct decreasing order of enthalpies of reaction for producing carbocation is :

(A) $\Delta H_1^\circ > \Delta H_2^\circ > \Delta H_3^\circ > \Delta H_4^\circ$ (B) $\Delta H_4^\circ > \Delta H_1^\circ > \Delta H_2^\circ > \Delta H_3^\circ$

(C) $\Delta H_3^\circ > \Delta H_2^\circ > \Delta H_1^\circ > \Delta H_4^\circ$ (D) $\Delta H_2^\circ > \Delta H_1^\circ > \Delta H_4^\circ > \Delta H_3^\circ$

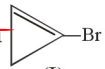
11  , which is not the correct statement :

(I)

(A) I is more soluble in water than bromocyclopropane

(B) I gives pale yellow ppt. on addition with aq. $AgNO_3$

(C) I is having lower dipole moment than bromocyclopropane

(D) I is more ionic than 

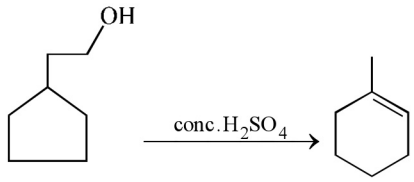
(I)

12. A solution of (–) –1–chloro–1–phenylethane in toluene racemises slowly in the presence of a small amount of $SbCl_5$, due to the formation of :-

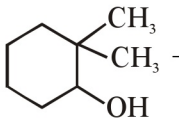
(A) carbanion (B) Beneze (C) carbocation (D) free radical

node06\B0BO-BA\Other center\JEE(Advanced)\Nature\Physics\Modules\01\Eng\01_Units & Dimension, Basic Math & Vector\01_Theory & Ex.

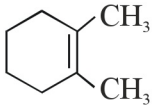
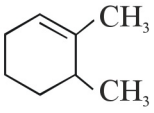
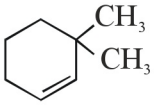
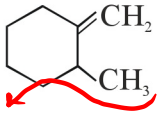
13. How many 1,2-Shifts of carbocation intermediate are involved during the course of following reaction :

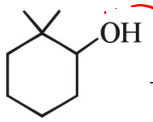


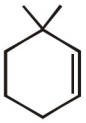
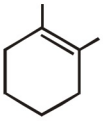
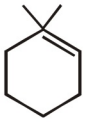
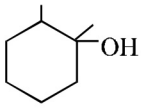
(A) 1 (B) 2 (C) 3 (D) 4

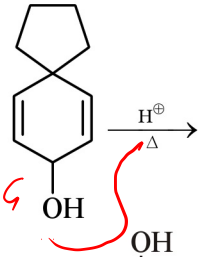
14.  (X) (Major product)

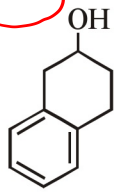
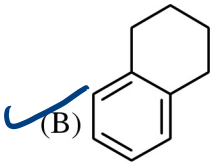
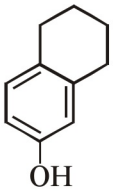
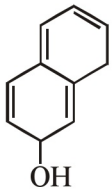
Major product (X) is :

(A)  (B)  (C)  (D) 

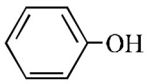
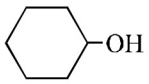
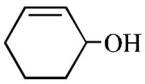
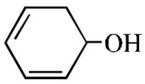
15.  P. The product P is :

(A)  (B)  (C)  (D) 

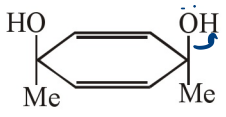
16.  Product ; Product is :

(A)  (B)  (C)  (D) 

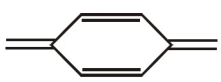
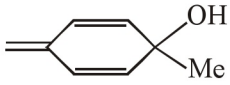
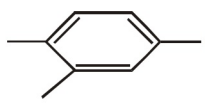
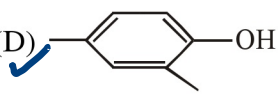
17. Among the given compounds, the correct order of rate of dehydration is :

(I)  (II)  (III)  (IV) 

(A) I < II < III < IV (B) II < III < IV < I (C) I < III < IV < II (D) I < II < III = IV

18  Major product -

Major product is :

(A)  (B)  (C)  (D) 

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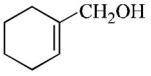
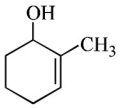
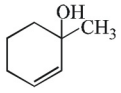
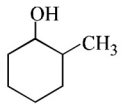
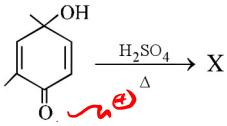
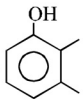
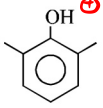
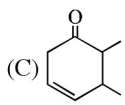
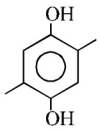
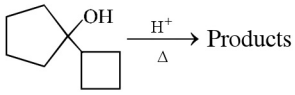
19. Identify the correct order of rate of dehydration when given compounds are treated with conc. H_2SO_4:

(P)  (Q)  (R)  (S) 

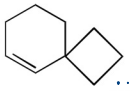
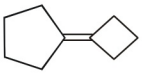
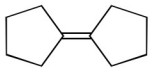
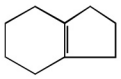
(A) P > Q > R > S (B) Q > P > R > S (C) R > Q > P > S (D) R > Q > S > P

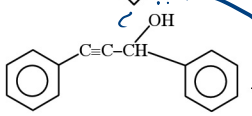
20.  ; X is :

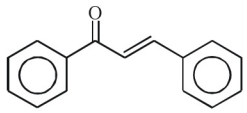
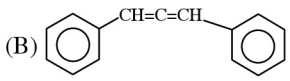
(A)  (B)  (C)  (D) 

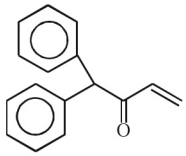
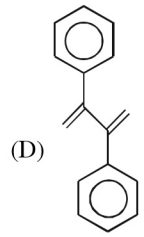
21. 

Major products is :

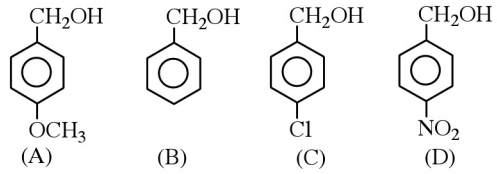
(A)  (B)  (C)  (D) 

22 

(A)  (B) 

(C)  (D) 

23. What is the decreasing order of rate of reaction with HBr for the following benzyl alcohol and its derivative :



(A) (B) (C) (D)

(A) A > C > D > B (B) A > B > D > C (C) D > C > B > A (D) A > B > C > D

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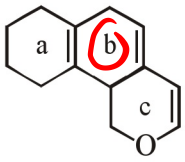
24. Which will dehydrate at fastest rate by H_3PO_4 :

(A) 2-methyl butan-2-ol (B) 3-methyl butan-2-ol

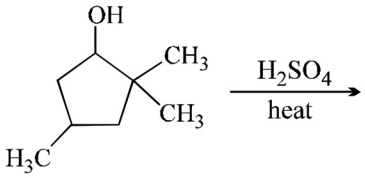
(C) Butan-1-ol (D) 2-methyl butan-1-ol

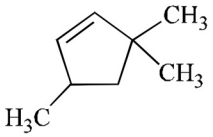
25. 

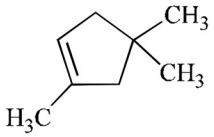
The double bond which is most reactive towards attack of electrophile :

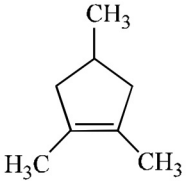
(A) a (B) b (C) c (D) None

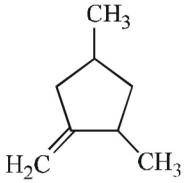
26. The major product formed in the following reaction is :



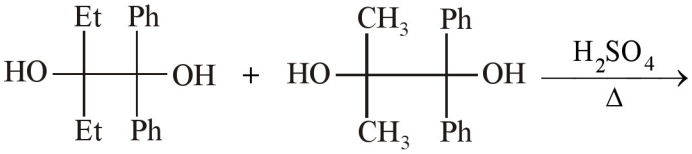
(A) 

(B) 

(C) 

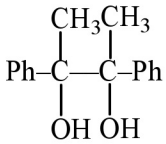
(D) 

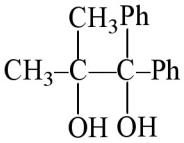
27. How many products are obtained in the given reaction :

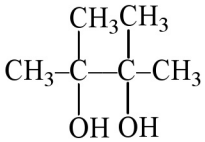


(A) 1 (B) 2 (C) 3 (D) 4

28. Compare rate of reaction towards pinacol pinacolone rearrangement.

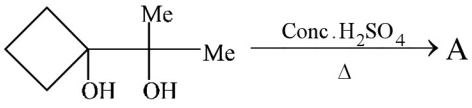


(I)


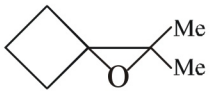
(II)


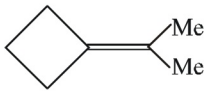
(III)

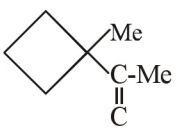
(A) II > III > I (B) III > II > I (C) II > I > III (D) I > II > III

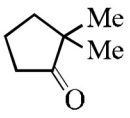
29. 

Product A is :

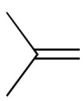
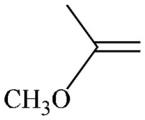
(A) 

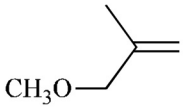
(B) 

(C) 

(D) 

30. What is the order of reactivity with HBr :

(x) 
(y) 

(z) 

(A) x > y > z (B) y > x > z (C) z > y > x (D) y > z > x

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Effect of substrate

(239) 3° R-X ⟶ S_N1

Alkyl halide

+ Cl ⟶ S_N1

3° Alkyl halide

1° R-X ⟶ S_N2

CH_3-Br ⟶ S_N2

1° Alkyl halide

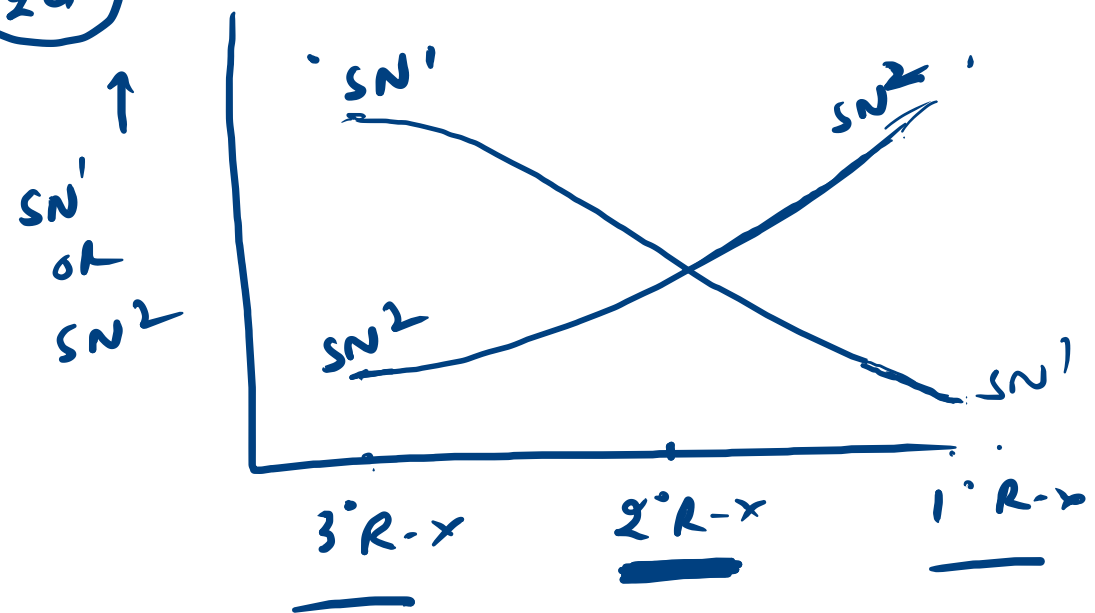
2° — Cl, 2° Alkyl, X

weak $Nu^{\ominus}$ eg. H_2O ⟶ S_N1

strong $Nu^{\ominus}$ eg. aq $K^{\oplus}OH^{\ominus}$ ⟶ S_N2

S_N2

(240)



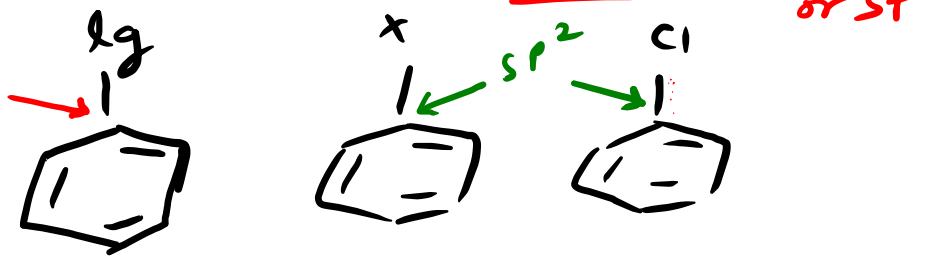
Note certain compounds do not give SN^1 & SN^2

---

(241) CH_3-F NO SN^1
NO SN^2
or
R-F -F is Poor lg.

lg attached to sp^3 carbon gives SN^1 & SN^2 (Not attached with sp^2 C or sp)

(242)



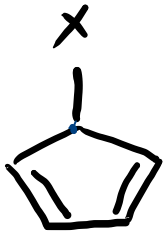
Note lg attached to sp^2 carbon do not SN^1 & SN^2.

(243) $CH_2=CH-X$ (sp^2) NO SN^1 NO SN^2

(244)

NO SN^1 (Bredt's rule)
NO SN^2 (Back side attack of $Nu^{\ominus}$ is not possible)

(245)



NO SN^1 ($\oplus$ Anti Aro)
NO SN^2
↳ (T.S has character of anti aromatic)

Compare Rate of S_N1

(246)

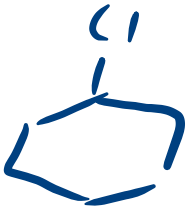
$\xrightarrow{NaOH}$ r1

$\xrightarrow{NaOH}$ r2

$CH_3-CH_2-Br \xrightarrow{NaOH}$ r3

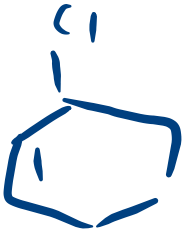
$r1 > r2 > r3$

(247)

(a) 

$\xrightarrow{CH_3OH,\ S_N1}$ r1

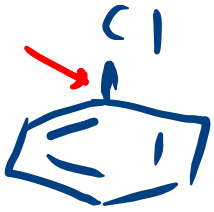
Methanolysis

(b) 

$\xrightarrow{"}$ r2

(c)

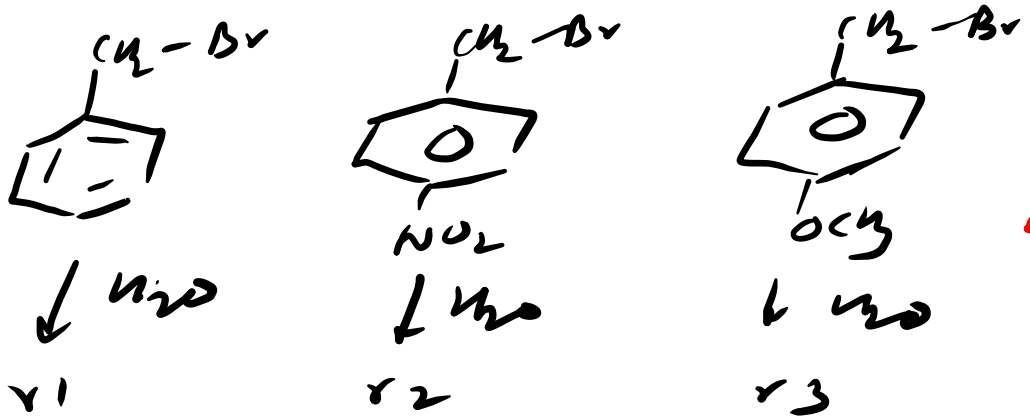
$\xrightarrow{"}$ r3

(d) 

$\xrightarrow{"}$ r4 S_N1 ✗

Rate of S_N1 $r3 > r2 > r1 > r4$

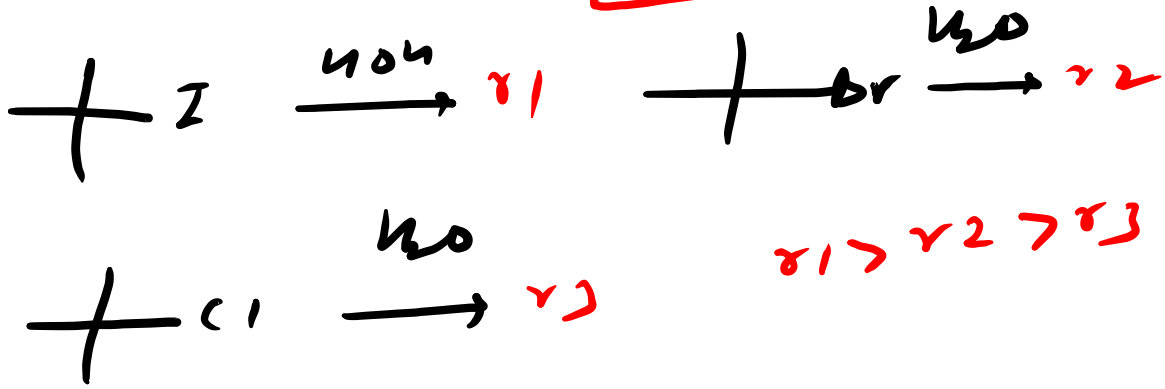
248



r1 r2 r3

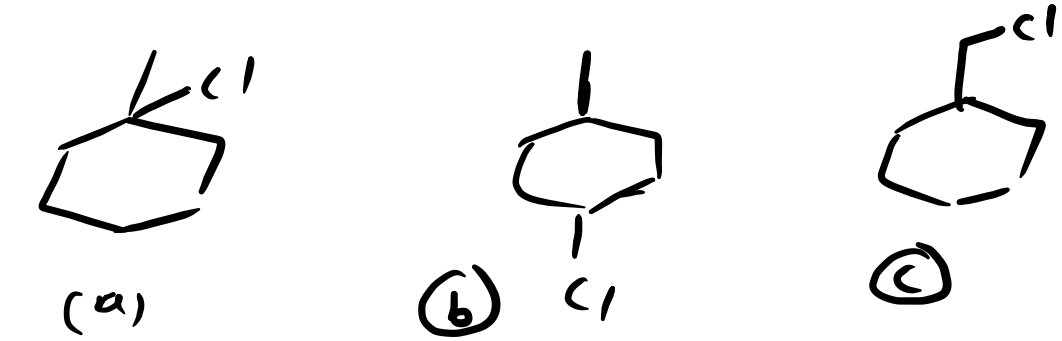
Rate of SN1 $r_3 > r_1 > r_2$

249



$r_1 > r_2 > r_3$

250

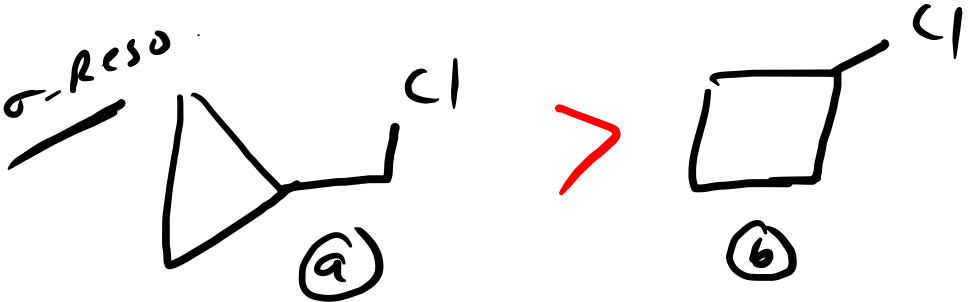


Rate of SN1 a > b > c

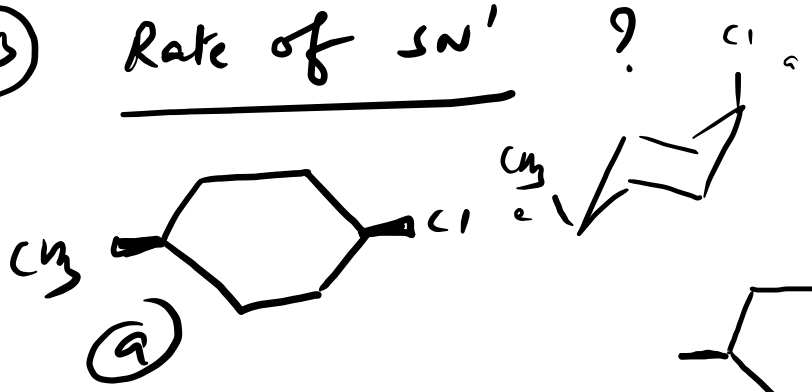
251

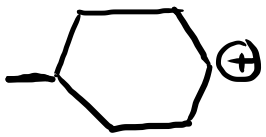
a > c > b (Rate of SN1)

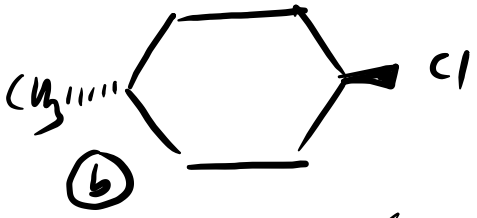
252

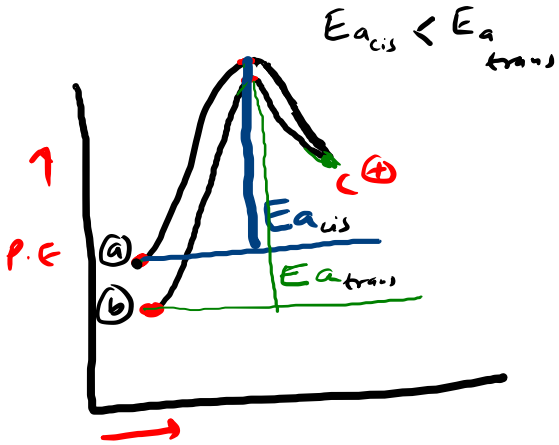


(253) Rate of SN1 ?



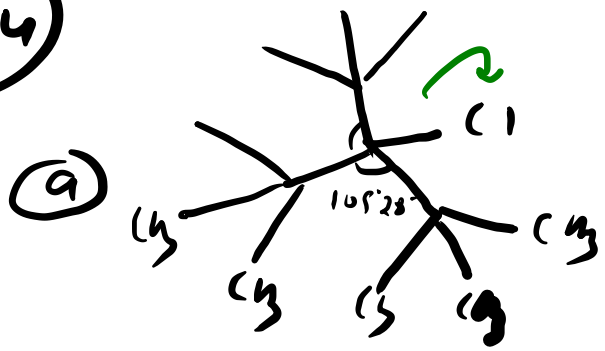


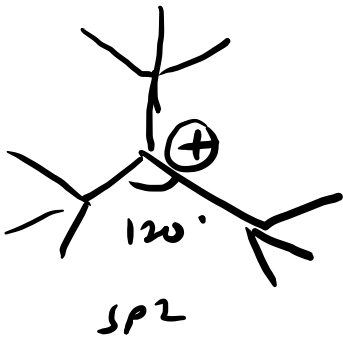


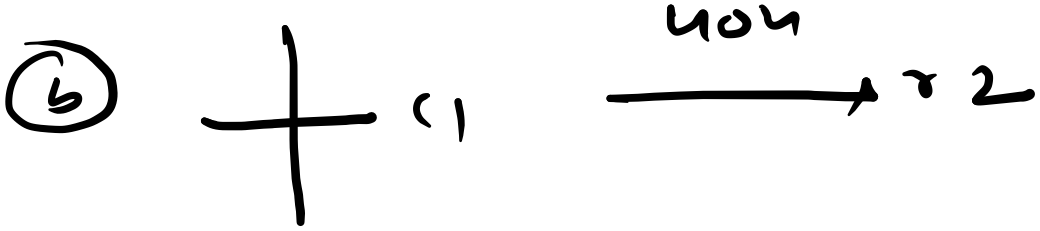


Rate of SN1 **a > b**

(254)



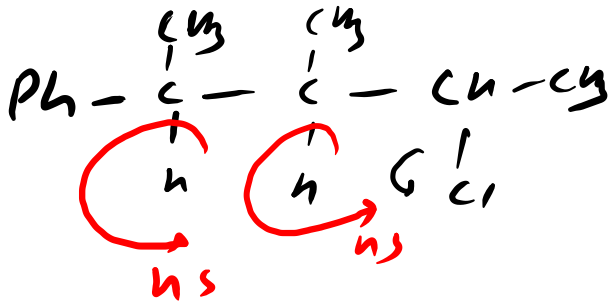




Rate of SN1 a > b.

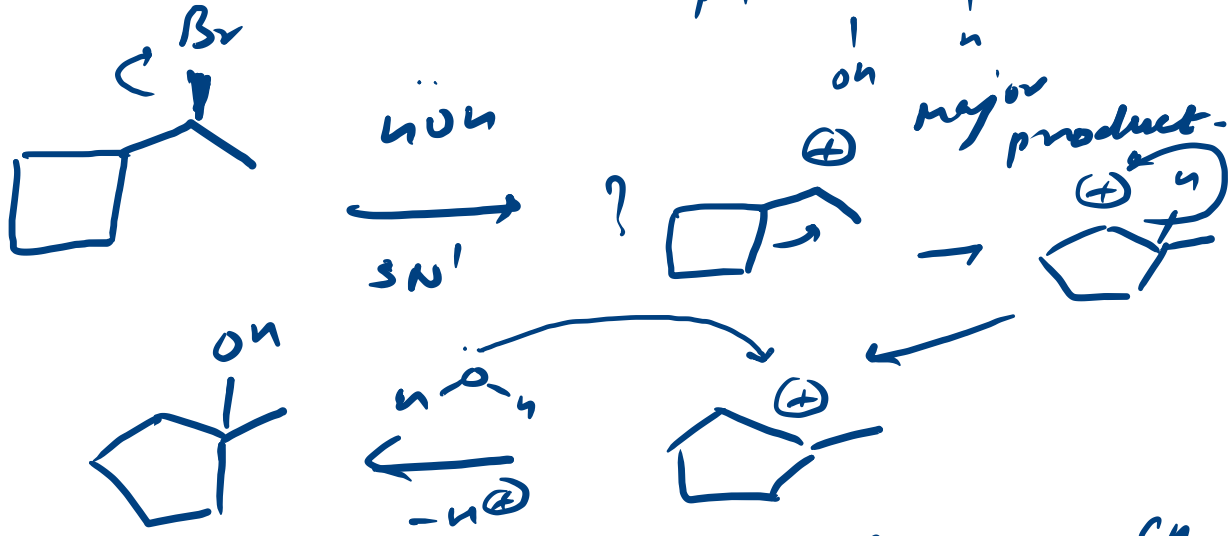
Steric repulsion

255



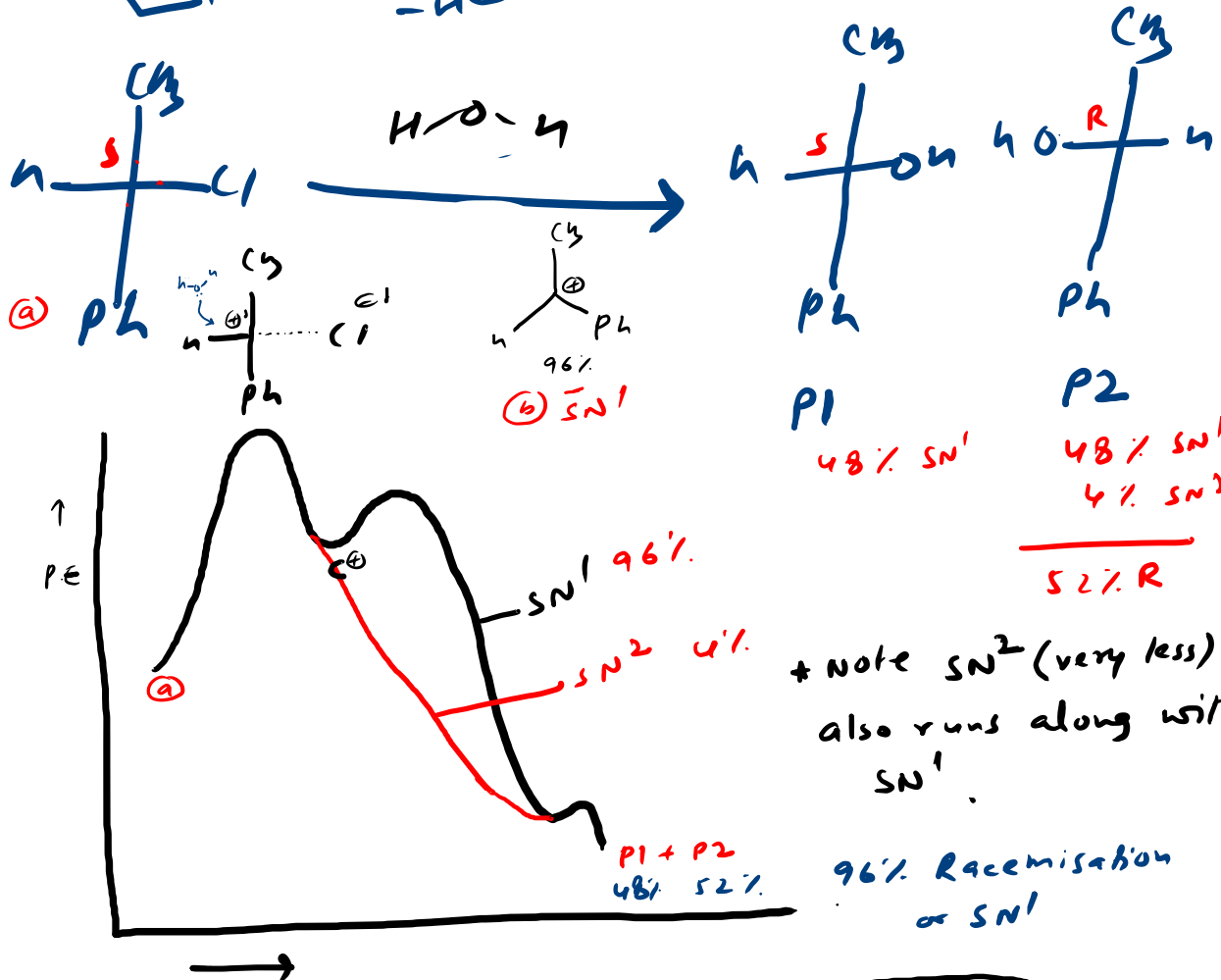
major product.

256



major product.

257



P1 48% SN1

P2 48% SN1 + 4% SN2 = 52% R

+ Note SN2 (very less) also runs along with SN1.

96% Racemisation of SN1

Racemisation & stability of ⊕ in SN1

Williamson Ether synthesis (WES) S_N2

* Reaction of alkyl halide with sodium alkoxide ($R-O^{\ominus}Na^{\oplus}$) gives Ether as product by S_N2 mechanism, which is known as WES.

Eq. (258) $R-\underset{\curvearrowright}{X} + R'-O^{\ominus}Na^{\oplus} \xrightarrow[S_N2]{WES} R'-O-R + NaX$

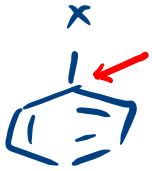
$X = -I, -Br, -Cl, -\cancel{F}$ Plg.

Leaving $Cl^- < Br^- < I^-$

Note: CH_3-I is the best alkyl halide to be used in WES.

Note certain Alkyl halide can not be use in WES.

Eq (259) $R-F$ S_N2 X

(260)  S_N2 X

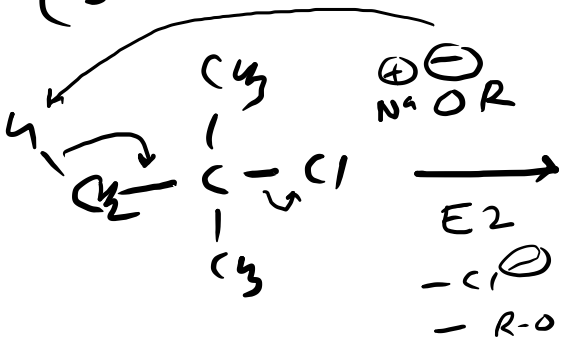
(261) $CH_2=CH-X$ (sp^2) S_N2 X

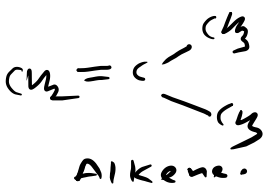
(262) S_N2 X

(263) S_N2 X

Note

* (264) 3° R–X also do not give WES (because it give alkene as major product by E-2 elimination)

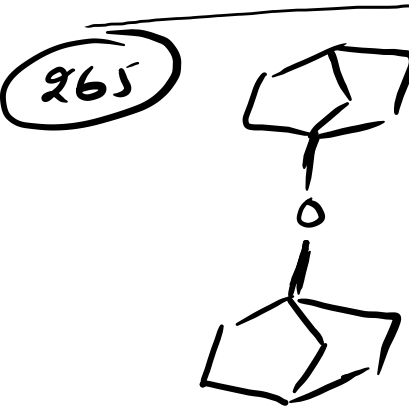


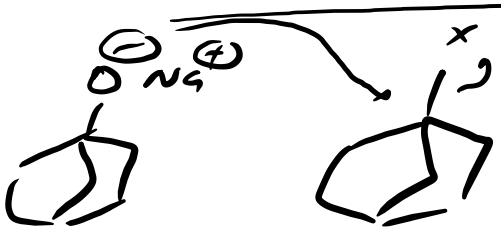


Alkene.

certain Ether can not be synthesize by WES

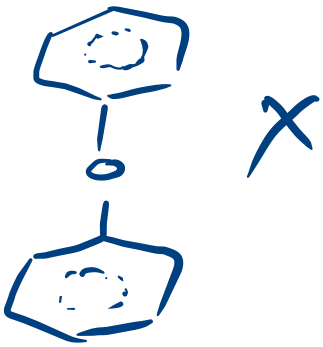
265





→ NO Reaction

266

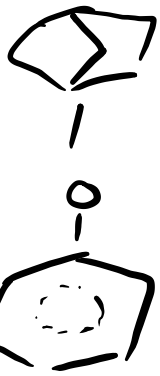


X

267

X

NO

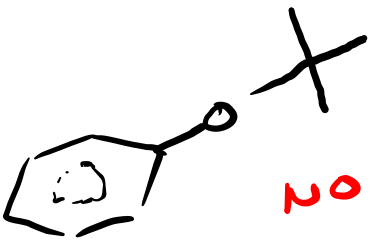


269

NO.

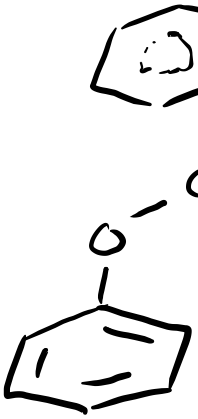
NO.

268



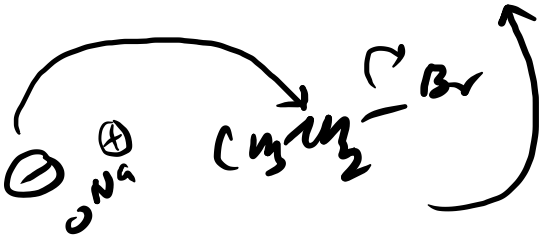
NO

270

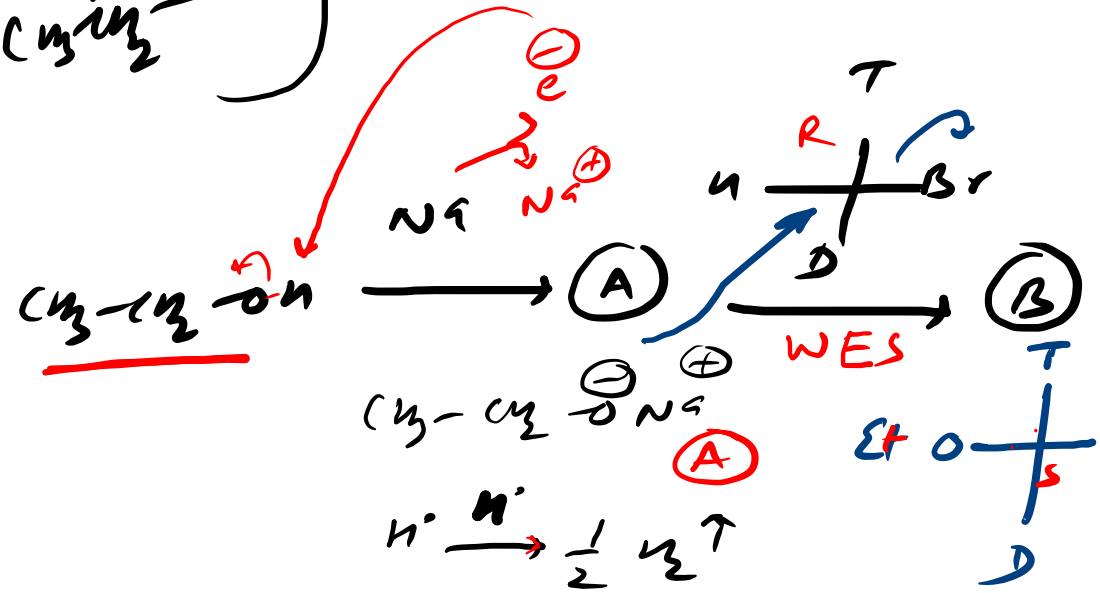


yes.

271



272



Lucas Test (LT)

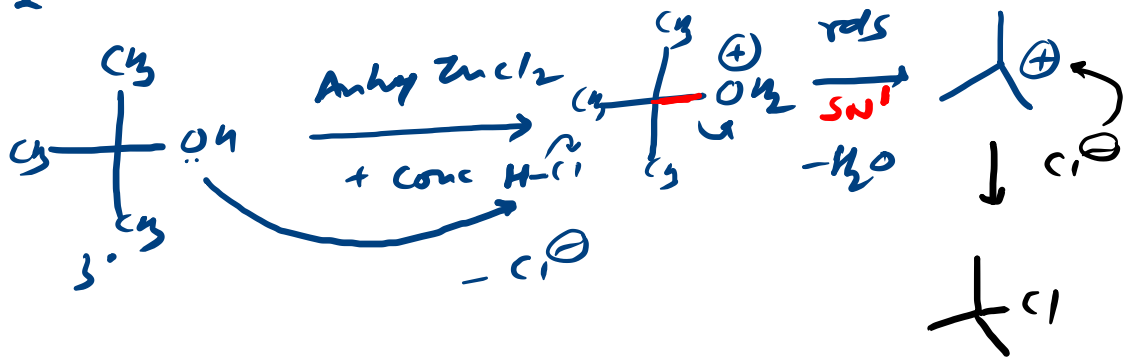
It is given by Lucas reagent

⇓

anhy. $ZnCl_2$ + conc. HCl

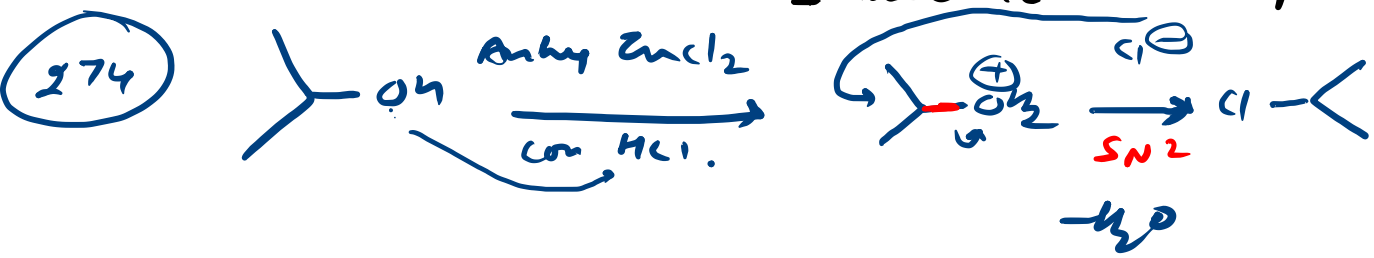
⇒ Lucas test is used to differentiate 1°, 2° and 3° Alcohols.

(273)



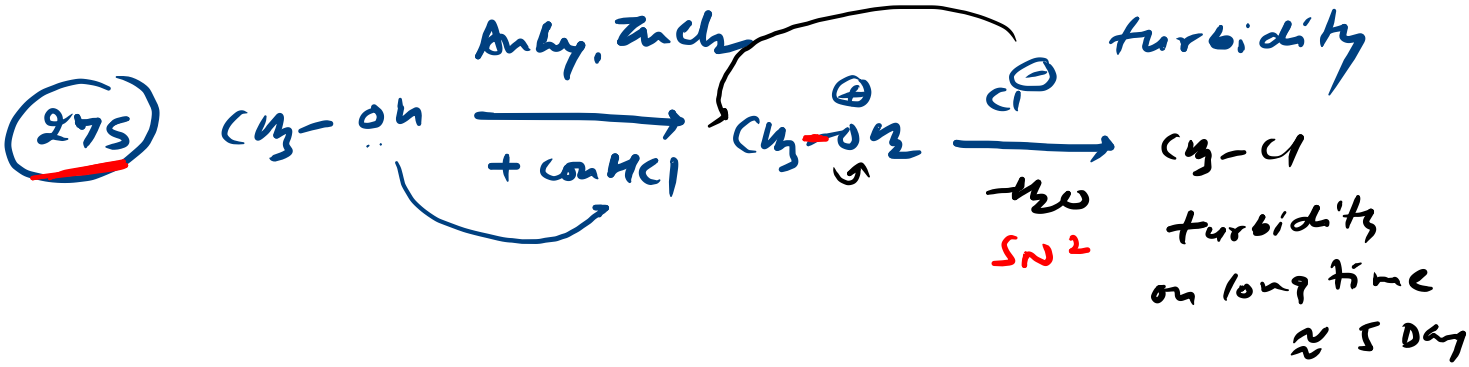
Immediate turbidity

(274)



Takes 5 to 10 mins for turbidity

(275)



turbidity on long time ≈ 5 Days.

or turbidity on heating.

Note: If alcohol gives resonance stabilized C⊕, Then it follow SN1 and gives Immediate turbidity

Note: The compounds which do not give SN1 & SN2 also do not give Lucas test.

HW Identify the mechanism (S_N1 or S_N2) also time taken for turbidity in lucas test.

276 Lucas Reagent (LR) ⟶

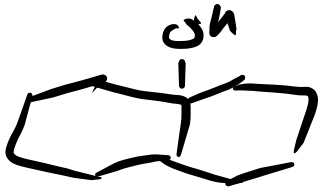
277 LR ⟶

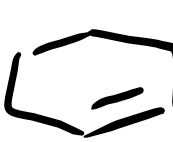
278 LR ⟶

279 LR ⟶

280 $CH_2=CH-OH$ LR ⟶

281 $CH_2=CH-CH_2-OH$ LR ⟶

282  LR ⟶

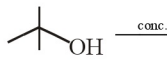
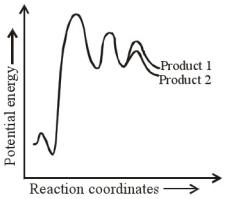
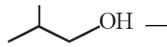
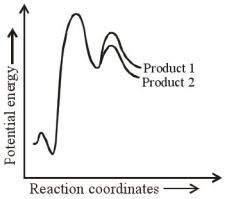
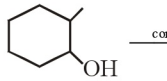
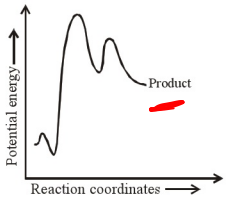
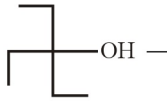
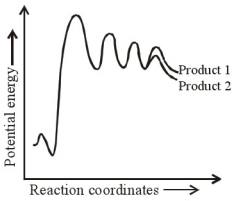
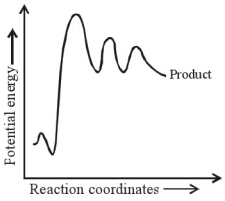
283  LR ⟶

284 CH_3-CH_2-OH LR ⟶

RACE # 19 — ORGANIC CHEMISTRY

M.M. : 47 — TIME : 30 Min.

1. Match the following reaction with their corresponding pattern of graph : **[8, 0]**

Column-I (Conversions)	**Column-II (Reagents)**
(A) 	(P) 
(B) 	(Q) 
(C) 	(R) 
(D) 	(S) 
	(T) 

2. **Statement 1 :** 1,1,1-trideutero-2-propanol reacts with conc. H_2SO_4 at high temperature to give only one alkene, 3,3,3-trideutero propene. **[3, –1]**

Statement 2 : C–D bond is stronger than C–H bond.

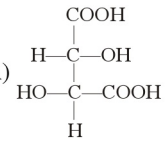
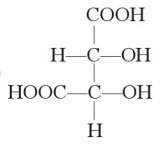
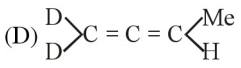
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

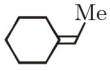
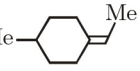
(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

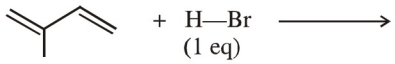
3. Identify optically active compound among following : **[3, –1]**

(A) 
(B) 
(C)
(D) 

4. Which of the following compound can show geometrical isomerism as well as optical isomerism? **[3, –1]**

(A)
(B) 
(C) 
(D)

5 Correct statement regarding product of reaction : **[3, –1]**



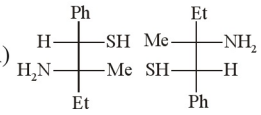
(A) Kinetic controlled product is a secondary bromide whereas thermodynamically controlled product is a tertiary bromide

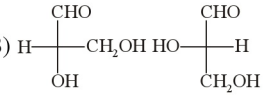
(B) Kinetic controlled product is a tertiary bromide whereas thermodynamically controlled product is a primary bromide

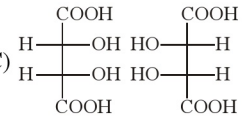
(C) Both kinetic & thermodynamic controlled product is secondary bromide

(D) Both kinetic & thermodynamic controlled product is a tertiary bromide

6. Which of the following is/are incorrect in the given isomeric pair : **[4, –1]**

(A)  (Enantiomer)

(B)  (Identical)

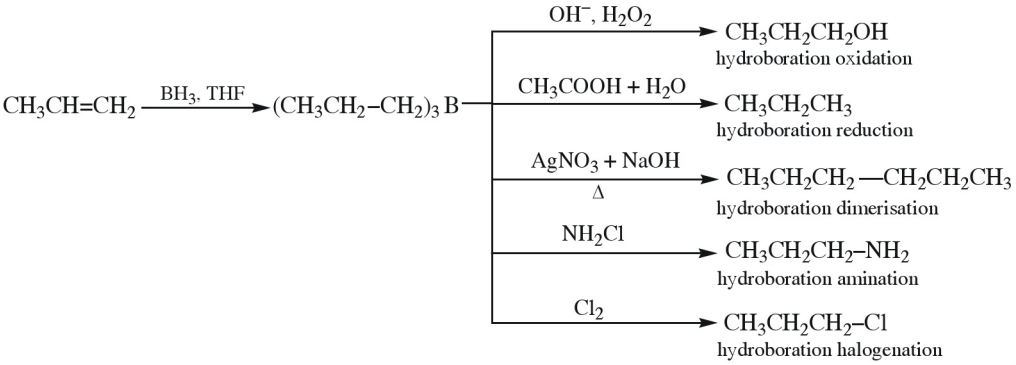
(C)  (Enantiomer)

(D) (Distereomer)

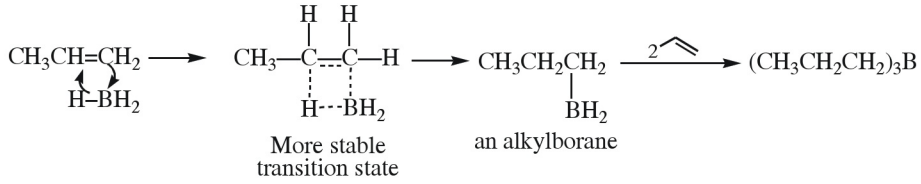
Read the following paragraph and answer the questiosn 7 to 9 given below :

HYDROBORATION-OXIDATION

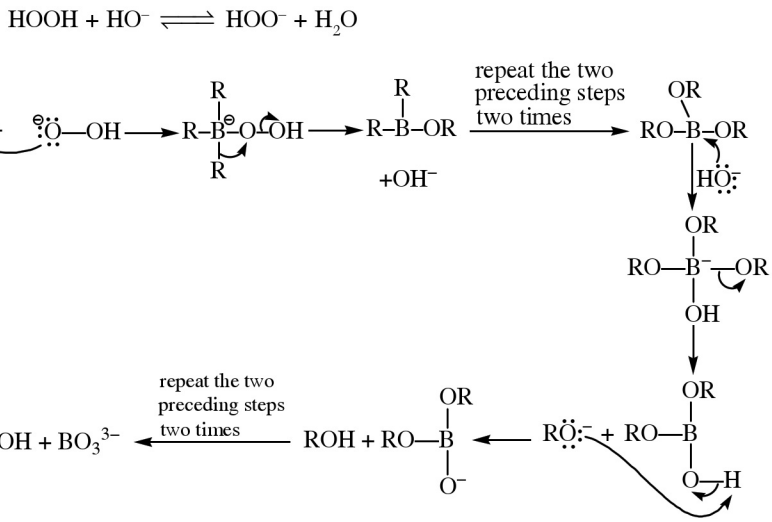
Hydroboration has been developed by brown as a reaction of tremendous synthetic utility because alkyl boranes are able to undergo a variety of transformation. Hydroboration is a one step, four centre, cis addition process in accordance with Markovnikov's rule but after oxidation it seems to be appear to violate Markovnikov's rule.



Mechanism of Hydroboration :



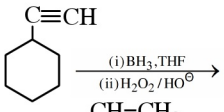
Mechanism of oxidation :

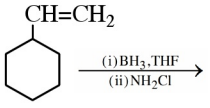


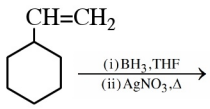
7. **I.** $CH_2 = CH_2 \xrightarrow[(ii)HO^-,H_2O_2]{(i)BH_3}$ **II.** $\xrightarrow[(ii)H_2O/AcOH]{(i)BH_3,THF}$ **III.** $\xrightarrow[(ii)HO^-,H_2O_2]{(i)BH_3}$ **[5,0]**

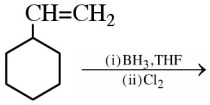
IV. $\xrightarrow[(ii)H_2O/AcOH]{(i)BH_3,THF}$ **V.** $\xrightarrow[(ii)CH_3COOH,H_2O]{(i)BD_3,THF}$ $\xrightarrow[(ii)\ CH_3COOD,D_2O]{(i)\ BH_3,\ THF}$ $\xrightarrow[(ii)\ CH_3COOD,D_2O]{(i)\ BT_3,\ THF}$

8. **[5, 0]**

I. 

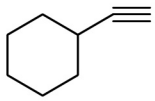
II. 

III. 

IV. 

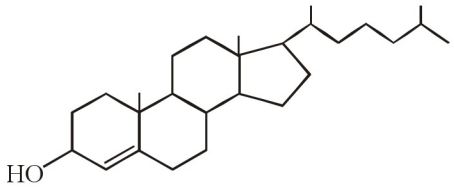
V. $CH_3–C\equiv CH \xrightarrow[(ii)AgNO_3+NaOH,\Delta]{(i)BH_3,THF}$

9. Which of the following will not produce same product when treated with dil. H_2SO_4 / $HgSO_4$ or when undergo hydroboration oxidation ? **[4, –1]**

(A)  (B) $CH_3–C\equiv CH$ (C) $Ph–C\equiv CH$ (D) $CH_3–C\equiv C–CH_3$

10(a). How many diastereomeric pair are possible with $CH_3–CH=CH–\underset{\displaystyle Br}{\underset{|}{CH}}–CH=CH–CH_3$. **[4, 0]**

10(b). Calculate total number of chiral centre present in following compound : **[4, 0]**



FILL THE ANSWER HERE

1.	A (P) (Q) (R) (S) (T) B (P) (Q) (R) (S) (T) C (P) (Q) (R) (S) (T) D (P) (Q) (R) (S) (T)	**2.**	(A) (B) (C) (D)	**3.** (A) (B) (C) (D) **4.** (A) (B) (C) (D)
5.	(A) (B) (C) (D)	**6.**	(A) (B) (C) (D)	
7.	**(I)**	**(II)**	**(III)**	**(IV)**
	(V)			
8.	**(I)**	**(II)**	**(III)**	**(IV)**
	(V)			
9.	(A) (B) (C) (D)			
10(a).	(0) (1) (2) (3) (4) (5) (6) (7) (8) (9)		**10(b).**	(0) (1) (2) (3) (4) (5) (6) (7) (8) (9)

276 S_N1 Immediate

277 LTX

278 S_N2 5 to 10 min.

279 LTX

280 LTX

281 S_N1 Immediate turbidity

282 S_N1 Immediate "

283 S_N1 Immediate "

284 S_N2 Long time for turbidity.

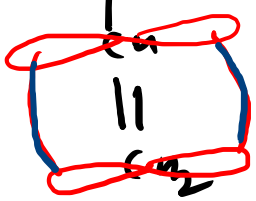
Special compounds for S_N2

285 compare Rate of S_N2

	compound		Rate of S_N2
(a)		KI / Acetone →	1
(b)	$CH_2=CH-CH_2-Cl$	" →	79 times
(c)	CH_3-Cl or	" →	200 times
(d)	CH_3-O-CH_2-Cl	" →	920 times
(e)	$Ph-\overset{O}{\overset{\|}{C}}-CH_2-Cl$	" →	1 lac times.

(285) Solution

(b)



SN2 ✓
SN1 ✓
(now)

T.S is stabilized by π electrons of p-orbital

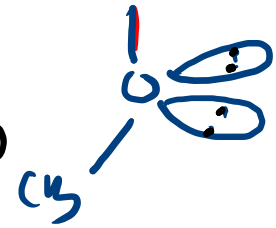
T.S

(c)

SN2 ✓
SN1 ✓
(now)

T.S is stabilized by π-electrons of p-orbital

(d)

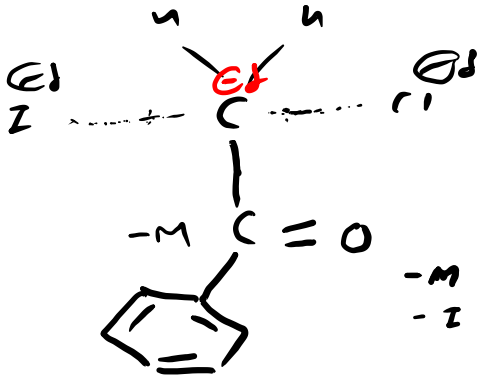


SN2 ✓
SN1 ✓
(π-e-n)

T.S is stabilized by ēles of Lone pair.

I Lac

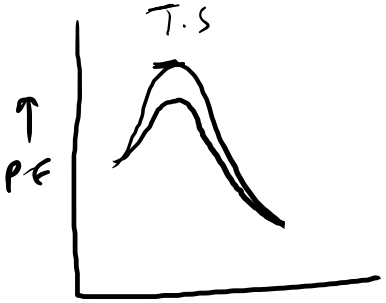
(e)



$-\overset{O}{\overset{\|}{C}}-$ -M

highly stabilized the TS of SN^2

-M
-I

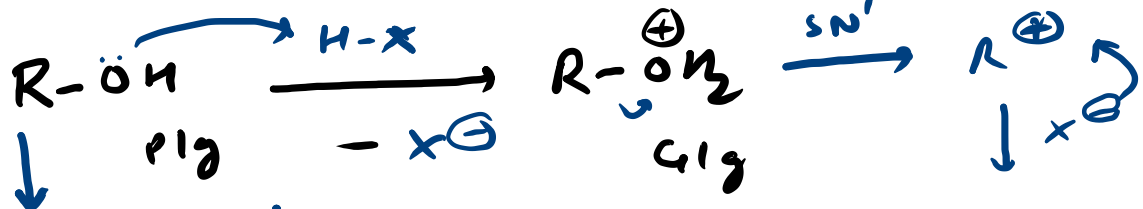


Note compound e gives exclusively SN^2 or do not give SN1

($Ph-\overset{O}{\overset{\|}{C}}-\overset{\oplus}{C}H_2$ unstable)

-M
-I

Reaction of Alcohol with H-X



$3^{\circ} \longrightarrow SN^1$

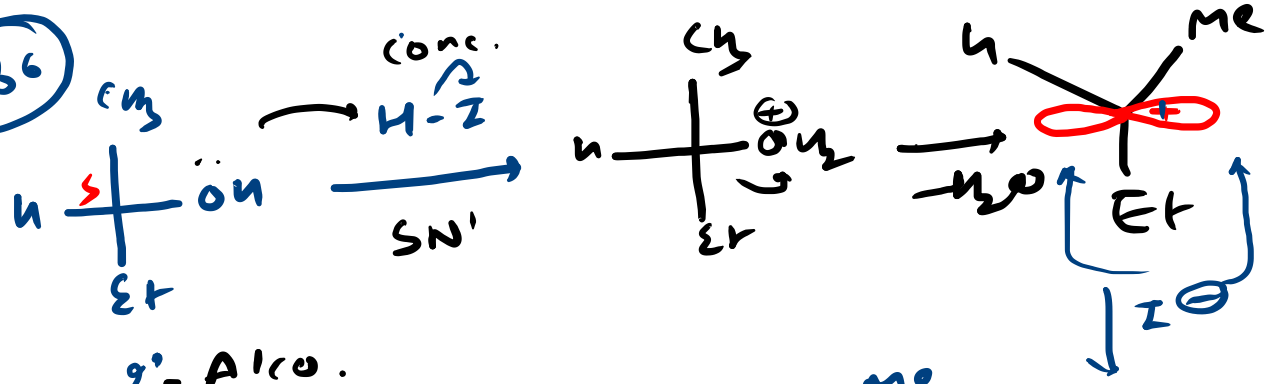
$2^{\circ} \longrightarrow SN^1$

$1^{\circ} \longrightarrow SN^2$

R = 3° or 2°

H-X (H-Cl, H-Br, H-I)

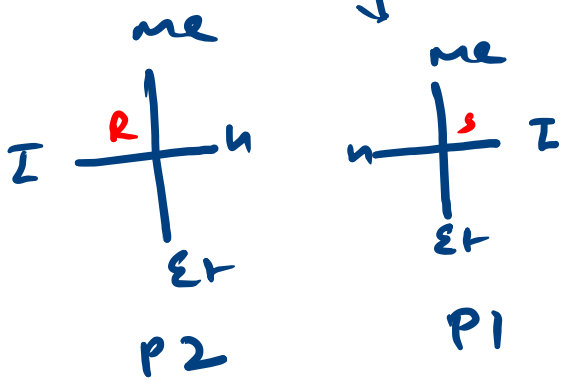
(286)



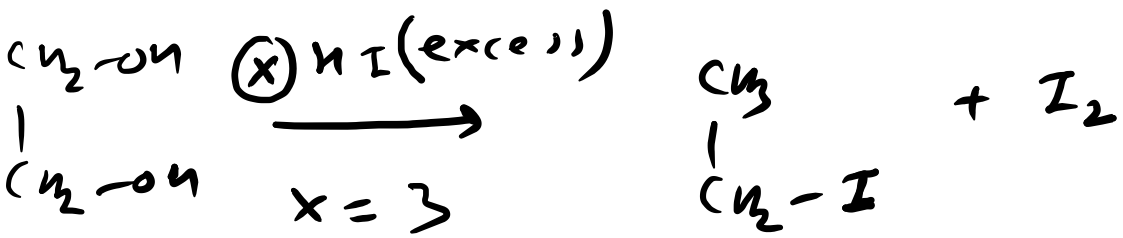
2°-Alco.

$P_2 \geq P_1$

product

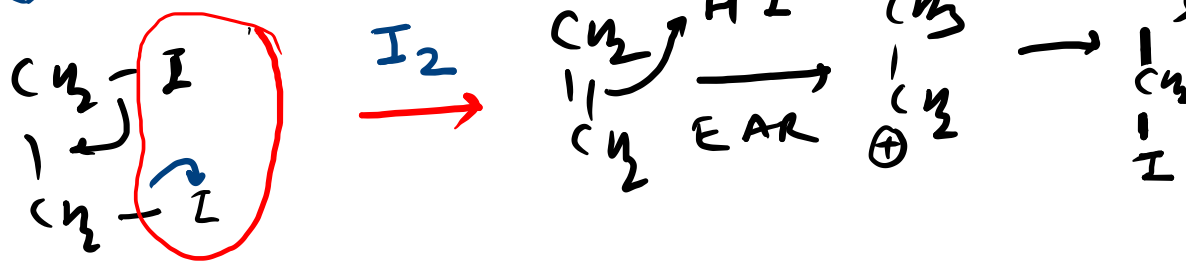


(287)



Mech

SN^2 ↓ 2HI



Reaction of Ether with HI

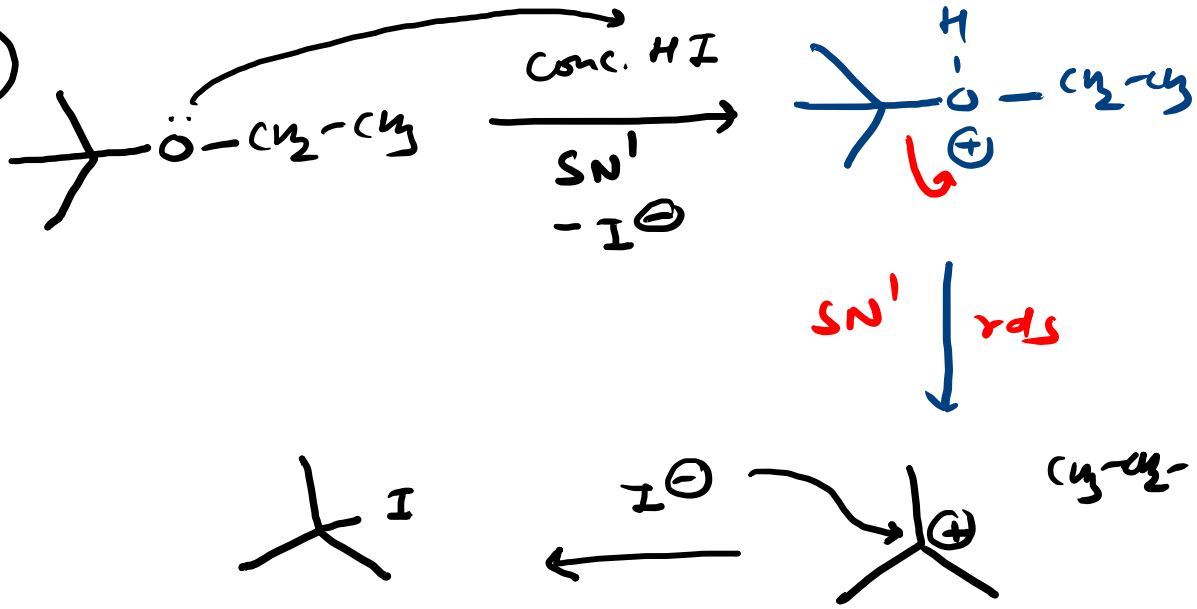
(288) $R-O-R$ —(Conc HI)→ $R-OH + R-I$

—(anhy. HI)→ $R-OH + R-I$

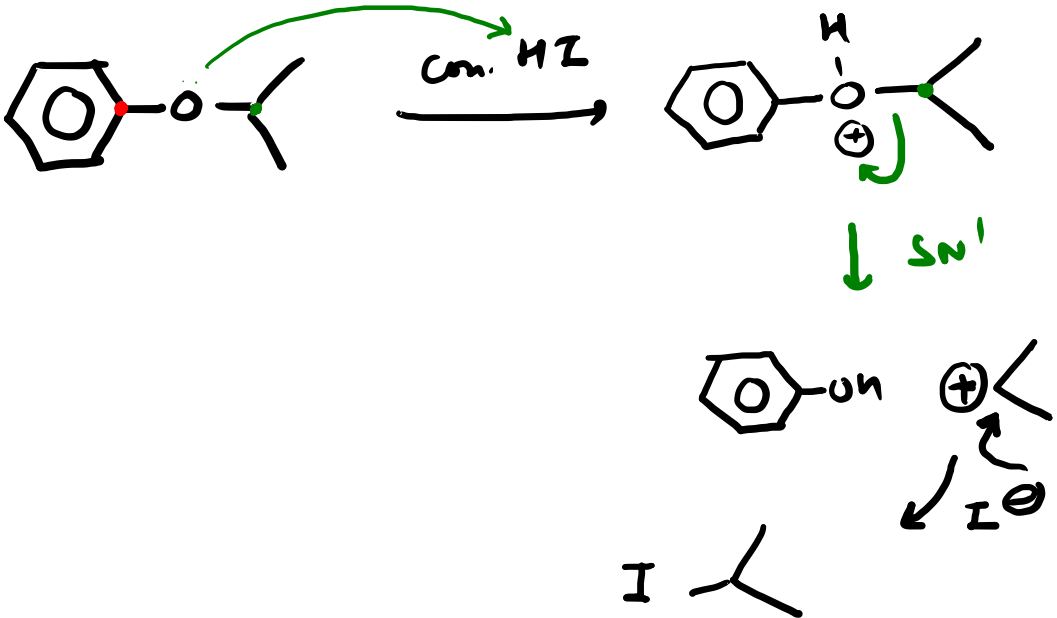
(289) $R-O-R$ —(conc. HI)→ $R-OH + R-I$

3°	3°	S_N1
3°	2°	
3°	1°	
2°	2°	
2°	1°	
1°	1°	→ S_N2

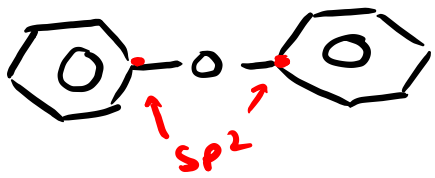
(290)



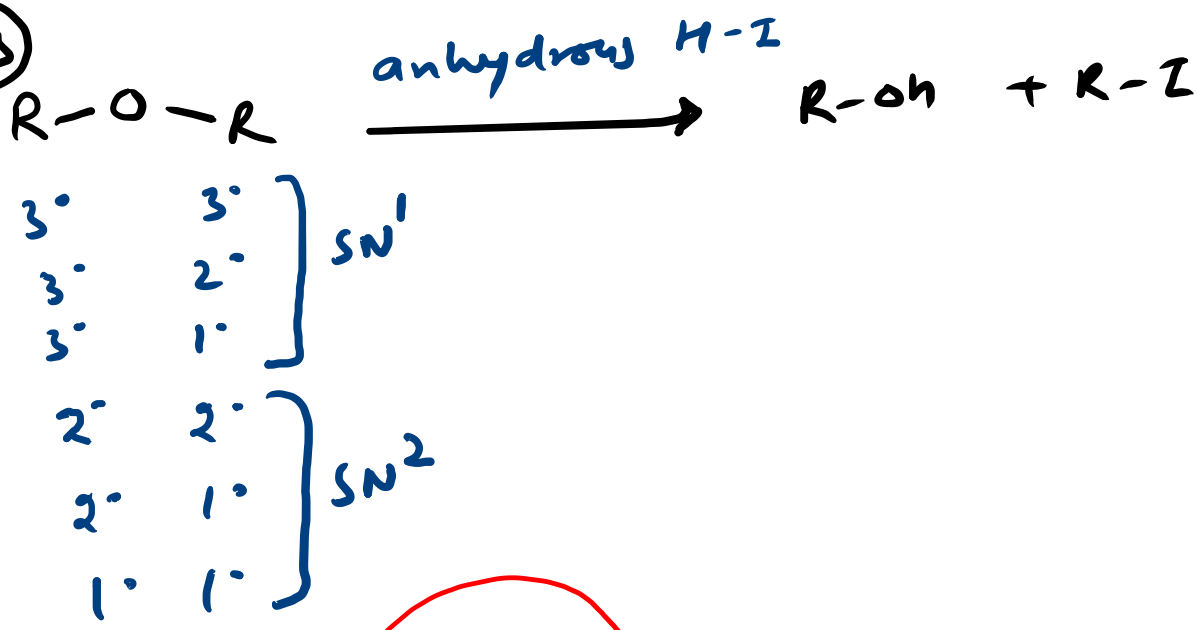
(291)



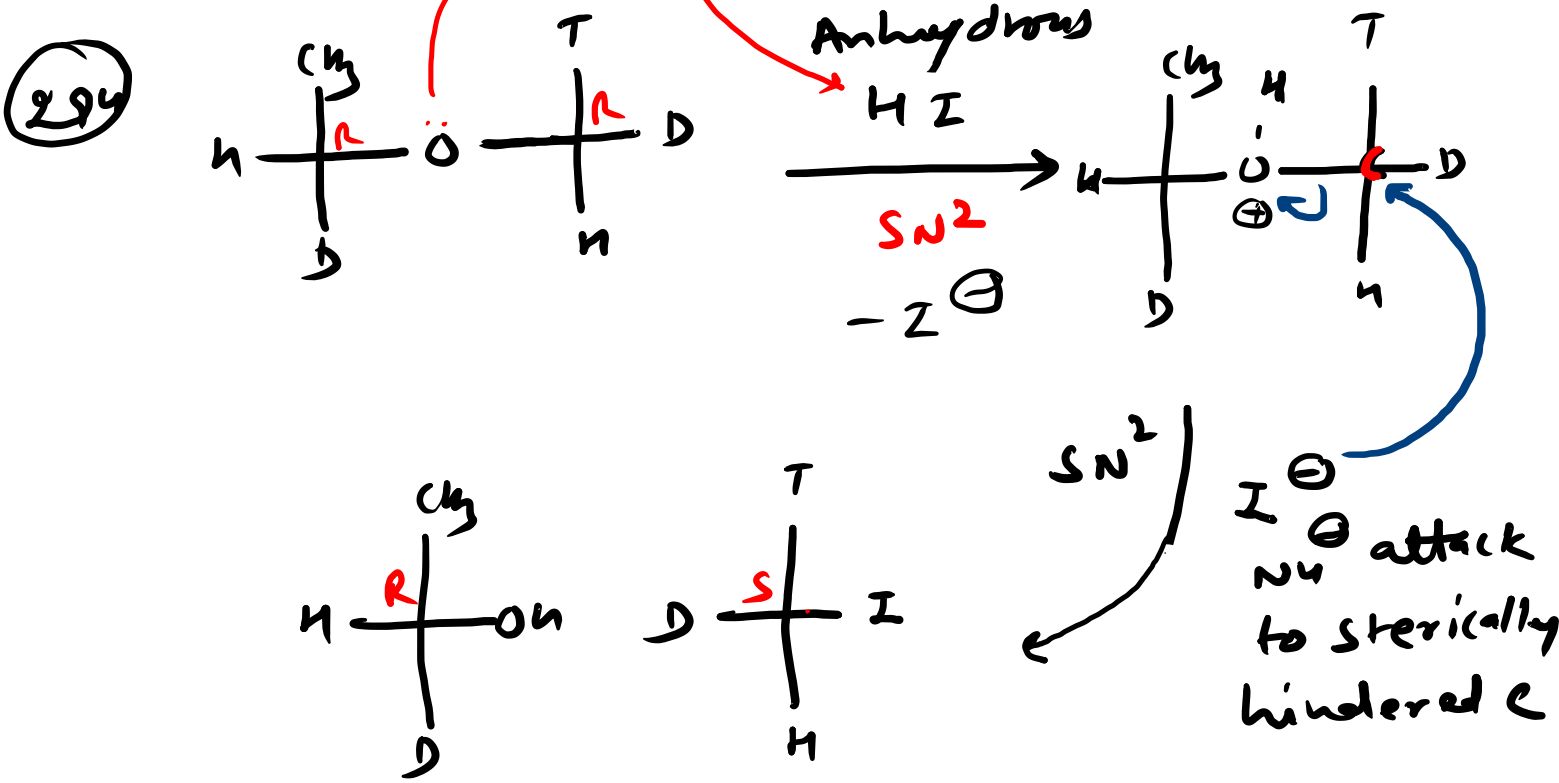
292


HI
No reaction.
sp^2

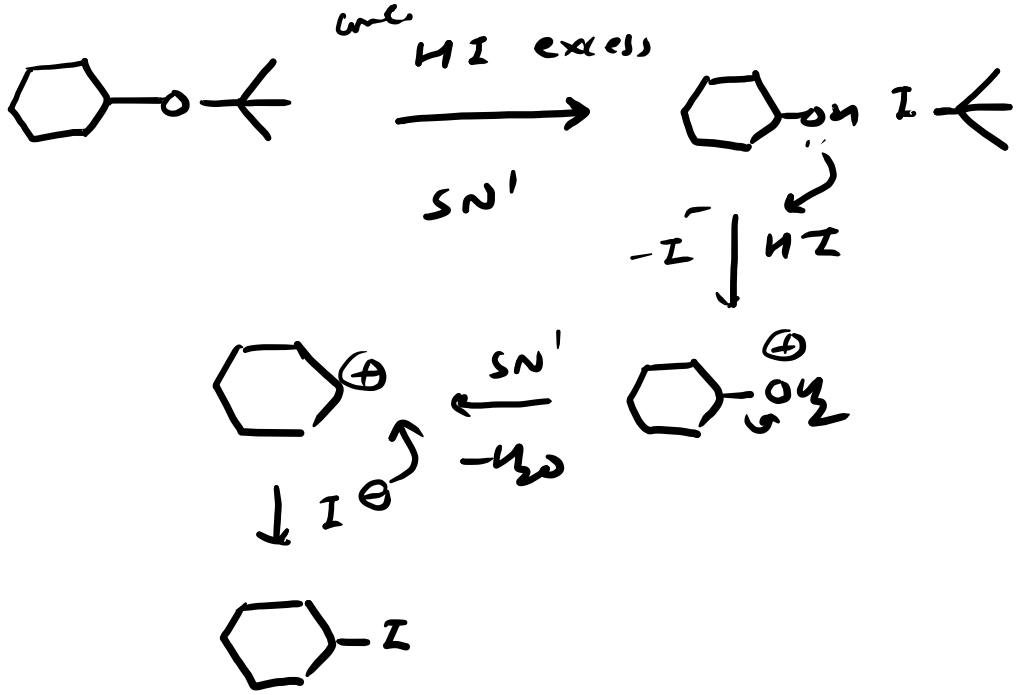
293


R–O–R
anhydrous H–I
R–OH + R–I
3° 3°
3° 2°
3° 1°
SN1
2° 2°
2° 1°
1° 1°
SN2

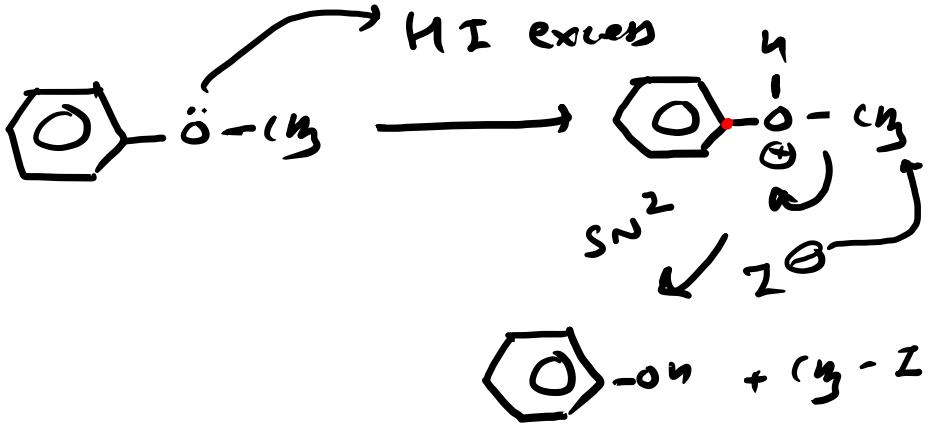
294


Anhydrous
HI
SN2
$-I^{\ominus}$
SN2
$I^{\ominus}$ attack
Nu to sterically
hindered C
R
S

(295)

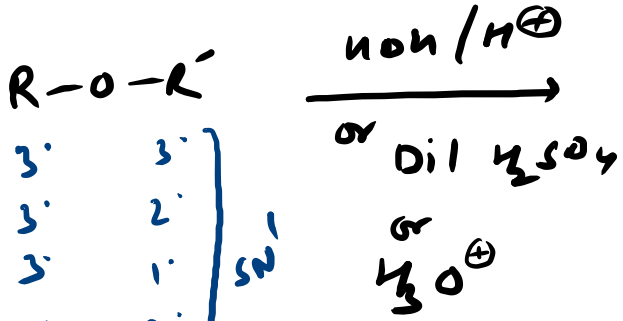


(296)

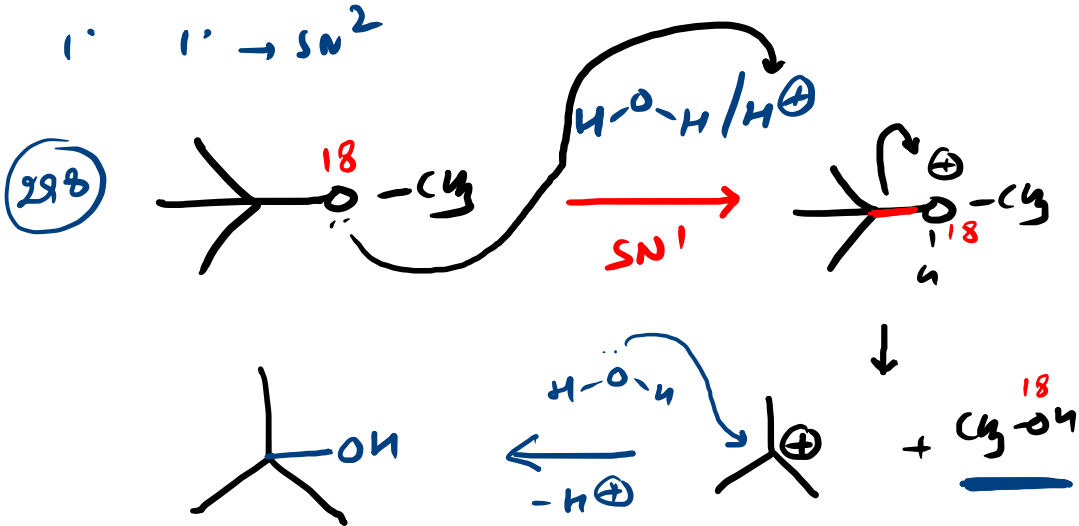


Acidic hydrolysis of Ether

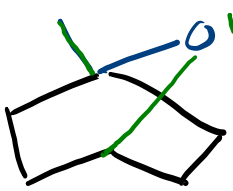
(297)



(298)



299 HW



$H_3O^{\oplus}$

300

$CH_2=CH-\ddot{O}-CH_3$ $\xrightarrow{H-O-H/H^{\oplus}}$

OH | HO

enol $CH_2=CH-OH$ ⇌ Tauto $HO-CH$

$CH_3-CH=O$

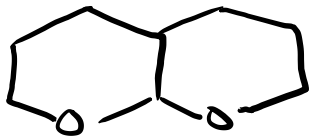
Gd $CH_2=CH-\ddot{O}-CH_3$ $\xrightarrow{H^{\oplus}}$ $CH_2-\overset{\oplus}{C}H-\ddot{O}-CH_3$ ⇌ $CH_3-CH=\overset{\oplus}{O}-CH_3$

$CH_3-CH=O$ + CH_3-OH ← $H-\ddot{O}-H$, $-H^{\oplus}$

$NH_2^{\ominus}$ → $CH_3-\overset{\oplus}{O}H_2$

HW

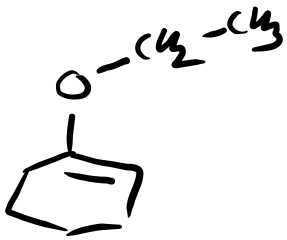
301



$HOH/H^{\oplus}$

HW

302



$H_3O^{\oplus}$ → ?

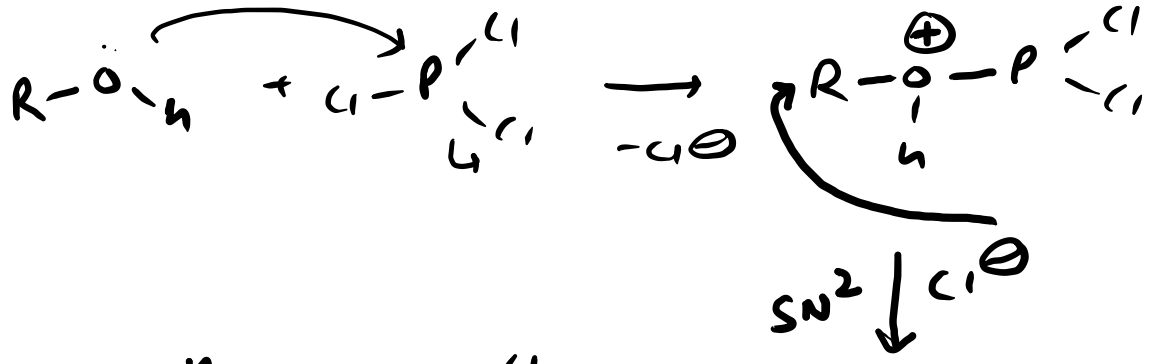
303

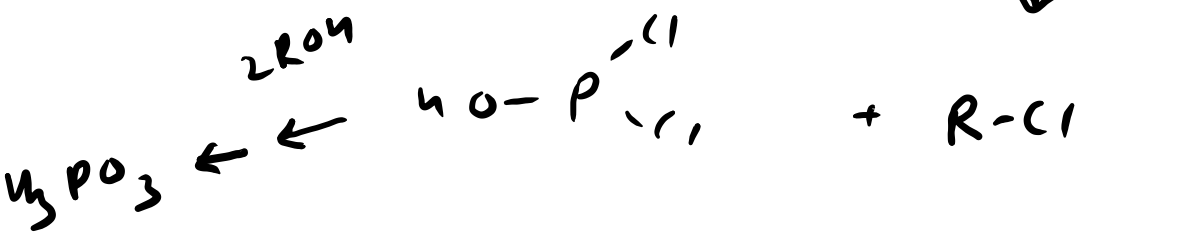
$H_3O^{\oplus}$ (excess) →

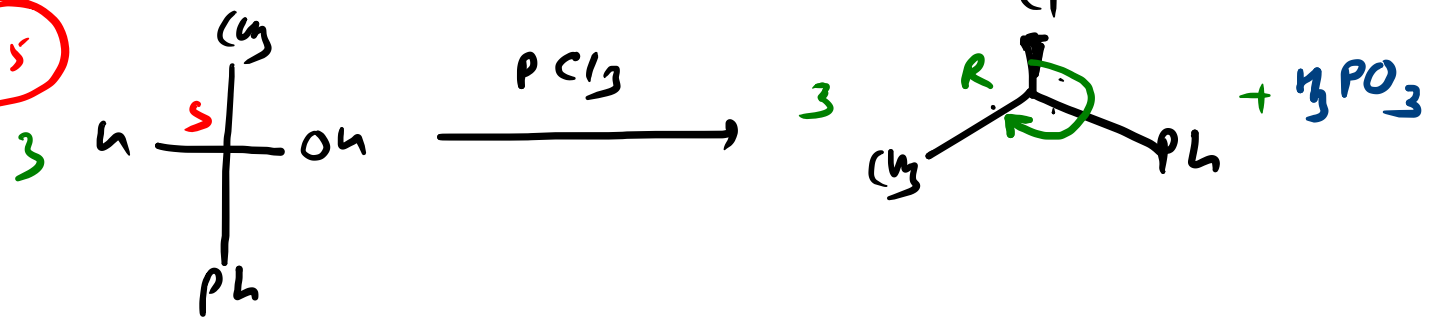
+ CH_3-OH + $Et-OH$

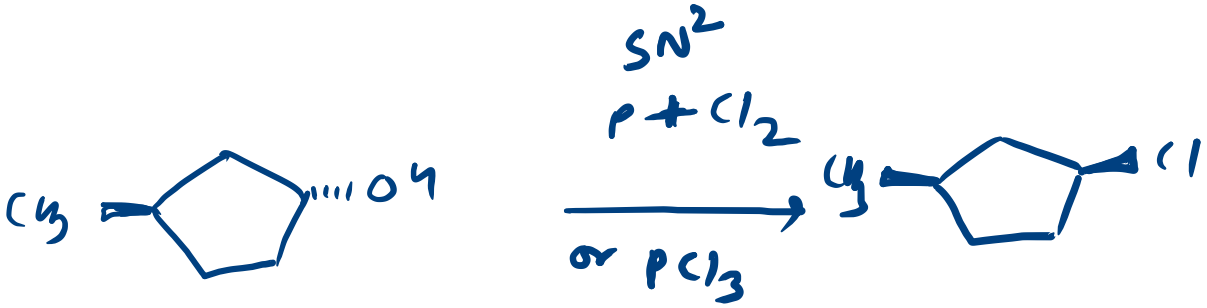
Reaction of alcohol with PCl_3 (SN^2)

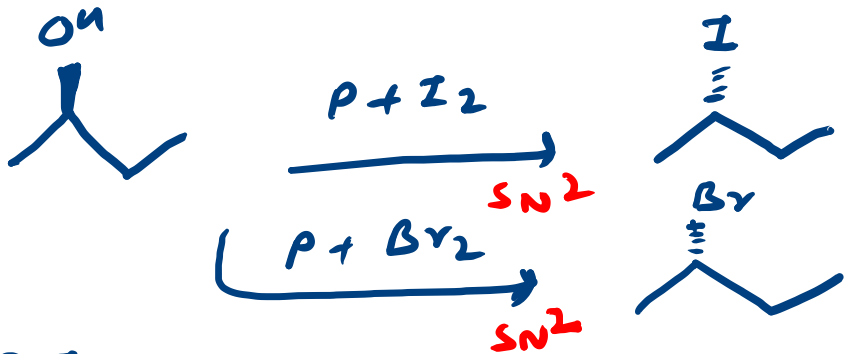
(304) $3\,R-OH + PCl_3 \xrightarrow{SN^2} 3\,R-Cl + H_3PO_3$

Mech 



(305) 

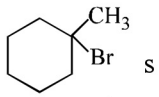
(306) 

(307) 

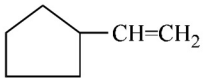
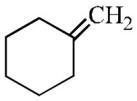
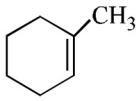
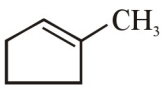
$2P + 3I_2 \rightarrow 2PI_3$

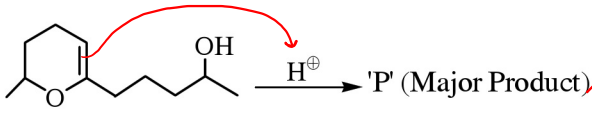
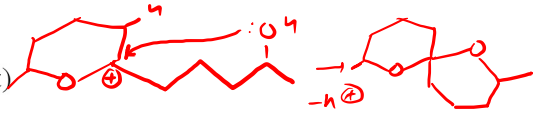
$2P + 3Br_2 \rightarrow PBr_3$

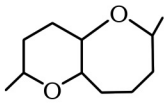
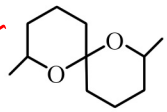
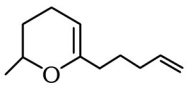
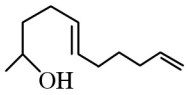
31. In the given reaction

C_7H_{12} (X) $\xrightarrow{HBr}$  as major product

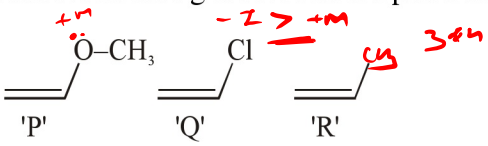
(X) can not be :

(A)  (B)  (C)  (D) 

32.  'P' (Major Product) 

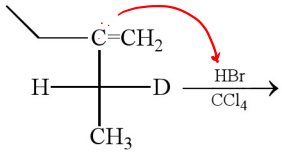
(A)  (B)  (C)  (D) 

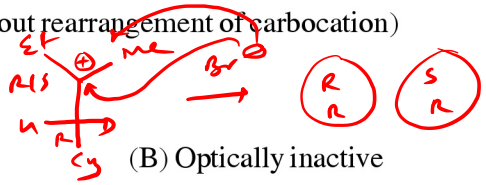
33. Arrange the following compounds in decreasing order of electrophilic addition :



'P' 'Q' 'R'

(A) P > Q > R (B) P > R > Q (C) R > P > Q (D) R > Q > P

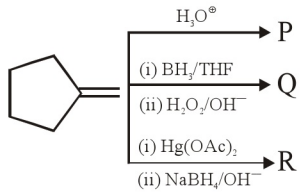
34.  Product (without rearrangement of carbocation)

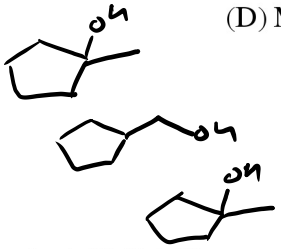


What is stereochemistry of product :

(A) Racemic mixture (B) Optically inactive

(C) Mixture of diastereomers (D) Meso product

35. 

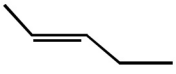


Correct statement regarding products P, Q & R

(A) Product P & R are same (B) Product Q & R are same

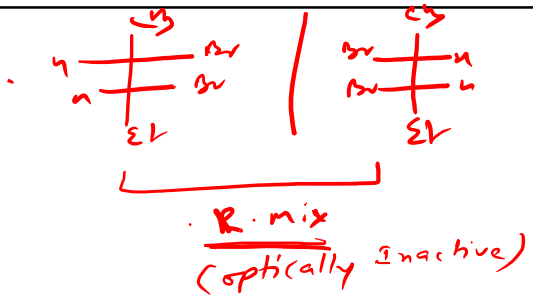
(C) P & Q are functional isomers (D) Product P, Q & R all are different

36. Select the incorrect statement about the product mixture in the following reaction :

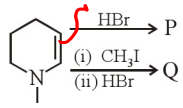
 Products

(A) It is optically active (B) It is racemic mixture

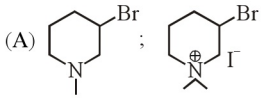
(C) It is a resolvable mixture (D) It is a mixture of erythro compounds

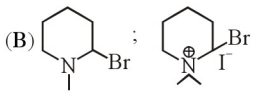


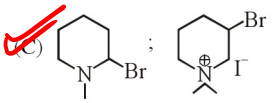
node06\B0B0-BA\Other center\JEE(Advanced)\Nurture\Physics\Module#01\Eng\01-Unit & Dimension, Basic Maths & Vector\01_Theory & Ex.

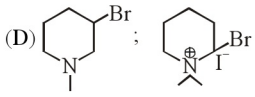
37. 

If P & Q are the major products then P & Q are respectively :

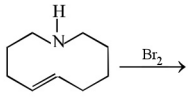
(A)  ;

(B) 

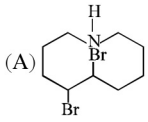
(C) 

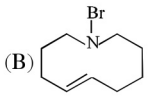
(D) 

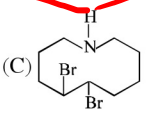
38. In the given reaction :

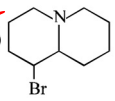


[X] , [X] is :

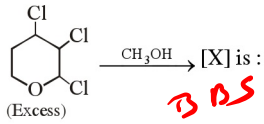
(A) 

(B) 

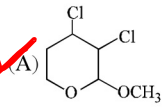
(C) 

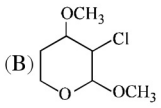
(D) 

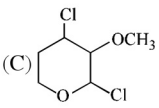
39. In the given reaction:

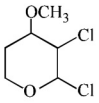


[X] is :

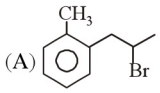
(A) 

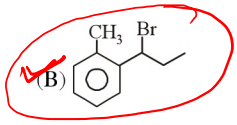
(B) 

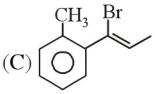
(C) 

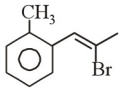
(D) 

40. Which compound undergoes hydrolysis by the S_N1 mechanism at the fastest rate?

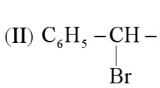
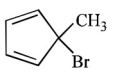
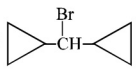
(A) 

(B) 

(C) 

(D) 

41. Arrange the following compounds in decreasing order of their reactivity for hydrolysis reaction

(I) $C_6H_5–CH_2–Br$ (II)  (III)  (IV) 

(A) I > II > III > IV (B) IV > II > I > III (C) III > IV > II > I (D) IV > III > II > I

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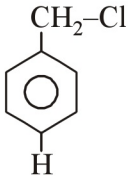
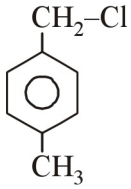
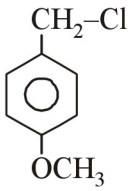
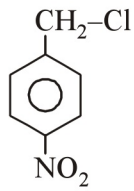
42. Which of the following is most reactive toward S_N1 reaction.

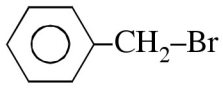
(A) 

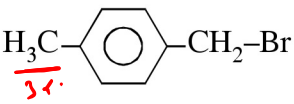
(B) 

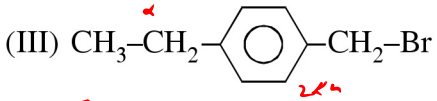
(C) 

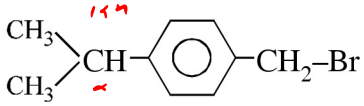
(D) 

43. Arrange the following compounds in order of decreasing rate of hydrolysis for S_N1 reaction:

(I) 

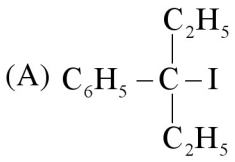
(II) 

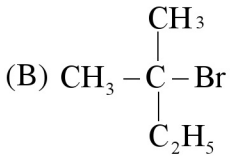
(III) 

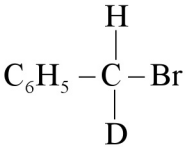
(IV) 

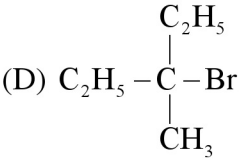
(A) II > III > IV > I (B) IV > III > II > I (C) III > IV > II > I (D) I > II > III > I

44. Which one of the following compounds will give enantiomeric pair on treatment with HOH?

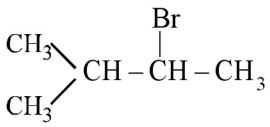
(A) 

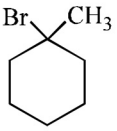
(B) 

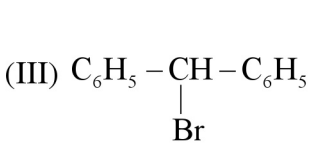
(C) 

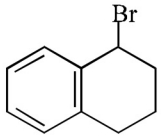
(D) 

45. Consider the S_N1 solvolysis of the following halides in aqueous formic acid:

(I) 

(II) 

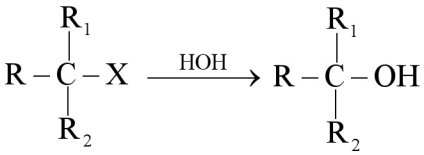
(III) 

(IV) 

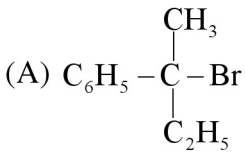
Decide decreasing order of reactivity of above alkyl halide?

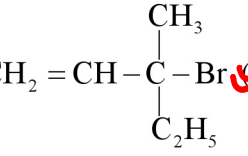
(A) III > IV > II > I (B) II > IV > I > III (C) I > II > III > IV (D) III > I > II > IV

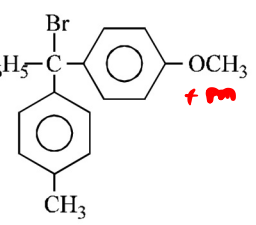
46 For the given reaction

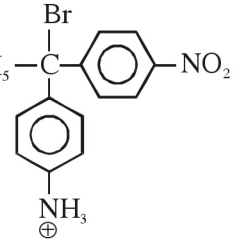


Which substrate will give maximum racemisation?

(A) 

(B) 

(C) 

(D) 

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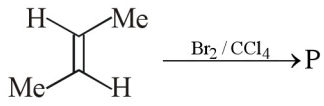
47. Select incorrect statements about the product (P) of the reaction :



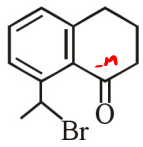
(A) P is optically inactive due to internal compensation

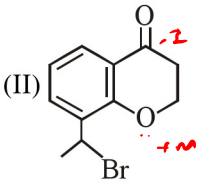
(B) P is optically inactive due to the presence of plane of symmetry in the molecule

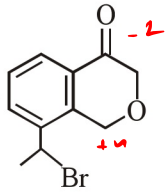
(C) The structure of P can have three optical isomers possible.

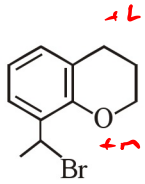
(D) P can have four possible optical isomers.

48. Consider the following molecules :

(I) 

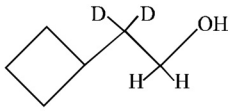
(II) 

(III) 

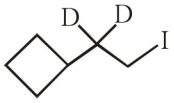
(IV) 

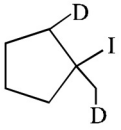
The correct decreasing ease of hydrolysis of alkyl halide is :

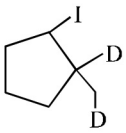
(A) II > III > IV > I (B) II > IV > III > I (C) II > I > III > IV (D) IV > II > III > I

49. 

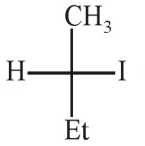
Major product is:

(A) 

(B) 

(C) 

(D) None of these

50.  Products. (If 96% racemisation takes place)

Find out the correct statement about the reaction.

(A) Among the products 48% S and 48% R configuration containing molecules are present

(B) Among the products 50% S and 50% R configuration containing molecules are present

(C) Among the products 48% S and 52% R configuration containing molecules are present

(D) Among the products 52% S and 48% R configuration containing molecules are present

51. In the given reaction the product [P] can be :

$CH_3–CH=CH–CH_2–OH \xrightarrow[S_N1']{HBr} [P]$

(A) $CH_3–CH=CH–CH_2–Br$

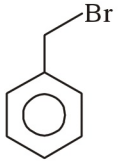
(B) $CH_3-\underset{\displaystyle Br}{\underset{|}{C}}H-CH=CH_2$

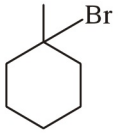
(C) $CH_2=CH–CH=CH_2$

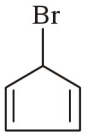
(D) $CH_3-\underset{\displaystyle Br}{\underset{|}{C}}H-CH_2-CH_2-OH$

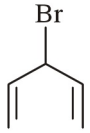
node06\B0B0-BA\Other center\JEE(Advanced)\Nurture\Physics\Module#01\Eng\01-Unit & Dimension, Basic Maths & Vector\01_Theory & Ex.

52. Which of the following can not give S_N1 reaction easily?

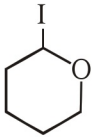
(A) 

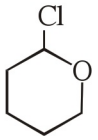
(B) 

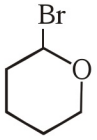
(C) 

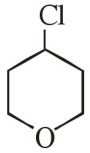
(D) 

53. Which one of the following compounds will be most reactive for S_N1 reactions ?

(A) 

(B) 

(C) 

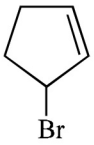
(D) 

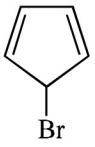
54. Which of the following compounds is most rapidly hydrolysed by S_N1 mechanism ?

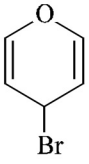
(A) C_6H_5Cl (B) $Cl–CH_2–CH=CH_2$

(C) $(C_6H_5)_3CCl$ (D) $C_6H_5CH_2Cl$

55. Among the bromides I–III given below, the order of reactivity in S_N1 reaction is:

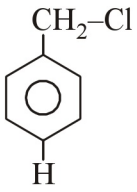
(I) 

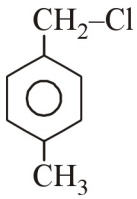
(II) 

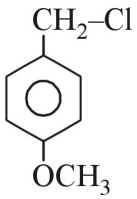
(III) 

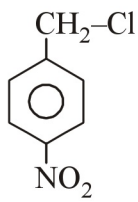
(A) III > I > II (B) III > II > I (C) II > III > I (D) II > I > III

56. Which of the following is most reactive toward S_N2.

(A) 

(B) 

(C) 

(D) 

57. For reaction $CH_3Br + OH^- \longrightarrow CH_3OH + Br^-$

the rate of reaction is given by the expression :

(A) Rate = k $[CH_3Br]$ (B) Rate = k $[OH^-]$

(C) Rate = k $[CH_3Br][OH^-]$ (D) Rate = k $[CH_3Br]^o$ $[OH^-]^o$

58. Select suitable reason for non-occurence of the following reaction.

$Br^- + CH_3OH \longrightarrow BrCH_3 + OH^-$

(A) Attacking nucleophile is stronger one

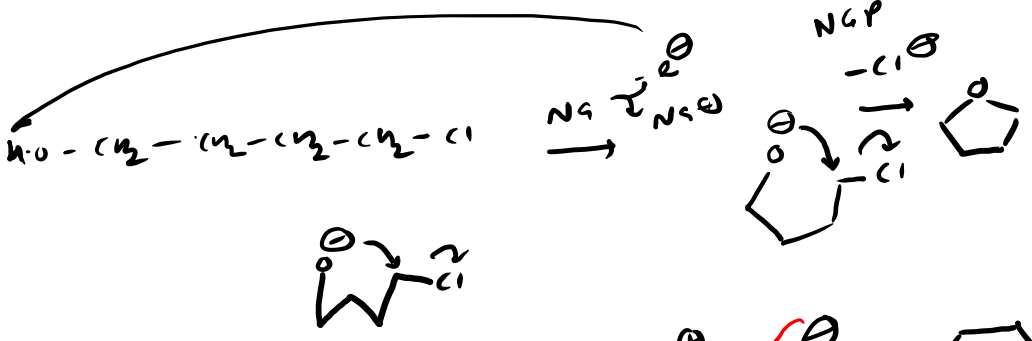
(B) Leaving group is a stronger base than nucleophile

(C) Alcohols are not good substrate for S_N reaction

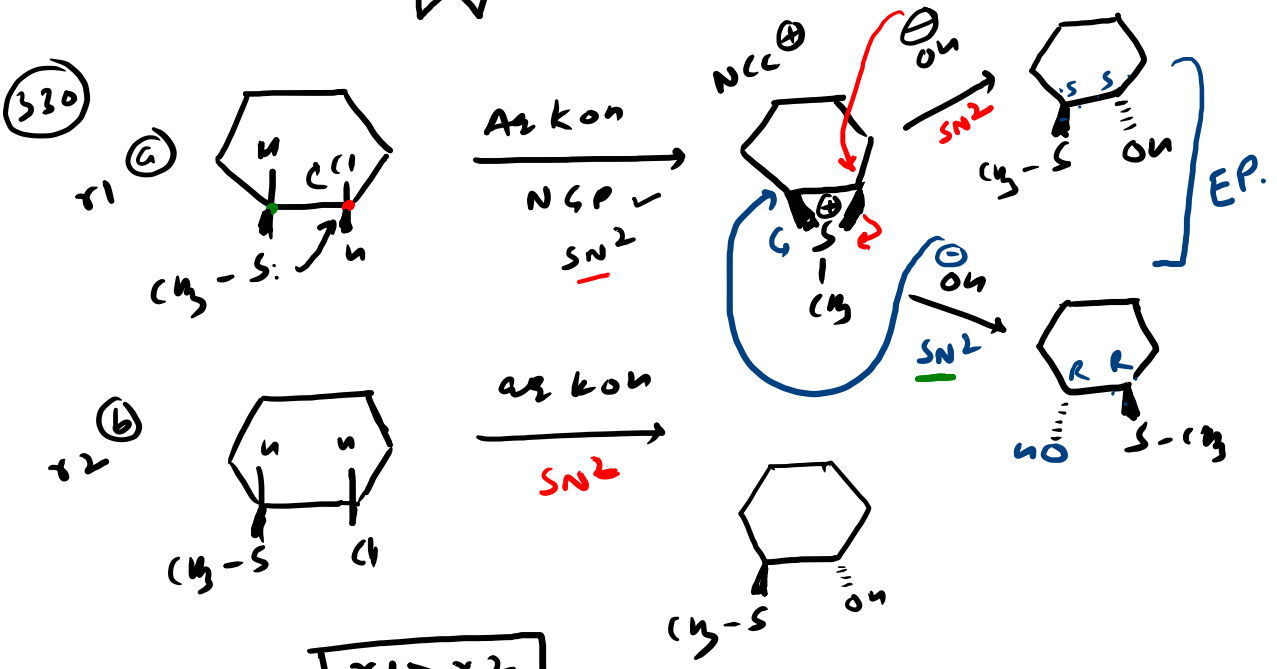
(D) Hydroxide ions are weak bases

node06\B0B0-BA\Other center\JEE(Advanced)\Nurture\Physics\Module#01\Eng\01-Unit & Dimension, Basic Maths & Vector\01_Theory & Ex.

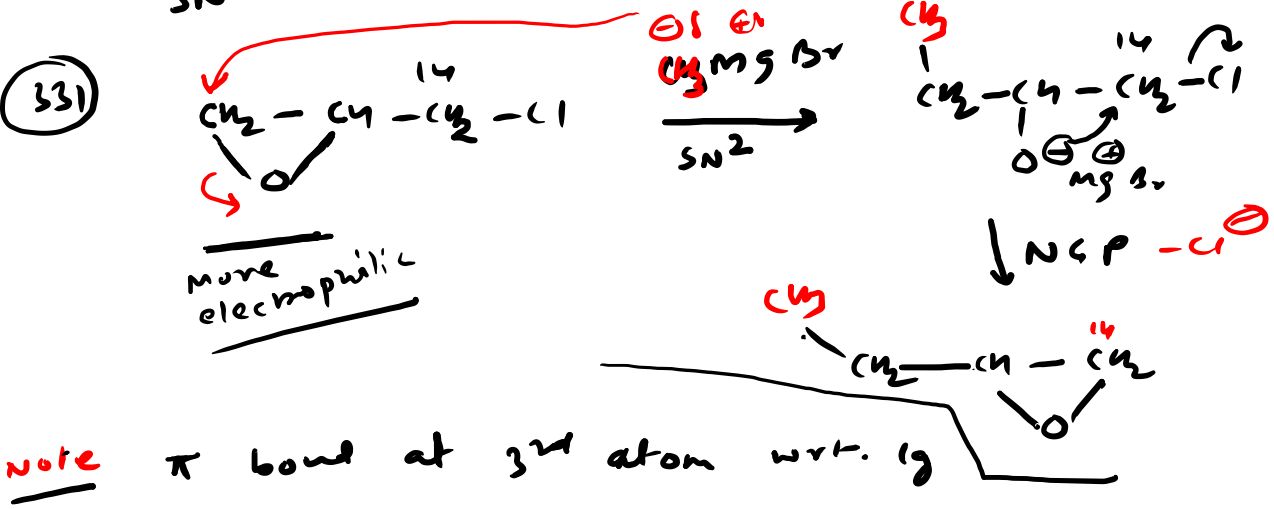
329



330

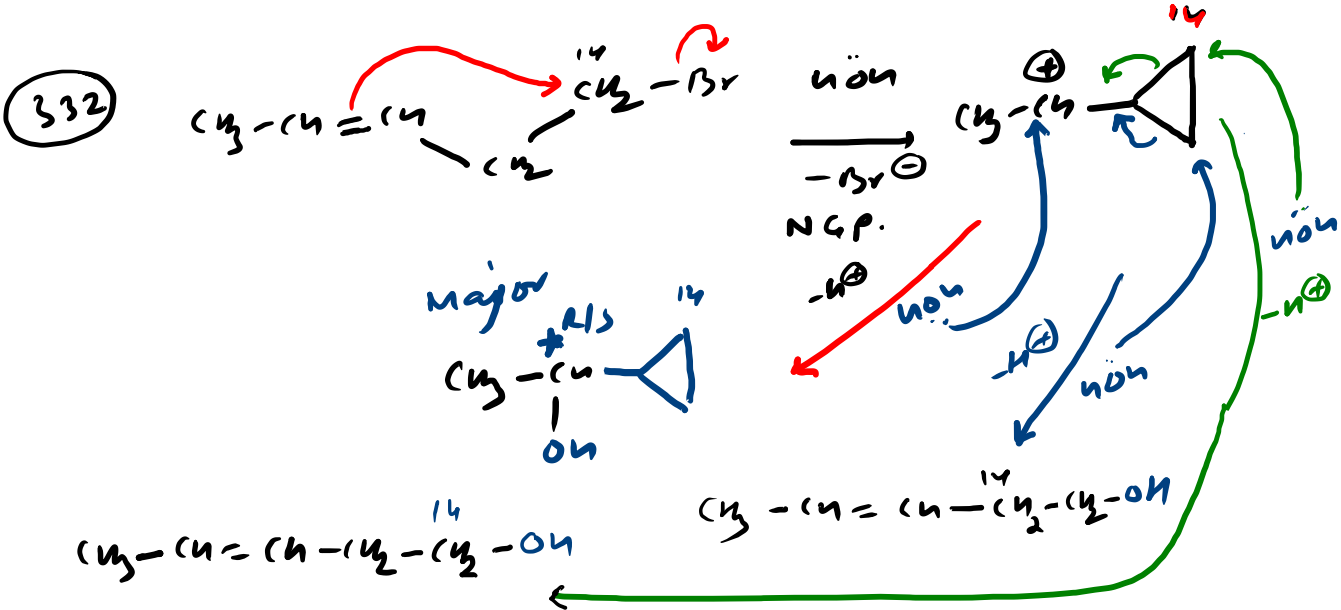


331

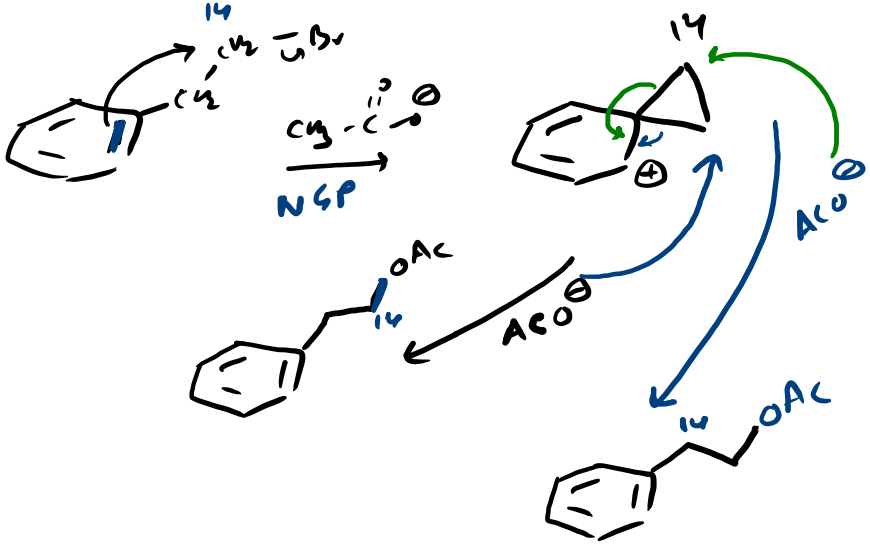


Note π bond at 3rd atom wrt. Cl also gives NGP step.

332

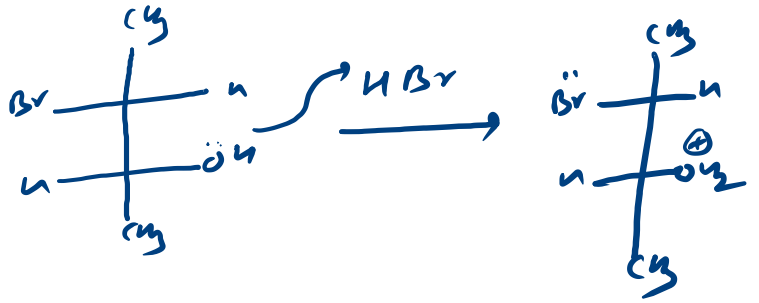


333



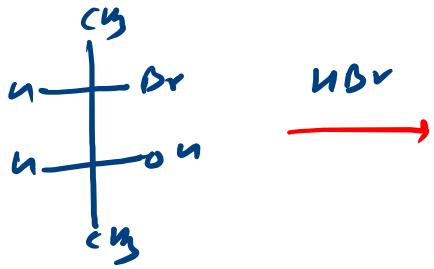
HW

333



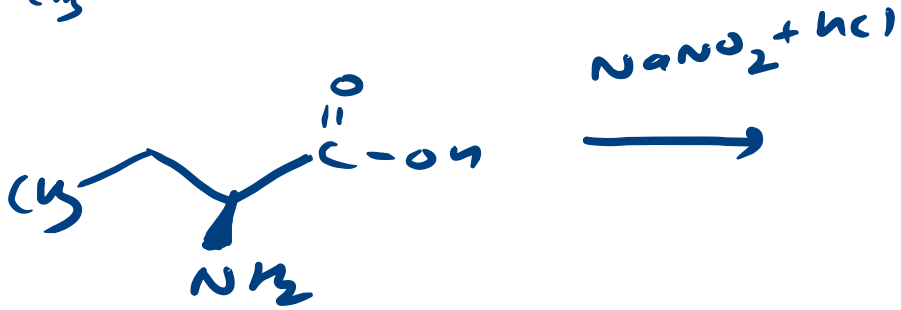
HW

334



HW

335



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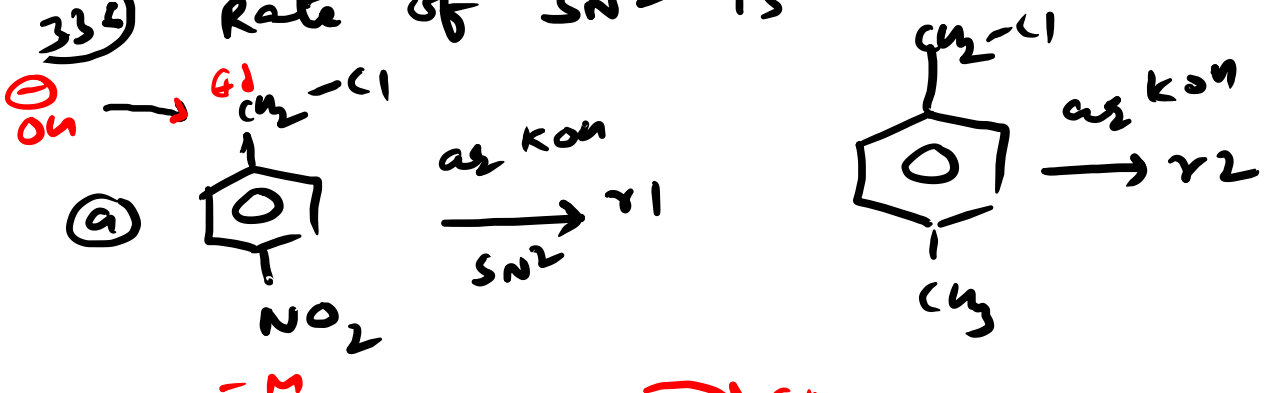
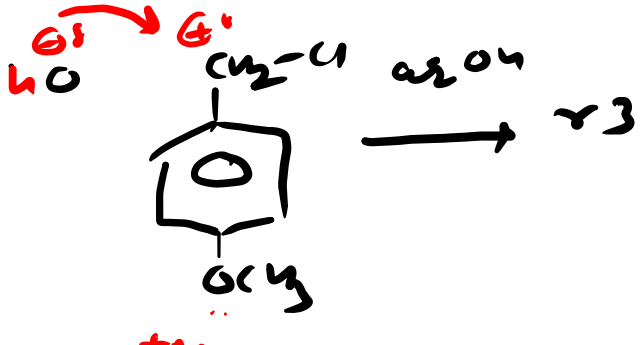
336 Rate of SN^2 is



(a) NO_2 $-M$

Rate of SN^2



$r1 > r2 > r3$

Elimination reactions

Saytzeff's major

(A) → E1 dehydration of alcohol

(B) → E2 (i) Dehydrohalogenation of R-X

(ii) Dehalogenation of vicinal di halide

Hoffmann's major

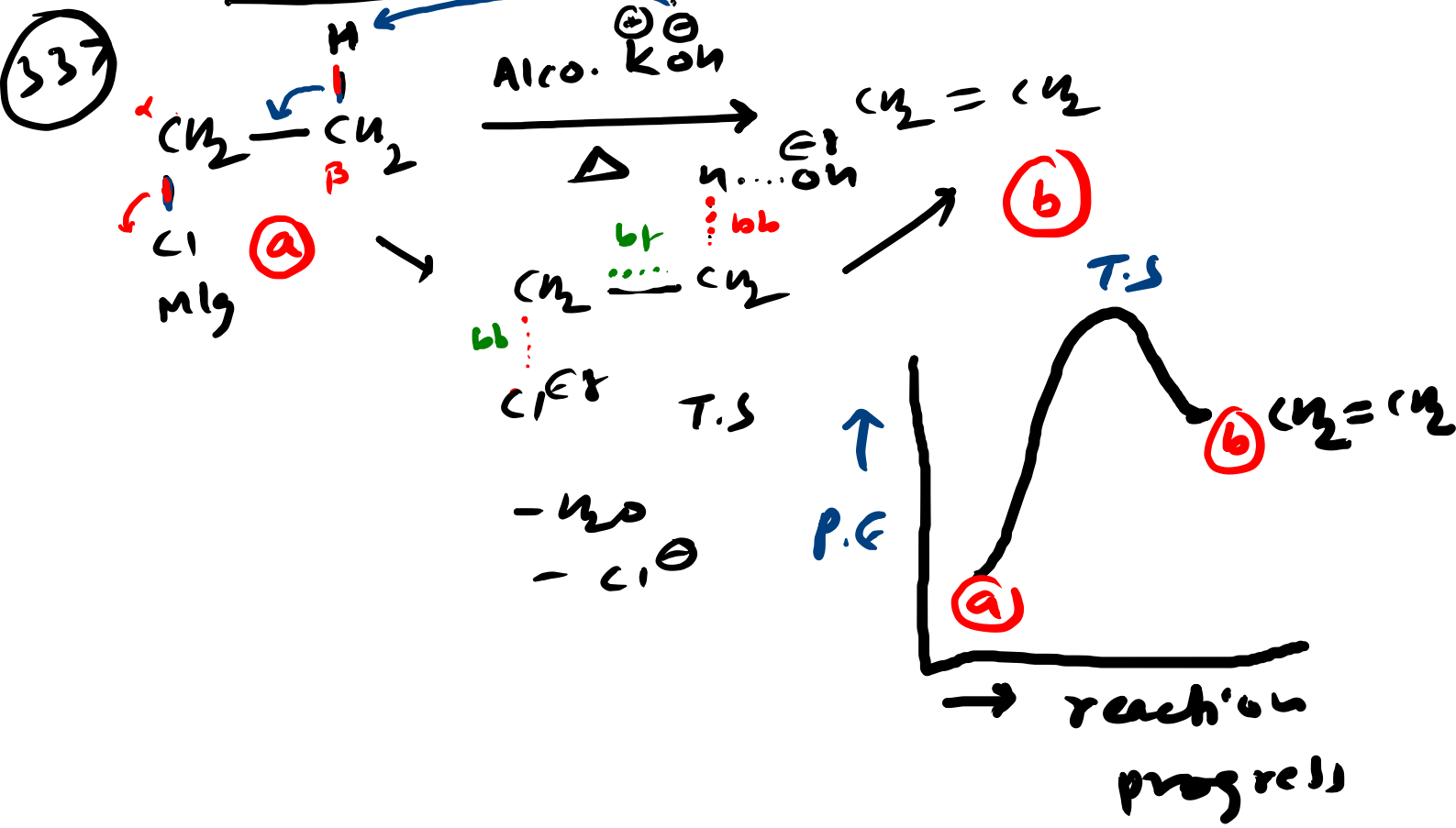
(C) → E1cb

(D) → E2 or Ei

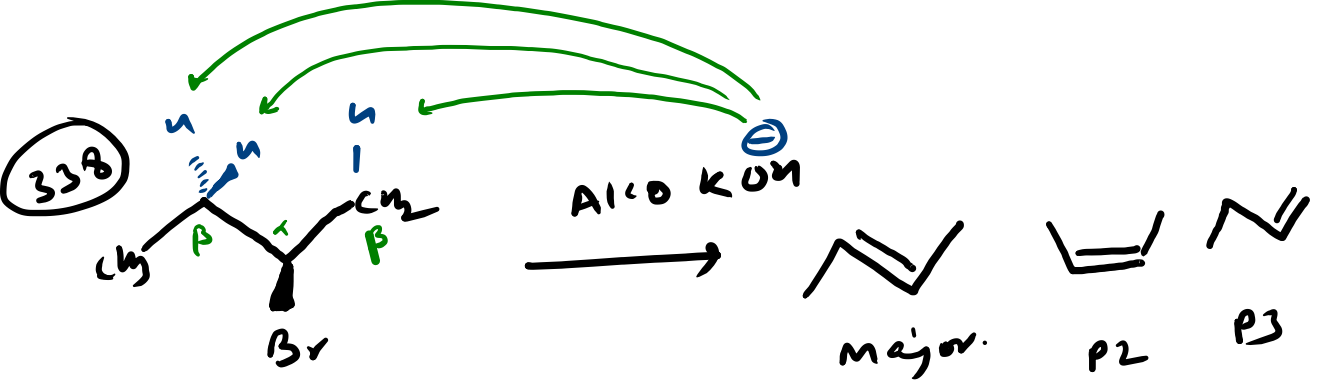
E2 - Elimination

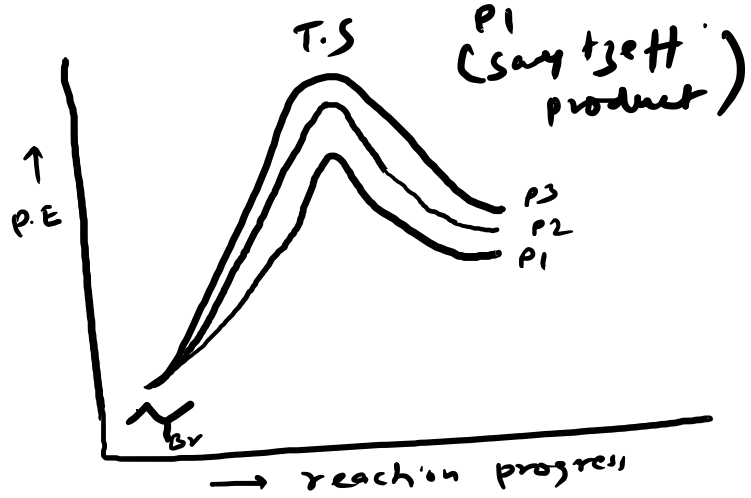
(1) Dehydrohalogenation of alkyl halide (Removal of H & X)

(33)



338





Strong Base

* Alcoholic $\overset{\ominus}{O}H$ Δ, (Alco KOH, Alco NaOH)
* Alcoholic $\overset{\ominus}{O}R$ Δ, (C_2H_5OH / $CH_3-CH_2-\overset{\ominus}{O}\overset{\oplus}{K}$)
  ↳ (+OH, $+\overset{\ominus}{O}Na^{\oplus}$)
* # $NaNH_2$ Δ

Characteristics of reaction

(1) It is one step reaction proceed via T.S

(2) Rate of E2 ∝ stability of Alkene

(3) Rate of E2 ∝ [R-X] [Base]

∴ of follow 2nd order kinetics
∴ called E2

(4) Generally saytzeff's product (more stable alkene) will be the major product

(5) E2 Elimination is anti Elimination

i.e lg and β-H must be anti

to each other eg (340)

(6) Kinetic isotopic effect are

observed in E2

eg (339)

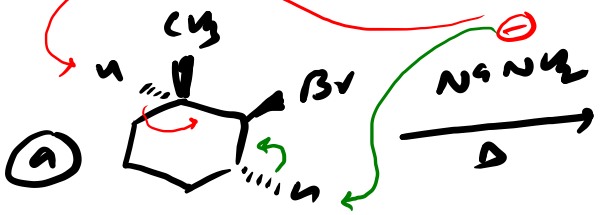
(339) Compare Rate of E2

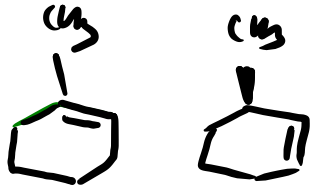
(a) $CH_2(H)-CH_2-Cl \xrightarrow[\Delta]{Alco\ KOH} r1\ CH_2=CH_2$

B.E → $CD_2(H)-CH_2-Cl \xrightarrow[\Delta]{Alco\ KOH} r2\ CD_2=CH_2$

$r1 > r2$

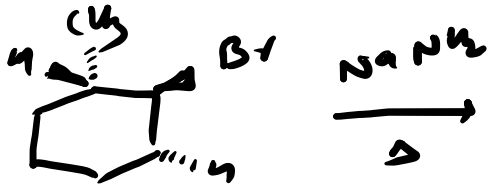
eg (340)

(a) 



P1 P2

major

(b) 

3°

$r1 > r2$

(7) (a) Structure which gives stable alkene favour E2

(b) Nlg favour E2

(c) strong Base favour E2

(d) higher temp favour Elimination reactions.

Compare Rate of E2

341

(a) $\xrightarrow[\Delta]{\text{Alc KOH}}$ r1

(b) $\xrightarrow[\Delta]{\text{Alc KOH}}$ r2 >

(c) $CH_3-CH_2-Cl \xrightarrow[\Delta]{\text{Alc KOH}}$ r3 $CH_2=CH_2$

r1 > r2 > r3

342

(a) $\xrightarrow[\Delta]{\text{Alc NaOH}}$ r1

(b) $\xrightarrow[\Delta]{\text{" "}}$ r2

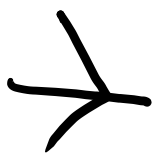
$\xrightarrow[\Delta]{\text{" "}}$ r3

r3 > r2 > r1

343

(a) $\xrightarrow[\Delta]{NaNH_2}$ r1

(b) $\xrightarrow[\Delta]{NaNH_2}$ r2

(c)  $\xrightarrow[\Delta]{NaNH_2}$ r3

r1 > r2 > r3

344

(a) $CH_3-CH_2-Br \xrightarrow[\Delta]{\text{Alco. KOH}} CH_2=CH_2$ r1

(b) $CH_3-CH_2-Br \xrightarrow[\Delta]{CH_3-CH_2-O^{\ominus}\,Na^{\oplus}} CH_3-CH_2-OH$ r2

(c) $CH_3-CH_2-Br \xrightarrow[\Delta]{NaNH_2}$ r3

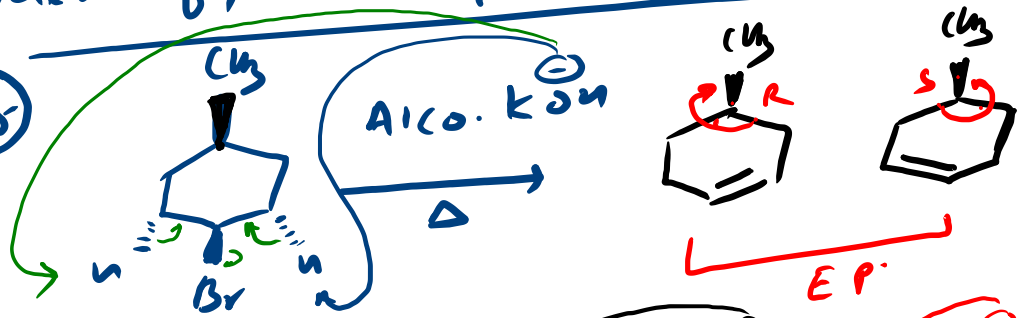
Basic strength

$NH_2^{\ominus} > OCH_2-CH_3^{\ominus} > OH^{\ominus}$

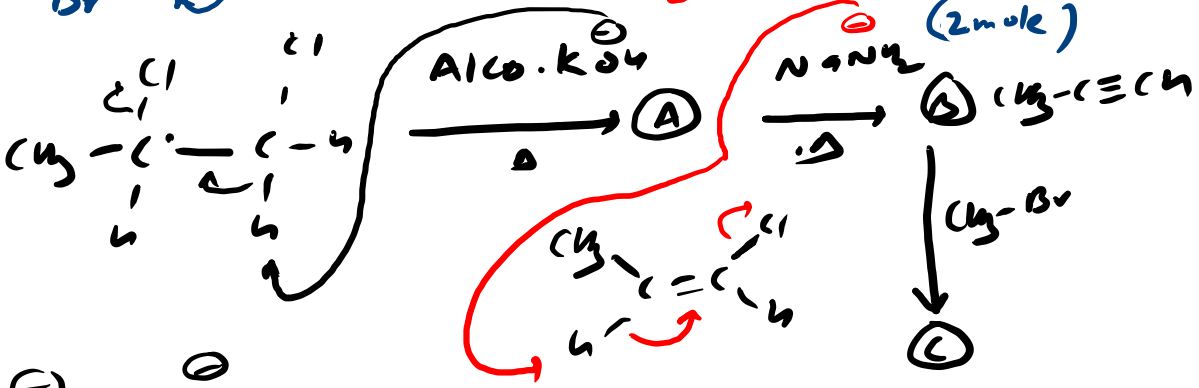
r3 > r2 > r1

Identify the product (s).

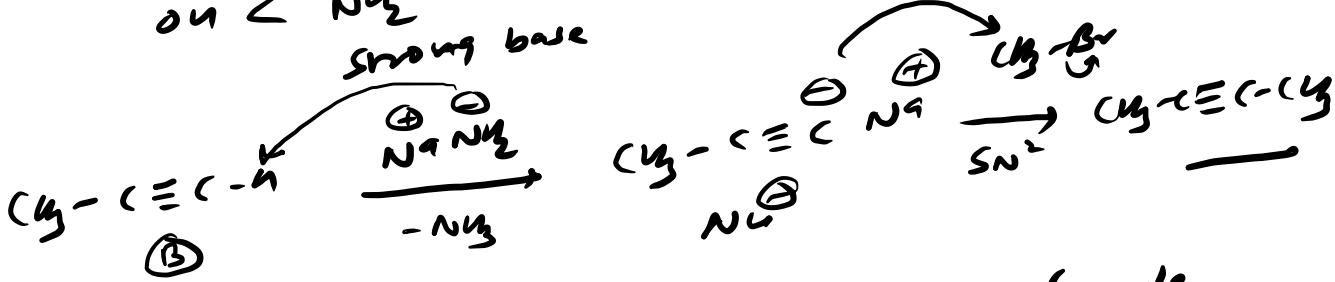
345



346

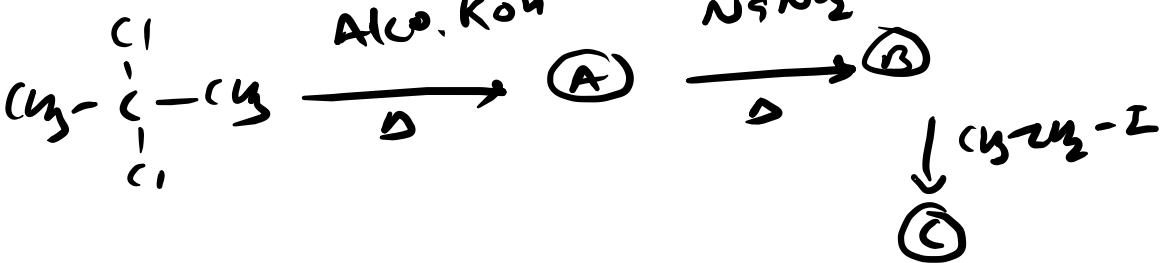


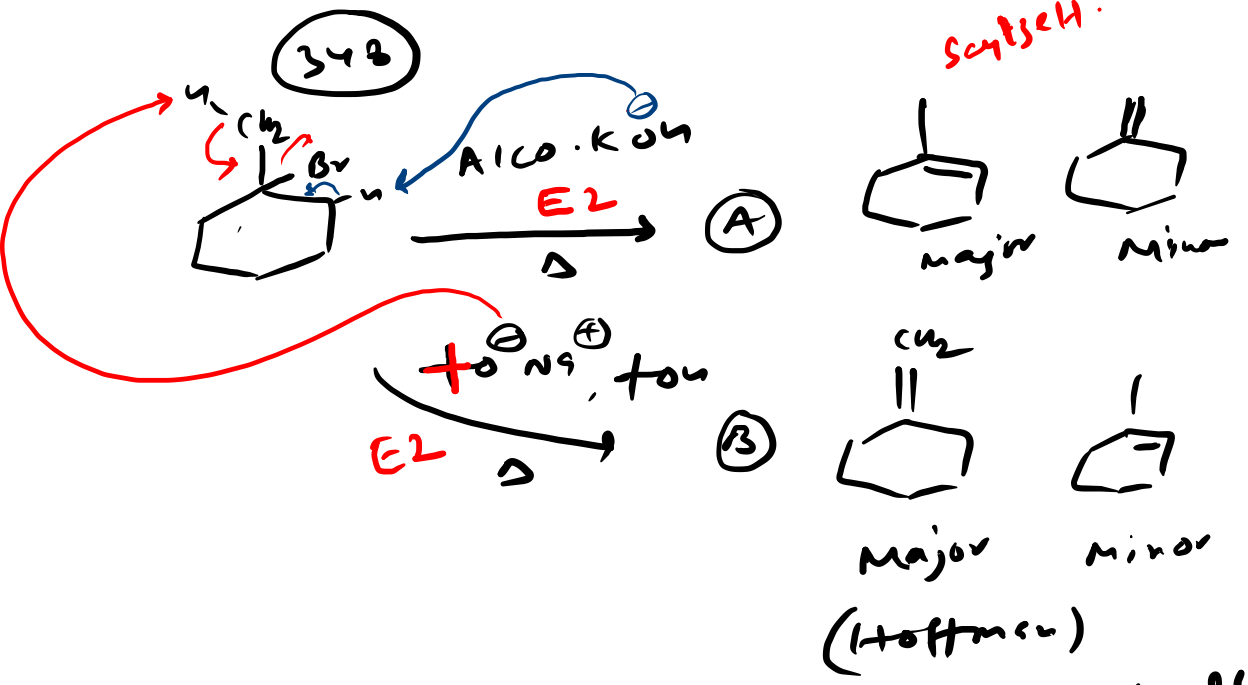
$OH^{\ominus} < NH_2^{\ominus}$



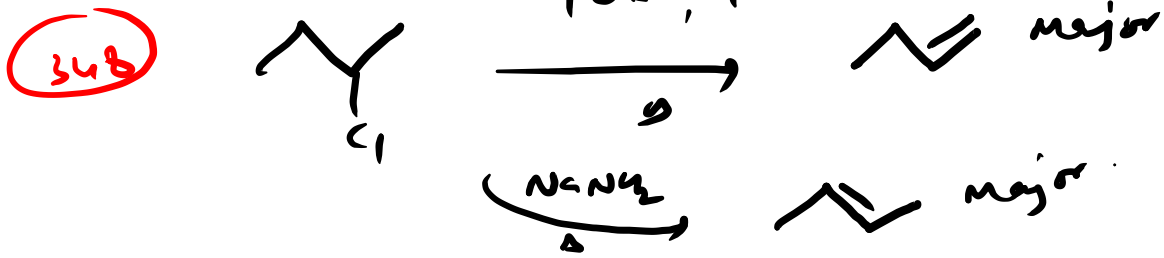
HW

347

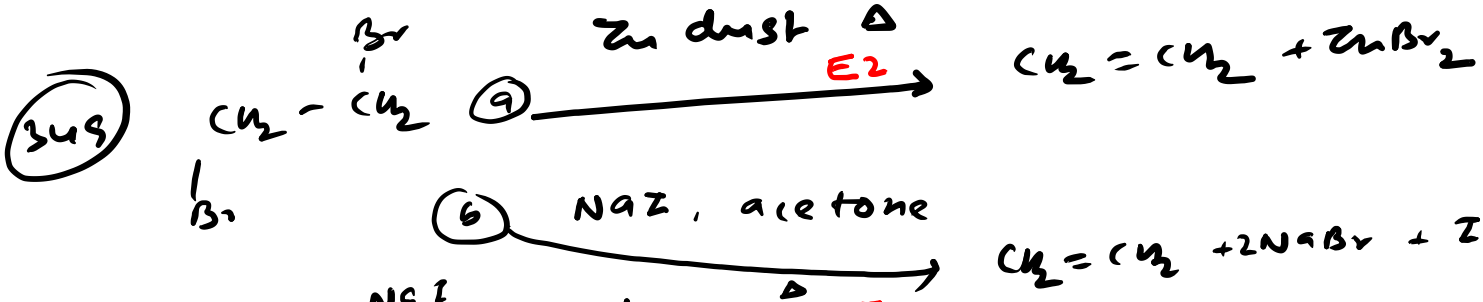




Bulky base remove, less sterically hindered β H for major product.



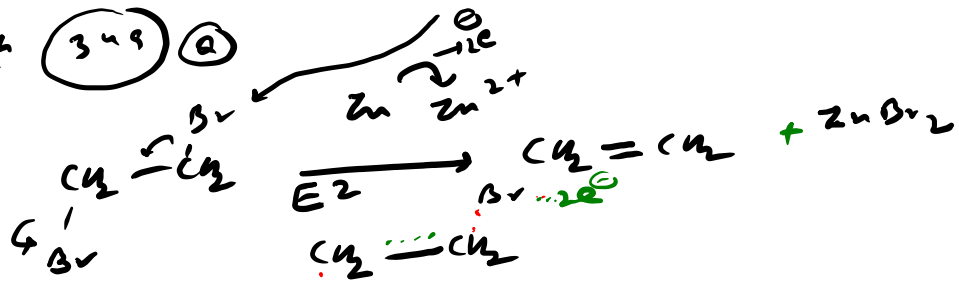
② Dehalogenation of viccinal di halide



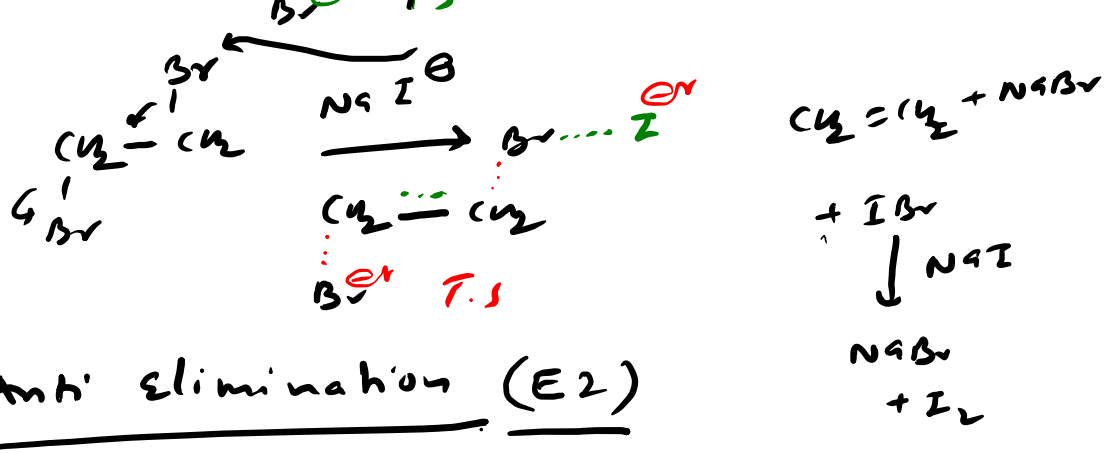
Mono halide $\xrightarrow[\text{acetone}]{NaI}$ SN^2

Vic. di halide $\xrightarrow{NaI\ \text{acetone}}$ E2

Mech (348) (a)



Mech (349) (b)



* Anti elimination (E2)

Stereo chemistry of E2

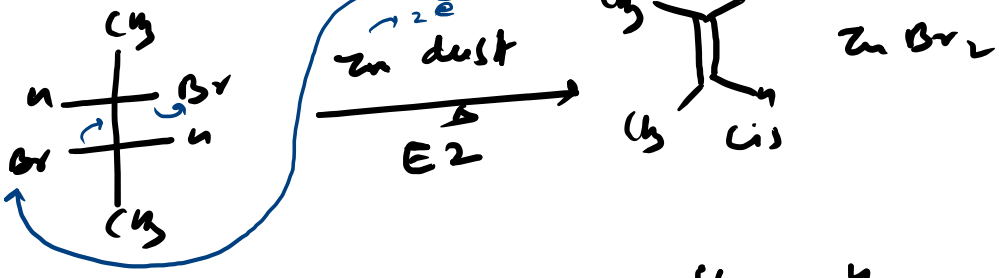
Addition

Cis Anti Threo

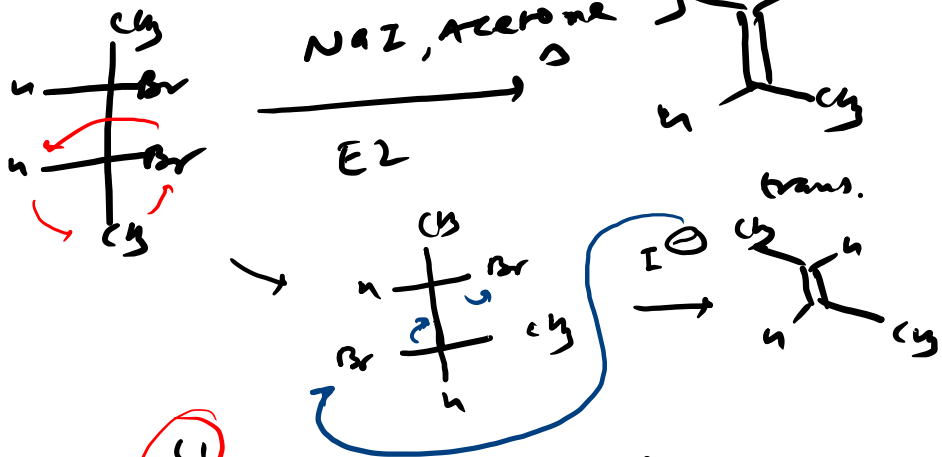
Trans Anti Erythro

Elimi

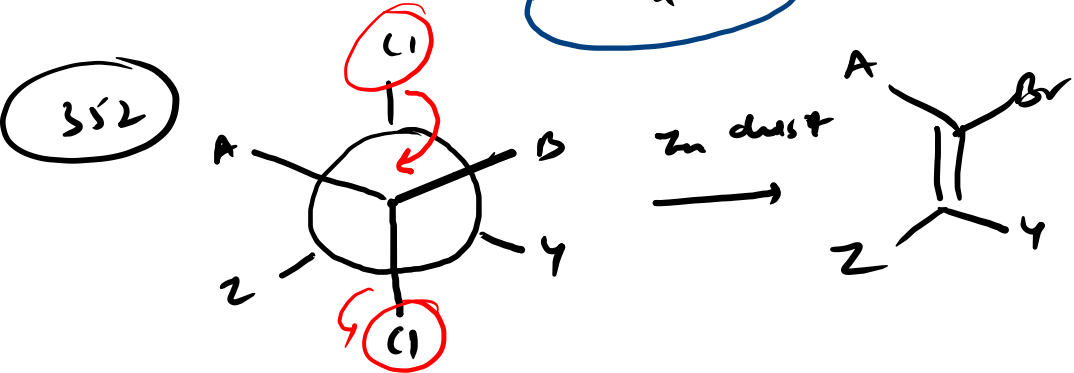
(350)



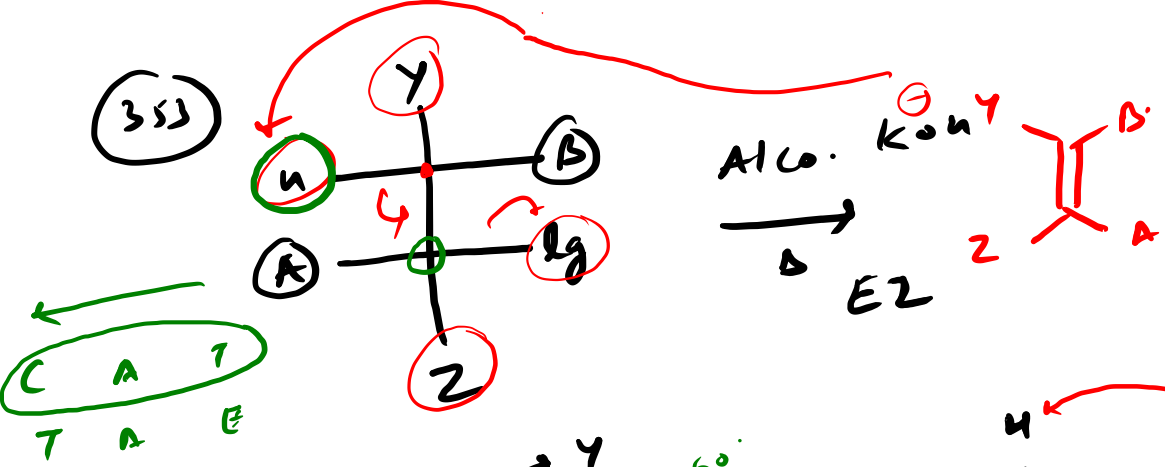
(351)

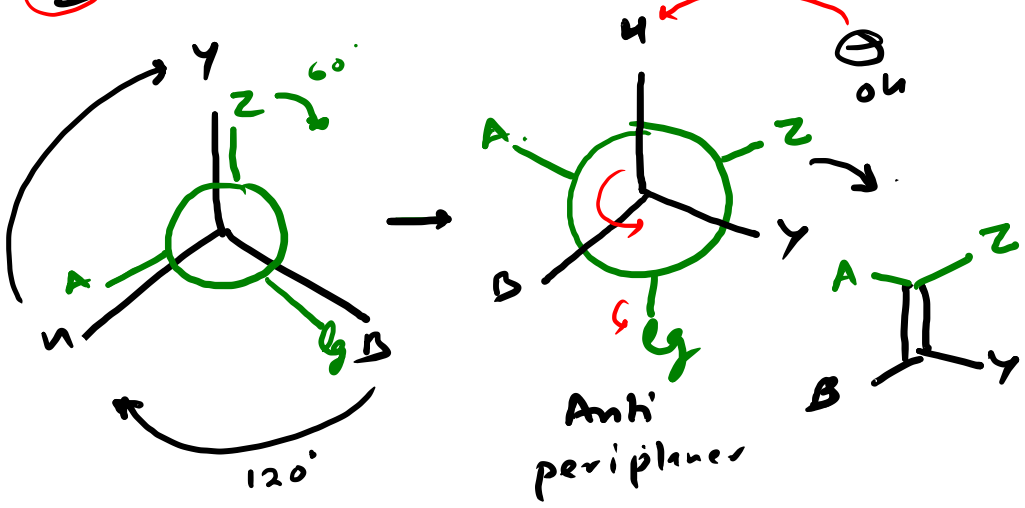


(352)

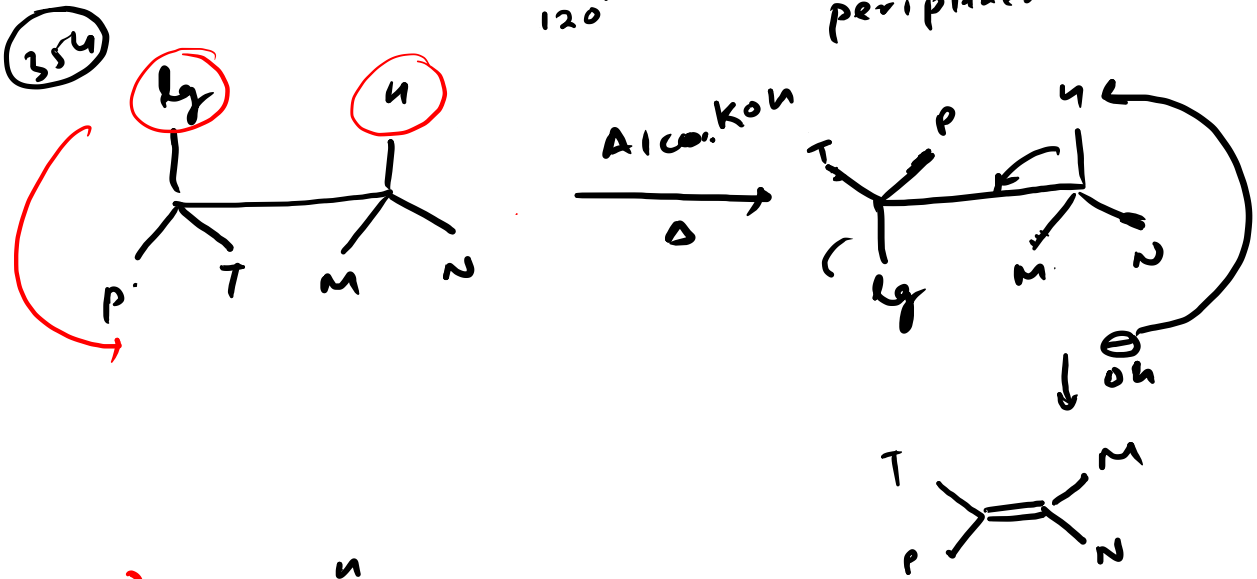


353

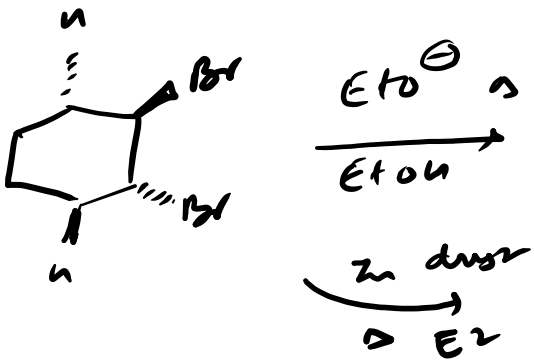




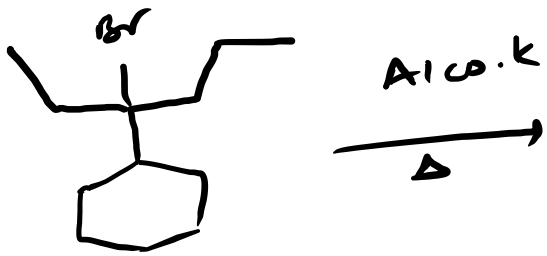
354



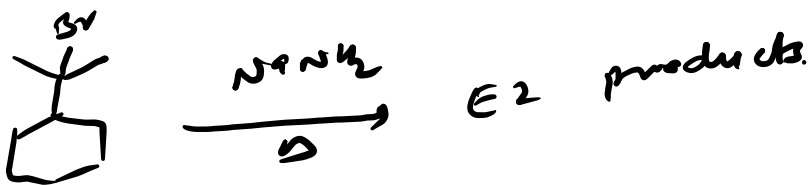
HW 354



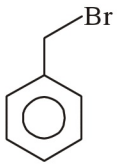
HW 355

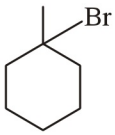


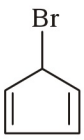
HW 356

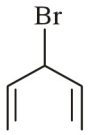


52. Which of the following can not give S_N1 reaction easily?

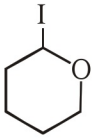
(A) 

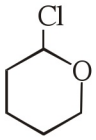
(B) 

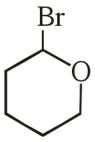
(C) 

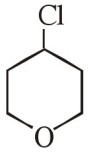
(D) 

53. Which one of the following compounds will be most reactive for S_N1 reactions ?

(A) 

(B) 

(C) 

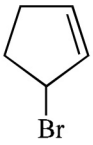
(D) 

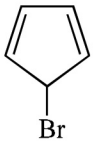
54. Which of the following compounds is most rapidly hydrolysed by S_N1 mechanism ?

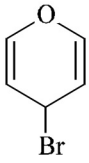
(A) C_6H_5Cl (B) $Cl–CH_2–CH=CH_2$

(C) $(C_6H_5)_3CCl$ (D) $C_6H_5CH_2Cl$

55. Among the bromides I–III given below, the order of reactivity in S_N1 reaction is:

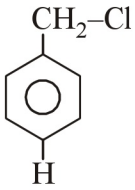
(I) 

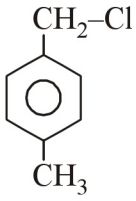
(II) 

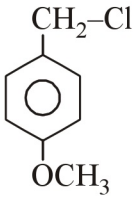
(III) 

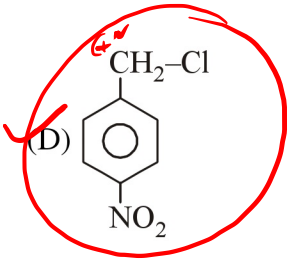
(A) III > I > II (B) III > II > I (C) II > III > I (D) II > I > III

56. Which of the following is most reactive toward S_N2.

(A) 

(B) 

(C) 

(D) 

57. For reaction $CH_3Br + OH^- \longrightarrow CH_3OH + Br^-$

the rate of reaction is given by the expression :

(A) Rate = k $[CH_3Br]$ (B) Rate = k $[OH^-]$

(C) Rate = k $[CH_3Br][OH^-]$ (D) Rate = k $[CH_3Br]^o [OH^-]^o$

58. Select suitable reason for non-occurence of the following reaction.

$Br^- + CH_3OH \longrightarrow BrCH_3 + OH^-$

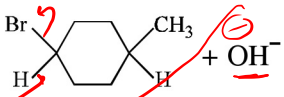
(A) Attacking nucleophile is stronger one

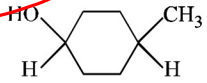
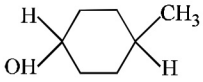
(B) Leaving group is a stronger base than nucleophile

(C) Alcohols are not good substrate for S_N reaction

(D) Hydroxide ions are weak bases

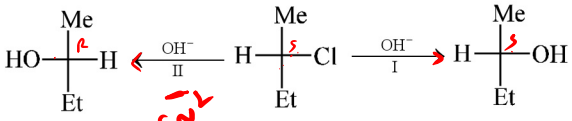
node06\B0BO-BA\Other center\JEE(Advanced)\Nurture\Physics\Module#01\Eng\01-Unit & Dimension, Basic Maths & Vector\01_Theory & Ex.

59.  ; A is :

(A)  (B)  (C) Both (D) None

60. The reactivity of 2-bromo-2-methylbutane (I), 1-bromopentane (II) and 2-bromopentane (III) towards S_N2 displacement is such that :

(A) I > II > III (B) I > III > II (C) II > III > I (D) II > I > III

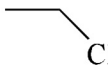
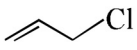
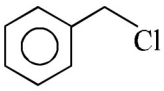
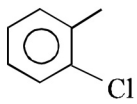
61. 

B A

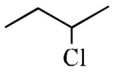
Mechanism involved :

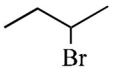
(A) I can't be S_N1 (B) II can't be S_N2 (C) I S_N1 & II S_N2 (D) I S_N2 & II S_N1

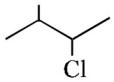
62. In which of the following replacement of Cl^- is most difficult ?

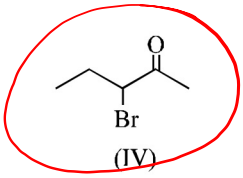
(A)  (B)  (C)  (D) 

63 Arrange these compounds in order of increasing S_N2 reaction rate :

 (I)

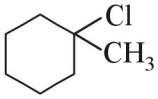
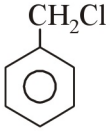
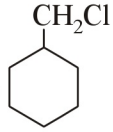
 (II)

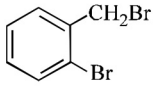
 (III)

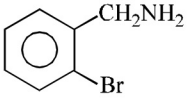
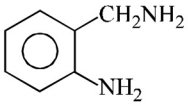
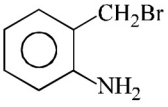
 (IV)

(A) III < I < II < IV (B) III < II < I < IV (C) IV < III < I < II (D) III < IV < I < II

64. Which reaction proceeds faster with NaI in DMSO ?

(A)  (B)  (C) (D) 

65. The major product in the given reaction:  $+ NH_3 \longrightarrow$

(A)  (B)  (C)  (D) All of these

66. The given compound $CH_3–O–CH_2–Br$ gives which one of the following reactions:

(A) Only S_N1 (B) Only S_N2

(C) S_N1 as well as S_N2 (D) E_1 only

node06\B0BO-BA\Other center\JEE(Advanced)\Nurture\Physics\Module#01\Eng\01-Unit & Dimension, Basic Maths & Vector\01_Theory & Ex.

Dehydration of alcohol

eg. (87) $CH_3-CH_2-OH \xrightleftharpoons{\text{conc } H_2SO_4,\ \Delta} CH_2=CH_2 + H_2O$

Mech

(A) Plg: $CH_3-CH_2-\ddot{O}H + H^{\oplus} \rightleftharpoons CH_3-CH_2-\overset{\oplus}{O}H_2$

(B) Glg

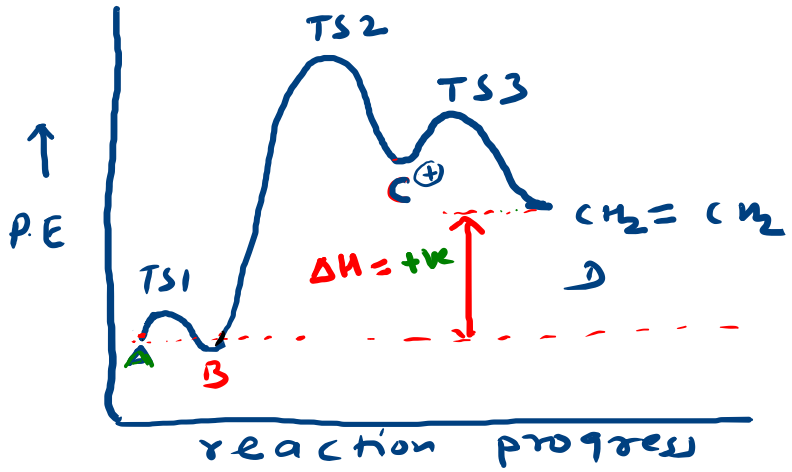
(C) $CH_3-\overset{\oplus}{C}H_2$ — carbocation (sp^2)

RDS: $CH_3-CH_2-OH_2^{\oplus} \xrightleftharpoons[-H_2O]{\Delta} CH_3-\overset{\oplus}{C}H_2$

(D) $CH_2=CH_2$ + $HSO_4^{\ominus}$ + $H^{\oplus}$

$H_2SO_4 \rightleftharpoons H^{\oplus} + HSO_4^{\ominus}$

Acid (strong) — C. base (weak)



TS1 = CH_3-CH_2 ... $H^{\oplus}$ ... $O-H$ ... $H^{\oplus}$

TS2 = CH_3-CH_2 ... O ... H

TS3 = HSO_4 ... H ... CH_2-CH_2

Characteristics of E1 Reaction (Elimination Unimolecular)

(1) Generally it is two step reaction

Step-I Formation of I^{st} carbocation which is RDS.

Step-II Removal of proton.

*(2) Rate of reaction ∝ stability of I^{st} carbocation

*(3) Rate of Reaction ∝ [Alcohol]

It follow I^{st} order kinetics ∴ called E1.

(4) Carbocation Intermediate undergoes rearrangement, if possible.

(5) Saytzeff's product (more stable alkene), will be the major product. (or zetsev)

(6) By fractional distillation Alkene & H_2O are separated from eachother.
(Otherwise reverse reaction takes place at low temp.)

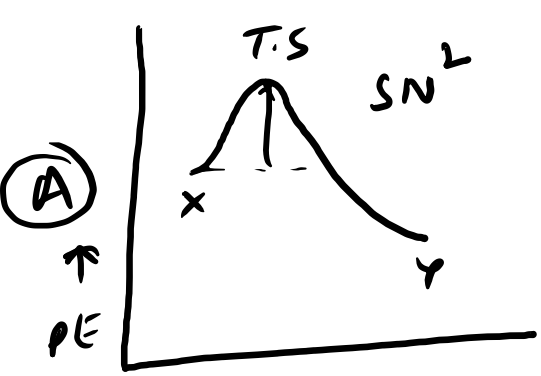
(7) factors which favour E1

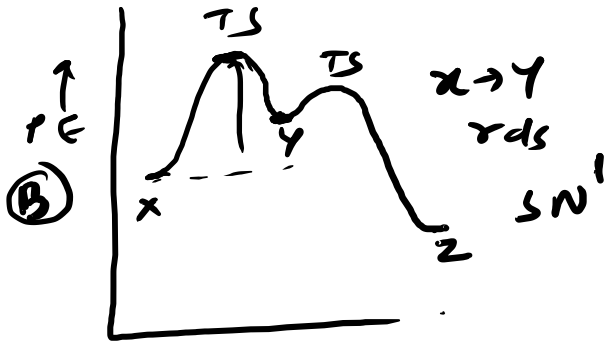
(a) structure of substrate
Structure which gives stable I^{st} $C^{\oplus}$ favour E1

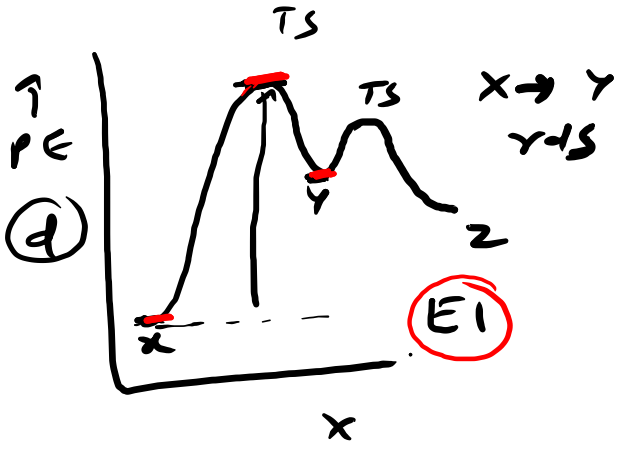
(b) Good leaving group favour E1

(c) weak Base

(d) Higher temperature favour Elimination reactions.







Hammond's postulate

Close the energy of TS,
close will be the characteristics.

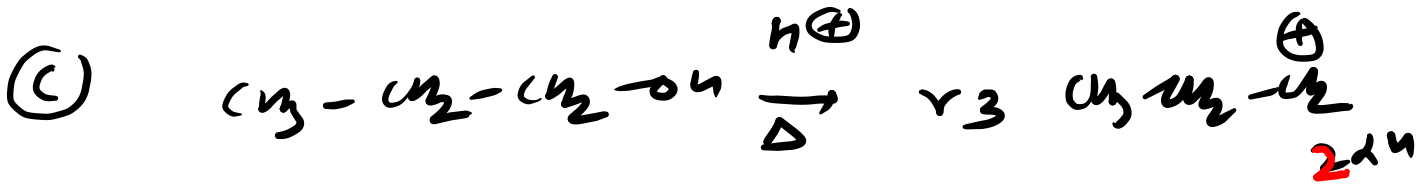
of graph (B)/(D) TS of X to Y is
close to Y in energy
so T.S has the character of Y
(i.e Y is $C^{\oplus}$)

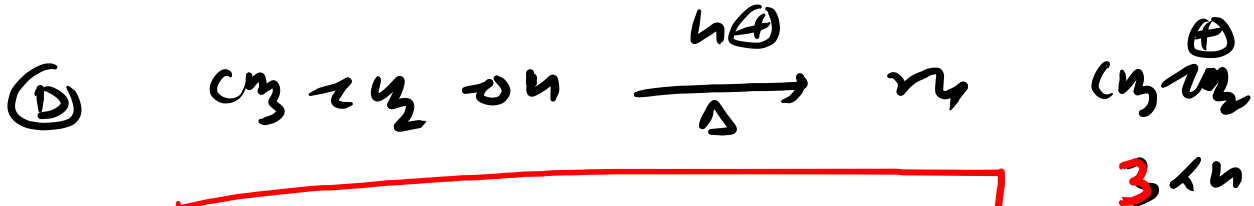
∴ Rate of E1 ∝ stability of $C^{\oplus}$

(88) Compare Rate of Dehydration

(a)

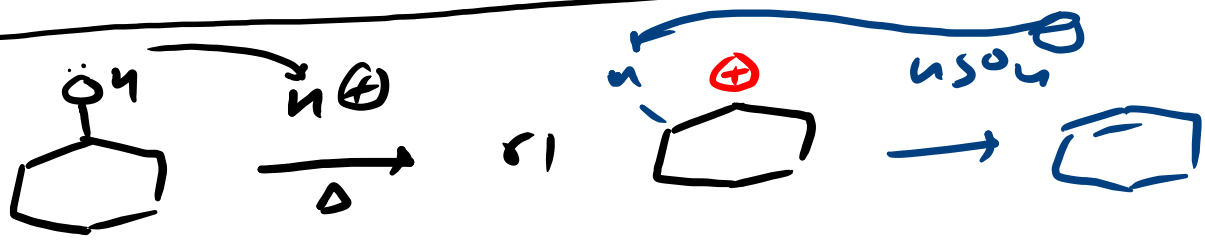
(b)

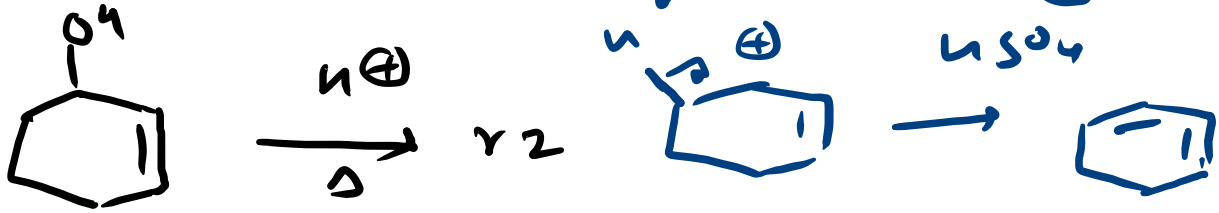
(c) 

(D) 

r1 > r2 > r4 > r3

(89) Compare Rate of Dehydration

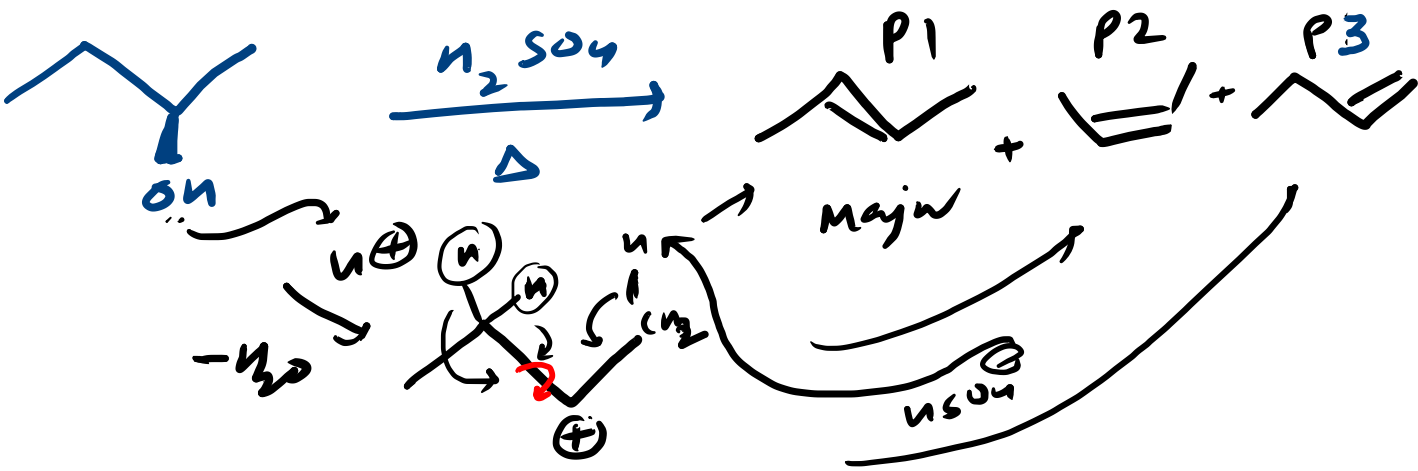
(a) 

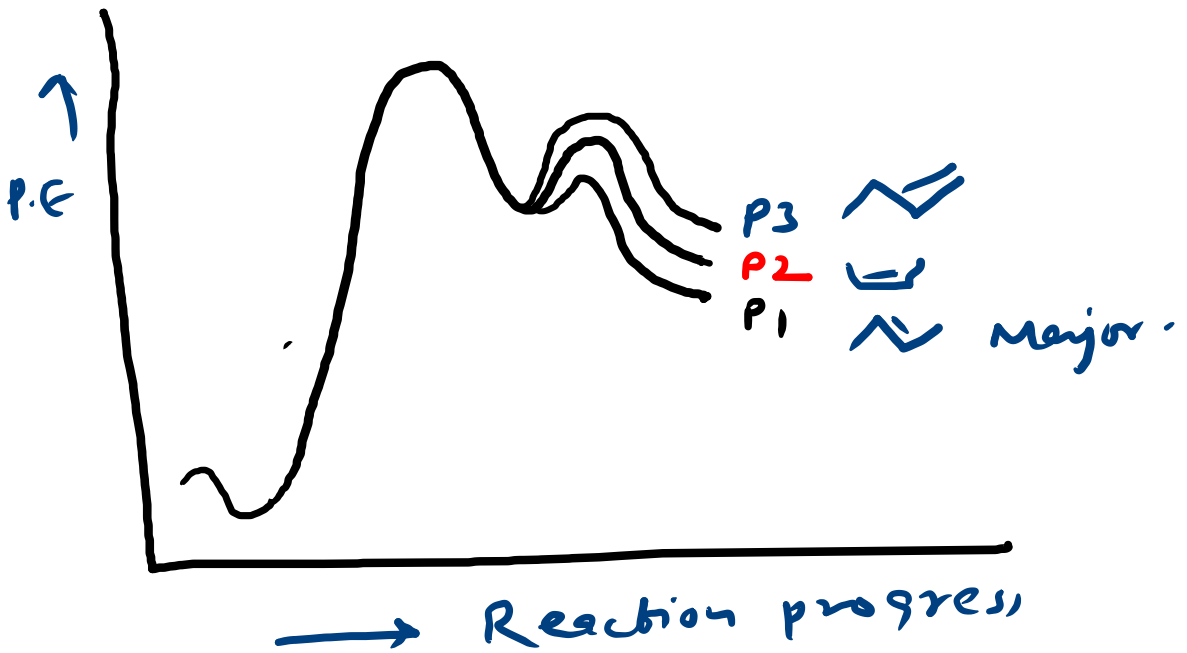
(b) 

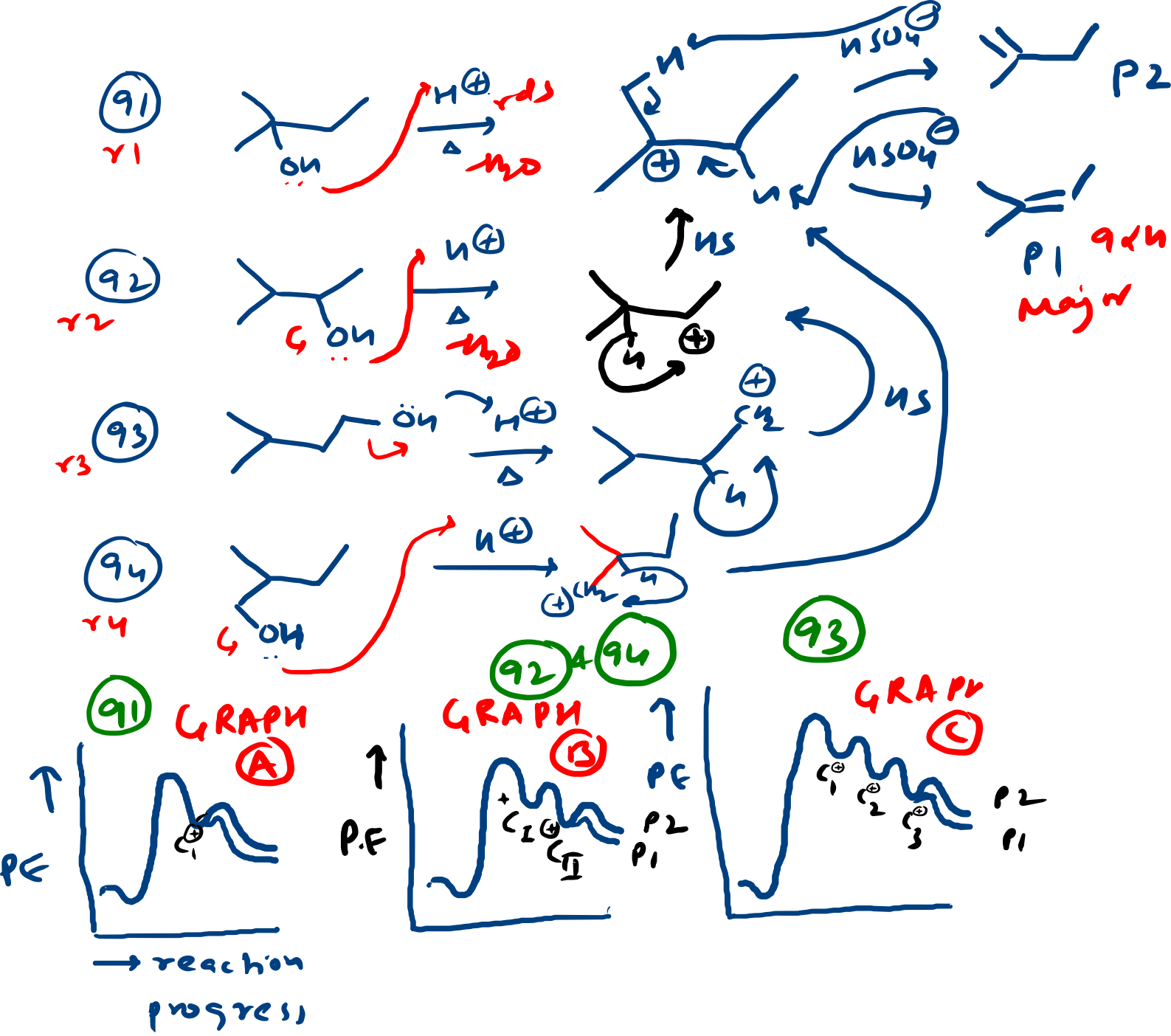
(c)

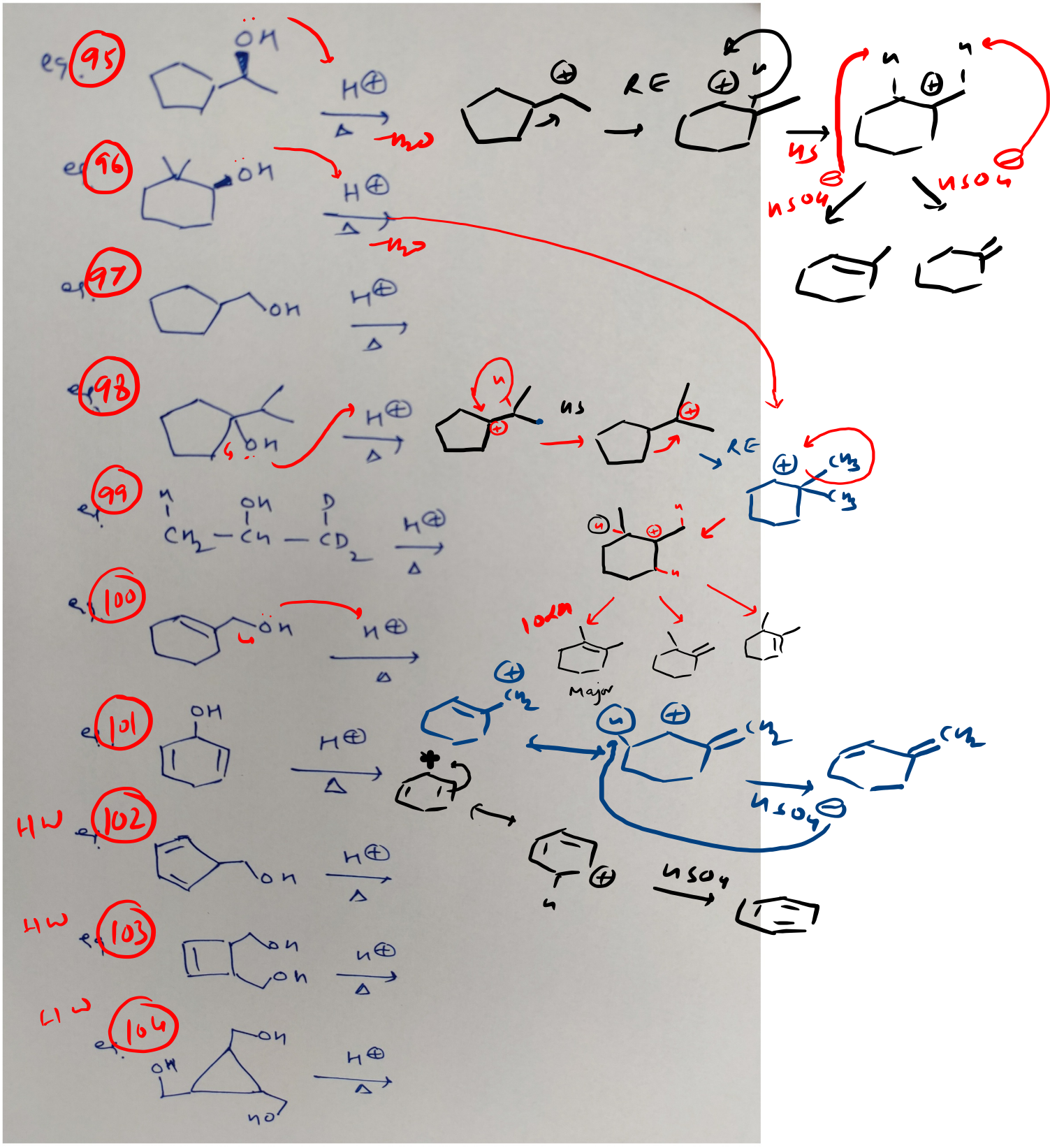
r3 > r2 > r1

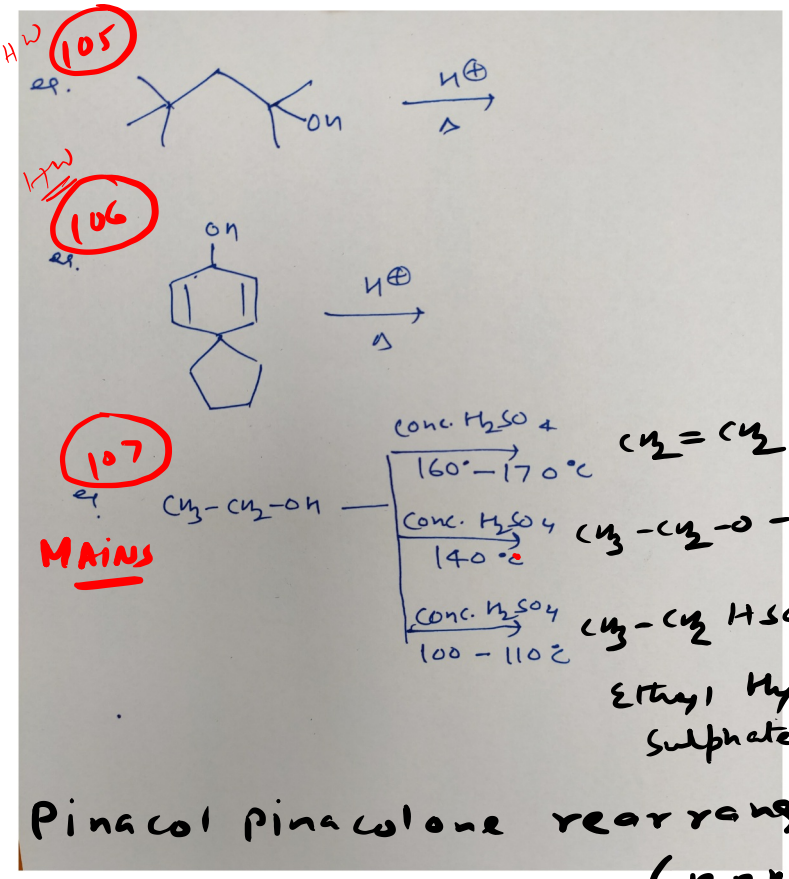
(90)

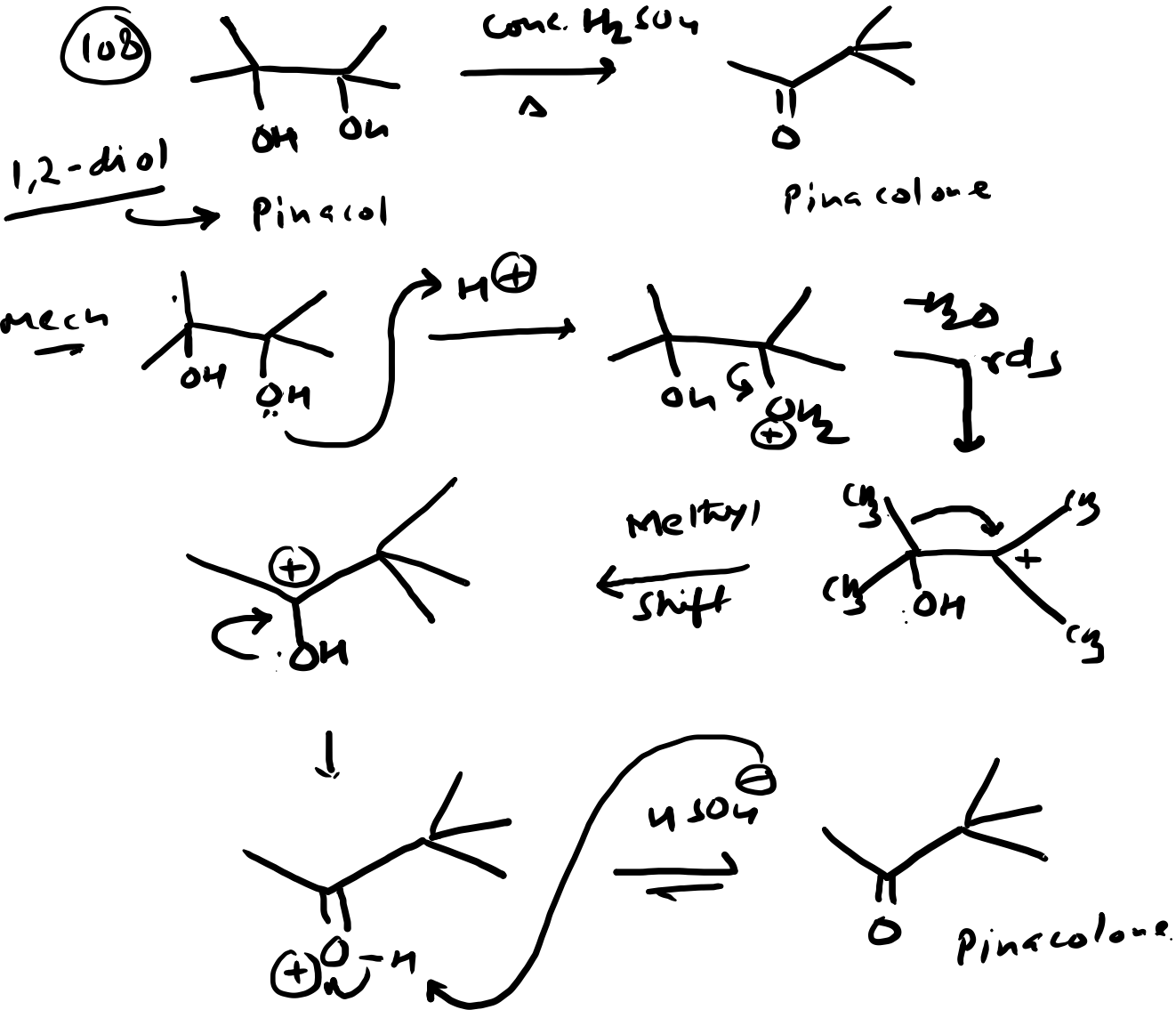

H_2SO_4
Δ
OH
P1
P2
P3
Major
$-H_2O$
HSO_4


P.E
P3
P2
P1
Major.
Reaction progress


91
92
93
94
r1
r2
r3
r4
rds
P2
P1
Major
GRAPH A
GRAPH B
GRAPH C
PE
P.E
reaction progress


95
96
97
98
99
100
101
102
103
104
RE
Major
HW
HW


105
eg.
OH
H⊕
Δ
106
eg.
OH
H⊕
Δ
107
eg.
CH3-CH2-OH
Conc. H2SO4
160°–170°C
CH2=CH2
Conc. H2SO4
140°C
CH3-CH2-O-CH2-CH3
Conc. H2SO4
100–110°C
CH3-CH2 HSO4
Ethyl Hydrogen Sulphate.
MAINS
Pinacol pinacolone rearrangement (PPR)


108
OH OH
Conc. H2SO4
Δ
1,2-diol
Pinacol
Pinacolone
mech
H⊕
-H2O
rds
Methyl
shift
HSO4⊖
Pinacolone

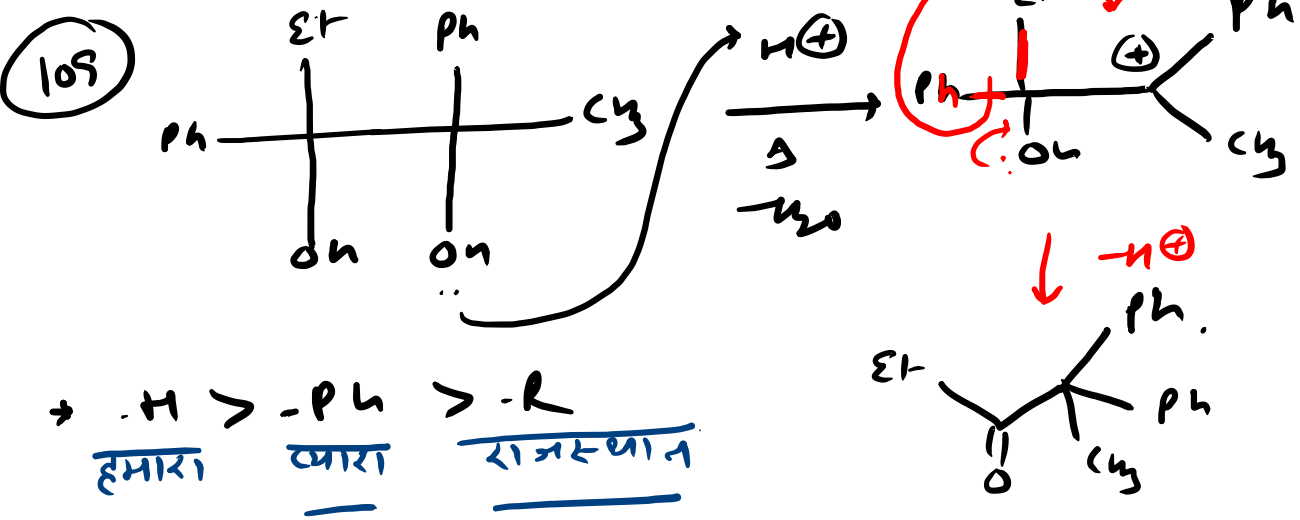
Characterishics

(1) formation of Ist $C^{\oplus}$ is rds

(2) out of 2 OH groups, that OH undergoes protonation, which gives more stable Ist $C^{\oplus}$, (2nd)

(3) PPR follow the migratory aptitude (the group which has more migratory aptitude undergoes migration)

(10g)

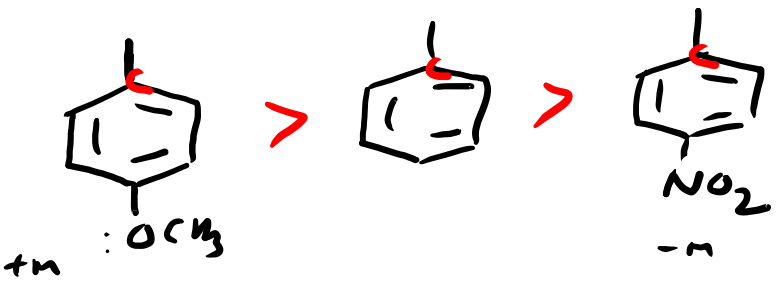


+ -H > -Ph > -R
हमारा प्यारा राजस्थान

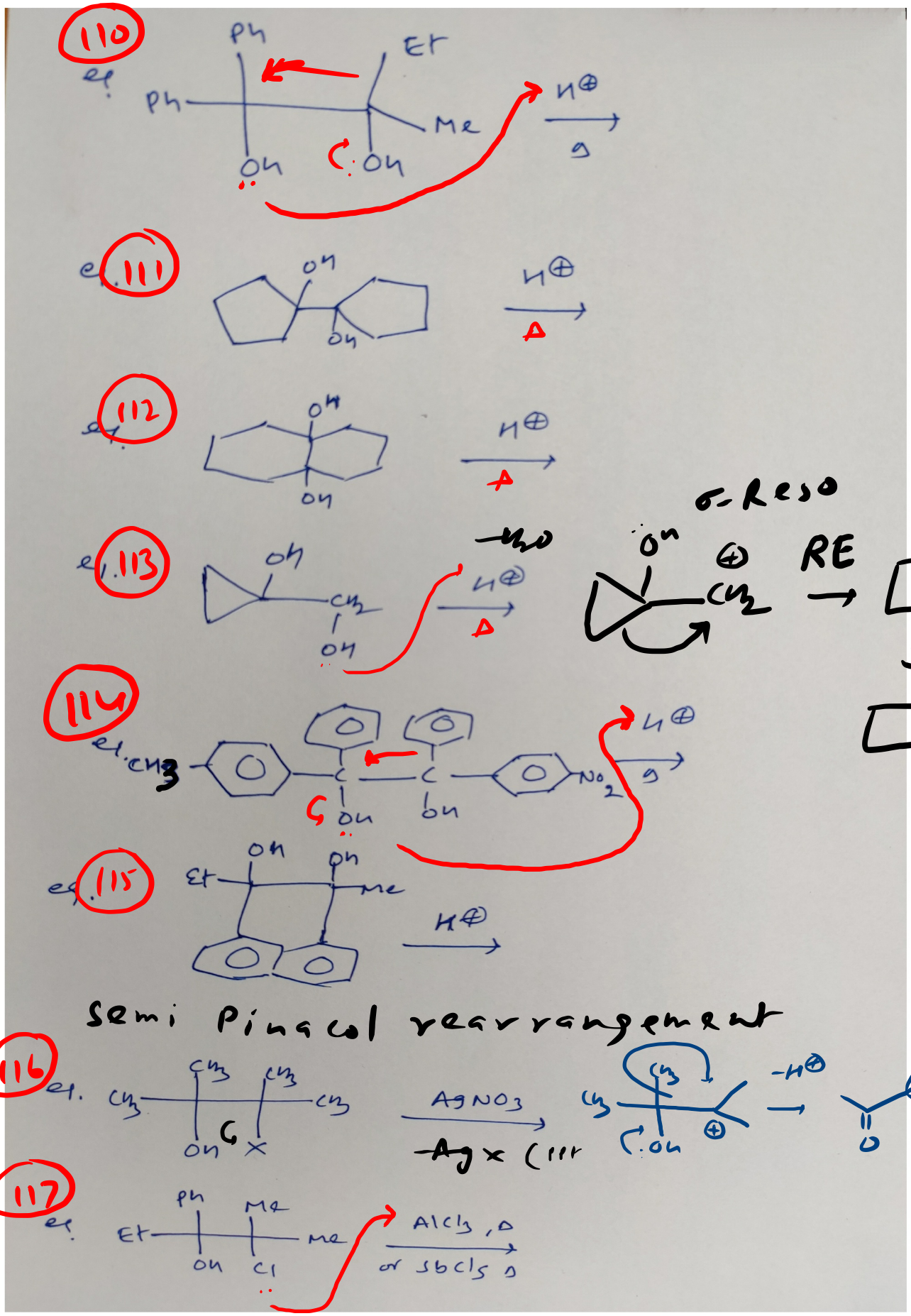
Migratory aptitude

+ $-CH_2-CH_2-CH_3 > -CH_2-CH_3 > -CH_3$

+ $-CH(CH_3)_2 > -CH_2-CH_2-CH_3$

+ 

+ -H > -D > -T (Bond energy C-H < C-D < C-T)


110
Ph
Et
Ph
Me
OH
OH
H⊕
Δ
111
OH
OH
H⊕
Δ
112
OH
OH
H⊕
Δ
113
OH
CH2
OH
H⊕
Δ
-H2O
σ-Reso
RE
-H⊕
114
CH3
NO2
OH
OH
H⊕
Δ
115
OH
OH
Et
Me
H⊕
Semi Pinacol rearrangement
116
CH3
CH3
CH3
CH3
OH
X
AgNO3
-AgX (ppt)
-H⊕
117
Ph
Me
Et
Me
OH
Cl
AlCl3, Δ
or SbCl5 Δ

EXERCISE # (J-MAINS)

1. Geometrical isomerism is not shown by- **[AIEEE-2002]**

(1) 1,1–dichloro–1–pentene (2) 1,2–dichloro–1–pentene

(3) 1,3–dichloro–2–pentene (4) 1,4–dichloro–2–pentene

2. Recemic mixture is formed by mixing two : **[AIEEE-2002]**

(1) Isomeric compounds (2) Chiral compounds

(3) Meso compounds (4) Enantiomers with chiral carbon

3. Following types of compounds I and II **[AIEEE-2002]**

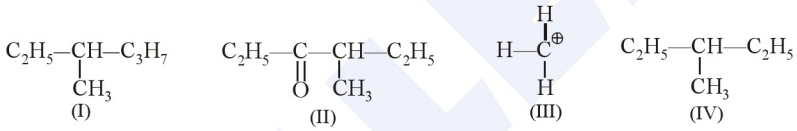
(I) $CH_3CH=CHCH_3$

(II) $CH_3-\underset{\displaystyle CH_2CH_3}{\overset{\displaystyle *}{C}H}-OH$, are studied in terms of isomerism in-

(1) Chain isomerism (2) Position isomerism

(3) Conformers (4) Stereo isomerism

4. Among the following four structures I to IV **[AIEEE-2003]**



(I) (II) (III) (IV)

It is true that-

(1) All four are chiral compounds

(2) Only I and II are chiral compounds

(3) Only III is a chiral compound

(4) Only II and IV are chiral compounds

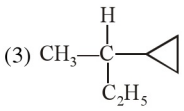
5. Which of the following will have a meso-isomer also- **[AIEEE-2004]**

(1) 2–chlorobutane (2) 2,3–dichlorobutane

(3) 2,3–dichloropentene (4) 2–hydroxy propanoic acid

6. Amongst the following compounds, the optically active alkane having lowest molecular mass is

(1) $CH_3-CH_2-C\equiv CH$ (2) $CH_3-CH_2-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{C}}H-CH_3$ **[AIEEE-2004]**

(3)  (4) $CH_3-CH_2-CH_2-CH_3$

node06\B0BO-BA\Kota\JEE(Advanced)\Enthuse\Physics\Sheet\Magnetic Effect of Current\Eng\00_Theory.p65

7. Which of following compounds is not chiral **[AIEEE-2005]**

(1) 1–chloropentane (2) 2–chloropentane

(3) 1–chloro–2–methyl pentane (4) 3–chloro–2–methyl pentane

8. Of the five isomeric hexanes, the isomer which can give two monochlorinated compounds is : **[AIEEE-2005]**

(1) 2-methyl pentane (2) 2,2–dimethyl butane

(3) 2,3–dimethyl butane (4) n-hexane

9. Which types of isomerism is shown by 2,3–dichloro butane- **[AIEEE-2005]**

(1) Structural (2) Geometric (3) Optical (4) Diastereo

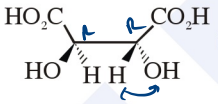
10. Increasing order of stability among the three main conformations (i.e. Eclipse, Anti, Gauche) of 2-fluoroethanol is **[AIEEE-2006]**

(1) Gauche, Eclipse, Anti (2) Eclipse, Anti, Gauche

(3) Anti, Gauche, Eclipse (4) Eclipse, Gauche, Anti

11. Which one of the following conformations of cyclohexane is chiral ? **[AIEEE-2007]**

(1) Twist boat (2) Rigid (3) Chair (4) Boat

12. The absolute configuration of  is : **[AIEEE-2008]**

(1) S, S (2) R, R (3) R, S (4) S, R

13. The alkene that exhibits geometrical isomerism is :- **[AIEEE-2009]**

(1) 2–butene (2) 2–methyl–2–butene

(3) Propene (4) 2–methyl propene

14. The number of stereoisomers possible for a compound of the molecular formula CH_3–CH=CH–CH(OH)–Me is :- **[AIEEE-2009]**

(1) 4 (2) 6 (3) 3 (4) 2

15. Out of the following, the alkene that exhibits optical isomerism is :- **[AIEEE-2010]**

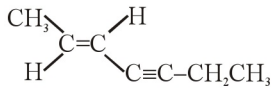
(1) 2-methyl-2-pentene (2) 3-methyl-2-pentene

(3) 4-methyl-1-pentene (4) 3-methyl-1-pentene

16. Identify the compound that exhibits tautomerism :- **[AIEEE-2011]**

(1) 2-Pentanone (2) Phenol (3) 2-Butene (4) Lactic acid

17. The IUPAC name of the following compounds is : **[JEE-MAIN-2012]**



(1) (Z) – 5 hepten – 3 – yne (2) (Z) – 2 hepten – 4 – yne

(3) (E) – 5 hepten – 3 – yne (4) (E) – 2 hepten – 4 – yne

18. Dipole moment is shown by :- **[JEE-MAIN 2012]**

(1) trans-2, 3-dichloro- 2-butene (2) 1, 2-dichlorobenzene

(3) 1, 4-dichlorobenzene (4) trans-1, 2-dinitroethene

19. Maleic acid and fumaric acids are :- **[JEE-MAIN 2012]**

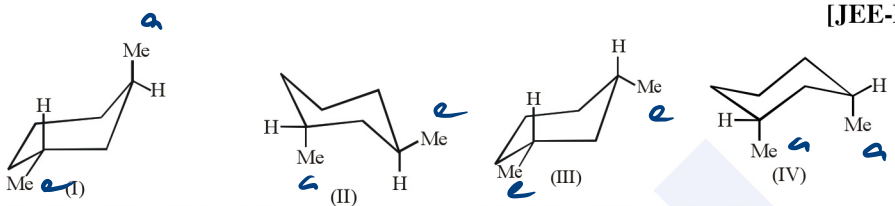
(1) Tautomers (2) Chain isomers

(3) Geometrical isomers (4) Functional isomers

20. Monocarboxylic acids are functional isomers of : **[JEE-MAIN 2013]**

(1) Esters (2) Amines (3) Ethers (4) Alcohols

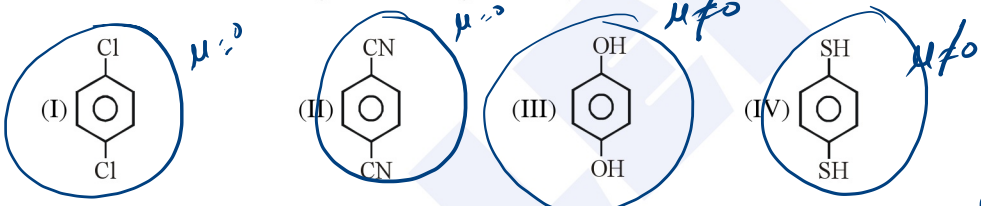
21. Arrange in the correct order of stability (decreasing order) for the following molecules: **[JEE-MAIN 2013]**



(1) (I) > (II) > (III) > (IV) (2) (IV) > (III) > (II) ≈ (I)

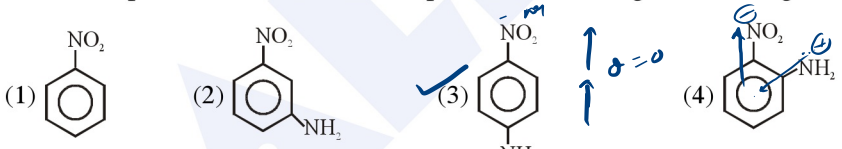
(3) (III) > (I) ≈ (II) > (IV) (4) (I) > (II) ≈ (III) > (IV)

22. For which of the following molecule significant $\mu \neq 0$ **[JEE-MAIN 2014]**



(1) Only (III) (2) (III) and (IV) (3) Only (I) (4) (I) and (II)

23. Which compound exhibits maximum dipole moment among the following :-**[JEE-MAIN 2015]**



24. The number of structural isomers for C_6H_{14} is: **[JEE-MAIN 2015]**

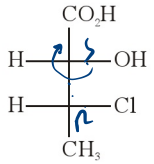
(1) 6 (2) 4 (3) 3 (4) 5

25. The optically inactive compound from the following is :- **[JEE-MAIN-2015]**

(1) 2-chloropropanal (2) 2-chlorobutane

(3) 2-chloro-2-methylbutane (4) 2-chloropentane

26. The absolute configuration of : **[JEE-MAIN-2016]**



(1) (2R, 3R) (2) (2R, 3S) (3) (2S, 3R) (4) (2S, 3S)

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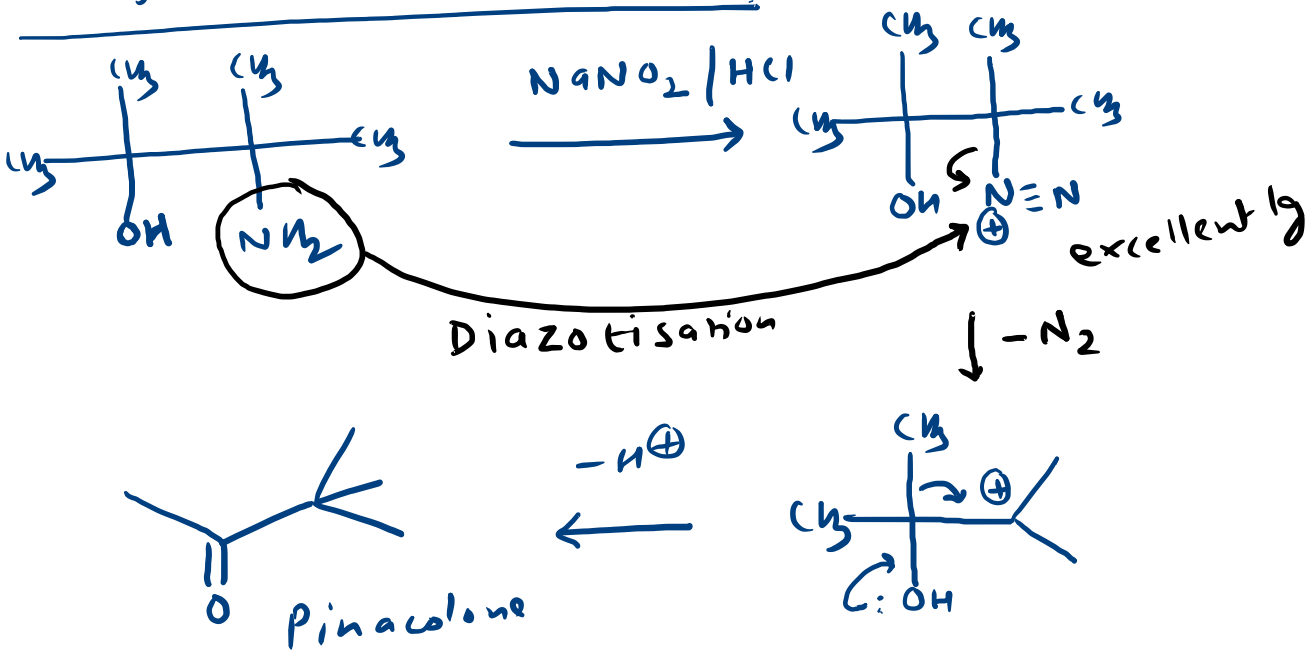
Demjanov Rearrangement

(118)



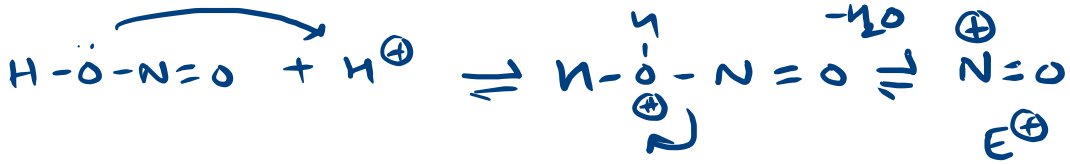
Diazotisation of primary amine

(119) $R-NH_2$ $\xrightarrow[\text{or } HNO_2 \quad 0-5^\circ C]{NaNO_2 + HCl}$ $R-N\equiv N\ Cl$

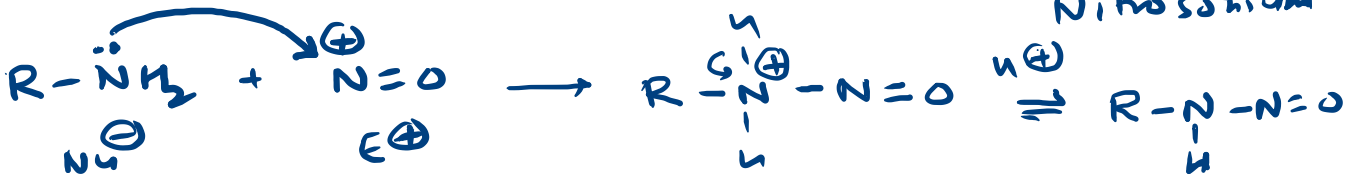
(120) $Ph-NH_2$ $\xrightarrow[0-5^\circ C]{}$

Mech (119)

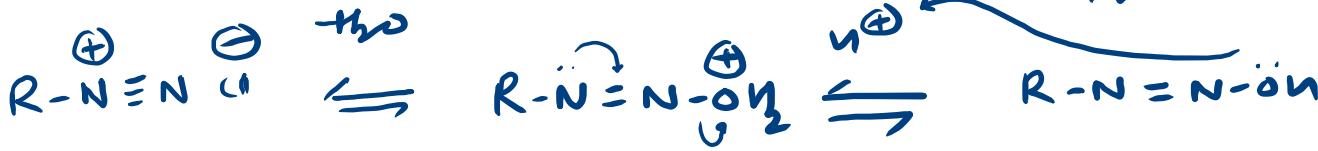
$$NaNO_2 + HCl \rightleftharpoons HNO_2 + NaCl$$



Nitrosonium

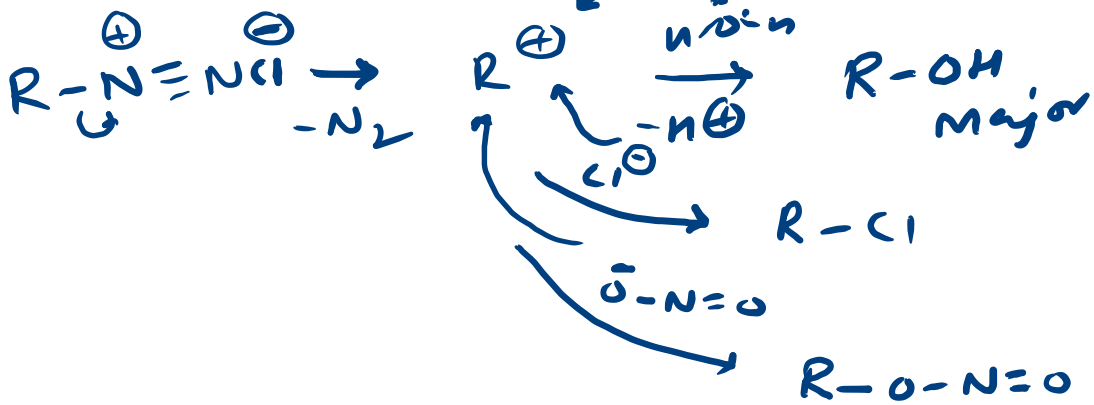


Tautomerism



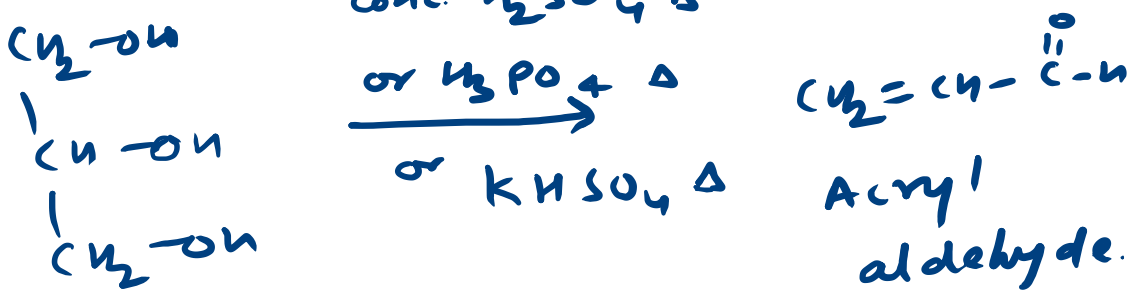
At Room temp

(121)

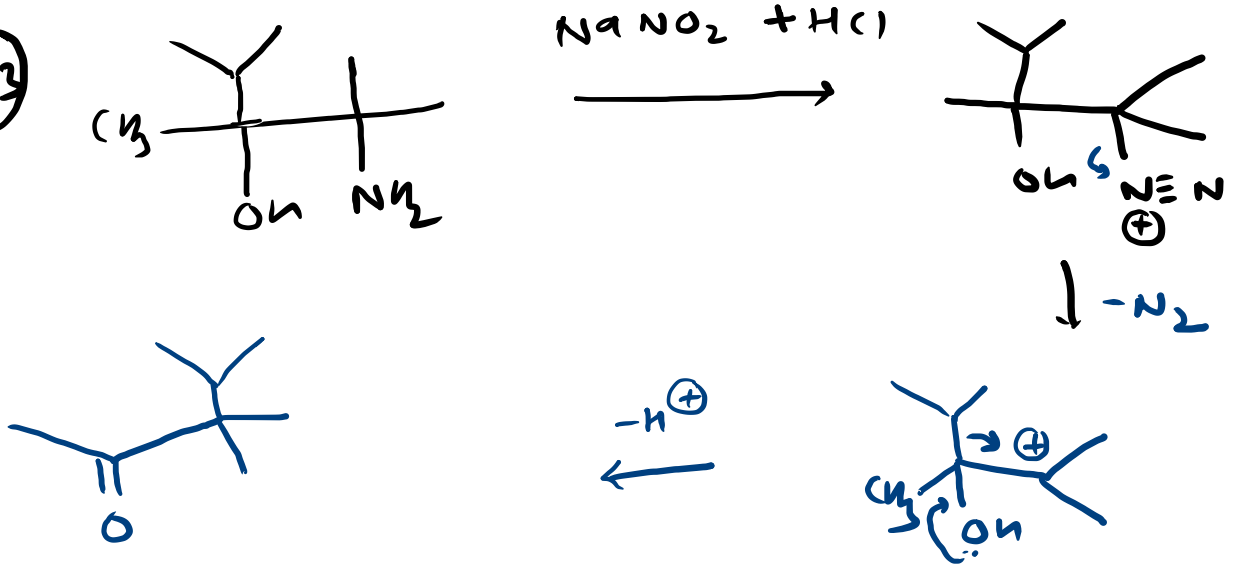


H.W

(122)

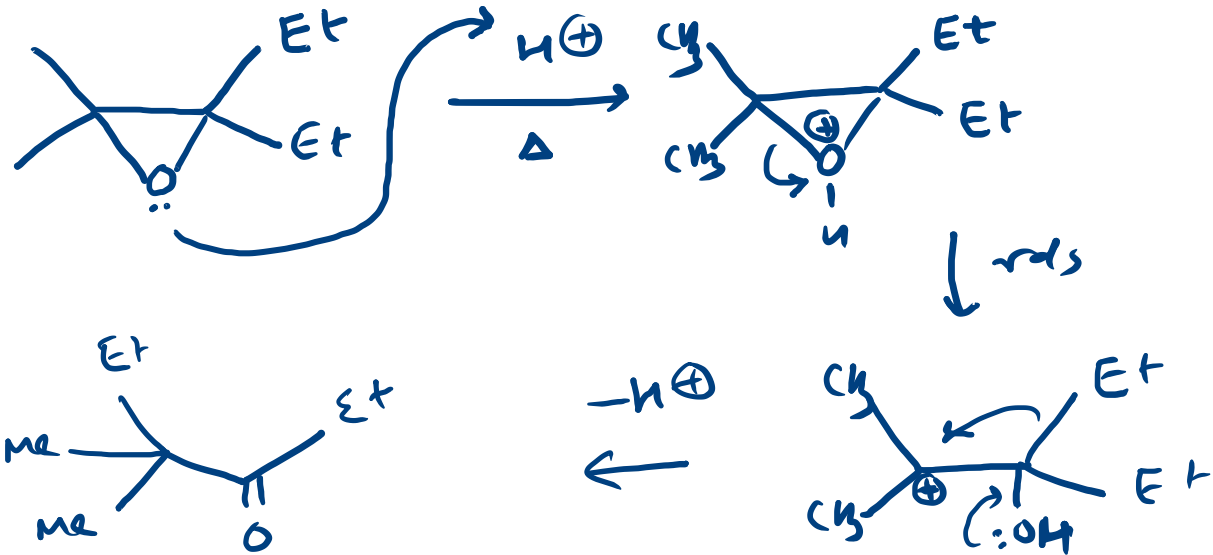


(123)



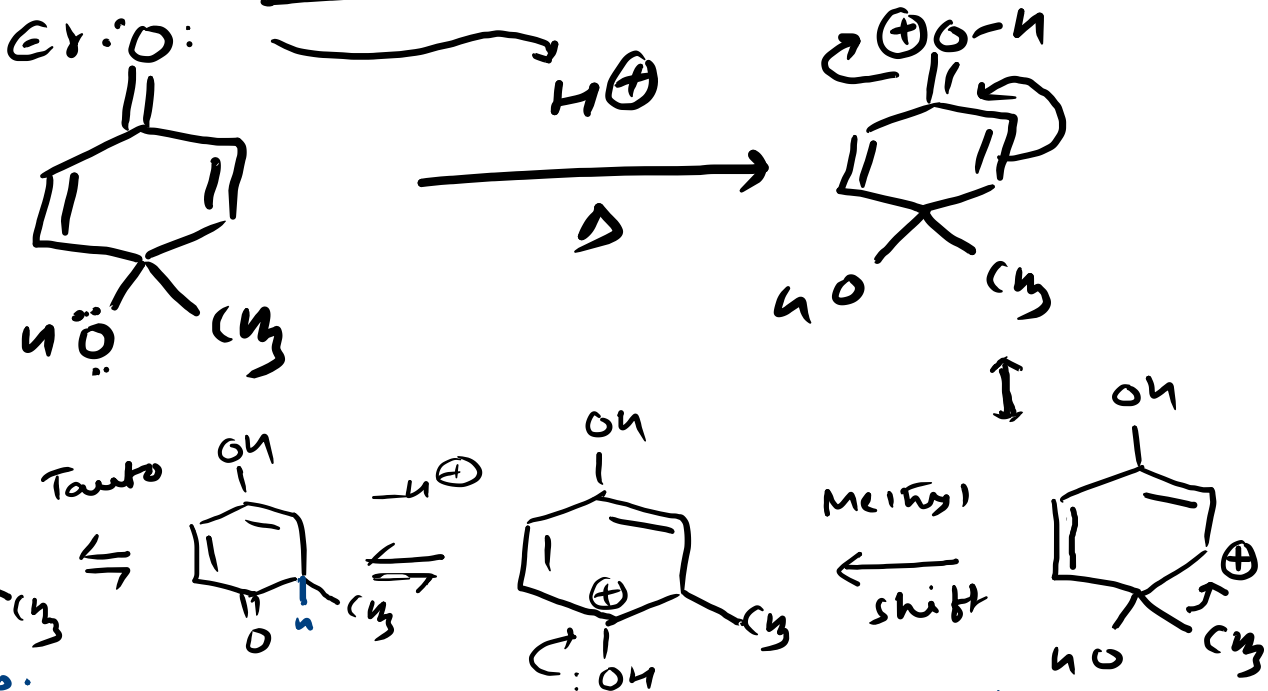
Pinacol fashion rearrangement

(124)

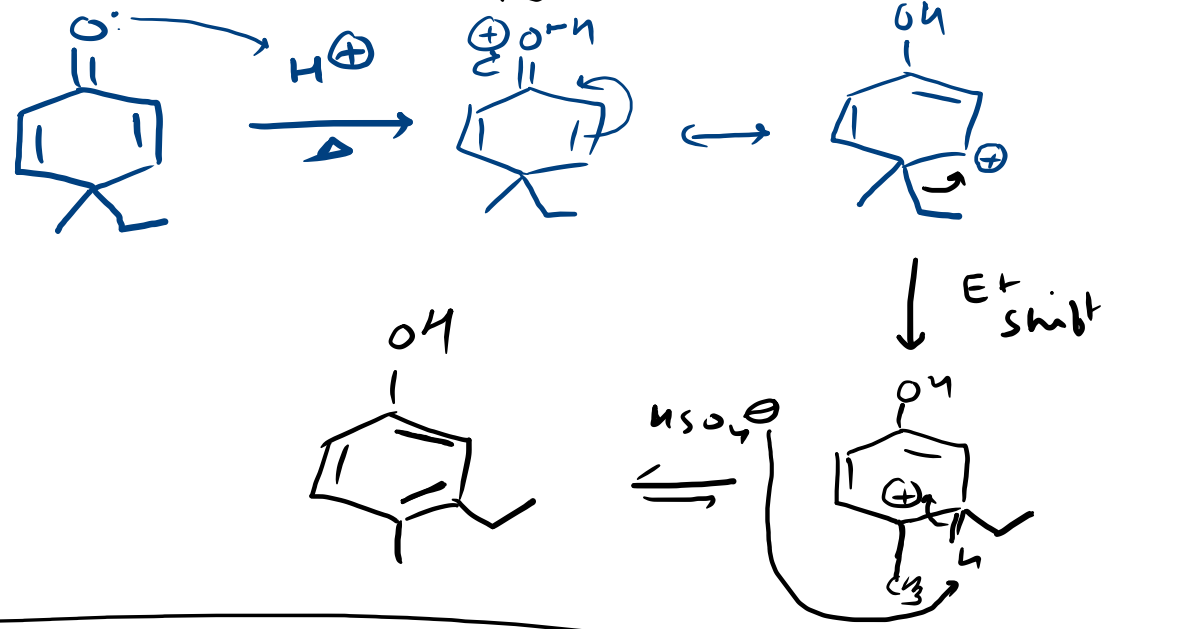


Di enone Phenol rearrangement

(125)



(126)



*** Migratory aptitude are followed, when migration step is rds or carbocation rerrange for BBS (Back bonding stability)

Electrophilic Addition Reactions

(A) EAR by classical carbocation

(I) Addition of H-X to alkene or Alkyne.

(II) Addition $H_2O/H^{\oplus}$ or $ROH/H^{\oplus}$ to Alkene.

(B) EAR by Non classical carbocation

(I) Addition of X_2 to Alkene/Alkyne

(II) Addition of $X_2 + H_2O$ to Alkene, Alkyne.

(III) Addition of Interhalogens (I-Cl, I-Br, Br-Cl) to Alkene, Alkyne.

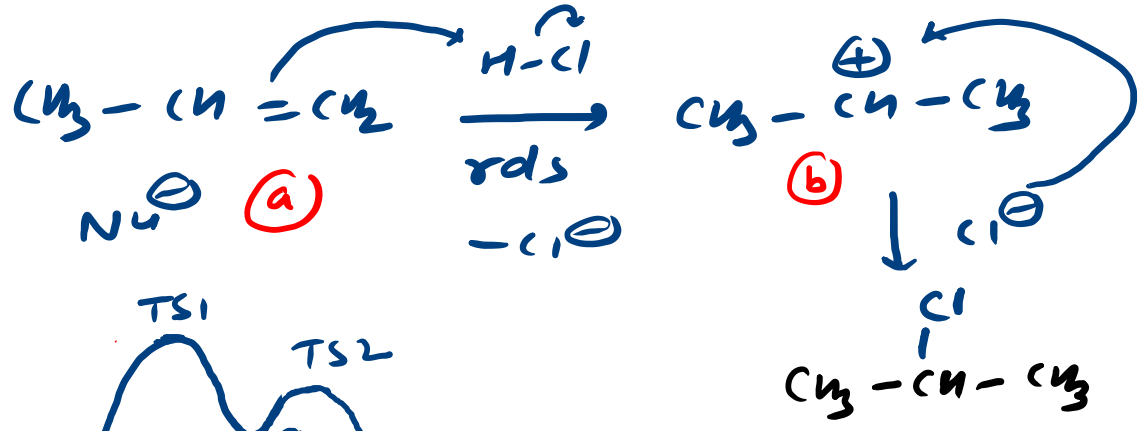
(IV) OMDM, Kucherov.

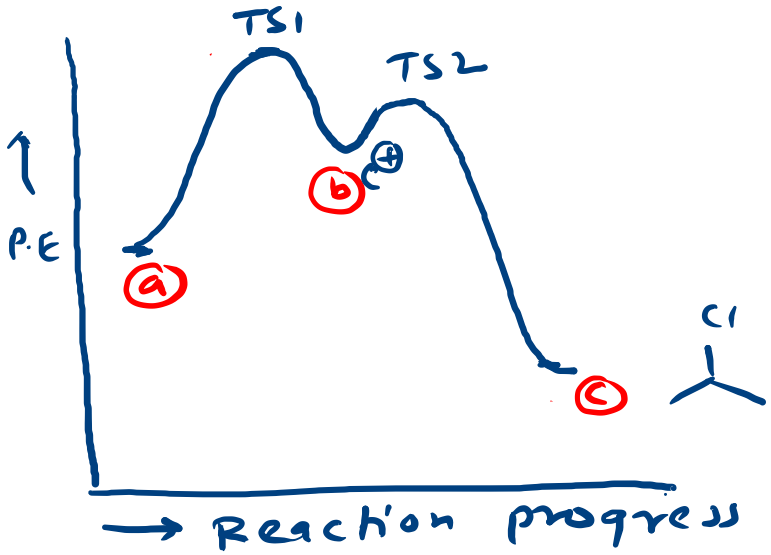
EAR by classical carbocation

(I) Addition H-X to Alkene or Alkyne

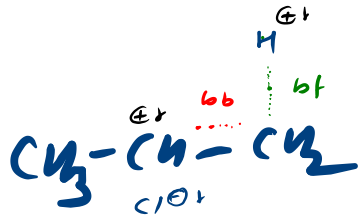
Q. (127) $CH_3-CH=CH_2 \xrightarrow{H-Cl} CH_3-\underset{Cl}{CH}-CH_3$

Mech

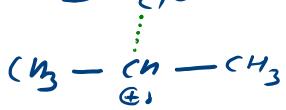




TS1 ⇒



TS2 ⇒



Characteristics of reaction

(1) it is two step reaction

Step I ⇒ formation of I^{st} $C^{\oplus}$ which is RDS

Step II ⇒ Attack of $Nu^{\ominus}$

(2) Rate of Reaction ∝ stability of I^{st} $C^{\oplus}$

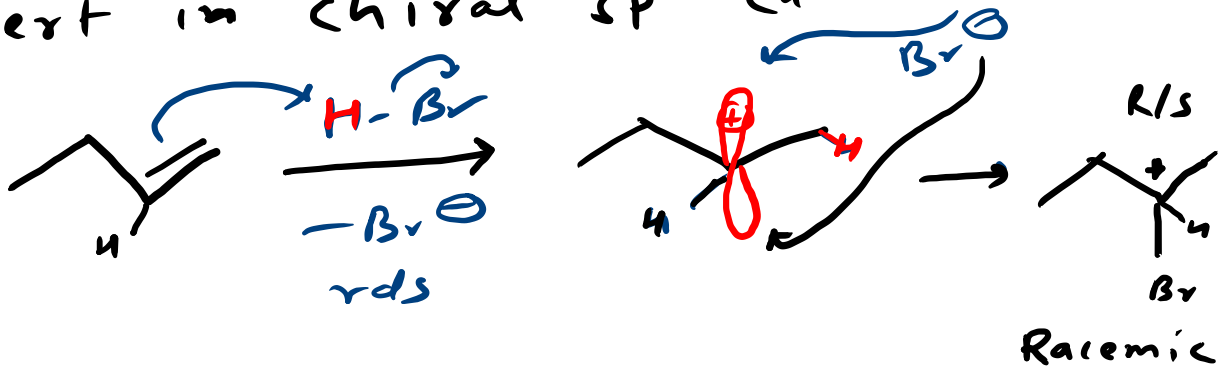
(3) Rate of Reaction ∝ [Alkene] $[H^+]$

∴ It follow 2^{nd} order kinetics.

(4) Carbocation intermediate undergoes rearrangement, if possible.

(5) p-orbital of carbocation intermediate can approach from either side to attacking $Nu^{\ominus}$, and gives two different configuration (i.e R/S), if sp^2 $C^{\oplus}$ convert in chiral sp^3 carbon.

eg (128)



(6) This Addition is Acc. to Markovnikov's rule.

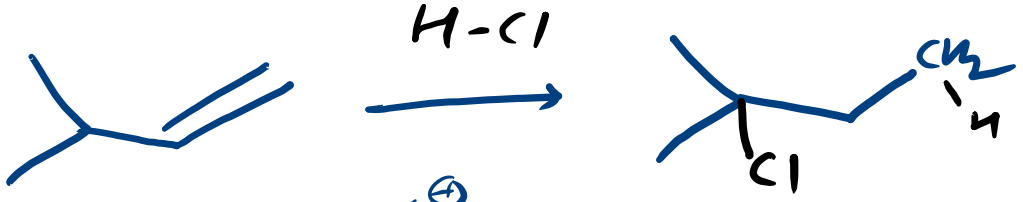
(Modified Markovnikov's rule)

MARCOVNIKOV'S RULE

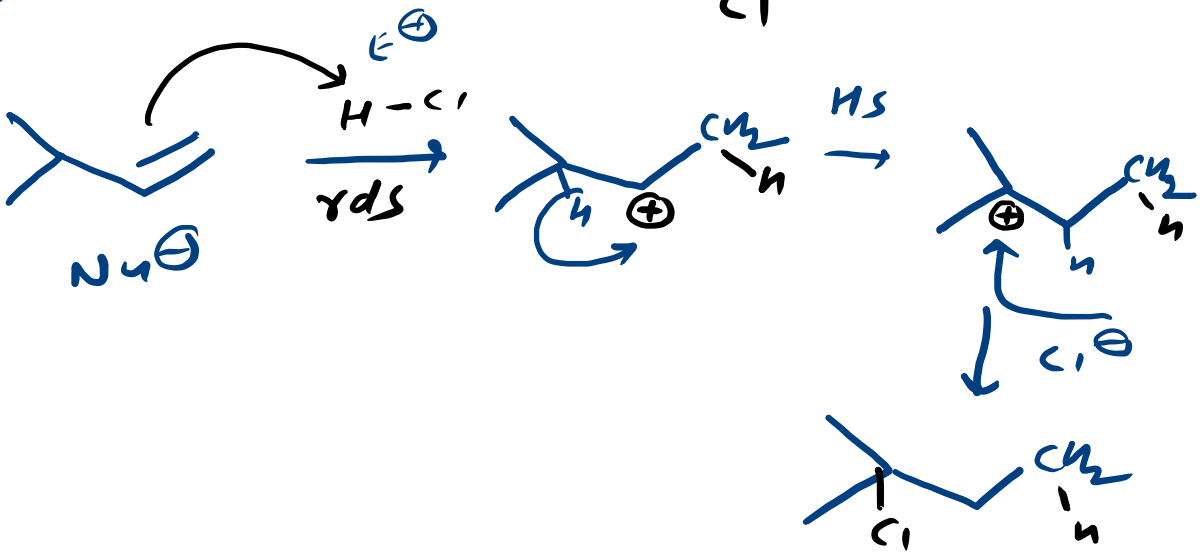
In Addition of H-X to unsymmetrical alkene or Alkyne, $H^{\oplus}$ attached to that double/tripple bonded carbon which has more No. of Hydrogens and $X^{\ominus}$ attached to that double or tripple bonded carbon, which has less No. of Hydrogen. (H Loves H)

Eg 127 & 128

(129)



Mech



Modified Markovnikov's rule. (MMR)

in EAR to unsymmetrical Alkene/Alkyne, $E^{\oplus}$ attached to that double bonded carbon which gives more stable Ist $C^{\oplus}$, & finally $Nu^{\ominus}$ attached to most stable carbocation.

Compare Rate of reaction towards EAR

(130) $CH_2=CH_2 \xrightarrow{HI} r1$ $CH_3\overset{\oplus}{C}H_2$

$CH_3-CH=CH_2 \xrightarrow{HI} r2$

$\gamma 3 > \gamma 2 > \gamma 1$

$\xrightarrow{HI} r3$

(131)

(a) $\xrightarrow{H-Cl} r1$

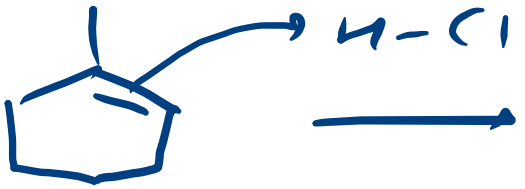
$\gamma 1 > \gamma 3 > \gamma 2$

(b) $\xrightarrow{HCl} r2$

(c) $\xrightarrow{HCl} r3$

write the product

(132)

(133) 

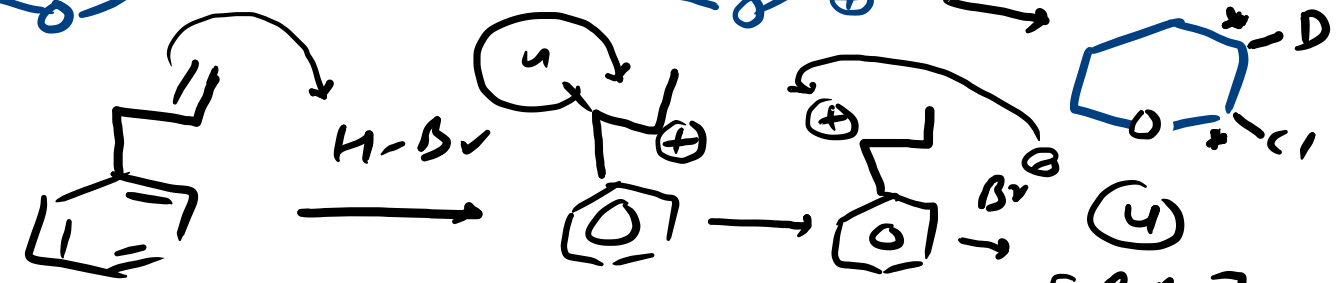
HW

(134)

135

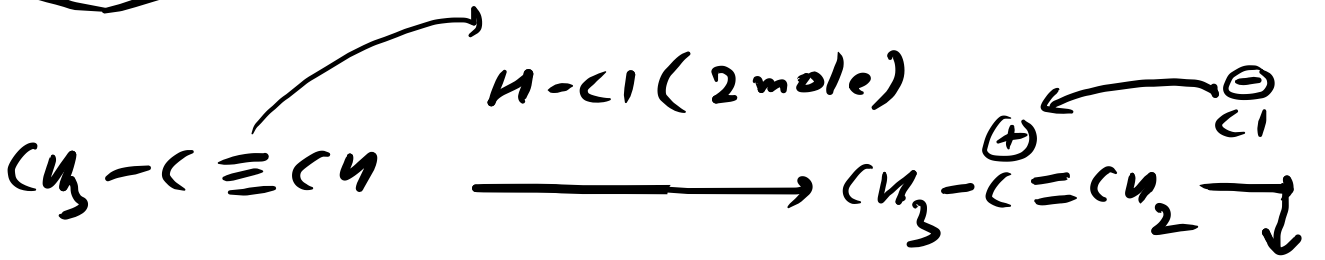
HW

136

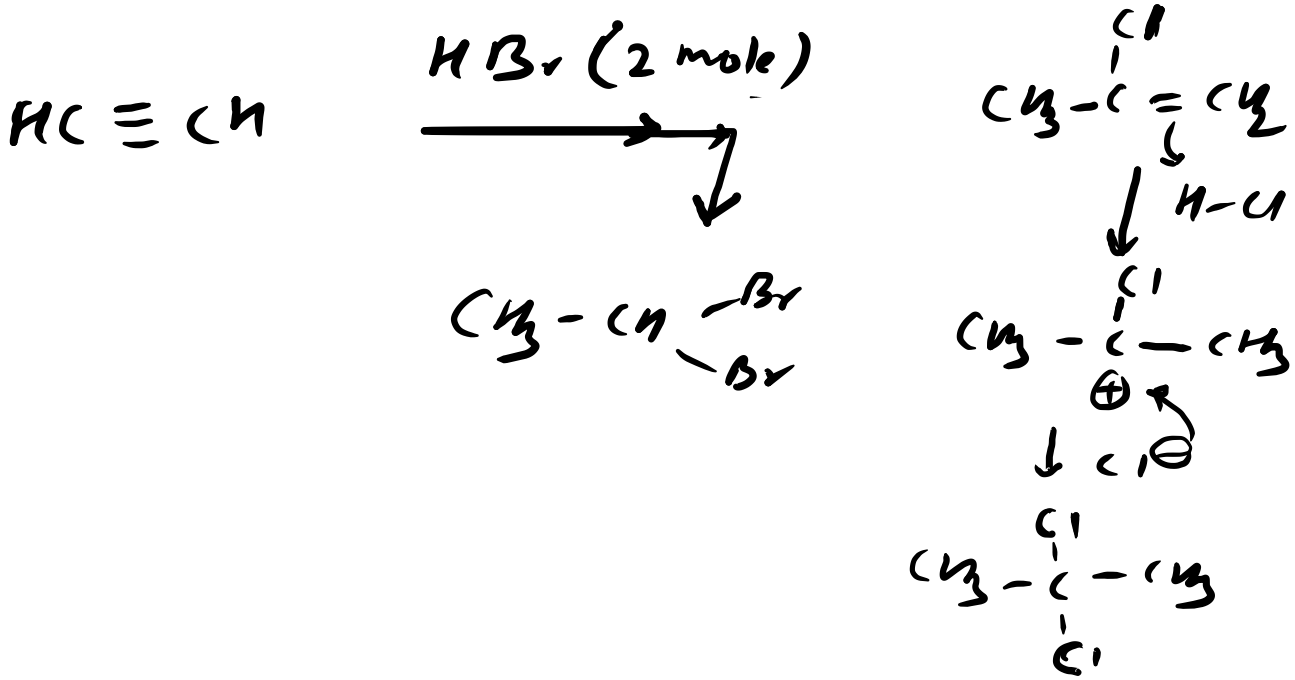


137

138



139



II Acidic hydration of Alkene

(140) $CH_3-CH=CH_2 \xrightarrow[\text{or dil } H_2SO_4]{H_2O/H^\oplus} CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3$

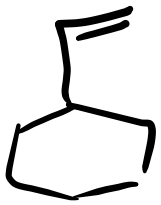
Mech

rds $\downarrow H^\oplus$

$CH_3-\overset{\displaystyle \oplus}{\underset{|}{CH}}-CH_3$ ($E^\oplus$) $\rightleftharpoons$ $H-\ddot{O}-H$ (:H_2O:) $\rightleftharpoons$ $H-\overset{\displaystyle \oplus}{O}(H)-H$

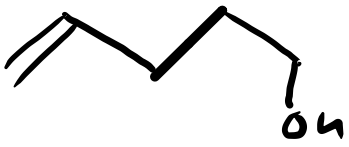
$CH_3-\underset{\displaystyle \overset{\displaystyle \oplus}{O}H_2}{\underset{|}{CH}}-CH_3 \downarrow -H^\oplus$

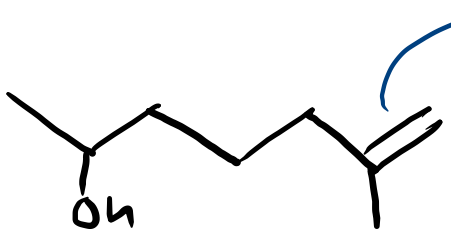
$CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3$.

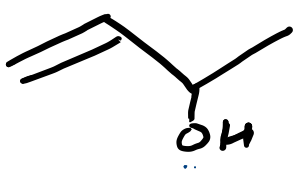
(141)  $\xrightarrow{H_2O/H^\oplus}$?

(142) $\xrightarrow{H_3O^\oplus}$?

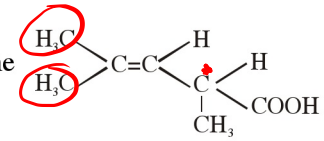
(143) $CH_3-CH=CH_2 \xrightarrow{CH_3-OH/H^\oplus}$?

(144)  $\xrightarrow{H^\oplus}$?

(145)  $\xrightarrow{H^\oplus}$?

(146)  $\xrightarrow{H^\oplus}$

EXERCISE #(J-ADVANCE OBJECTIVE)

1. The  shows : **[IIT-1995]**

(A) Geometrical isomerism (B) Optical isomerism

(C) Geometrical & optical isomerism (D) tautomerism

2. When cyclohexane is poured on water, it floats, because : **[IIT-1997]**

(A) cyclohexane is in 'boat' form (B) cyclohexane is in 'chair' form

(C) cyclohexane is in 'crown' form (D) cyclohexane is less dense than water

3. How many optically active stereoisomers are possible for butane –2,3–diol : **[IIT-1997]**

(A) 1 (B) 2 (C) 3 (D) 4

4. The number of possible enantiomeric pairs that can be produced during monochlorination of 2–methyl butane is : **[IIT-1997]**

(A) 2 (B) 3 (C) 4 (D) 1

5. Which of the following compounds will show geometrical isomerism : **[IIT-1998]**

(A) 2-butene (B) propene

(C) 1-phenylpropene (D) 2-methyl-2-butene

6. Which of the following compound will exhibits geometrical isomerism : **[IIT-2000]**

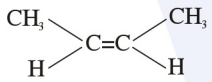
(A) 1–phenyl–2–butene (B) 3–phenyl–1–butene

(C) 2–phenyl–1–butene (D) 1, 1–diphenyl–1–propene

7. The number of isomers for the compound with molecular formula $C_2BrClFI$ is : **[IIT-2000]**

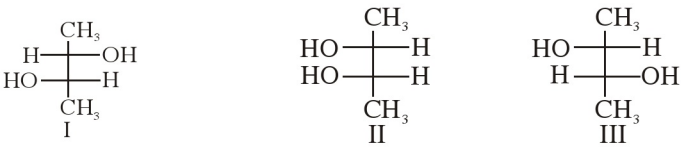
(A) 3 (B) 4 (C) 5 (D) 6

8. Which of the following has the lowest dipole moment : **[IIT-2000]**

(A)  (B) $CH_3–C\equiv C–CH_3$

(C) $CH_3CH_2C\equiv CH$ (D) $CH_2=CH–C\equiv CH$

9. Identify the pairs of enantiomers and diastereomers from the following compounds I, II and III **[IIT-2000]**

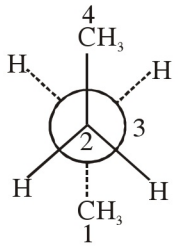


10. Which of the following compounds exhibits stereoisomerism– **[IIT-2002]**

(A) 2–Methylbutene–1 (B) 3–Methylbutyne–1

(C) 3–Methylbutanoic acid (D) 2–Methylbutanoic acid

11. In the given conformation, If C_2 is rotated about C_2–C_3 bond anticlockwise by an angle of 120° then the conformation obtained is : **[IIT-2004]**



(A) Fully eclipsed conformation (B) Partially eclipsed conformation

(C) Gauche conformation (D) Staggered conformation

12. On monochlorination of 2-methylbutane, the total number of chiral compounds formed is :

(A) 2 (B) 4 (C) 6 (D) 8 **[IIT-2004]**

13. (i) $\mu_{obs} = \sum_i \mu_i x_i$

where μ_i is the dipole moment of a stable conformer of the molecule, Z–CH_2–CH_2–Z and x_i is the mole fraction of the stable conformer. **[IIT- 2005]**

Given : μ_{obs} = 1.0 D and x (Anti) = 0.82

Draw all the stable conformers of Z–CH_2–CH_2–Z and calculate the value of $\mu_{(Gauche)}$.

(ii) Draw the stable conformer of Y–CHD–CHD–Y(meso form), when Y = CH_3 (rotation about C_2–C_3) and Y = OH (rotation about C_1–C_2) in Newmann projection.

14. The number of structural isomers of C_6H_{14} is : **[IIT-2007]**

(A) 3 (B) 4 (C) 5 (D) 6

15. **Statement-I :** Molecules that are not superimposable on their mirror images are chiral

Because

Statement-II : All chiral molecules have chiral centres. **[IIT-2007]**

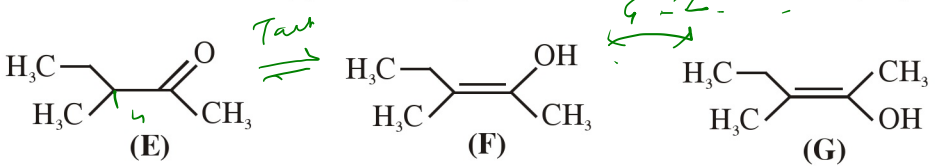
(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

16. The correct statement(s) concerning the structures E, F and G is (are) **[IIT-2008]**



(A) **E, F** and **G** are resonance structures (B) **E, F** and **E, G** are tautomers

(C) **F** and **G** are geometrical isomers (D) **F** and **G** are diastereomers

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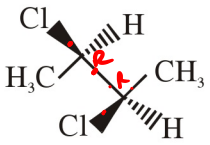
17. The correct statement(s) about the compound given below is (are) : **[IIT-2008]**



(A) The compound is optically active

(B) The compound possesses centre of symmetry

(C) The compound possesses plane of symmetry

(D) The compound possesses axis of symmetry

18. The correct statement(s) about the compound $H_3C(HO)HC-CH=CH-CH(OH)CH_3$ **(X)** is (are) :

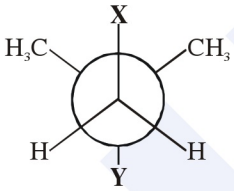
(A) The total number of stereoisomers possible for X is 6 **[IIT-2009]**

(B) The total number of diastereomers possible for X is 3

(C) If the stereochemistry about the double bond in X is trans, the number of enantiomers possible for X is 4

(D) If the stereochemistry about the double bond in X is cis, the number of enantiomers possible for X is 2

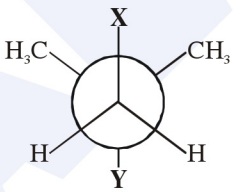
19. In the Newman projection for 2,2–dimethylbutane **[IIT-2010]**



X and **Y** can respectively be –

(A) H and H (B) H and C_2H_5 (C) C_2H_5 and H (D) CH_3 and CH_3

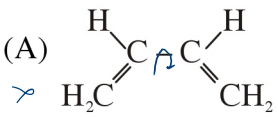
20. In the Newman projection for 2,2–dimethylbutane **[IIT-2010]**



X and **Y** can respectively be –

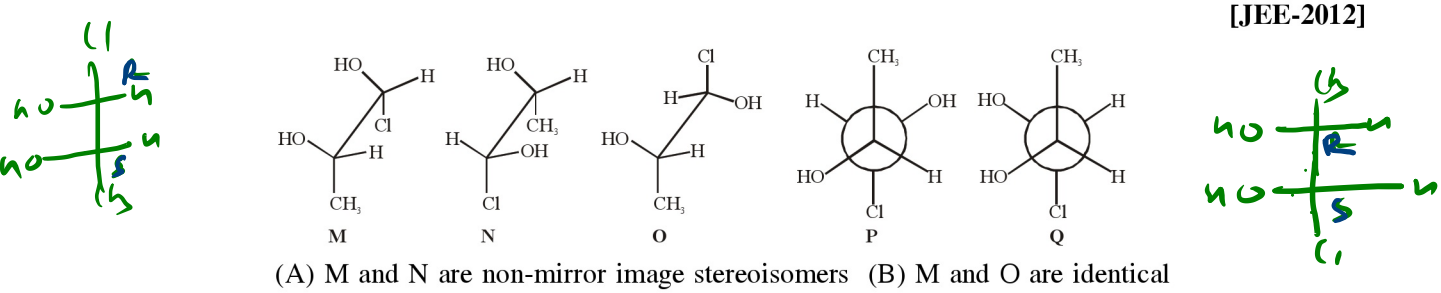
(A) H and H (B) H and C_2H_5 (C) C_2H_5 and H (D) CH_3 and CH_3

21. Amongst the given option, the compound(s) in which all the atoms are in one plane in all the possible conformations (if any), is (are) - **[IIT-2011]**

(A)  (B) $H-C\equiv C-C$ (C) $H_2C=C=O$ (D) $H_2C=C=CH_2$

model06\B0B0-BA\Kota\JEE(Advanced)\Unifinse Physics\Sheet\Magnetic Effect of Current\Eng\00_Theory.p65

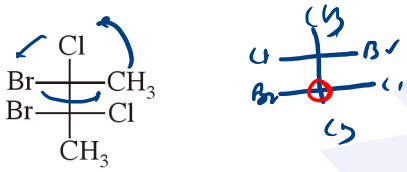
22. Which of the given statement(s) about N,O,P and Q with respect to M is (are) correct ? **[JEE-2012]**



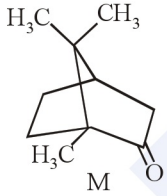
(A) M and N are non-mirror image stereoisomers (B) M and O are identical

(C) M and P are enantiomers (D) M and Q are identical

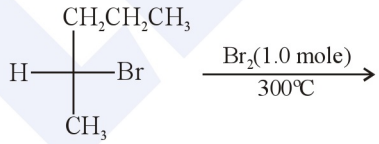
23. The total number(s) of **stable** conformers with **non-zero** dipole moment for the following compound is (are) **[JEE-2014]**



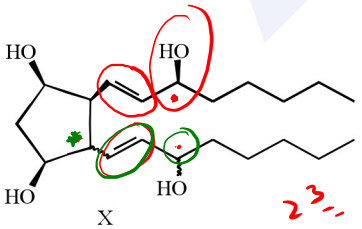
24. The total number of stereoisomers that can exist for M is : **[JEE-2015]**



25. In the following monobromination reaction, the number of possible chiral products is : **[JEE-2016]**



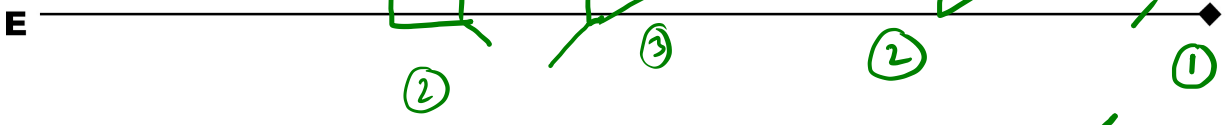
26. For the given compound X, the total number of optically active stereoisomers is____. **[IIT-JEE 2018]**



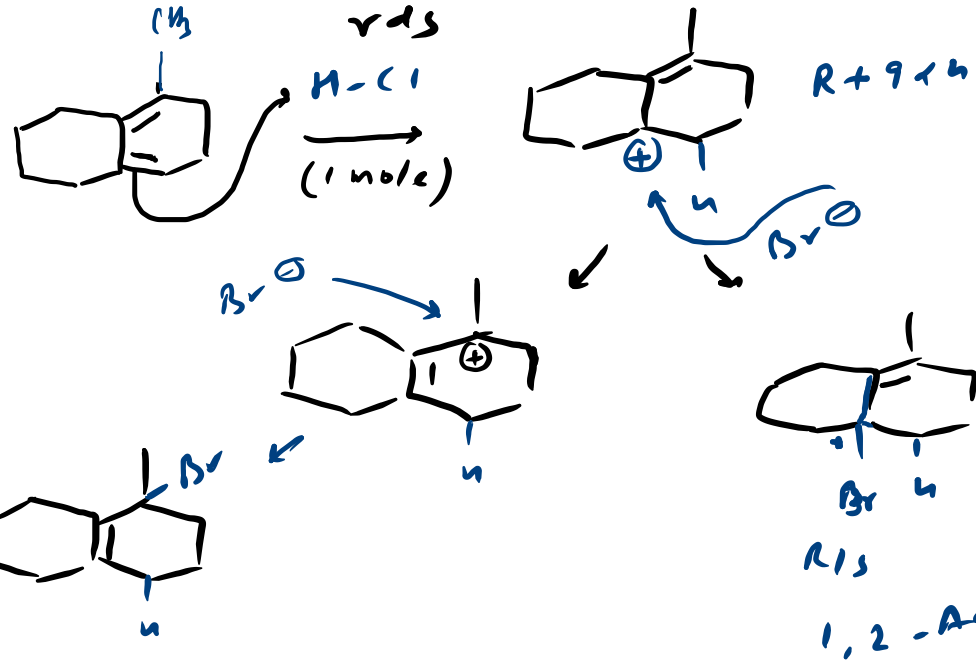
This type of bond indicates that the configuration at the specific carbon and the geometry of the double bond is fixed

This type of bond indicates that the configuration at the specific carbon and the geometry of the double bond is NOT fixed

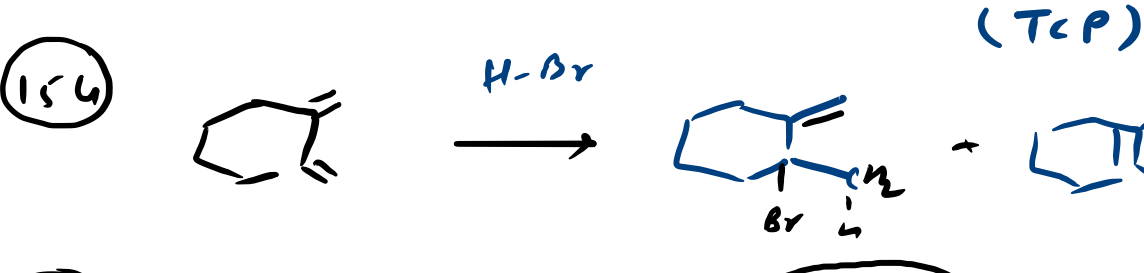
27. Total number of isomers, considering both structural and stereoisomers, of cyclic ethers with the molecular formula C_4H_8O is ___ **[IIT-JEE 2019]**



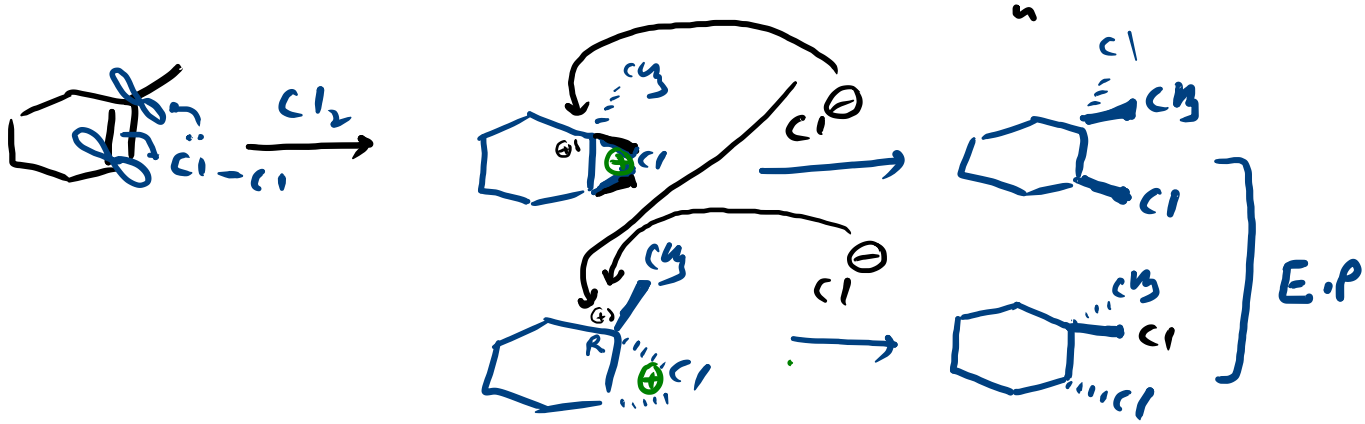
(153)



(154)



(160)



Note:

* :E⊕ (Electrophile with lone pair) form Non classical carbocation with alkene & Alkyne

eg. :Br_2 , :Cl_2 , :I-Cl , :I-Br, :I_2 , :Br-Cl etc.

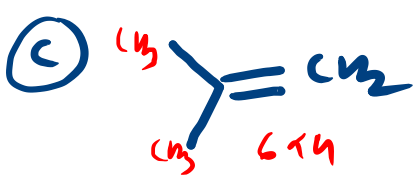
(16) Compare Rate of Reaction towards EAR

(a) $CH_2 = CH_2 \xrightarrow[CCl_4]{Br_2} r_1$

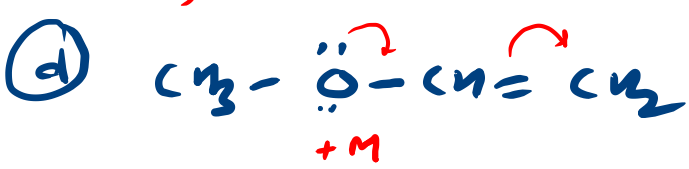
0 alk

(b) $CH_3 - CH = CH_2 \xrightarrow[\quad]{\quad} r_2$

3 alk

(c) 

$\xrightarrow[CCl_4]{Br_2} r_2$

(d) 

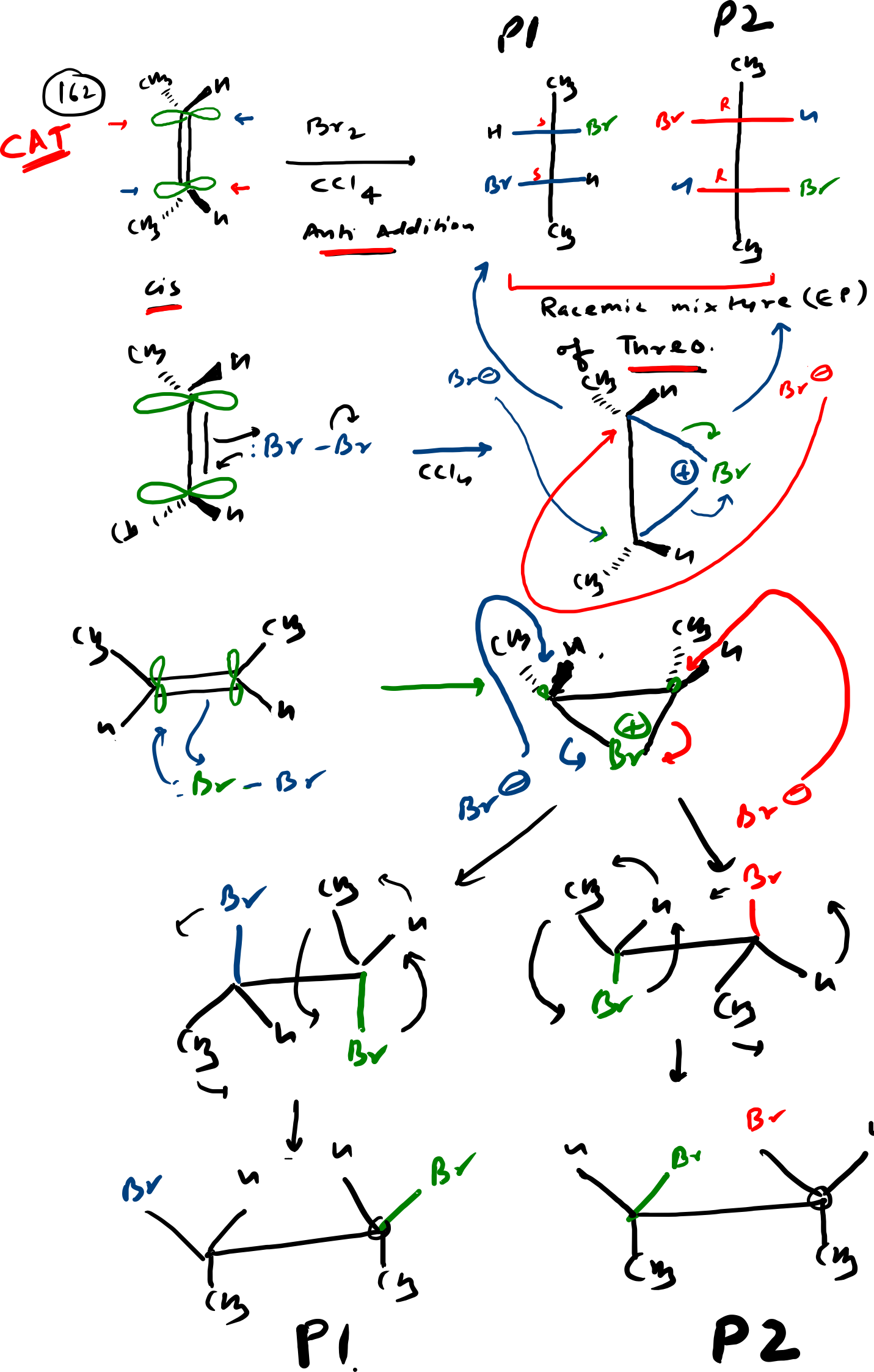
$\xrightarrow[CCl_4]{Br_2} r_4$

Rate of reaction ∝ Nucleophilicity of Alkene

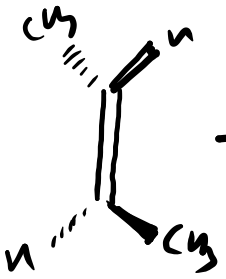
$r_4 > r_3 > r_2 > r_1$

Reactant	Addition	product
Cis	Anti	Threo
Trans	Anti	Erythro.

Cis	Syn	Erythro
Trans	Syn	Threo


P1
P2
162
CAT
Br_2
CCl_4
Anti Addition
cis
Racemic mixture (EP)
of Threo.
CCl_4
P1
P2

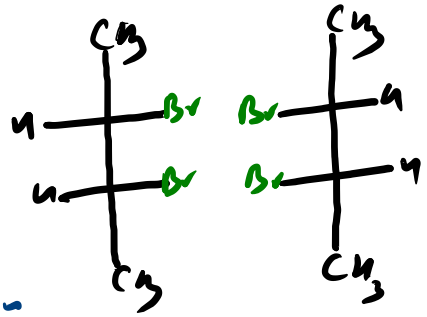
(163) TAE



Trans

Br_2 / CCl_4 — Anti Addition

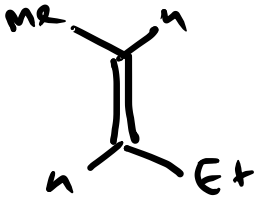
identical



Erythro

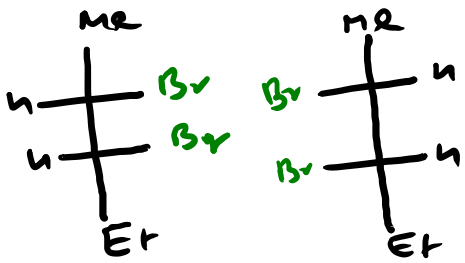
Meso (mirror image)

(164) TAE



Trans

Br_2



Racemic mix of Erythro.

for symmetrical only (2-butene, or 3-hexene)

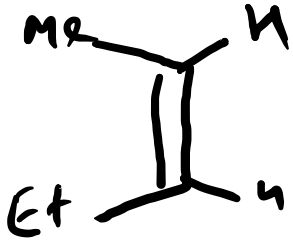
	Add	
Cis	Cis	Meso
T	T	Meso
Cis	Trans	Racemic mix.
T	C	Racemic mix

Cis reactant
Cis addition
meso product

Cis → syn

HW

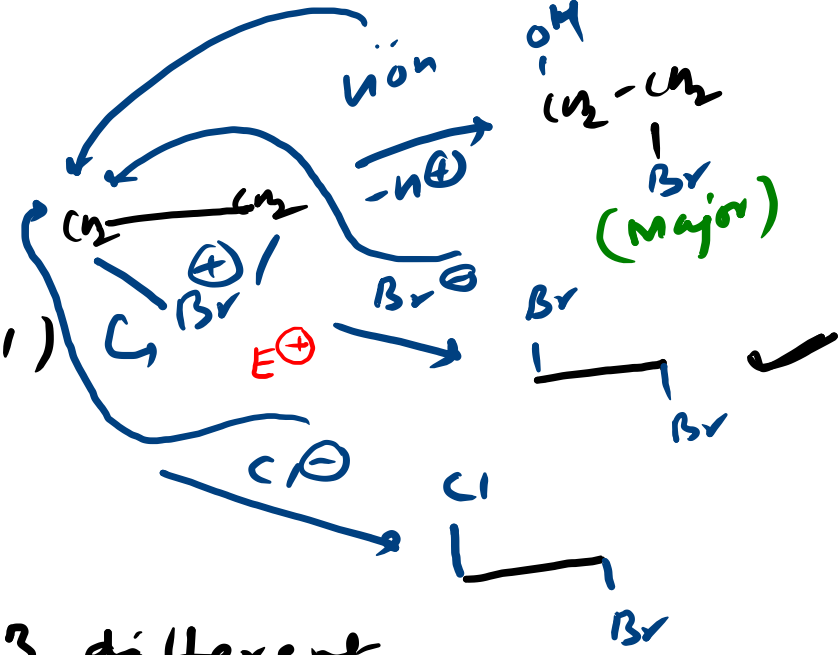
(165)



Cl_2 / CCl_4 → ?

Francis Experiment

(166) $CH_2 = CH_2 \xrightarrow[\text{aq NaCl}]{Br_2 \text{ rds}}$ (5% aq NaCl) Brine

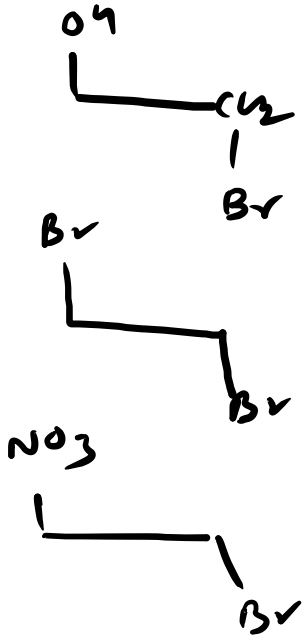


Francis got 3 different product, so he concluded that intermediate is in Electrophilic nature so Available 3 different $Nu^{\ominus}$ gives 3 different product.

∴ This reaction is EAR (NOT NAR)

or Addition of $E^{\oplus}$ participate in RDS.

(167) $CH_2 = CH_2 \xrightarrow[\text{aq } NaNO_3]{Br_2}$



HW ⇒ 168 to 179

168 Br_2 / CCl_4

169 Cl_2 / CCl_4

170 Br_2, HOH

$CH_3-CH=CH_2$

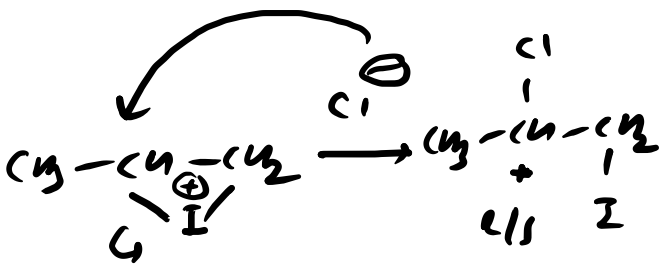
171 Cl_2, HOH

172 Br – OH

173 Cl – OH

174 I – OH

175 $\overset{\oplus}{I}-Cl$

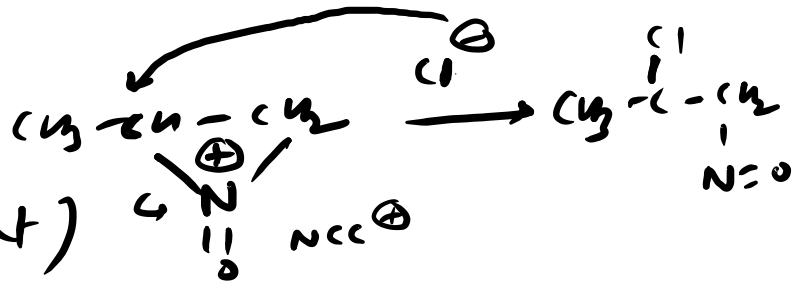


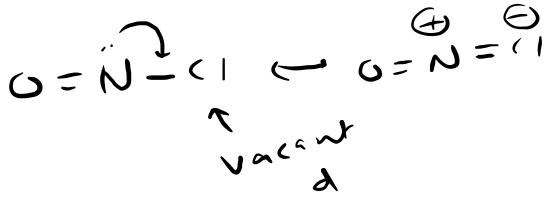
176 I – Br

177 Br – Cl

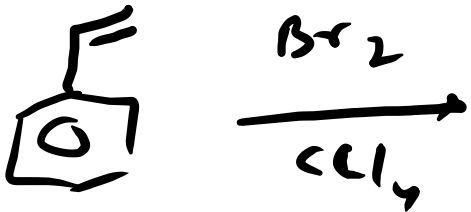
178 $O=\ddot{N}-Cl$ (NOCl)

(Tilden reagent)





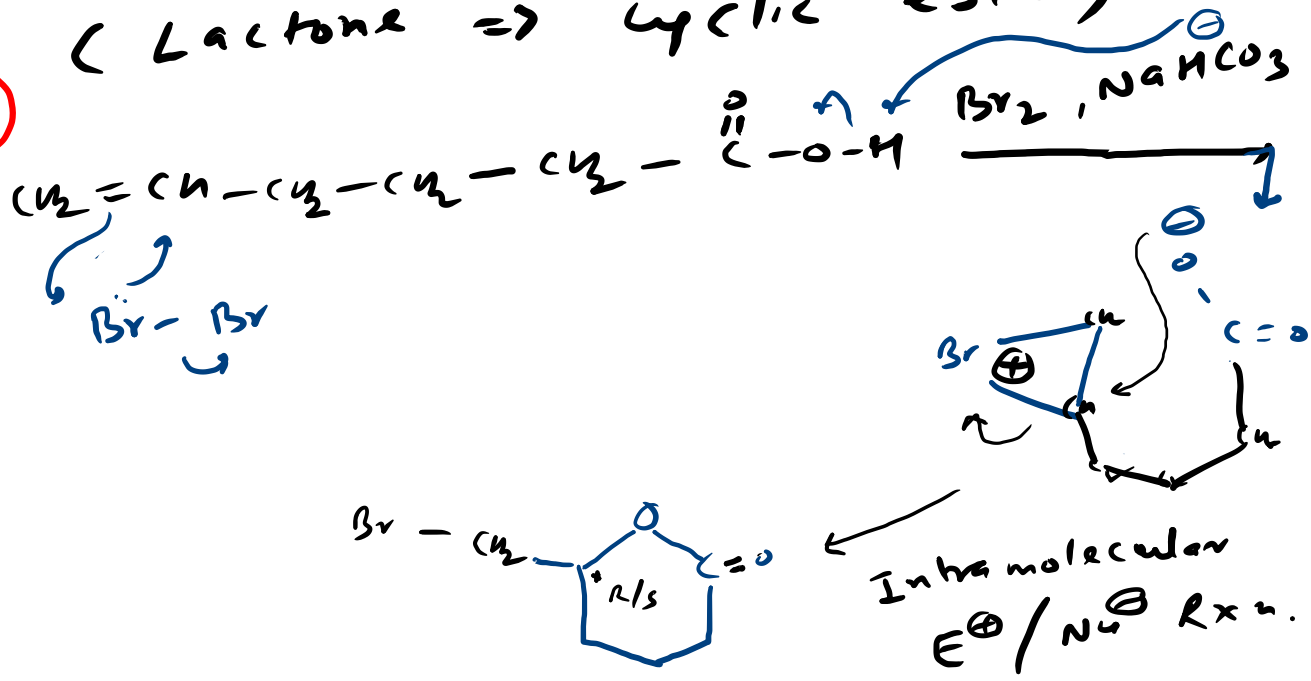
179



Bromo lactonisation

(Lactone => cyclic ester)

180



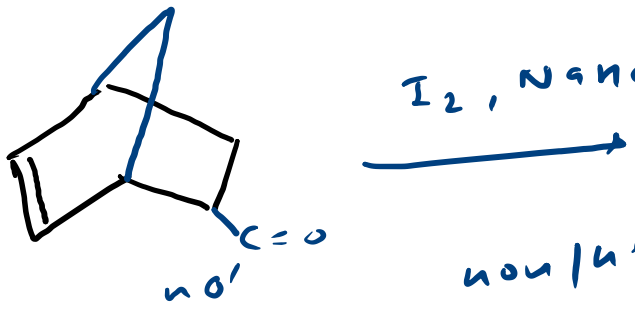
Iodo lactonisation

HW

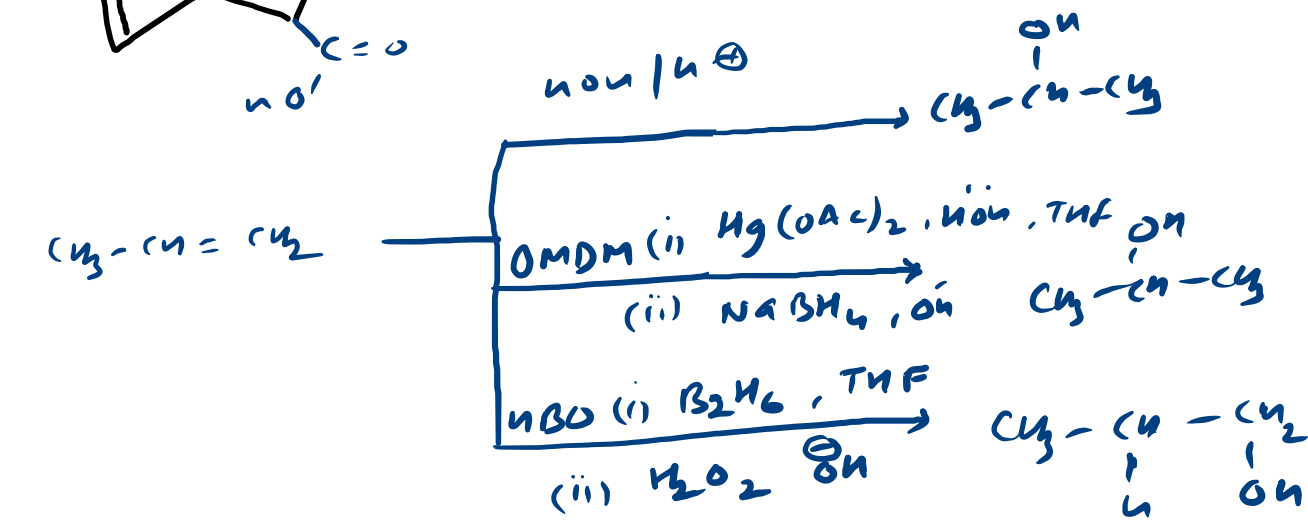
181

HW.

182



183

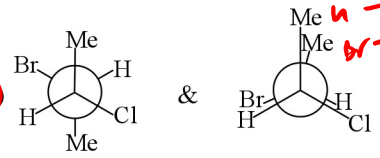


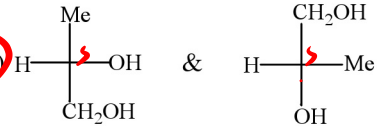
RACE # 16 — ORGANIC CHEMISTRY

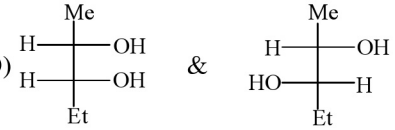
M.M. : 41 — TIME : 30 Min.

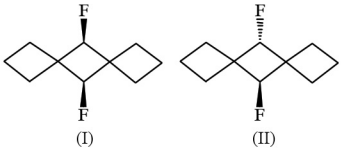
1. **Column - I** **Column - II** **[8, 0]**

(A) (P) Identical

(B)  (Q) Geometrical isomers

(C)  (R) Diastereoisomers

(D)  (S) Enantiomers

2. **Statement-1 :** 

(I) and (II) are optically active molecules. **[3, –1]**

Statement-2 : Molecules containing plane of symmetry or centre of symmetry or both cannot be optically active.

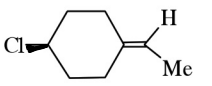
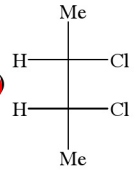
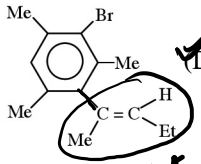
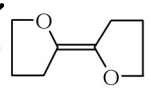
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

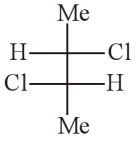
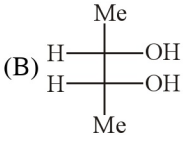
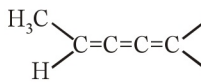
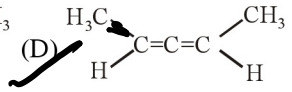
(C) Statement-1 is false, statement-2 is true

(D) Statement-1 is true, statement-2 is false

3. Which of the following can not show optical isomerism– **[3, –1]**

(A)  (B)  (C)  (D) 

4. Compound which never exist in diastereomeric pair **[3, –1]**

(A)  (B)  (C)  (D) 

5. Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false. **[3, –1]**

I. Me–CH=C=C=CH–Br is optically active.

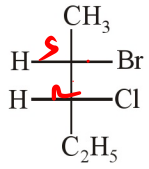
II. All optically active compound are chiral.

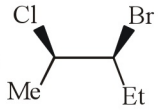
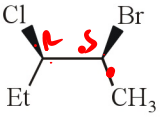
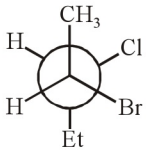
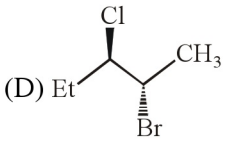
III. All chiral pyramidal molecules are optically inactive.

IV. $CH_3-\overset{1}{C}H_2-\overset{2}{C}H_2-\overset{3}{C}OOH$ and $CH_3-\underset{\underset{COOH}{|}}{\overset{2}{C}H}-CH_3$ are positional isomers.

(A) FTFF (B) FTFT (C) TFTF (D) FTTF

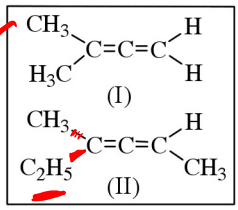
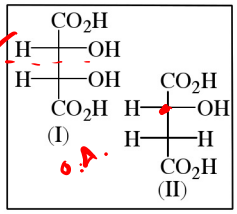
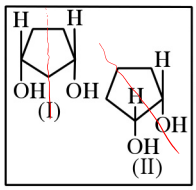
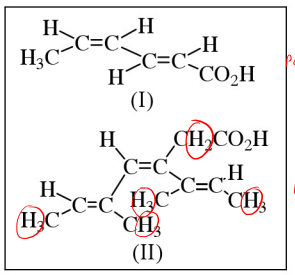
6. Which of the following molecule(s)/orientation(s) are identical to the [X] **[4, –1]**

[X] = 

(A)  (B)  (C)  (D) 

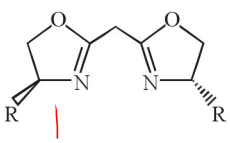
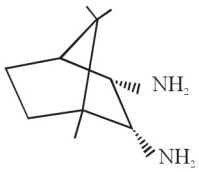
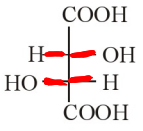
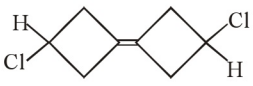
7. Which mixture of structure in each container would rotate plane polarised light : **[4, –1]**

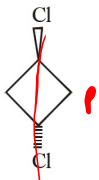
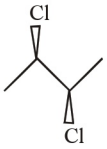
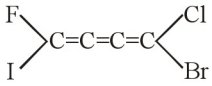
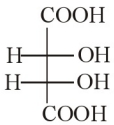
[Assuming each component does not react with each other]

(A)  (B)  (C)  (D) 

Paragraph for Q.No. 08 to 09

Among the following structures ?

(I)  (II)  (III)  (IV) 

(V)  (VI)  (VII)  (VIII) 

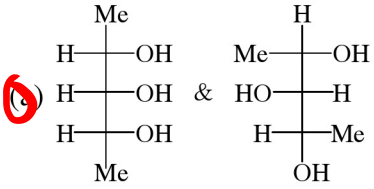
8. Optically active compound is : **[3, –1]**

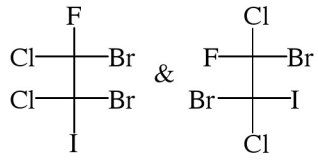
(A) III (B) IV (C) V (D) VII

9. Which of the following will not show optical isomerism - **[3, –1]**

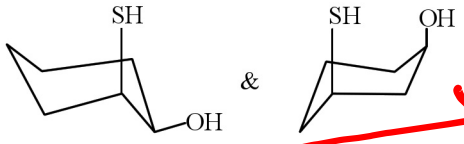
(A) I (B) II (C) V (D) VIII

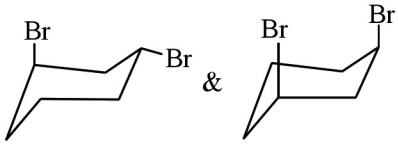
10 Find relationship between following pairs. **[6, 0]**

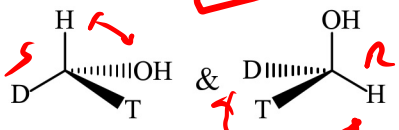
(a) 

(b) 

(c) $CH_3CH_2CH_2$–CHO & $CH_3CH(CHO)CH_3$

(d) 

(e) 

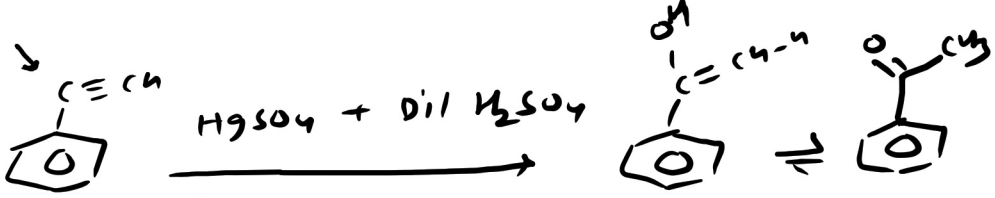
(f) 

Position

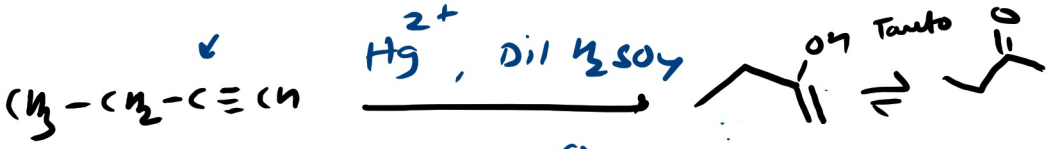
Enantiomers.

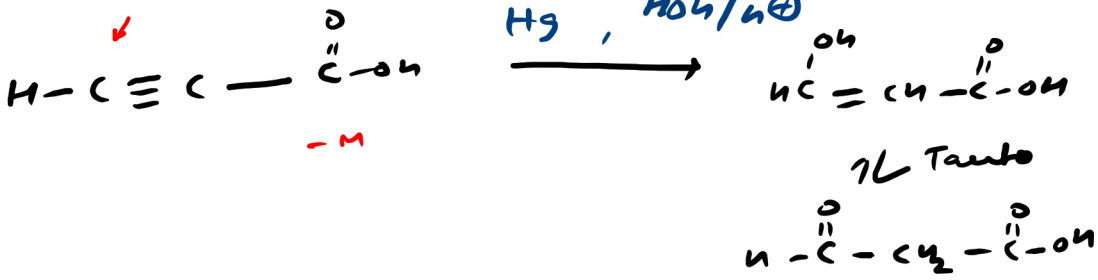
FILL THE ANSWER HERE

1.	A (P) (Q) (R) (S) (T)	2.	(A) (B) (C) (D)	3.	(A) (B) (C) (D)	4.	(A) (B) (C) (D)
	B (P) (Q) (R) (S) (T)						
	C (P) (Q) (R) (S) (T)						
	D (P) (Q) (R) (S) (T)						
5.	(A) (B) (C) (D)	6.	(A) (B) (C) (D)	7.	(A) (B) (C) (D)	8.	(A) (B) (C) (D)
9.	(A) (B) (C) (D)						
10(a).		(b)		(c)		(d)	
(e)		(f)					

(187) 

(188)

(189) 

(190) 

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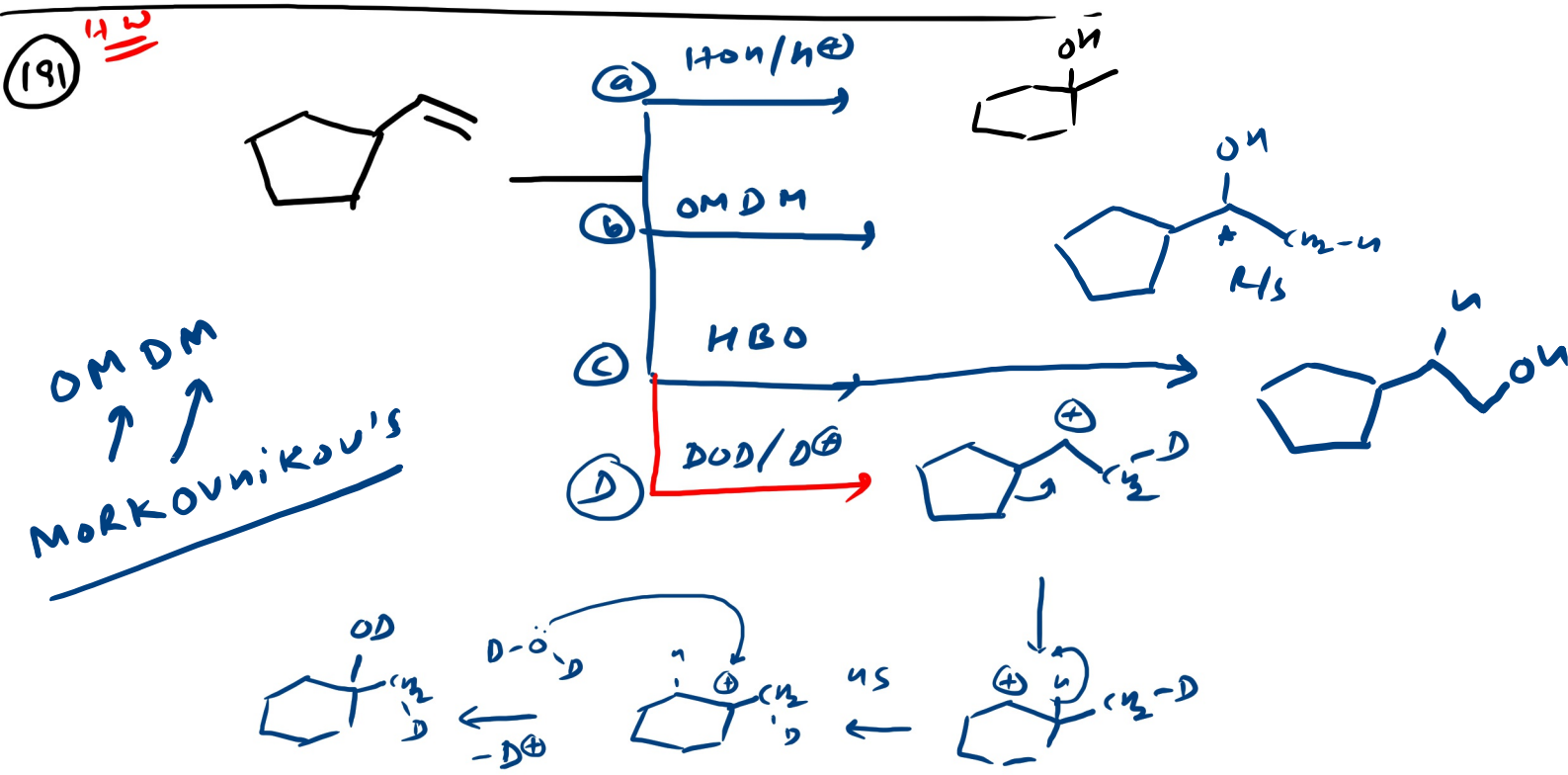
(191) 

Some examples of S_N2

HW

193 aq KOH or aq NaOH →

194 $Na^{\oplus}\,{}^{\ominus}S-R$ →

195 $CH_3\,Mg^{\delta\oplus}Br^{\delta\ominus}$ →

196 $CH_3-C\equiv C^{\ominus}\,Na^{\oplus}$ →

197 $LiAlH_4$ →

198 $K^{\oplus}\,{}^{\ominus}CN$ →

199 $Ag-CN$ →

200 $NaCN$ →

201 $K^{\oplus}\,{}^{\ominus}O-N=O$ or $NaNO_2$ →

202 $Ag-O-N=O$ →

203 $CH_3-\overset{O}{\overset{\|}{C}}-O^{\ominus}\,Na^{\oplus}$ →

204 $Na\,N_3$ →

205 $Na\,OR$ (WES) →

Swarts 206 $KF/NaF/AgF$ → DMF

Finkelstein 207 NaI, Acetone →

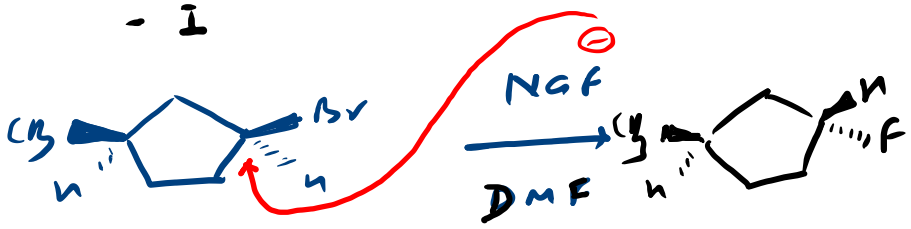
208 NH_3 →

(209) SWARTZ REACTION (halogen Exchange Rxn)

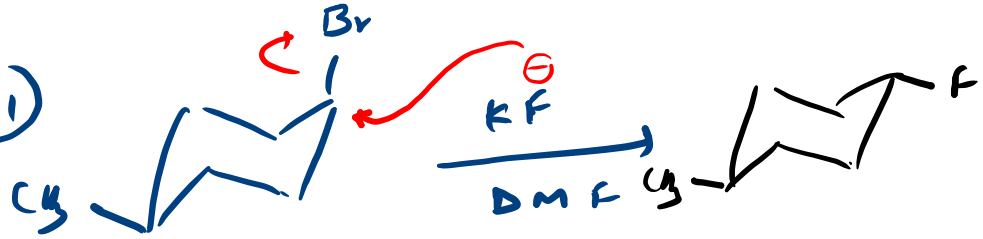
X = -Cl
-Br
-I

Stability $I^{\ominus} > Br^{\ominus} > Cl^{\ominus} > F^{\ominus}$

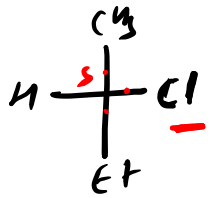
(210)



(211)



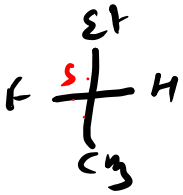
(212)



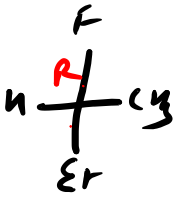
KF, DMF, S_N2 → (R) config product is

(A) ✗

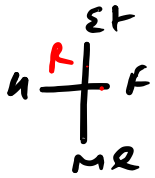
(B)



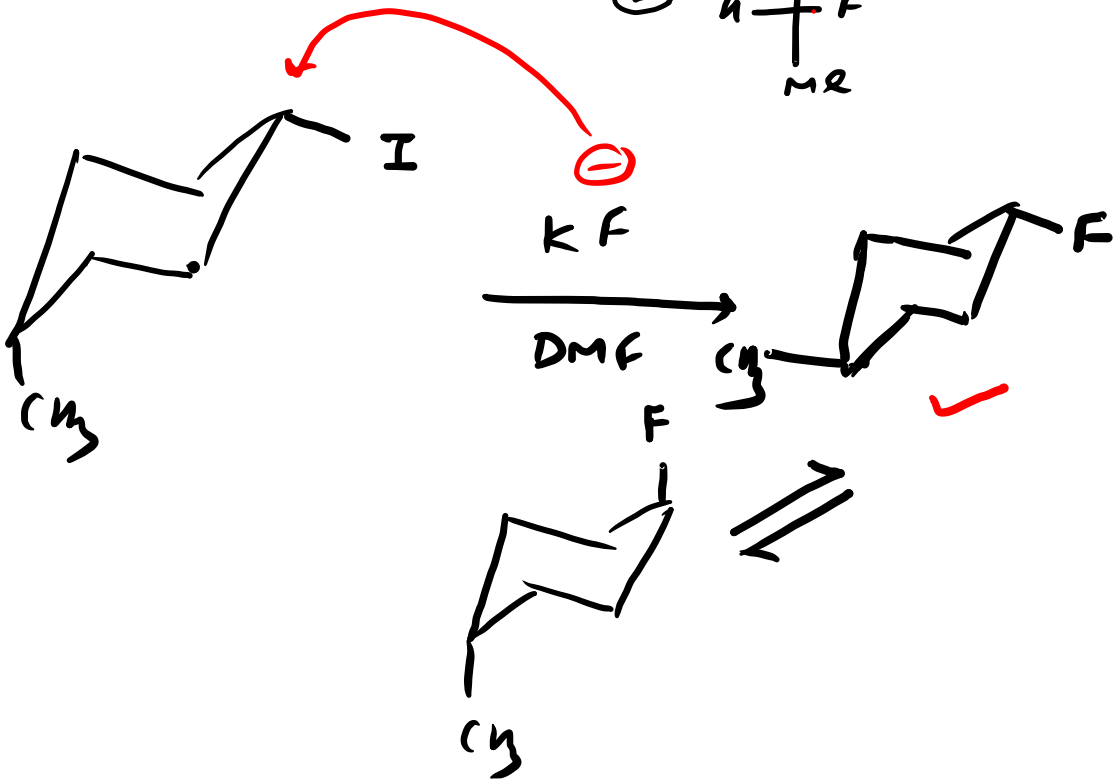
(C) ✓



(D) ✗

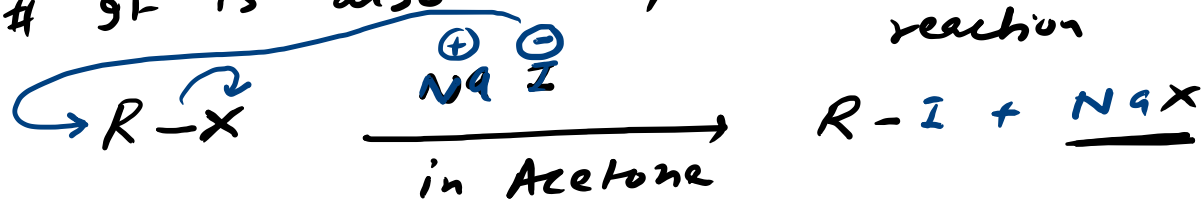


(213)



Finkelstein Reaction

It is also halogen exchange reaction



X = -Br ✓
-Cl ✓
~~-F~~

-F is Plg which do not give SN1 & SN2

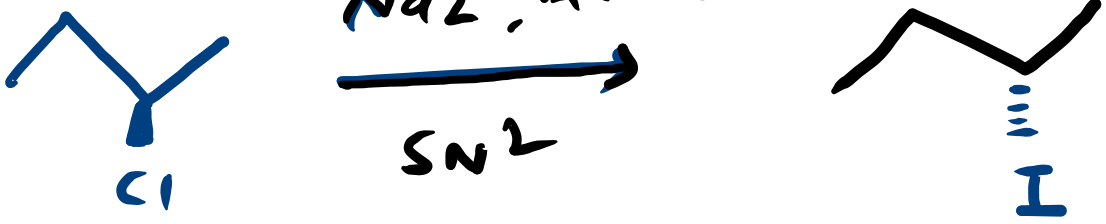
Lattice Energy is the driving force in Finkelstein reaction

Solubility in Acetone

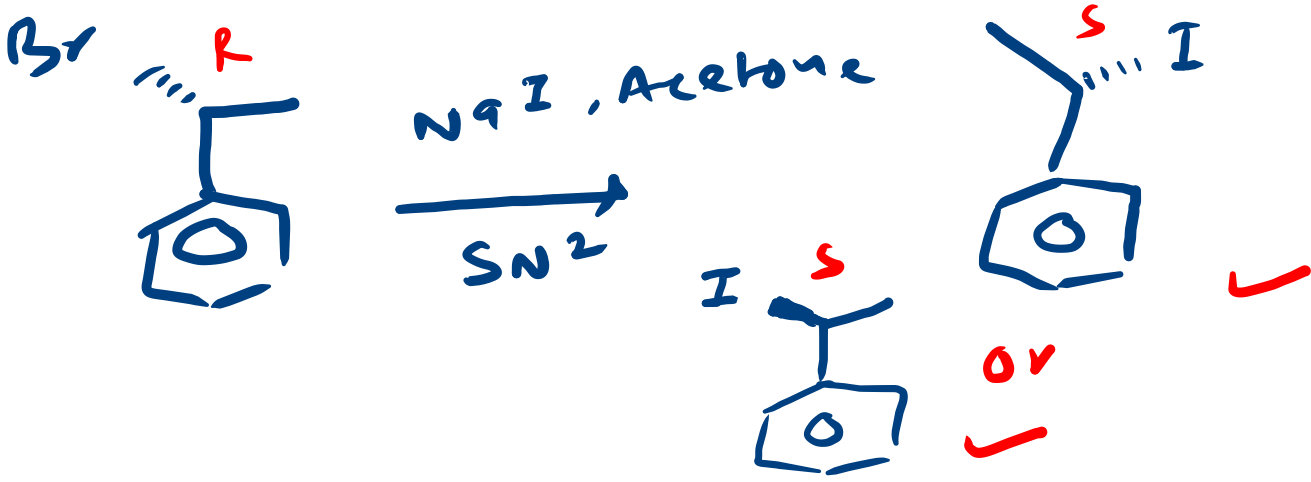
NaI > NaBr > NaCl

Soluble Insoluble Insoluble

21(i)

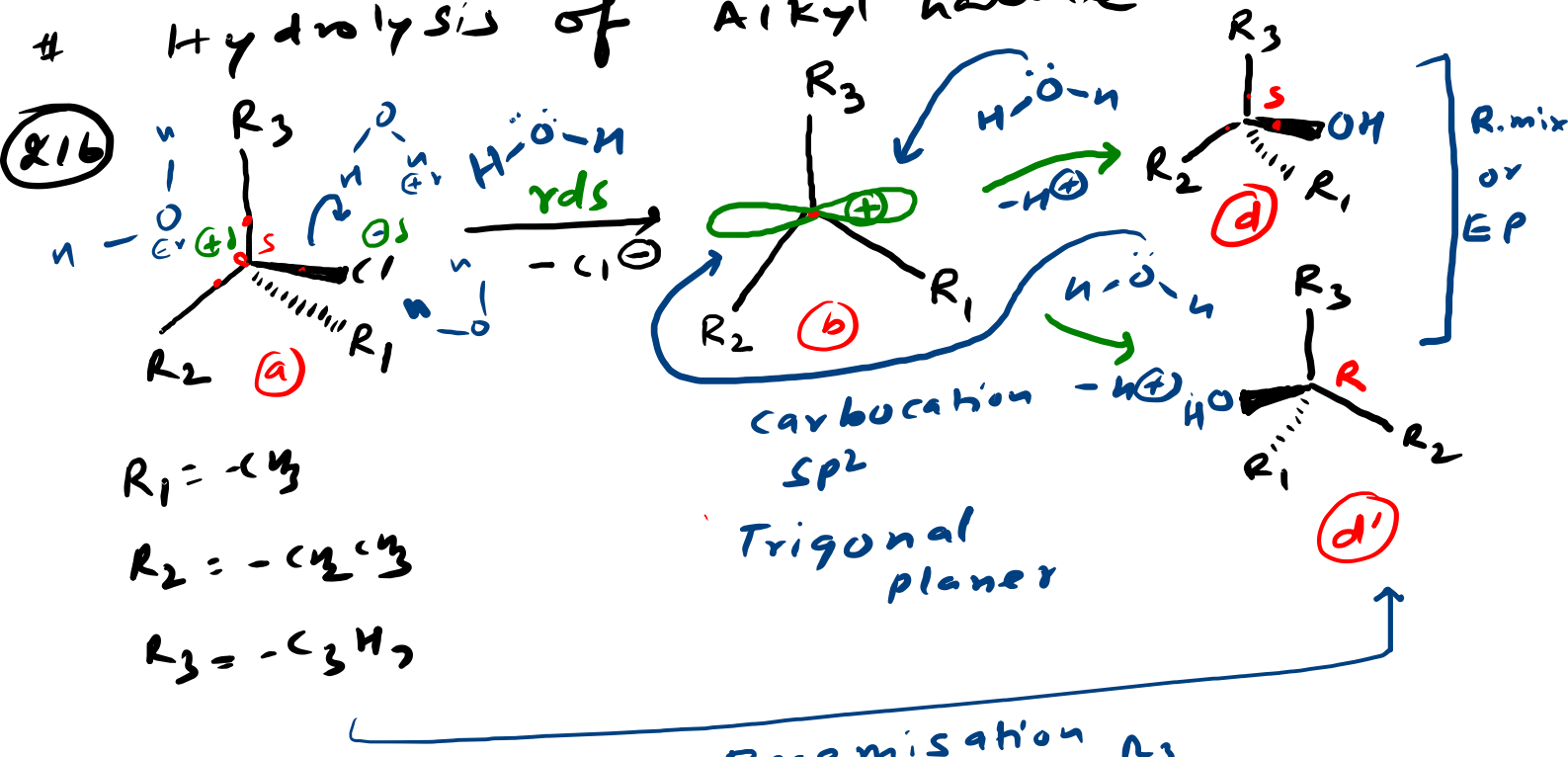


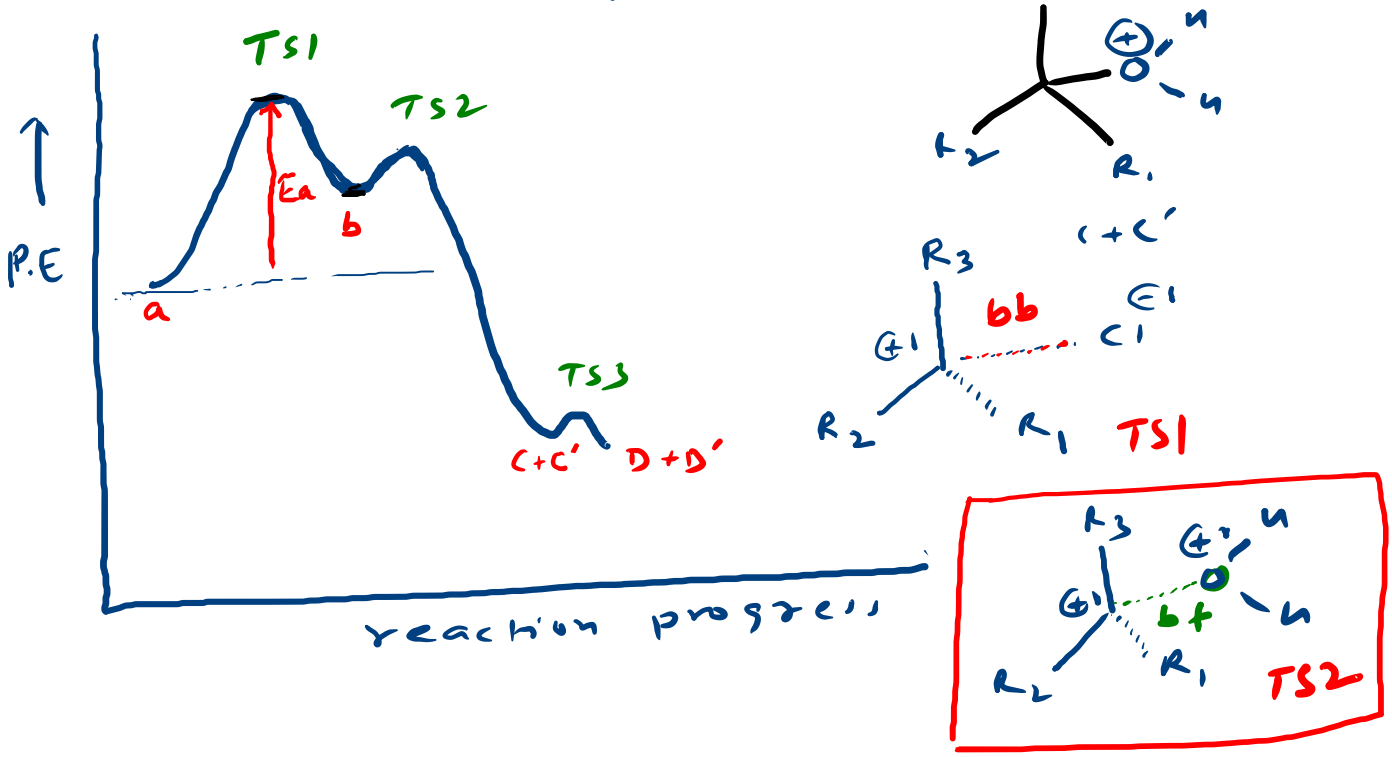
21(ii)



S_N1 (Nucleophilic substitution unimolecular)

Hydrolysis of Alkyl halide





Characteristics of reaction

(1) Generally it is two step reaction

Step-1 formation of carbocation which is rds

Step II Attack of $Nu^{\ominus}$

(2) **Rate of S_N1 $\propto$ stability of I^{st} carbocation**

(3) Rate of S_N1 $\propto$ [substrate] i.e. R-X

It follow I^{st} order kinetics
∴ called S_N1

(4) Carbocation intermediate undergoes rearrangement, if possible

(5) Solvation energy is the driving force in S_N1

(6) Racemisation takes place in S_N1 (But not always).

(7) factors which favour S_N1

(a) Structure of substrate

the structure which gives more stable I^{st} $C^{\oplus}$, favours S_N1

3° R-X > 2° R-X > 1° R-X > CH_3-X

(b) Effect of leaving group:

* Rate of $SN^1 \propto$ leaving tendency of lg

Rate of $SN^2 \propto$ leaving tendency of lg

Note poor lg do NOT give SN^1 and SN^2

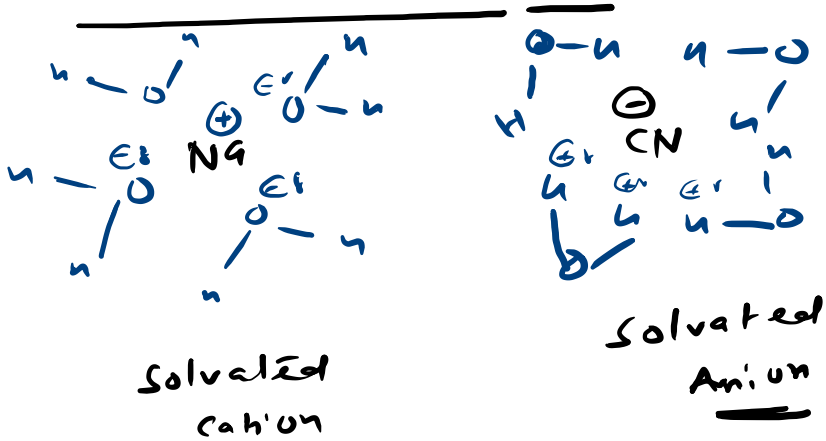
(c) Effect of Nucleophile

* Weak $Nu^{\ominus}$ favour SN^1
* Strong $Nu^{\ominus}$ favour SN^2

(d) Effect of solvent

* Polar protic solvent favour SN^1 (PPS) $\rightarrow$ SN^1
* Polar Aprotic solvent favour SN^2 (PAS) $\rightarrow$ SN^2

(21) NaCN in H_2O (PPS)



Solvated cation

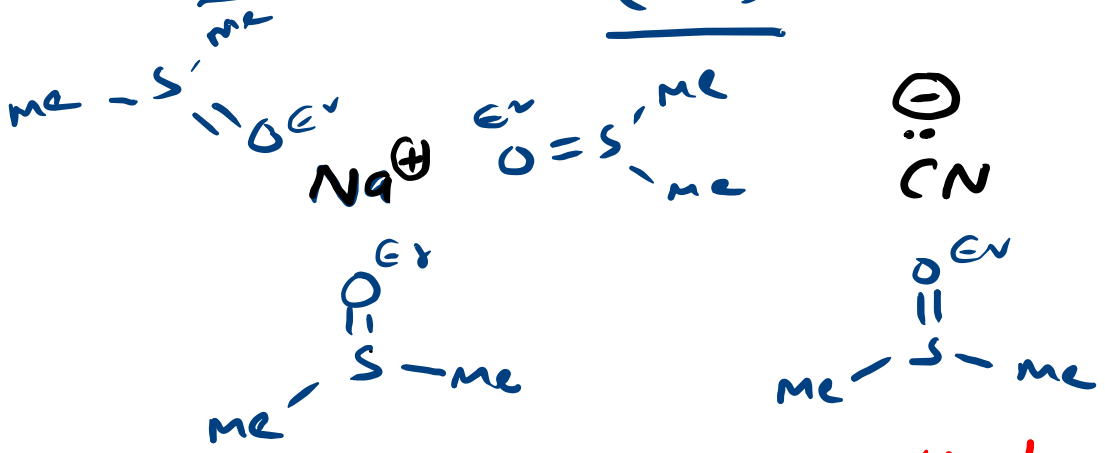
Solvated Anion

Nucleophilicity decreases

* PPS molecules have effective +δ and −δ charge, They can solvate cation as well as anion.

On the other hand molecules of PAS has only effective $\in$s charge, so PAS molecules solvate cations and repel anion, & increases nucleophilicity of Nucleophile.

Eg. (218) NaCN in DMSO (PAS)



Nucleophilicity in PPS

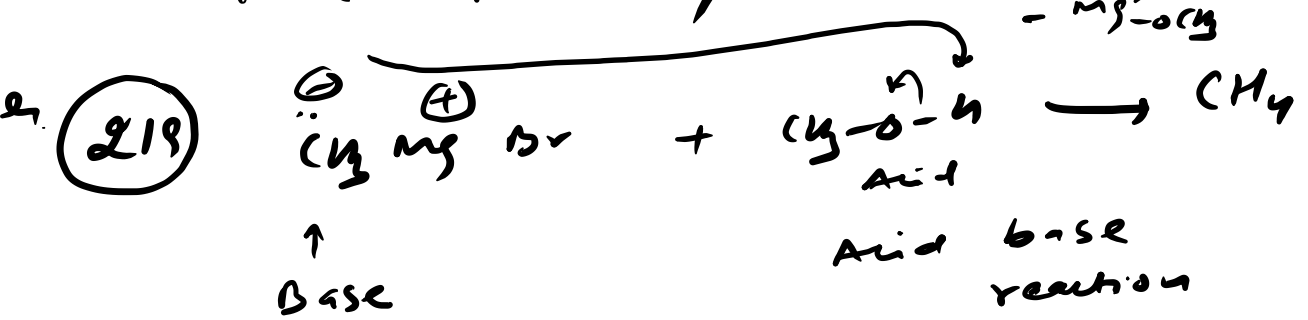
Case - 1 If Nucleophilic atom is same or belongs to same period,

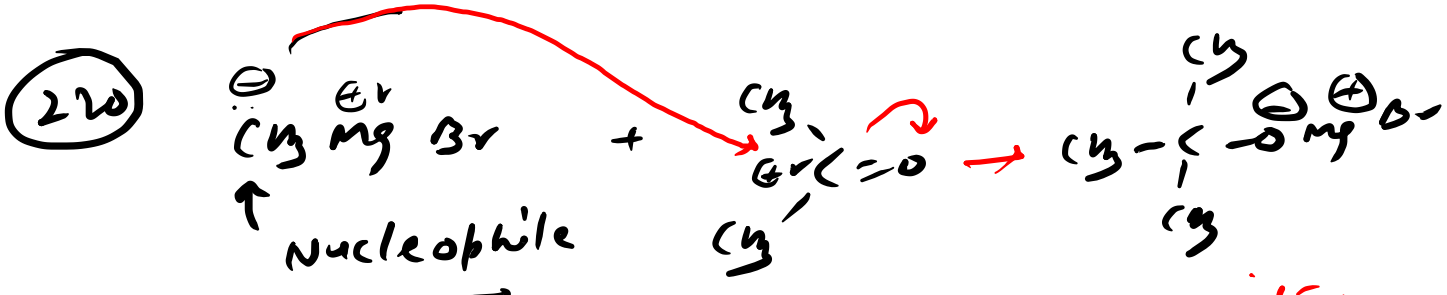
Then order of basicity & Nucleophilicity is same

> Basicity or Nucleophilicity $\propto \frac{1}{\text{stability}}$

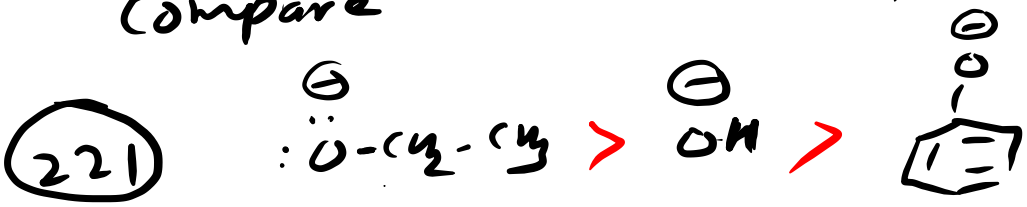
Basicity : The Tendency to donate the pair of electrons to $H^{\oplus}$ is called Basicity.

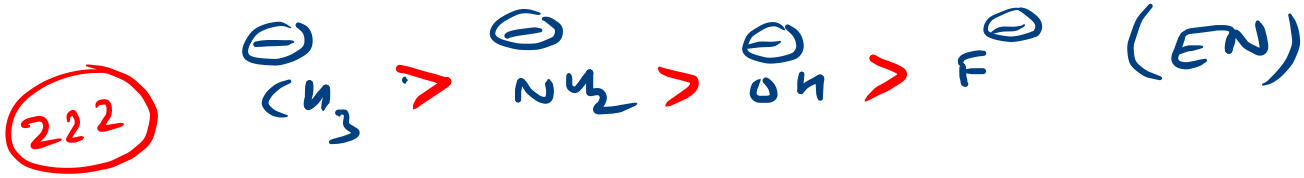
Nucleophilicity : The Tendency to donate the pair of electrons to other than $H^{\oplus}$ is called Nucleophilicity





Compare Nucleophilicity or basicity.





Case II If Nucleophilic atoms belongs to different period

Size ↑, solvation ↓, Nucleophilicity ↑

Compare Nucleophilicity in PPS

VVV 2nd

224 $F^{\ominus} < Cl^{\ominus} < Br^{\ominus} < I^{\ominus}$ (in PPS)

↑

More charge density = $\frac{\text{charge}}{\text{volume}}$ ∴ Effective solvation of $F^{\ominus}$

∴ less Nucleophilicity

VVV 2nd

225 $F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$ (in PAS)

226 $^{\ominus}OH < SH$ (in PPS)

Case III -ve charged $Nu^{\ominus}$ is more Nucleophilic than its corresponding Neutral Nu

227 $^{\ominus}OH > H-\ddot{O}-H$

228 $^{\ominus}OR > R-\ddot{O}-H$

229 $^{\ominus}SH > H-\ddot{S}-H$

230 $^{\ominus}SR > R-\ddot{S}-H$

231 $H_2N^{\ominus} > NH_3$

232 $R-NH^{\ominus} > R-NH_2$

Case IV Lone pair containing adjent atom increases Nucleophilicity

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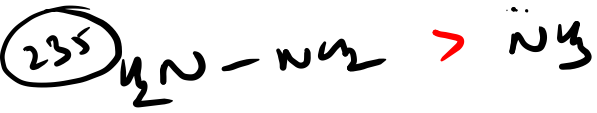
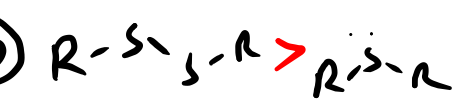
Compare Nucleophilicity

(233) (234)

(235) 

(236)

(237) 

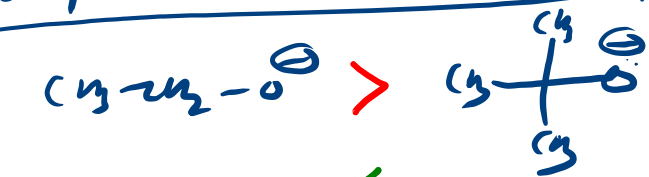
Nucleophilicity in PAS

Nucleophilicity $\propto \frac{1}{\text{Stability}}$

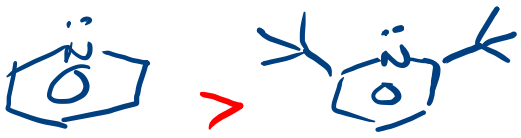
Note : Steric hindrance play important role to decide Nucleophilicity.

Compare Nucleophilicity

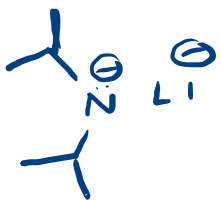
eg.

(236) 

(order of Nucleo.) < order of Basicity.

(237) 

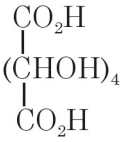
Note LDA (Lithium Di iso propyl Amide) acts as strong Base not Nucleophile.

(238) 

RACE # 18 ORGANIC CHEMISTRY

M.M. : 42 TIME : 30 Min.

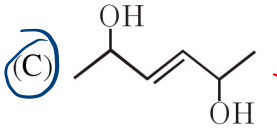
1. **Column-I** **Column-II** **[8, 0]**

(A) 

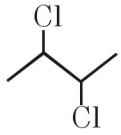
(P) Ten stereoisomers are possible out of these, two are optically inactive & eight are optically active.

(B)

(Q) Four stereoisomers are possible out of these, two are optically inactive & two are optically active.

(C) 

(R) Six stereoisomers are possible out of these, four are optically active & two are optically inactive.

(D) 

(S) Three stereoisomers are possible out of these, two are optically active and one is optically inactive.

(T) At least one chiral centre is present in any one of the stereoisomers of the compound.

2. **Statement-1 :** Me–$\overset{\oplus}{C}H_2$ is more stable than MeO–$\overset{\oplus}{C}H_2$ **[3, –1]**

Statement-2 : Me is a +I group where as MeO is a –I group.

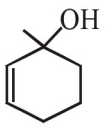
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

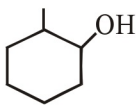
(C) Statement-1 is false, statement-2 is true.

(D) Statement-1 is true, statement-2 is false.

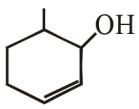
3. Find the correct order of rate of dehydration for given compounds with conc. H_2SO_4: **[3, –1]**



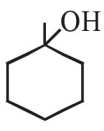
(I)



(II)



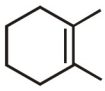
(III)

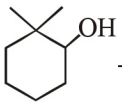
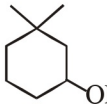
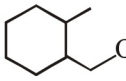
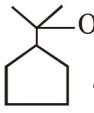


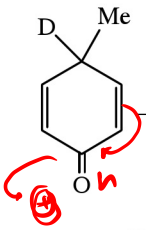
(IV)

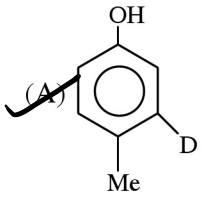
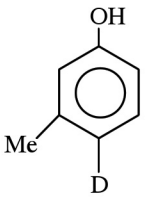
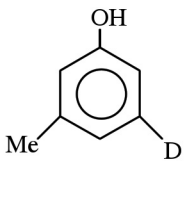
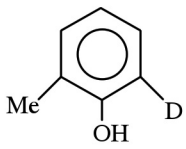
(A) I > II > III > IV (B) I > III > II > IV (C) I > III > IV > II (D) III > I > II > IV

4. Which of the following reaction will not produce given alkene as major product ? **[3, –1]**

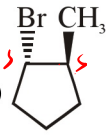
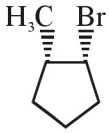
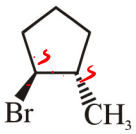


(A)  $\xrightarrow[\Delta]{H^+}$ (B)  $\xrightarrow[\Delta]{H^+}$ (C)  $\xrightarrow[\Delta]{H^+}$ (D)  $\xrightarrow[\Delta]{H^+}$

5.  $\xrightarrow[\Delta]{H^{\oplus}}$ Product is **[3, –1]**

(A) 
(B) 
(C) 
(D) 

6. Find out the correct statements about (I), (II), (III) & (IV)- **[4, –1]**

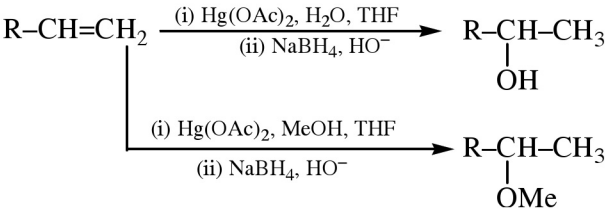
(I)
(II) 
(III) 
(IV) 

(A) I, III are diastereomer
(B) I & II are enantiomer
(C) II, III are identical
(D) II & IV are identical

Read the following paragraph and answer the questiosn 7 to 9 given below :

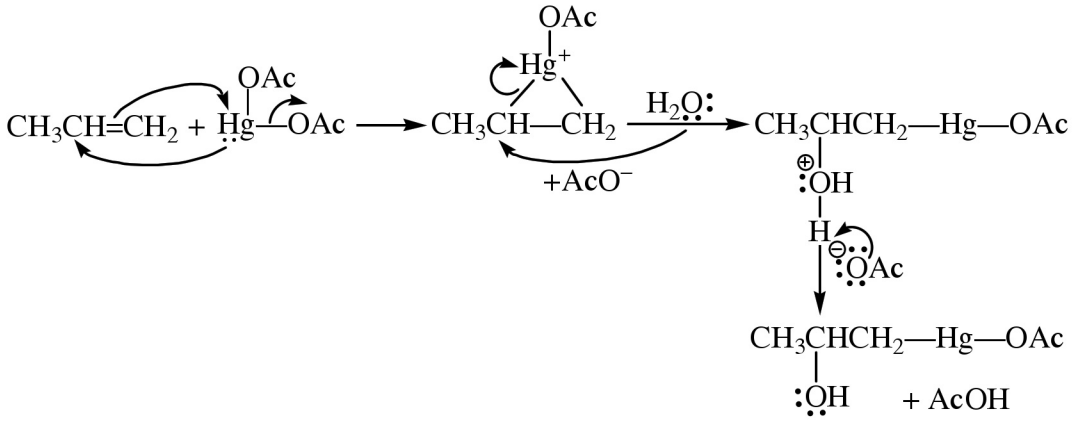
Oxymercuration-demercuration (OMDM) :

OMDM is a hydration process of alkene according to Markovnikov's rule with no rearrangement of cyclic mercuinium ion. In oxymercuration, the alkene is treated with mercuric acetate in aqueous tetrahydrofuran (THF). When reaction with that reagent is complete, sodium borohydride and hydroxide ion are added to the reaction mixture

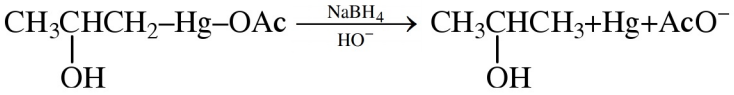


$AcO^- = CH_3COO^-$

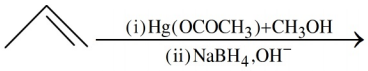
Mechanism for oxymercuration :

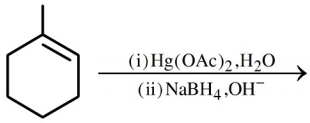


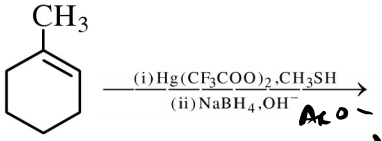
Sodium borohydride ($NaBH_4$) converts the carbon-mercury bond into a carbon-hydrogen bond. Because the reaction results in the loss of mercury, it is called demercuration.

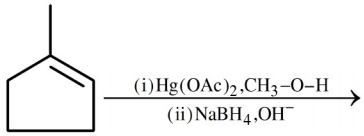


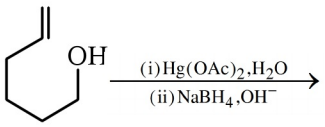
7. **[6, 0]**

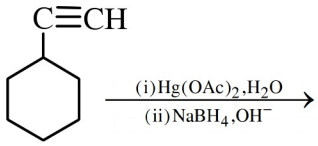
I. 

II. 

III. 

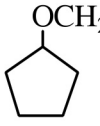
IV. 

V. 

VI. 

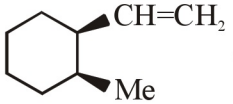
8. How could each of the following compounds be synthesized from an alkene by OMDM ?

(i)

(ii) 

(iii) $CH_3CCH_2CH_3$ (OH, CH_3)

(iv) $CH_3CCH_2CH_3$ (OCH_3, CH_3) **[4, 0]**

9. 

$\xrightarrow[NaBH_4]{Hg(OAc)_2/H_2O}$ Product mixture (P). **[4, 0]**

(a) Number of alcohol product formed in the above reaction.

(b) Number of stereoisomers theoretically possible for the major product

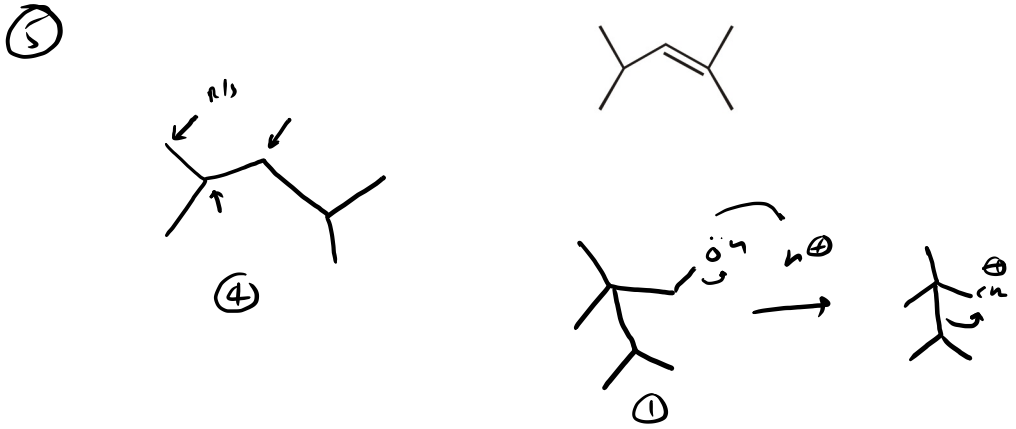
(c) Number of enantiomeric pairs formed in the product mixture

(d) Number of fractions obtained by fractional distillation of product mixture

Write answer of part (a), (b), (c) & (d) in the same order and present the four digit number as answer in OMR sheet. For example : If all these answer are 9 then fill 9999 in OMR sheet.

Ans. 2802

10. Number of alcohols which on H^+/Δ gives following alkene as a major product. **[4, 0]**



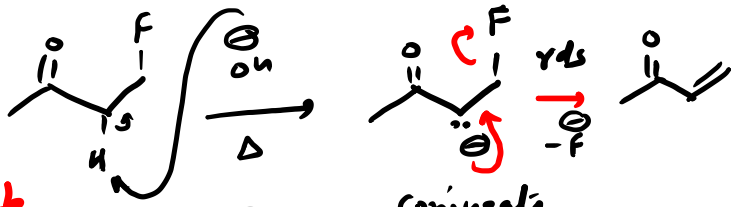
FILL THE ANSWER HERE

1.	A (P) (Q) (R) (S) (T) B (P) (Q) (R) (S) (T) C (P) (Q) (R) (S) (T) D (P) (Q) (R) (S) (T)	**2.**	(A) (B) (C) (D)	**3.**	(A) (B) (C) (D)	**4.**	(A) (B) (C) (D)
5.	(A) (B) (C) (D)	**6.**	(A) (B) (C) (D)				
7.	**(I)**	**(II)**		**(III)**		**(IV)**	
	(V)	**(VI)**					
8.	**(I)**	**(II)**		**(III)**		**(IV)**	
9.	(0) (1) (2) (3) (4) (5) (6) (7) (8) (9)			**10.**	(0) (1) (2) (3) (4) (5) (6) (7) (8) (9)		

E1cb (Elimination by conjugate base mech)

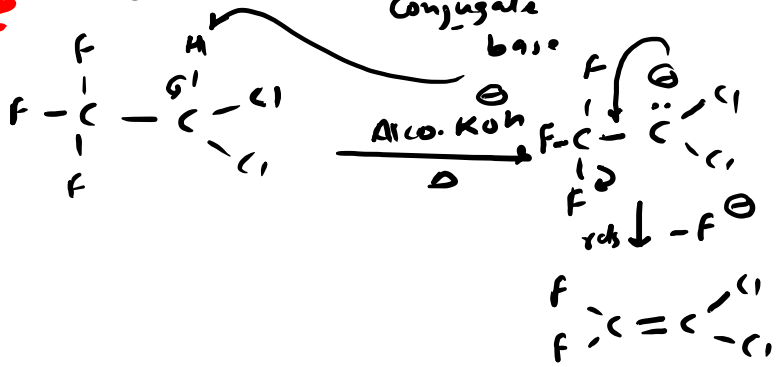
357 Pure E1cb Mech

Plg

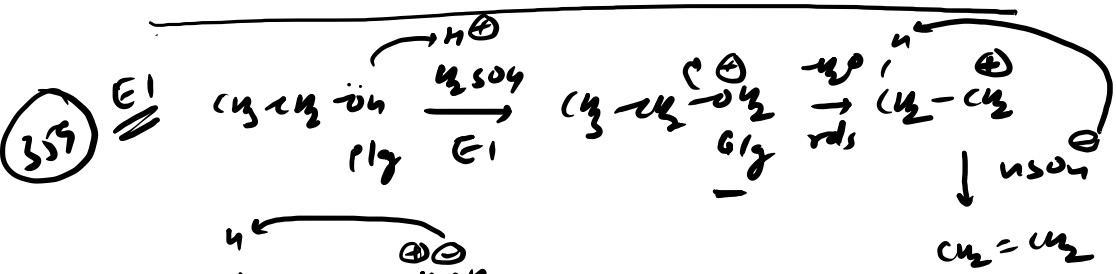


Pure E1cb

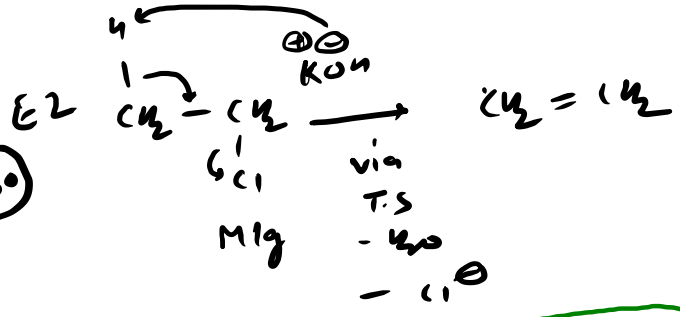
358



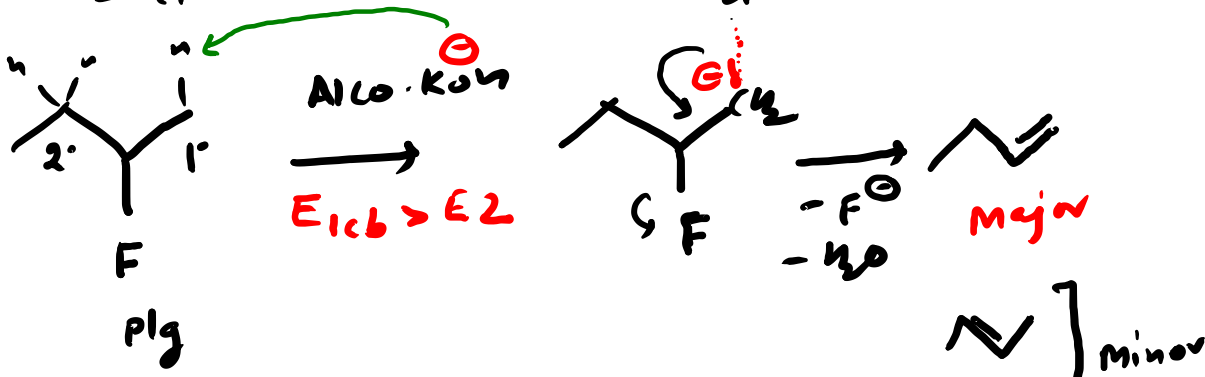
359 E1



360 E2



361



-ve on 1° C is more stable.

Characteristics of E1cb reaction

① Generally 9r is two step reaction

Step ① formation of carbanion (or removal of H⊕)

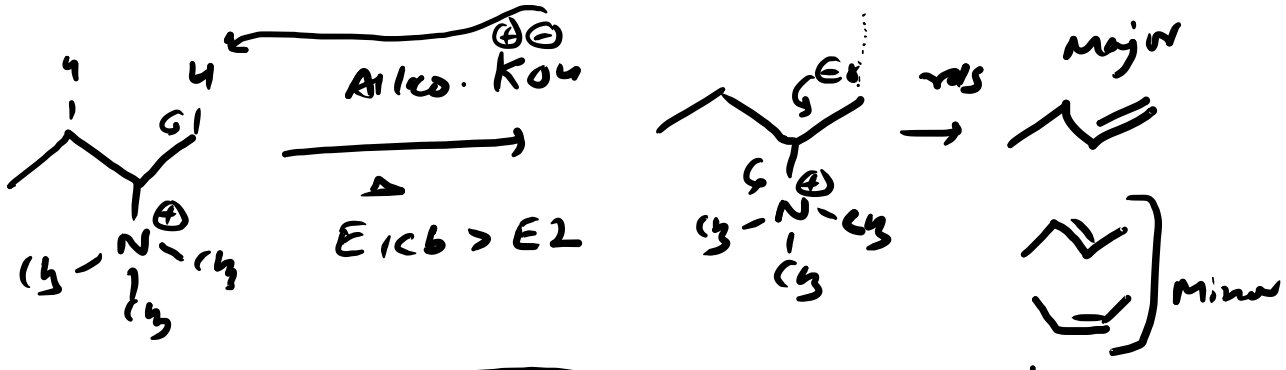
Step ② removal of lg, which is rds.

② Base remove that β-H which is more acidic for major product.

③ Generally Hoffman's product will be the major product.

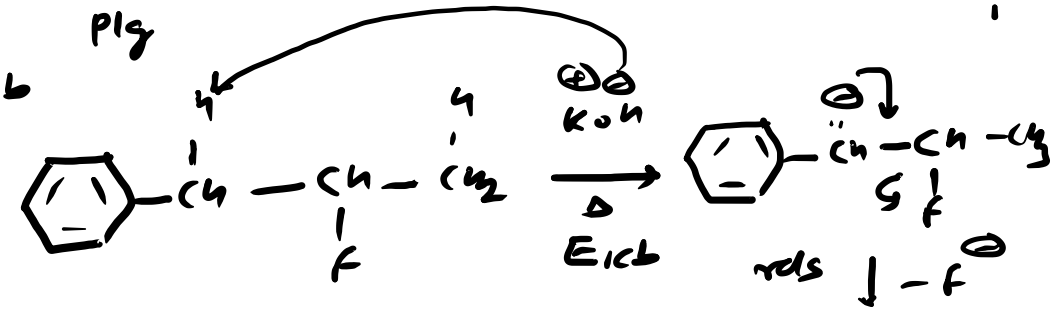
④ Plg (eg $-F$, $-\overset{\oplus}{N}R_3$, $-\overset{\oplus}{S}R_2$, $-OH$) favour E1cb.

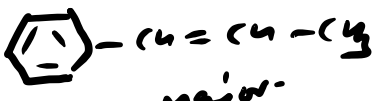
(362)



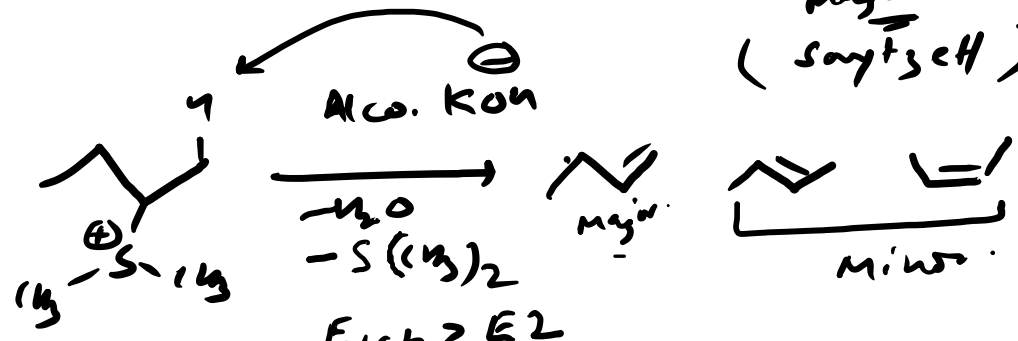
Pure E1cb

(363)



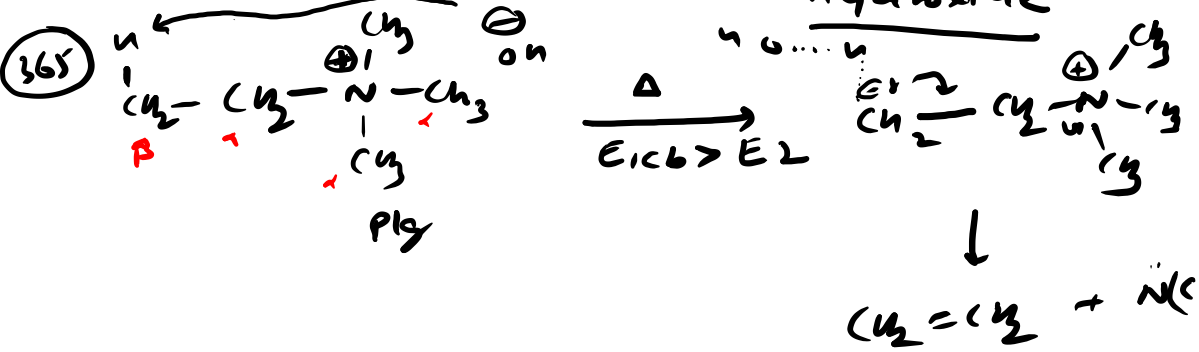


(364)



Heating of tetra alkyl ammonium Hydroxide

(365)

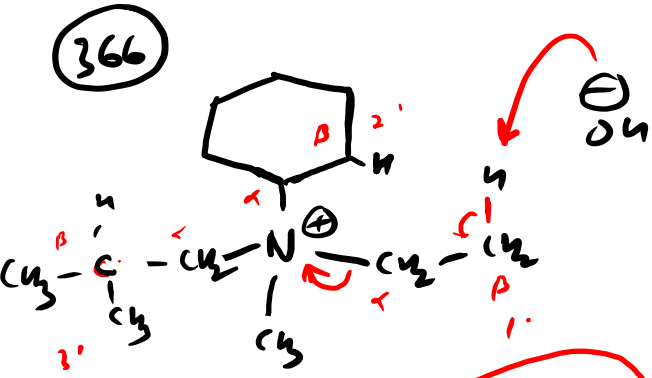


$\xrightarrow{\Delta}$ E1cb > E2

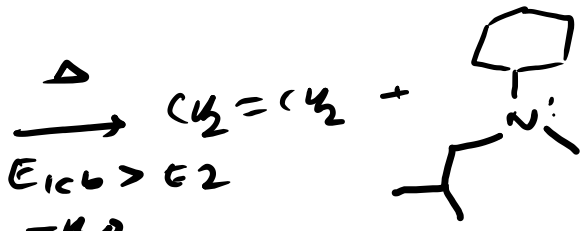
$\downarrow$

$CH_2=CH_2 + N(CH_3)_3$

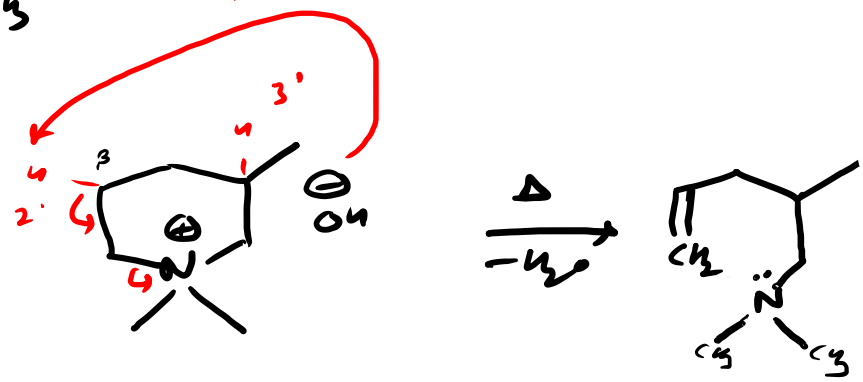
(366)



$\xrightarrow{\Delta}$ E1cb > E2, $-H_2O$



(367)



$\xrightarrow[-H_2O]{\Delta}$

Hoffman's Exhaustive methylation followed by Elimination

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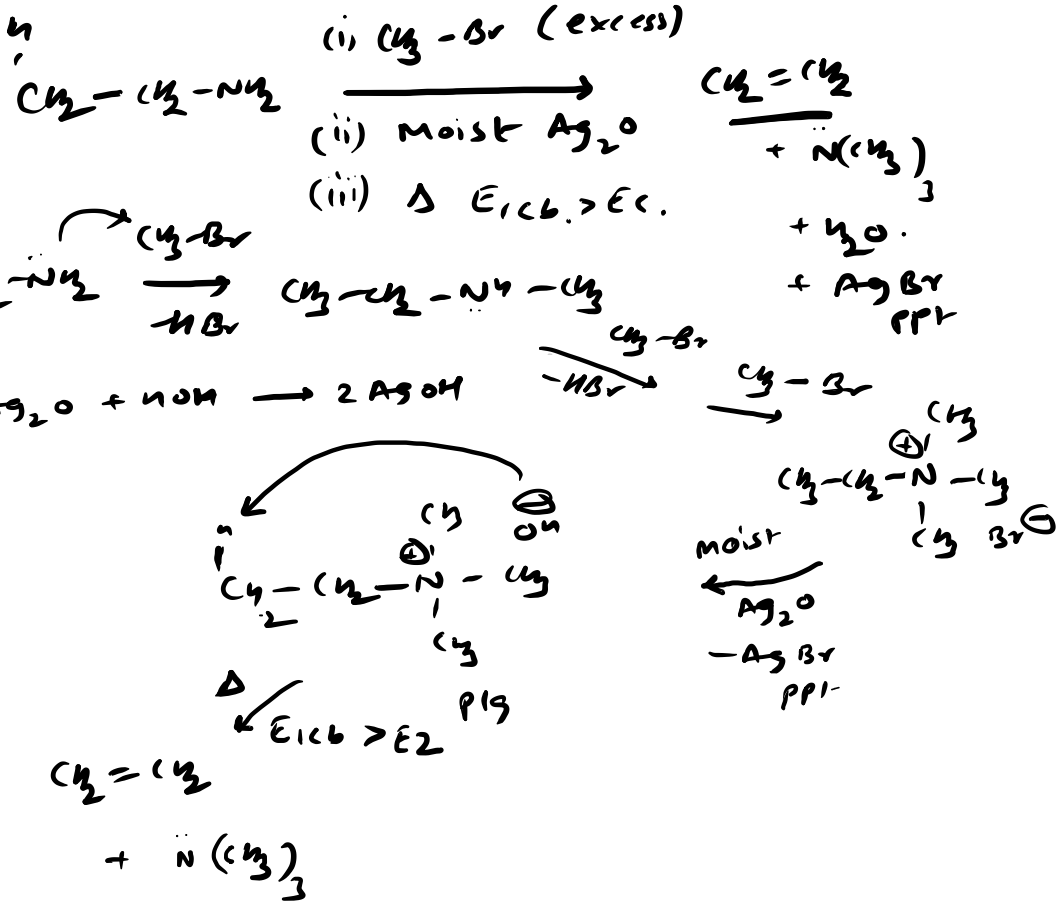
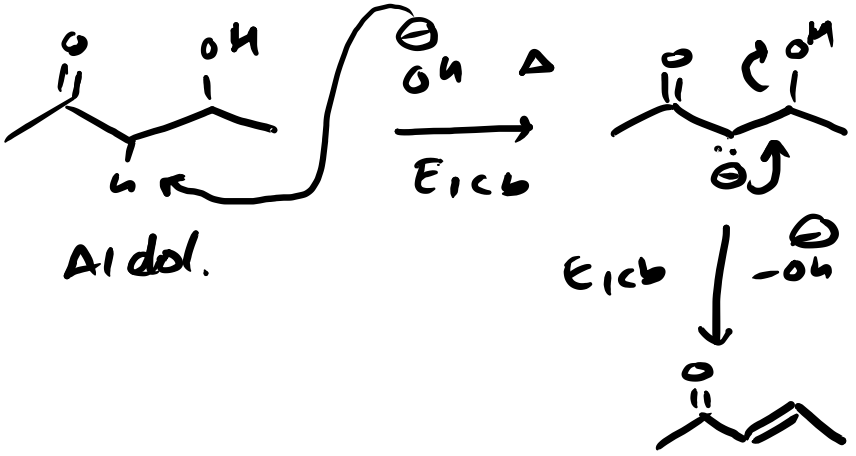
(368)



(369)

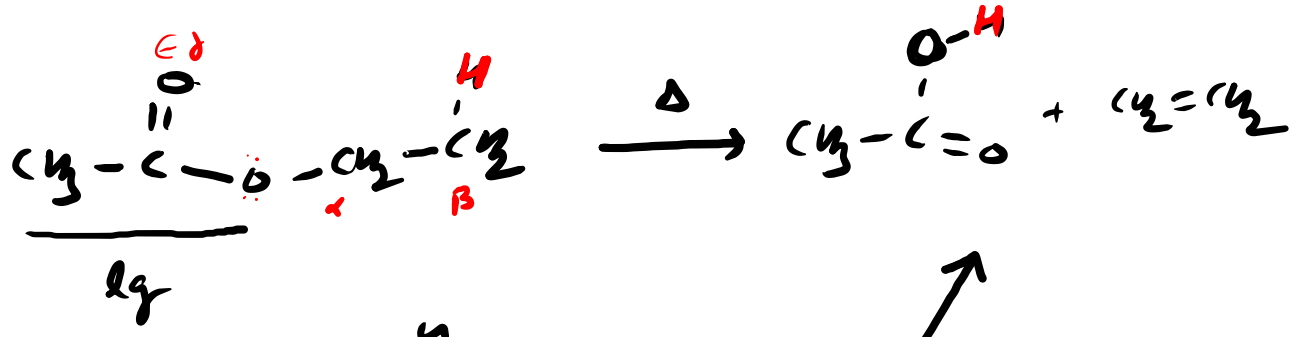


Ec/Ei (cyclic Elimination)
or (Elimination intramolecular)

* Pyrolysis of ester

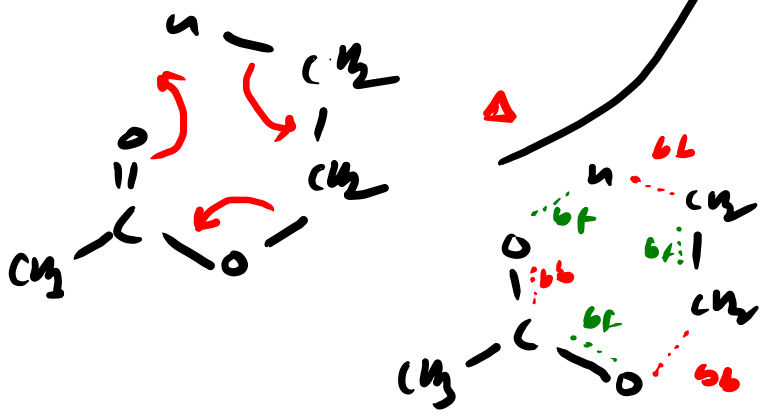
Py → temp
rolysis → to break

(370)



mech

pericyclic mech
⇑
mech having
cyclic T.S



6MCTS

characteristics of reaction

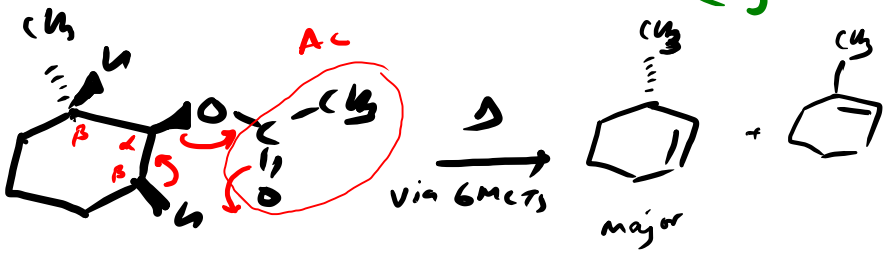
1. It is one step reaction proceed via T.S.
2. Compound having lg, β-H and base within the same molecule gives Ec on heating

③ Base remove less hindered β-H for major product.

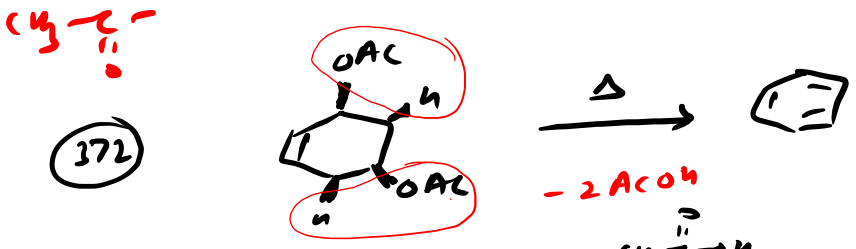
∴ ④ Generally Hoffman's product will be major product in Ec.

Ref: ⑤ Ec is the example of syn elimination (lg & β-H must be toward same side)

371



Ac = Acyl



372

- 2 $CH_3-\overset{O}{\overset{\|}{C}}-OH$

Addition

C A T

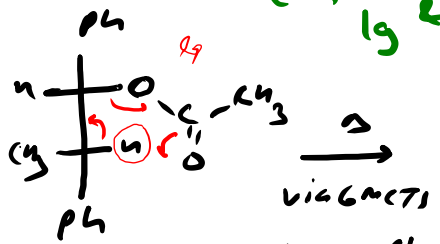
T A E

C S E

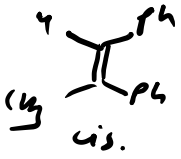
T S T

Elimination.

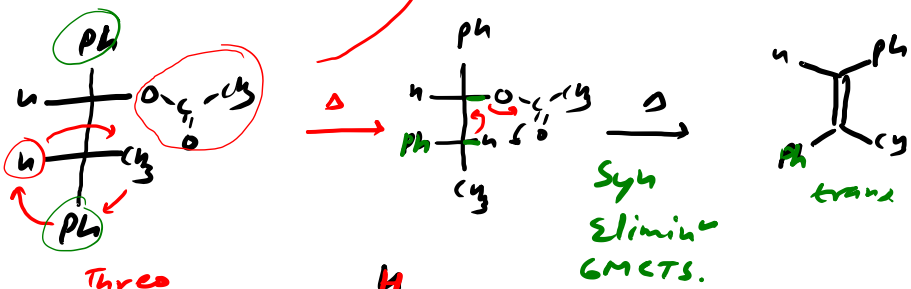
373



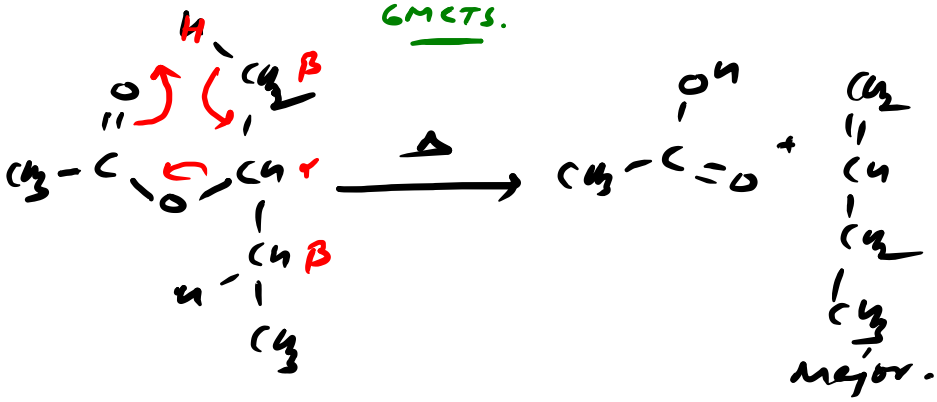
Emphasis lg & β-H

AcOH + 

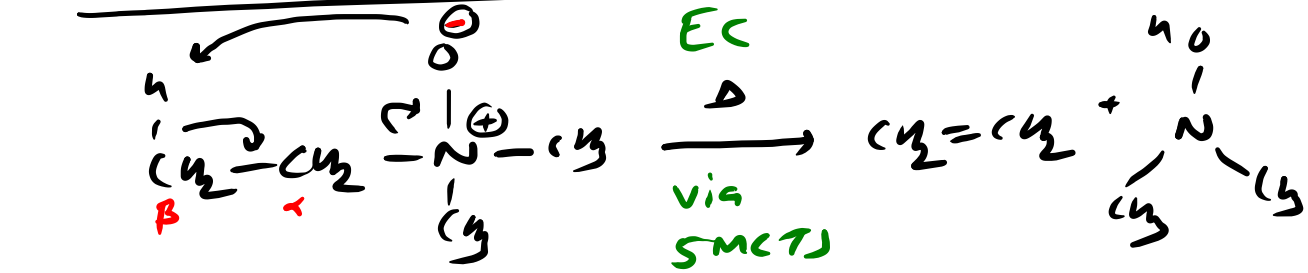
374

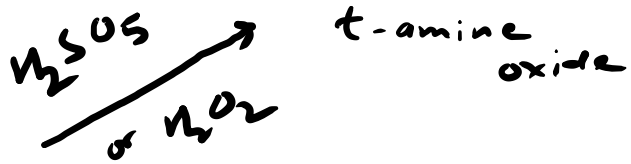


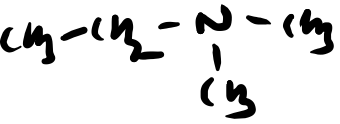
374 (A)



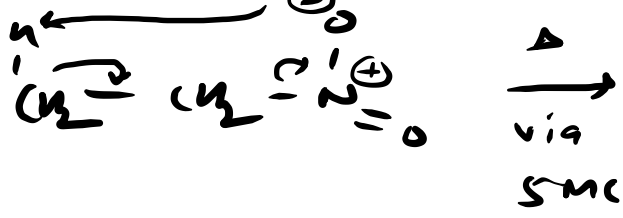
(375) **Pyrolysis of ter-amine oxide (Cope Elimination)**



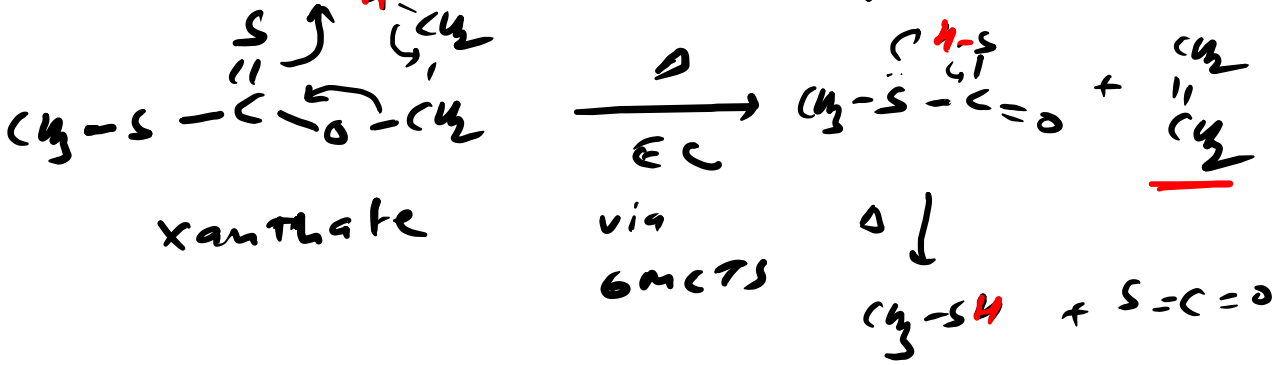




(376) **Pyrolysis of Nitro alkane**



(377) **Pyrolysis of xanthate (Chugav elimination reaction)**



FREE RADICAL

(378) **Stability order of FR**

$Ph_3\dot{C} > Ph_2\dot{C}H > Ph-\dot{C}H_2 > CH_2=CH-\dot{C}H_2 >$

$> CH_3-\dot{C}H-CH_3 > CH_3-\dot{C}H_2 > \dot{C}T_3 > \dot{C}D_3 > \dot{C}H_3 >$

$\dot{C}H=CH_2 >$ etc.

Generation of FR

(A) By hν or Δ

(379) $X-X \xrightarrow[\text{or } \Delta]{h\nu} X^{\cdot} + X^{\cdot}$

(380) $I-I \xrightarrow{\quad} 2\,I^{\cdot}$

(381) $Br-Br \xrightarrow{\quad} 2\,Br^{\cdot}$

(382) $Cl-Cl \xrightarrow{\quad} 2\,Cl^{\cdot}$

(383) $F-F \xrightarrow{\quad} 2\,F^{\cdot}$

(384) $H-O-O-H \xrightarrow[\text{or } \Delta]{h\nu} \dot{O}H + \dot{O}H$ (peroxide)

(385) $R-O-O-R \xrightarrow{\quad} \dot{O}R + \dot{O}R$

(386) $R-\overset{\overset{O}{\|}}{C}-O-O-\overset{\overset{O}{\|}}{C}-R \xrightarrow{\quad} R-\dot{C}(=O)-O^{\cdot}$

(387) $R-O-Cl \xrightarrow{\quad} R-\dot{O} + Cl^{\cdot}$

(388) $O=N-Cl \xrightarrow{\quad} \dot{N}=O + Cl^{\cdot}$

(B) Free Radical generate Rx

(388) $R-H + Cl^{\cdot} \longrightarrow R^{\cdot} + H-Cl$

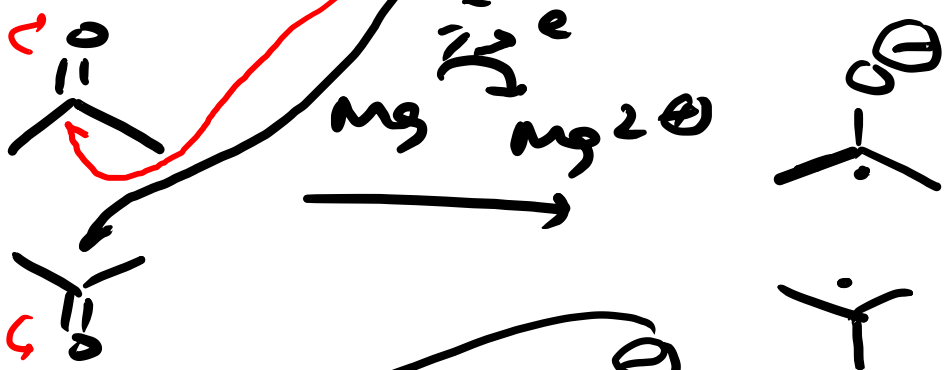
(390) $Cl-Cl + R^{\cdot} \longrightarrow R-Cl + Cl^{\cdot}$

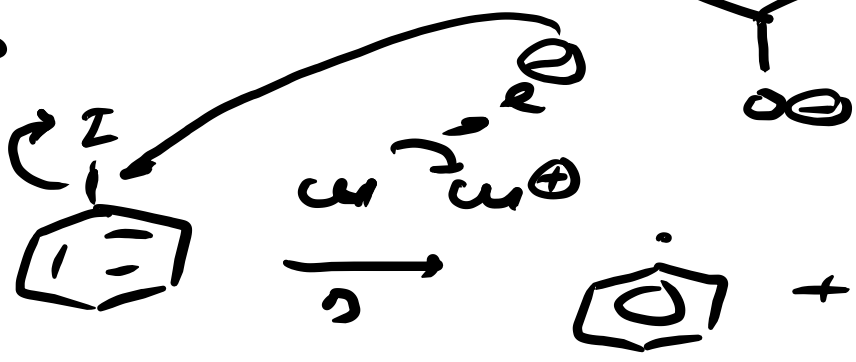
(391) $H-Br + \dot{O}H \longrightarrow Br^{\cdot} + H_2O$

(392) $H-I + \dot{O}R \longrightarrow I^{\cdot} + ROH$

(C) By metal

(393) $R-X \xrightarrow{Na \; Na^{\oplus}} R^{\cdot} + NaX$

(394) 

(395) 

Reactions of free Radical

(1) Photohalogenation of alkane

(396) $R-H + X_2 \xrightarrow{h\nu} R-X + H-Cl$

(397) $CH_3-H + Cl_2 \xrightarrow{h\nu} CH_3-Cl + H-Cl$.

(excess)

Mech (396) step I initiation step

$X-X \xrightarrow{h\nu} X^{\cdot} + X^{\cdot}$

step II propagation step

$R-H + X^{\cdot} \xrightarrow{rds} R^{\cdot} + H-X$

≃ 100 k.cal.

exo: H-F = 129 kcal, H-Cl = 103 kcal

endothermic: H-Br = 82 kcal, H-I = 74 kcal.

$R^{\cdot} + X-X \longrightarrow R-X + X^{\cdot}$

Termination step

$FR + FR \longrightarrow$ Neutral molecule.

$R^{\cdot} + R^{\cdot} \longrightarrow R-R$

$R^{\cdot} + X^{\cdot} \longrightarrow R-X$

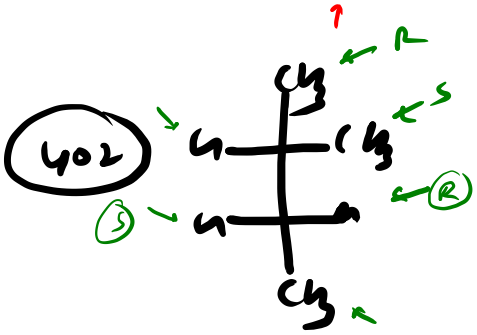
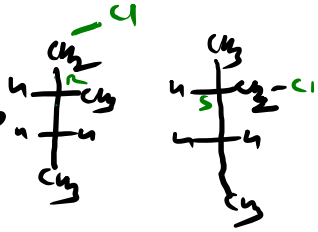
$X^{\cdot} + X^{\cdot} \longrightarrow X_2$

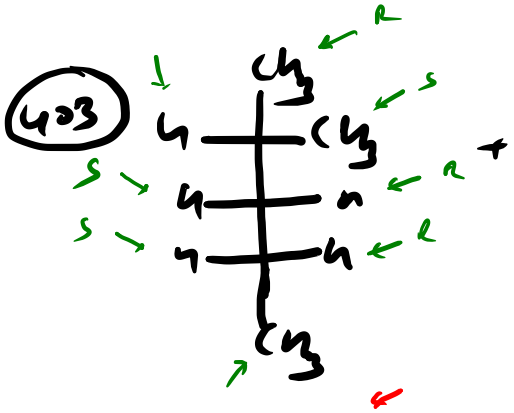
398 $CH_3-CH_3 + Cl_2 \xrightarrow{h\nu}$ mcr CH_3-CH_2-Cl (1)

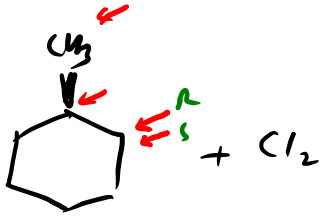
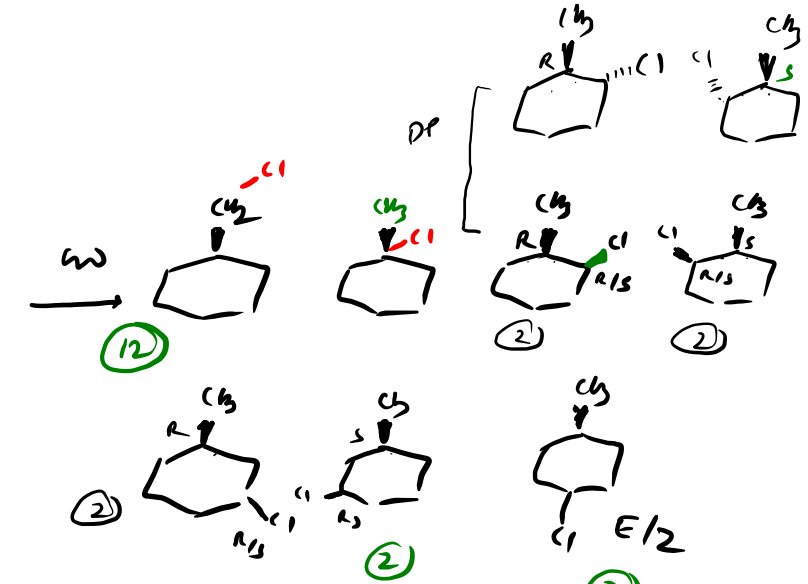
399 $CH_3-CH_2-CH_3 + Cl_2 \xrightarrow{h\nu}$ $CH_3-CH_2-CH_2Cl$ + (2)

400 $+ Cl_2 \xrightarrow{h\nu}$ mcr (3)

401 $+ Cl_2 \xrightarrow{h\nu}$ mcr (4)

402  $+ Cl_2 \xrightarrow{h\nu}$ mcp (6) 

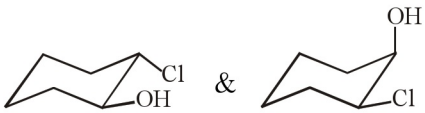
403  $+ Cl_2 \xrightarrow{h\nu}$ mcp (8)

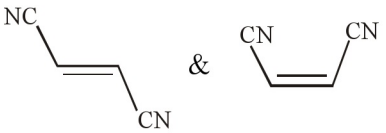
404  $+ Cl_2 \xrightarrow{h\nu}$ (12) 

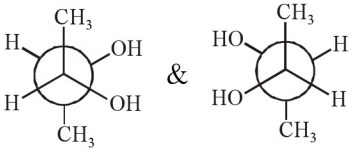
RACE # 21 — ORGANIC CHEMISTRY

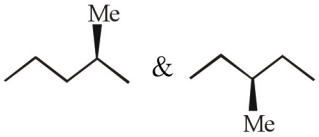
M.M. : 38 — TIME : 30 Min.

1. **Column - I (Compounds / Orientations)** **Column - I (Relation)** **[8, –0]**

(A)  (P) Position isomers

(B)  (Q) Diastereomers

(C)  (R) Geometrical isomers

(D)  (S) Enantiomers

(T) Stereoisomers

2. ***Assertion :*** Ethylidene chloride on treatment with aqueous KOH yield ethanal. **[3, –1]**

Reason : Ethylidene chloride is a vicinal dihalide.

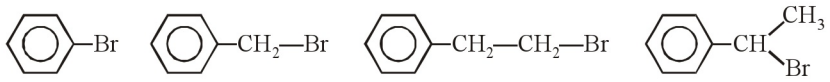
(A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(C) If Assertion is True but the Reason is False.

(D) If both Assertion & Reason are false.

3. Rate of S_N1 reaction is : **[3, –1]**



(P) (Q) (R) (S)

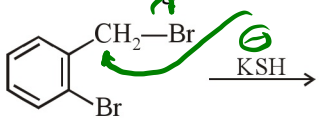
(A) S > Q > R > P (B) S > R > P > Q (C) P > Q > R > S (D) S > R > Q > P

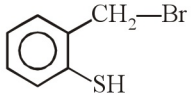
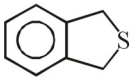
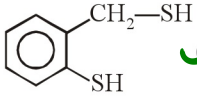
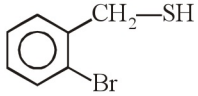
4. Which one of the following statement is correct about SN^1 reaction **[3, –1]**

(A) Perfect racemisation is observed (B) Only Walden inversion is observed

(C) Total retention of configuartion is observed (D) Polar protic solvent is preferred

5 Find the major product of the following reaction : **[3, –1]**



(A)  (B)  (C)  (D) 

6. Which of the following option is/are **not correct** regarding $CH_3S^{\ominus}$ and CH_3O^- : **[4, –1]**

(A) CH_3O^- is stronger base than $CH_3-S^{\ominus}$.

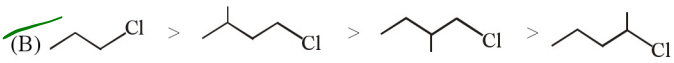
(B) $CH_3O^{\ominus}$ is stronger nucleophile than $CH_3S^{\ominus}$ (in H_2O).

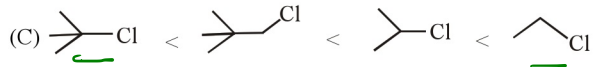
(C) CH_3O^- is weaker base than $CH_3S^{\ominus}$.

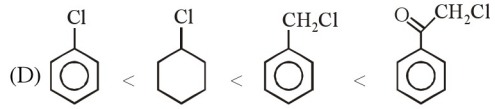
(D) CH_3O^- & $CH_3S^{\ominus}$ both give major elimination product when react with $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle Cl}{\underset{|}{\overset{|}{CH}}}}-Cl$

7. Correct order of rate of S_N2 reaction : **[4, –1]**

(A) $CH_3Cl > CH_3CH_2Cl > CH_3CH_2CH_2Cl > CH_3-\underset{\displaystyle Cl}{\overset{|}{CH}}-CH_3$

(B) 

(C) 

(D) 

Paragraph for Q.8 to Q.9

$Nu^- + Me\text{–}OTs \xrightarrow[25^\circ]{DMF} Me\text{–}Nu + TsO^-$ **[3, –1]**

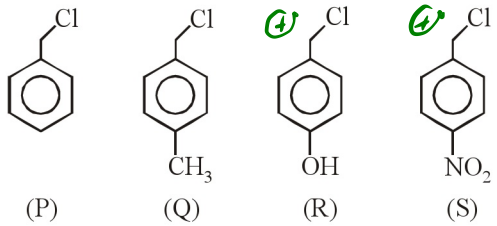
This is a S_N2 reaction where nucleophile attack Me–OTs in the rate determining step to give the product. Rate of this reaction increases with concentration as well as nucleophilicity of the nucleophile. Rate of S_{N^2} mechanism increases with increase in +δ on carbon of C–Lg bond.

8. $Nu^- + Me\text{–}OTs \xrightarrow[25^\circ]{DMF} Me\text{–}Nu + TsO^-$

(Nucleophile)		**(Relative rate)**	
(P)	F^-	(I)	3.25
(Q)	Cl^-	(II)	6.25
(R)	Br^-	(III)	1.0
(S)	I^-	(IV)	7.75

	P	**Q**	**R**	**S**
(A)	I	II	III	IV
(B)	IV	II	I	III
(C)	IV	I	II	III
(D)	IV	III	II	I

9. Correct order of S_N2 reaction is : **[3, –1]**



(P) (Q) (R) (S)

(A) S > P > Q > R (B) R > Q > P > S (C) Q > R > P > S (D) S > Q > P > R

10. (i) Cis - 2-Butene $\xrightarrow[CCl_4]{Br_2}$ X **[4, –0]**

(ii) Trans-cyclo octene $\xrightarrow[CCl_4]{Br_2}$ Y

(a) Number of organic products obtained in X

(b) Number of organic products obtained in Y

(c) Number of possible stereoisomer of product of X

(d) Number of possible stereoisomer of product of Y

Write answer of part (a), (b), (c) & (d) in the same order and present the four digit number as answer in OMR sheet. For example : If all these answer are 9 then fill 9999 in OMR sheet.

FILL THE ANSWER HERE

1.	A (P) (Q) (R) (S) (T) B (P) (Q) (R) (S) (T) C (P) (Q) (R) (S) (T) D (P) (Q) (R) (S) (T)	**2.**	(A) (B) (C) (D)	**3.**	(A) (B) (C) (D)	**4.**	(A) (B) (C) (D)
5.	(A) (B) (C) (D)	**6.**	(A) (B) (C) (D)	**7.**	(A) (B) (C) (D)	**8.**	(A) (B) (C) (D)
9.	(A) (B) (C) (D)	**10**	(0) (1) (2) (3) (4) (5) (6) (7) (8) (9)				